Beating Monte Carlo on rare circuit outputs: envelopes, proof-carrying sampling, and their limits

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Abstract

Given a Boolean circuit C and a product distribution D, how can one estimate a tiny $p = \Pr_D[C(x) = 1]$ faster than Monte Carlo (cost $\approx 1/p$) without ever being wrong? We seek procedures sound on every circuit and efficient on wide classes, alone or with one argument from an unbounded prover. Upper bounds need deduction, lower bounds need proposals, and sampling pays for the surprise that is left: an *envelope*, a tractable $\varphi \geq \mathbf{1}_C$ with computable mass, turns a deduction into an importance sampler whose cost is the looseness $K = \mathbb{E}_D\varphi/p$. Without a prover, a Decompose–Force–Sample–Branch procedure is sound everywhere and costs poly $\cdot K$ when its ideal explanation tree is polynomial: read-once formulas, read-once compositions of sub-circuits component-dense in their polarity, also under the literals they imply, propagation-visible triggers, small cutsets; envelope leaves and a meta-procedure add DNFs and DNNFs. Circuits whose threshold gates have polynomially bounded integer weights and whose $\wedge$-gates are bracketed with logarithmic *context width* have polynomial-size DNNFs, hence FPRASs (e.g. ORs of bounded-pathwidth CNFs, #P-hard to count). Further modules certify the presumption of independence, make annealing sound, tilt thresholds of overlapping local sub-circuits, and factor out affine structure; yet for events of polynomial density in the solution set of $m \leq n/6$ random parity checks, decomposable envelopes with small leaves cost, with high probability, at least the square root of Monte Carlo's price up to polynomial factors. With a prover, proof-carrying importance sampling over any Cook–Reckhow proof system is sound against every argument and complete when the rarity has a short proof of confinement to a tractable region, at cost poly$(|\pi|) + 2^{\sigma_u}$ for deductive content of size $|\pi|$ and unexplained surprise σ_u. Limits: no procedure beats sampling on all circuits unless NP $\subseteq$ BPP; no argument system is complete for upper bounds unless coNP $\subseteq$ MA; the prover strictly helps for upper bounds whenever slices of its proof system's canonical pair are not separable in randomized polynomial time (e.g. for Frege, if Blum integers cannot be factored by polynomial-size circuits); and such separations cannot rest on NP-hardness via randomized many-one reductions with gap at least $1 + 1/\text{poly}$. On ReLU networks trained to output a constant, fixed-level multilevel splitting, sound for lower bounds with no mixing assumption, certifies lower bounds of 0.865–0.973 times reference estimates of p at up to $d = 128$ inputs (references checked independently only down to $p \approx 6.3 \cdot 10^{-9}$), while for squared-loss networks with $p > 0$ at $d \geq 40$ no sound two-sided method we found beats Monte Carlo, for reasons that box, cube and decoupling barriers make precise.

Contents

1 Introduction **3**
1.1 The picture 3
1.2 Results 4
1.3 Related work 7

2 Setting **7**

3 Limits **9**
3.1 Without a prover: structure is necessary . . . 9
3.2 With a prover: upper bounds need deduction . . . 10
3.3 Interaction trivialises the problem . . . 11

4 Envelopes: the common mechanism **11**
4.1 Envelope circuits . . . 12
4.2 The envelope estimator . . . 13
4.3 Dense leaves . . . 14
4.4 Soft envelopes: Chernoff bounds as certificates . . . 15
4.5 Certified presumption of independence . . . 16

5 Without a prover: sound procedures and their classes **17**
5.1 Decompose–Force–Sample–Branch . . . 17
5.2 Envelope leaves and sound rewriting rules . . . 20
5.3 Context width: compiling any circuit into a DNNF . . . 21
5.4 Modules and quotients . . . 23
5.5 The presumption of independence, certified . . . 25
5.6 Certified telescoping with Las Vegas samplers . . . 28
5.7 Local threshold circuits . . . 31
5.8 Affine structure and linear thresholds . . . 37
5.9 Specialised modules . . . 45
5.10 What remains out of reach . . . 46

6 With a prover: proof-carrying importance sampling **47**
6.1 The argument language . . . 47
6.2 The two-sided workhorse: certified envelopes . . . 49
6.3 Affine arguments . . . 53
6.4 Lower bounds need no deduction . . . 55
6.5 Multilevel splitting needs no mixing for lower bounds . . . 56
6.6 Surprise accounting . . . 58
6.7 Gibbs envelopes . . . 59
6.8 How much stronger is the prover? . . . 60

7 Experiments **65**

8 Case study: networks trained to output a constant **70**
8.1 Certified evaluation . . . 71
8.2 The architecture is as hard as circuits . . . 73
8.3 What trained failure sets look like . . . 74
8.4 Sound lower bounds: splitting and betting . . . 75
8.5 Results for splitting . . . 77
8.6 Exact counts and certificates of zero: certified partitions . . . 78
8.7 Why upper bounds fail on trained squared-loss nets . . . 82
8.8 Regimes, heuristics, and what remains open . . . 86

9 Discussion and open problems **89**

A The second-order independence sandwich **91**

B Local threshold circuits: proofs **92**
B.1 The sampler . . . 92
B.2 The normaliser . . . 93
B.3 Anticoncentration . . . 94
B.4 Completeness and rarity . . . 95
B.5 The certification gap . . . 96

C Proofs for affine structure **96**
C.1 XOR-DNNF envelopes (Proposition 5.25) . . . 96
C.2 The parity barrier (Theorem 5.26) . . . 96
C.3 The corollaries . . . 98
C.4 Bounded width (Proposition 5.30) . . . 98
C.5 Biased parity events (Theorem 5.31) . . . 99
C.6 Linear thresholds . . . 99

1 Introduction

Let $C : \{0,1\}^n \to \{0,1\}$ be a Boolean circuit and D a product distribution on its inputs (Section 2). We want $p = \Pr_{x\sim D}[C(x) = 1]$ when p is tiny. Monte Carlo sampling finds it to relative error ε with $\Theta(\log(1/\delta)/(p\varepsilon^2))$ evaluations of C: polynomial in $1/p$, so exponential in the *surprise* $\log_2(1/p)$ of the event. This note asks two questions.

1. *Without help.* For which circuits can a polynomial-time procedure, one that may check some property of the circuit and then sample, estimate p to relative error at a cost polynomial in $\log(1/p)$ instead of $1/p$, while never being wrong (beyond probability δ) on any circuit?

2. *With a prover.* An unbounded prover sends one argument; a polynomial-time verifier checks it and samples. No argument may make the verifier accept a false claim with probability above δ, on any circuit. For which circuits do convincing arguments exist?

These are certified versions of the low-probability estimation problems studied by the Alignment Research Center [WH24, CNX22], where the aim is to estimate the probability of a rare behaviour of a model faster than by sampling, using the model's internals.

1.1 The picture

One idea organises almost everything: **upper bounds need deduction; lower bounds need proposals; sampling pays for whatever surprise is left**.

- A lower bound $p \geq L$ can be certified by exhibiting mass: a proposal distribution concentrated where $C = 1$, with computable likelihood ratios. Clipped importance weights make this sound for any proposal whose sampler and likelihoods the verifier derives from one object (Proposition 6.8).

- An upper bound below the sampling scale cannot be certified by evaluating C, however cleverly: under the uniform distribution (or any D whose collision probability $\sum_x D(x)^2$ is much smaller than $1/T$), a verifier that makes T evaluations cannot certify $p < 1/T$ (up to constants), whatever argument it is given (Theorem 3.2). It must read the circuit and *deduce* that $C(x) = 1$ confines x to a region of small, exactly computable measure.
- Combining the two, an *envelope*, a tractable function φ with $\varphi \geq \mathbf{1}_C$ whose mass $Z = \mathbb{E}_D\varphi$ is computable and from which one can sample, yields an importance sampler with bounded weights, hence two-sided intervals, after $O(K(\log(1/\delta) + \log\log K)\varepsilon^{-2})$ samples, where $K = Z/p$ (Theorem 4.3). The envelope explains $\log_2(1/Z)$ bits of the surprise; the remaining $\log_2 K$ bits are paid for by sampling, and this cannot be avoided: under the uniform distribution, once the envelope is certified, a verifier that accesses the rest of C only through evaluations needs about K of them on some circuits inside the envelope, whatever further argument it reads (Theorem 6.12(b)).

Karp–Luby–Madras estimation of DNF probabilities [KLM89], exact counting for decision-DNNFs and bounded treewidth, exponential tilting for large deviations [Sie76] and the Chernoff bound are all instances.

1.2 Results

Limits (Section 3). Beating sampling on all circuits is impossible: it would put NP in BPP, and with exponentially hard pseudorandom generators no procedure beats sampling by more than a $p^{-o(1)}$ factor even on explicit small circuits (Proposition 3.1). With a prover, no argument system is complete for upper bounds unless coNP $\subseteq$ MA (Theorem 3.4), and under the uniform distribution (or any D of small collision probability) a verifier that only evaluates C cannot certify upper bounds below about $1/T$ with T evaluations (Theorem 3.2). Interaction would trivialise the problem (sum-check gives exact p for every circuit), so the one-message model is the interesting one.

Without a prover (Section 5). The procedure DFSB (Decompose–Force–Sample–Branch) recursively splits the event by exact rules (independence of variable-disjoint parts, literals forced by propagation and Gaussian elimination, Shannon expansion) and samples only at leaves that are dense or have tight envelopes. It is sound on every circuit, and complete on every circuit whose ideal K-explanation tree has m nodes, with at most d sampled leaves whose estimates are multiplied together, using $\tilde{O}(m\,K\,d^2\varepsilon^{-2}\log(m/\delta))$ evaluations besides its pilots (Theorem 5.2). This covers read-once formulas (exactly); read-once compositions of non-constant sub-circuits each of which, in the polarity in which it occurs and after restriction by any literals it implies, is *component-dense*: unit propagation and independence split it into exact or K-dense pieces (Corollary 5.3); triggers that propagation or Gaussian elimination exposes; circuits with small cutsets (exactly, by cutset conditioning); and versions of all of these disguised by rewriting cones of logarithmic support, which are computed exactly. DNFs and DNNFs, whose ideal trees can be exponential, are covered instead by envelope leaves built from their own envelopes (K at most the number of terms for a DNF), and by the #DNNF FPRAS, which the meta-procedure runs (Proposition 5.4 and Theorem 5.36). Beyond DFSB:

- *Context width* (Theorem 5.6): every circuit whose NNF N has no opaque gates compiles into an equivalent DNNF of size $O(|N|(\mathrm{cw}+1)2^{\mathrm{cw}})$ ($|N| = O(|C|)$ for $\{\wedge, \vee, \neg, \oplus\}$-circuits), where cw measures, for the given bracketing of the $\wedge$-gates, how many variables a gate must

be conditioned on to make every $\wedge$ decomposable; with the #DNNF FPRAS [MdC26] this gives an FPRAS whenever $\mathrm{cw} = O(\log |N|)$, e.g. for ORs of bounded-pathwidth CNFs, each bracketed along a path decomposition (Lemma 5.7), whose exact counting is #P-hard.

- *Modules* (Lemma 5.9): functions of independent sub-circuits (thresholds, majorities) are estimated through their quotients, walking the module tree. The log-ratio of the final interval is at most the sum S of those of the quotient estimates, in the polarities the parents use, when each parent is monotone or antitone in each input or the interval for that input's used polarity lies in $[0, \frac{1}{2}]$, and at most $e^{O(dS)}S$ in general, for $dS = O(1)$ and d the depth of the module tree; so small relative errors in the used polarities suffice.

- *The presumption of independence, certified* (Theorems 4.7 and 5.10): for unate $\wedge$'s of arbitrary sub-circuits A_i, Harris and Janson inequalities give $M \le p \le Me^{\Delta/(1-\max_i b_i)}$ for the independence estimate $M = \prod_i D(A_i)$ ($b_i = D(\neg A_i)$, Δ the probability carried by dependent pairs). For unate CNFs both ends are computed exactly. In general a sampling procedure certifies p to ratio $1+\varepsilon$ whenever $\Delta \le \ln(1+\varepsilon)/64$ and $\max_i b_i \le \frac{1}{4}$, with samples linear in $\sum_i b_i \le \log(1/p) + \Delta/(1-\max_i b_i)$ plus $O(\varepsilon^{-1}\log(1/\delta))$ per dependent pair, independent of $1/p$. A covariance correction reduces the error to third order, and pairwise data provably cannot do better than $e^{0.29\Delta}$ (Proposition 5.11).

- *Certified telescoping* (Theorem 5.14): annealing estimators become sound when the intermediate samplers are exact Las Vegas samplers (coupling from the past) whose truncation is certified empirically; this handles unate circuits with strong, extensive correlations whose bounding chains coalesce quickly.

- *Local thresholds* (Section 5.7): for $C = [\sum_j g_j(x_{A_j}) \ge t]$ with arbitrary sub-circuits g_j of bounded arity and degree, whose scopes may overlap like an expander, exponential tilting with a heat-bath sampler whose mixing is certified by a locally checked Dobrushin condition, and a normaliser certified by thermodynamic integration, is sound on every input (Theorem 5.18); if the marginals are bounded away from 0 and 1 and a constant fraction of the g_j have private pivotal inputs, it is complete for $m \ge m_0$ with $\mathrm{poly}(n)/\varepsilon^2$ samples (constants depending on the arity, the degree and these bounds) at every threshold up to the mean under the crude certified tilt θ_0, a range that contains thresholds with $p = e^{-\Omega(m)}$. For supermodular statistics coupling from the past removes the certificate (Proposition 5.20).

- *Affine structure* (Section 5.8): certified affine gates and an exact affine and triangular presolve (Proposition 5.23) factor out the parity-check part of the surprise exactly, in front of any sound estimator, when the pivots can be chosen on uniform coordinates (a pivot of bias $|q_i - \frac{1}{2}| \le \eta$ costs a factor up to $(1+2\eta)/(1-2\eta)$ in samples), and unions of affine subspaces give envelopes (Theorem 5.24). Parity checks are also where decomposition fails: for $m \le n/6$ random checks and events that fill a $1/\mathrm{poly}(n)$ fraction of their solution set, with high probability envelopes with small leaves, context-width compilation and DFSB's completeness guarantee cost Monte Carlo's price up to polynomial factors, and any use of a DNNF envelope at least its square root (Theorem 5.26 and Corollary 5.29). Under biased product distributions, small $\wedge$'s of parity checks on N inputs admit no sound procedure complete within a factor $2^{N^{1-\gamma}}$ unless $\mathsf{NP} \subseteq \mathsf{BPP}$, and no argument system complete within a factor 2 unless $\mathsf{coNP} \subseteq \mathsf{MA}$ (Theorem 5.31).

- Bounded treewidth (exact), thresholds $[S \ge t]$ of k unit-weight independent blocks (exponential tilting with $K \le \frac{4}{3}(2\sqrt{k}+1)$ for $\mathbb{E}_D S \le t < k_+$, k_+ the number of blocks that can

be 1, Proposition 4.6), and the known FPRAS classes [Moi19, GKM10].

With a prover (Section 6). Proof-carrying importance sampling: the argument is a DAG of deductive steps checked by any Cook–Reckhow proof system (implications, exclusivity, independence) and sampling steps with declared confidence budgets. It is sound on every circuit and every argument (Theorem 6.2), and complete on every circuit that has a short proof of being confined to a tractable region in which it is not rare (Theorem 6.5). Lower bounds need no proof at all. XOR-propagation-tree arguments add parity reasoning by gate propagation and Gaussian elimination, with affine pieces under product distributions (Theorem 6.7), and Gibbs envelopes extend the soft envelopes to statistics with overlapping scopes, with the mixing certificate recomputed by the verifier (Proposition 6.14). The verifier's cost is $\mathrm{poly}(|\pi|) + 2^{\sigma_u}$ (the second term times $\varepsilon^{-2}(\log(1/\delta)+\log(1+\sigma_u))$ and the cost of a sample), where $|\pi|$ is the size of the argument's deductive part (envelopes, proofs, circuits, but not its unary sample caps) and σ_u is the surprise it leaves unexplained. Under the uniform distribution both terms have a sharp meaning for verifiers that use (part of) C only through evaluations: in the worst case (planted subcubes), each bit of argument reduces the number of evaluations needed for a lower bound by at most a factor 2, and once an envelope is certified, a verifier that accesses the rest of C only through evaluations needs about 2^{σ_u} of them on some circuits for a two-sided interval, whatever further argument it reads, which the envelope estimator achieves (Theorem 6.12). The prover helps strictly: for lower bounds unless $\mathsf{NP} \subseteq \mathsf{BPP}$; for upper bounds on circuits with short refutations in the verifier's proof system P (and, for resolution, cutting planes and Frege-type systems, on padded versions with exactly certified $p > 0$) whenever the corresponding slice of the canonical pair of P is not separable in randomized polynomial time (Proposition 6.17), which holds for TC^0-Frege, Frege and extended Frege if Blum integers cannot be factored by polynomial-size circuits (Corollary 6.18); and two-sidedly, for a verifier that checks proofs in a theory such as ZFC, if factoring is not in BPP. Whether weak automatability of P removes the prover's advantage on general envelope classes is open. Upper-bound and two-sided separations cannot be derived from NP-hardness by randomized many-one reductions to gap problems with gap at least $1+1/\mathrm{poly}$ unless $\mathsf{coNP} \subseteq \mathsf{AM}$.

Trained networks (Section 8). On ReLU networks trained to output a constant on uniform inputs $x \in \{-1,+1\}^d$, the rare event being a failure, fixed-level multilevel splitting gives lower bounds that are sound with no assumption on mixing (Theorem 6.10), and a betting e-value turns its replicates into a confidence bound without clipping (Proposition 8.3): it certifies 0.920–0.942 of the exact p on the suite nets with $d \le 28$ and $p \ge 10^{-5}$, and 0.865–0.973 of a reference value p_{ref} at $d = 32$–128, for p_{ref} down to $5.2 \cdot 10^{-21}$, with $4.1 \cdot 10^6$–$5.7 \cdot 10^7$ evaluations. The reference was checked independently on 8 of these 15 nets (exactly, or by Monte Carlo down to $p \approx 6.3 \cdot 10^{-9}$); on the others it is the mean of the same estimator and could miss a mode that the chains never reach. Certified partitions give exact values at $d \lesssim 30$ and certificates of $p = 0$ for long-trained nets. For squared-loss networks with $p > 0$ at $d \ge 40$ no sound two-sided method we found beats Monte Carlo: the architecture is as hard as circuits (Proposition 8.2), and on the trained nets failures are joint small-ball events at the centre of the box, where box, cube and decoupling barriers (Propositions 8.6 to 8.8) block every deduction we could make cheap.

Experiments (Section 7). A Python implementation (`rareprob`) computes, e.g., $p \approx 4 \cdot 10^{-25}$ for an 80-bit trigger exactly, with no samples, and $p \approx 2.7 \cdot 10^{-25}$ for an 80-bit trigger on a dense, non-decomposable 20-variable body to ratio 1.03 with $1.8 \cdot 10^5$ evaluations; it estimates $p \approx 4 \cdot 10^{-46}$ for an OR of eight window-CNFs on 800 variables to 0.4% from 1,024 samples (by

context-width compilation), and certifies $p \in [6.18, 7.35] \cdot 10^{-22}$ for a unate CNF with $2 \cdot 10^5$ clauses deterministically and $p \in 10^{[-20.31,-20.08]}$ (exact $10^{-20.19}$) for a chain of 1600 overlapping random local detectors with about $7 \cdot 10^4$ Glauber samples, where Monte Carlo would need about $2 \cdot 10^{22}$. When d dense leaves must be sampled and multiplied, the cost grows like d^3, and Monte Carlo is cheaper on $\wedge$'s of up to three dense 16-variable blocks. With SAT-solver proofs as arguments the verifier certifies a 20-bit SPN-hidden trigger on which Part 1, which sees such tests only by tabulating small cones and branching, gives no answer within 900 seconds. In 2,250 random and adversarial trials, 750 of them with dishonest arguments, no interval missed p.

1.3 Related work

Approximate counting. Karp–Luby–Madras [KLM89] and the stopping rule of [DKLR00] are the prototypes; for self-reducible classes, approximate counting and approximate sampling are equivalent [JVV86], and Karger's unreliability FPRAS [Kar99] is an early instance of a combinatorial theorem (few near-minimum cuts) serving as the envelope. FPRASs exist for DNNF [MdC26] (and before it nFBDD [MdC25] and NFA [ACJR21]), for CNFs in the local-lemma regime [Moi19, FGYZ21, FHY21, JPV21, HWY22, WY24], and for knapsack [GKM+11, GKM10]; d-DNNF, OBDD, SDD and bounded treewidth admit exact counting [DM02, Dar11, SS10]. Hashing-based counters [CMV13] use an NP oracle; partial compilation with sampling [LMY23] and AND/OR importance sampling [GD08] combine decomposition with sampling for variance reduction, without relative-error guarantees for rare events. Certified exact model counting [BNAH23] checks compilation by proofs. Affine decision trees [KLMT13] and the elimination of defined variables [LLM16] use parity and functional structure in exact counting, and tilting with local central limit theorems counts independent sets of a given size [DP21, JPSS21]. *Rare-event simulation* [AG07, Buc04] studies importance sampling with bounded relative error; the soundness requirement here (valid intervals on every input) is the computer-science twist. *Interactive proofs* certify exact counts [LFKN92] and lower bounds [GS86]; the Merlin–Arthur setting connects to proof complexity [CR79, KP98, BPR00, Pud03] and to the class SBP [BGM06]. *Heuristic arguments.* ARC's programme [CNX22, Hil24, WH24] seeks estimators that use a model's internals to beat sampling, judged by "surprise accounting"; this note isolates the part of that programme that can be made sound, and shows where soundness forces deduction. ARC's "matching sampling principle" [NLW+25] asks for estimators that do at least as well as sampling (on average over a distribution of circuits, without soundness), and its "computational no-coincidence principle" [Ney25] conjectures that a polynomial-time verifier can check short advice certifying a certain property of reversible circuits whenever the property holds, while accepting no advice for 99% of random reversible circuits, which almost never have it; our Part 2 is a worst-case, statistically sound relative of the latter, for probabilities instead of properties.

2 Setting

Circuits. A circuit C on inputs $x_1, \ldots, x_n$ is a directed acyclic graph whose leaves are the inputs and the constants 0 and 1, and whose internal nodes (gates) compute $\wedge$, $\vee$ (any fan-in), $\neg$, $\oplus$ or a threshold function $[\sum_j w_j y_j \geq t]$ of their inputs; one gate is the output. The weights w_j and the threshold t are rationals given in binary (a floating-point number stands for its exact binary value), and the comparison is exact, with no tolerance (our implementation evaluates threshold gates the same way). $|C|$ is the number of wires. For a gate g, $\mathrm{var}(g) \subseteq [n]$ is the set of input variables in its cone (its *support*; always the syntactic one). Pushing negations to the inputs

gives the negation normal form (NNF) of C: an $\{\wedge,\vee\}$-circuit over literals and constants. For $\{\wedge,\vee,\neg\}$-circuits it has at most two copies of every gate, one per polarity; a $\oplus$-gate of fan-in k expands to $O(k)$ gates per polarity, a threshold gate whose weights, scaled to integers, have polynomially bounded total $W=\sum_j|w_j|$ expands by a sequential counter with $O(kW)$ wires (a negative weight acts on the complemented input), and other gates stay opaque. An $\wedge$-gate of an NNF is *decomposable* if its children have pairwise disjoint supports, and an $\vee$-gate is *deterministic* if its children are pairwise inconsistent; an NNF all of whose $\wedge$-gates are decomposable is a DNNF, and a d-DNNF if moreover all its $\vee$-gates are deterministic [DM02].

Distributions. D is a product distribution on $\{0,1\}^n$ with rational marginals $\Pr[x_i=1]=q_i$, again given in binary (a floating-point marginal stands for its exact value). The *input size* of (C,D) is $|C|$ plus the total bit length of the q_i and of the weights and thresholds of C; running times that mention only $|C|$ count arithmetic operations on such numbers. Product structure is what makes "disjoint supports" mean "independent". A distribution given by a sampler circuit $S:\{0,1\}^m\to\{0,1\}^n$ reduces to the uniform case: $\Pr_{x\sim S(U_m)}[C(x)=1]=\Pr_{u\sim U_m}[(C\circ S)(u)=1]$. (The reduction can destroy structure; we do not rely on it.) We write

$$p \;=\; p_D(C) \;=\; \Pr_{x\sim D}[C(x)=1]$$

and think of p as tiny: 2^{-50}, $n^{-\omega(1)}$, or $2^{-\Omega(n)}$. For a set or function E we write $D(E)=\mathbb{E}_D[E]$.

What counts as an answer. Because p can be arbitrarily small, additive error is useless; we ask for relative error. To let a procedure decline, we let it output intervals.

Definition 2.1 (sound interval estimation)**.** An *estimation procedure* gets (C,D,ε,δ) and outputs an interval $[L,U]\subseteq[0,1]$ (possibly the trivial $[0,1]$, meaning "no verdict"). It is *sound* if for every circuit C and every D,

$$\Pr\left[p_D(C)\notin[L,U]\right]\le\delta.$$

It is *complete* on a class $\mathcal{C}$ if for all $C\in\mathcal{C}$, with probability at least $1-\delta$ the interval is tight: $U\le(1+\varepsilon)L$. It is *efficient* if it runs in time polynomial in the input size, $1/\varepsilon$ and $\log(1/\delta)$.

A sound procedure that is complete on $\mathcal{C}$ gives, on $\mathcal{C}$, a $(1\pm\varepsilon)$-approximation of p with probability $1-2\delta$; outside $\mathcal{C}$ its output is still a valid (if loose) interval. Soundness everywhere is what makes the "check a property, then sample" procedures of the problem statement safe to use on circuits we know nothing about.

The baseline. Monte Carlo draws $x^{(1)},\dots,x^{(N)}\sim D$ and reports the fraction of ones. A relative-error interval needs $N=\Theta\big(\log(1/\delta)/(p\varepsilon^2)\big)$ samples, so its cost is

$$\mathrm{cost}_{\mathrm{MC}}=\Theta\Big(\frac{|C|\log(1/\delta)}{p\,\varepsilon^2}\Big),$$

exponential in the *surprise* $\log_2(1/p)$. The Dagum–Karp–Luby–Ross stopping rule achieves this without knowing p in advance, and their algorithm $\mathcal{AA}$ is optimal up to a constant factor among procedures that only see i.i.d. samples of a $[0,1]$-valued variable [DKLR00] (see [Che07] for a repaired proof of the stopping-rule guarantee); for 0/1 samples of mean $p\ll1$ the optimal cost is $\Theta(\log(1/\delta)/(p\varepsilon^2))$. Beating sampling therefore requires looking at the circuit.

Definition 2.2 (beating sampling)**.** A procedure beats sampling on a class $\mathcal{C}$ if it is sound, efficient, and complete on $\mathcal{C}$, and $\mathcal{C}$ contains circuits with $p = n^{-\omega(1)}$ (for most of our classes, $p = 2^{-\Omega(n)}$). Its cost is then polynomial in the input size whatever p is, whereas that of Monte Carlo grows like $1/p$, exponentially in the surprise.

Prover-assisted estimation. In the second setting an unbounded prover sends one string π (an *argument*) to a randomized polynomial-time verifier V, which then flips coins (for instance to sample) and outputs an interval $[L, U]$.

Definition 2.3 (argument system)**.** V is *sound* if for every circuit C, every D and every string π, $\Pr_{\text{coins of } V}\left[p_D(C) \notin V(C, D, \pi, \varepsilon, \delta)\right] \le \delta$. It is *complete* on $\mathcal{C}$ if for every $C \in \mathcal{C}$ there is a π of size polynomial in the input size such that with probability $1 - \delta$ the output is tight. One-sided variants (lower-bound or upper-bound arguments) are defined in the obvious way.

This is the non-interactive, statistically sound (Merlin–Arthur) model. We will see in Section 3 that interaction (or computational soundness in the random-oracle model) would trivialise the problem, while in this model upper bounds on p cannot be certified for all circuits unless $\mathsf{coNP} \subseteq \mathsf{MA}$.

3 Limits

This section fixes what cannot be done, so that the positive results of Sections 4 to 6 can be read against it.

3.1 Without a prover: structure is necessary

Proposition 3.1 (worst case)**.** *(a) If a sound efficient procedure is complete (for some fixed $\varepsilon > 0$; any finite ε works) on all 3-CNF formulas under the uniform distribution, then* $\mathsf{NP} \subseteq \mathsf{BPP}$.

(b) Let $G_k : \{0,1\}^k \to \{0,1\}^{2k}$ be computable by $\mathrm{poly}(k)$*-size circuits and suppose every circuit of size $T(k)$ distinguishes $G_k(U_k)$ from U_{2k} with advantage less than $1/2$. For $y \in \{0,1\}^{2k}$ let $C_y(x) = [G_k(x) = y]$ on inputs $x \in \{0,1\}^k$, $k \ge 3$. Then $p(C_{G(s)}) \ge 2^{-k}$ for every seed s, while $\Pr_y[p(C_y) > 0] \le 2^{-k}$ for uniform y; and any sound procedure with $\delta \le 1/8$ that is complete on $\{C_{G(s)}\}$ within time t (on that family) satisfies $t \log t \ge T(k)/\mathrm{poly}(k)$.*

Proof. (a) Answer "satisfiable" iff $L > 0$. If $p = 0$, then $L > 0$ means $p \notin [L, U]$, which has probability at most δ. If $p > 0$ (so $p \ge 2^{-n}$), then $L = 0$ means the output is not tight or does not contain p, which has probability at most 2δ. Run with $\delta = 1/8$. (b) The bounds on p are immediate ($\mathbb{E}_y |G^{-1}(y)| = 2^{k-2k}$). Let A be the procedure, t its running time on the family, and consider the test "$A(C_y)$, stopped after t steps, outputs $L > 0$" (stopping only lowers the firing probability on other inputs). On $y = G(s)$ it fires with probability at least $1 - 2\delta$ (a tight interval containing $p > 0$ has $L > 0$; tightness and containment each fail with probability at most δ). On uniform y, with probability at least $1 - 2^{-k}$ we have $p = 0$, and soundness forces $L \le p = 0$ except with probability δ. The advantage is at least $1 - 3\delta - 2^{-k} \ge 1/2$ for $\delta \le 1/8$, $k \ge 3$. Fixing A's coins to their best value and simulating t steps by a circuit of size $O(t \log t) + \mathrm{poly}(k)$ gives the claim. □

Monte Carlo on $C_{G(s)}$ costs $O(2^k \mathrm{poly}(k))$. So if G is exponentially hard ($T(k) = 2^{(1-o(1))k}$), no sound procedure beats sampling on this family by more than a $2^{o(k)}$ factor ($p^{-o(1)}$ when $p = 2^{-\Theta(k)}$), even though every instance is a small, explicit circuit. Beating sampling is a matter of structure: the classes of Section 5 are where the structure is visible to a polynomial-time algorithm, and those of Section 6 are where it is visible to a polynomial-time verifier given a hint.

3.2 With a prover: upper bounds need deduction

Lower and upper bounds behave very differently. A lower bound $p \geq L$ can be witnessed by *showing* probability mass: a proposal distribution Q concentrated on $C^{-1}(1)$ (Section 6.4). An upper bound must exclude mass that the verifier has not seen. The next theorem makes this precise: under the uniform distribution (or any D of small collision probability), a verifier that only evaluates C cannot certify an upper bound much below the reciprocal of its number of evaluations, whatever argument it is given.

Theorem 3.2 (sampling cannot certify upper bounds)**.** *Let V be a verifier that receives an argument π, has oracle access to $f : \{0,1\}^n \to \{0,1\}$, flips coins, makes at most T oracle queries (at adaptively chosen points, from any distributions it likes), and outputs a number U. Suppose V is sound for upper bounds under the uniform distribution: for all f, π, $\Pr[U < p(f)] \leq \delta$. Then for every f, every π and every $k \leq n$,*

$$\Pr\left[V^f(\pi) \text{ outputs } U < 2^{-k}\right] \;\leq\; \delta + T\,2^{-k}.$$

In particular, for $2^{-(n+1)} \leq \eta \leq 1$, to output an upper bound below η with probability $1 - \delta'$ a black-box verifier needs $T \geq (1 - \delta - \delta')/(2\eta)$ queries.

Proof. Fix f, π, k. For $t \in \{0,1\}^k$ let $H_t = \{x : x_{1..k} = t\}$ and $f_t = f \vee \mathbf{1}_{H_t}$, so $p(f_t) \geq 2^{-k}$. Fix the coins r of V. The run of $V^f(\pi; r)$ queries a fixed sequence of points $z_1, \ldots, z_T$. The run of $V^{f_t}(\pi; r)$ is identical unless some $z_i \in H_t$; for each fixed z_i this happens for exactly one t. Hence for uniform t, $\Pr_{r,t}[\text{the runs differ}] \leq T2^{-k}$, and

$$\mathbb{E}_t \Pr_r\left[V^{f_t}(\pi; r) < 2^{-k}\right] \;\geq\; \Pr_r\left[V^f(\pi; r) < 2^{-k}\right] - T2^{-k}.$$

Some t attains at least the average; for it, V outputs $U < 2^{-k} \leq p(f_t)$ with that probability, which soundness bounds by δ. For the last claim take $k = \max(0, \lceil \log_2(1/\eta) \rceil - 1) \leq n$, so that $\eta \leq 2^{-k} \leq 2\eta$. □

Remark 3.3. The same argument works for any distribution D with enough collision entropy, planting $H = \{x : Ax = b\}$ for a random $k \times n$ matrix A over $\mathbb{F}_2$ and random b: every fixed point lies in H with probability 2^{-k}, and by pairwise independence $D(H) \geq 2^{-k-1}$ except with probability $4 \cdot 2^k \sum_x D(x)^2$ (for a product distribution, $\sum_x D(x)^2 = \prod_i (q_i^2 + (1-q_i)^2)$). The bound becomes $\Pr[V^f(\pi) \text{ outputs } U < 2^{-k-1}] \leq \delta + T2^{-k} + 4 \cdot 2^k \sum_x D(x)^2$. It also extends to verifiers that see a circuit for f and its size but use the circuit only through evaluations, provided the planted gadget $G_{A,b} = [Ax = b]$ is padded to a topology independent of (A, b) (each row as $\bigoplus_j (x_j \wedge a_{ij}) \oplus b_i$ with constant leaves a_{ij}, b_i): the planted circuits $C \vee (1 \wedge G_{A,b})$ then all have the topology and size of $C \vee (0 \wedge G_{A_0,b_0})$, so the bound holds for *some* circuit for f of size $|C| + O(nk)$, namely the latter. It cannot hold for every circuit computing f: under the uniform distribution, a verifier that sees $|C| = 3$ knows $p \in \{0\} \cup [1/8, 1]$.

The verifiers of Section 6 escape Theorem 3.2 by reading the circuit: they check a proof that $C(x) = 1$ forces x into a region of small, exactly computable measure. Short such proofs cannot always exist unless $\mathsf{coNP} \subseteq \mathsf{MA}$:

Theorem 3.4 (no complete upper-bound arguments)**.** *Suppose a sound argument system (soundness error $\delta \le 1/3$) is complete for upper bounds on all 3-CNF formulas with $p = 0$ under the uniform distribution, i.e. for each such C some π of polynomial size makes V output $U < 2^{-n}$ with probability at least $2/3$. Then $\mathsf{coNP} \subseteq \mathsf{MA}$, and the polynomial hierarchy collapses to its second level.*

Proof. Merlin sends π; Arthur runs V and accepts iff $U < 2^{-n}$. If C is unsatisfiable he accepts with probability $\ge 2/3$. If C is satisfiable then $p \ge 2^{-n}$, and accepting means V output $U < p$, which happens with probability at most $1/3$. So 3-UNSAT$\in \mathsf{MA} \subseteq \mathsf{AM}$, and $\mathsf{coNP} \subseteq \mathsf{AM}$ collapses PH to Σ_2^p [BHZ87] (in fact $\mathsf{coNP} \subseteq \mathsf{MA}$ gives $\mathsf{PH} = \mathsf{MA}$). □

The lower-bound side is weaker but not free: a system complete for lower bounds within a constant factor on all circuits under the uniform distribution exists iff $\mathrm{prSBP} \subseteq \mathrm{prMA}$ for the class SBP of [BGM06] (amplify the SBP gap by conjunctions of copies; conversely send the scale with an MA proof), and in particular it would give $\mathsf{SBP} \subseteq \mathsf{MA}$. Since $\mathsf{MA} \subseteq \Sigma_2^p$ relativizes and there is an oracle relative to which $\mathsf{SBP} \not\subseteq \Sigma_2^p$ [BGM06], no relativizing argument shows that such a system exists. It does exist if $\mathrm{prAM} = \mathrm{prNP}$, which follows from standard derandomisation hypotheses (Section 6.8); its soundness then rests on the hypothesis.

3.3 Interaction trivialises the problem

The restriction to one message is essential:

Proposition 3.5. *(a) (Sum-check [LFKN92].) For every circuit C and product D, let B be the total bit length of the denominators of the marginals, and let C' be C with its threshold gates compiled into $\{\wedge, \vee, \neg, \oplus\}$-subcircuits (scale the weights to integers and add them in binary), so that $|C'|$ is at most quadratic in the input size. The exact value of p has an interactive proof over a prime field with $|\mathbb{F}| > 2^{B+1}$, with a polynomial-time verifier, $O(n+|C'|)$ rounds and soundness error $O((n+|C'|)\, d/|\mathbb{F}|)$, where $d \le |C'|+2$ bounds the individual degrees of the arithmetised Tseitin summand.*

(b) (Set lower bounds [GS86].) For uniform D (or dyadic product D, via the sampler reduction), two-round public-coin protocols, repeated in parallel, certify $p \ge L/c$ for a constant c for every circuit.

Applying the Fiat–Shamir transform to (a) gives, in the random-oracle model (or heuristically with a fixed hash), a one-message argument that is only computationally sound, which is not the model asked for. So the interesting question is the Merlin–Arthur one, where by Theorems 3.2 and 3.4 an argument certifying upper bounds below the sampling scale must contain *deductive* content about the circuit, and, unless $\mathsf{coNP} \subseteq \mathsf{MA}$, can be complete for upper bounds only on a restricted class.

4 Envelopes: the common mechanism

Most of our upper-bound-capable methods, with and without a prover, are instances of one estimator (the exceptions are the correlation-inequality bounds of Sections 4.5 and 5.5, the product-of-ratios estimators of Section 5.6, and two variants of it: the counter-guided sampler of Theorem 6.4,

and the tilted Gibbs estimator of Section 5.7, whose mass and proposal are themselves certified statistically). It turns a *deductive* fact ("$C(x) = 1$ implies $\varphi(x) \geq 1$") and an *exact* computation ($\mathbb{E}_D\varphi$) into an importance sampler whose cost depends only on how loose the deduction is.

4.1 Envelope circuits

Definition 4.1 (envelope)**.** An *envelope circuit* Φ is an arithmetic circuit whose leaves are literal indicators $[x_i = b]$, nonnegative rational constants, and (Section 4.3) *boxes* $[B(x) = 1]$ for circuits B; whose gates are SUM and PROD; and whose PROD gates are *decomposable*: their children have pairwise disjoint supports, where the support $\mathrm{var}(v)$ of a node is the set of variables occurring at the leaves of its cone ($\mathrm{var}(B)$ for a box). We also require that every PROD-child with empty support be a constant leaf (fold constant subcircuits first). Φ computes $\varphi : \{0,1\}^n \to \mathbb{R}_{\geq 0}$. It *dominates* C if $\varphi(x) \geq 1$ whenever $C(x) = 1$. Its *mass* is $Z = \mathbb{E}_D\varphi$; if $p > 0$, its *looseness on* C is $K = Z/p$, and domination gives $Z \geq p$, i.e. $K \geq 1$.

A *derivation* of Φ is a proof tree of its tree unfolding: keep, top-down, one child of every SUM occurrence and all children of every PROD occurrence. By decomposability a derivation visits each node of nonempty support at most once. With literal leaves and unit constants, $\varphi(x)$ is the number of derivations all of whose leaves are satisfied by x; with general constants it is the corresponding weighted count. A DNF $\bigvee_j T_j$ with consistent, duplicate-free terms gives the envelope $\sum_j \prod_{\ell \in T_j} [\ell]$, and a DNNF gives the envelope obtained by reading $\vee$ as SUM and $\wedge$ as PROD.

Lemma 4.2 (exact mass, exact sampling)**.** *Let Φ have literal and constant leaves only, D be a product distribution with rational marginals, and $Z > 0$. Then $Z = \mathbb{E}_D\varphi$ is computed by one bottom-up pass (SUM $\mapsto$ sum of children's masses, PROD $\mapsto$ product) with $O(|\Phi|)$ operations on rationals of* $\mathrm{poly}(|\Phi|, b)$ *bits (b the total bit length of the marginals and of the constants of Φ), and the following top-down procedure samples exactly from $Q_\varphi(x) = D(x)\varphi(x)/Z$: at a SUM gate choose a child with probability proportional to its mass; at a PROD gate visit all children; at a literal leaf set the literal; finally draw every unset variable from D.*

Proof. By decomposability and independence, $\mathbb{E}_D \prod_c \varphi_c = \prod_c \mathbb{E}_D\varphi_c$; linearity handles SUM. The procedure selects a derivation $\mathcal{T}$ with probability $w(\mathcal{T})/Z$, where $w(\mathcal{T})$ is the product of the literals' probabilities and the constants in $\mathcal{T}$ (the choice probabilities telescope), and then draws $x \sim D$ conditioned on all literals of $\mathcal{T}$, whose density is $D(x)\mathbf{1}[x \models \mathcal{T}]\, c(\mathcal{T})/w(\mathcal{T})$ with $c(\mathcal{T})$ the product of its constants. Summing over derivations, $\Pr[x] = \sum_{\mathcal{T}} c(\mathcal{T})D(x)\mathbf{1}[x \models \mathcal{T}]/Z = D(x)\varphi(x)/Z$. (A literal is set at most once per derivation: two literals on the same variable in one derivation would lie below different children of a decomposable PROD.) Bit lengths: the mass of a node is a sum of at most $2^{|\Phi|}$ products of at most $|\Phi|$ constants and marginals. □

In floating point, masses must be kept in the log domain and the final interval widened by the accumulated rounding error. Our implementation does exactly this: along every derivation the computed weight and the sampler's choice probabilities telescope, so the relative error of $\hat{Z} \cdot \mathbb{E}_{\hat{Q}} W$ is at most $e^{\pm\eta}$, where η is the largest sum of local rounding errors on a derivation (a max-sum dynamic program over Φ); the discreteness of the categorical draws at SUM gates adds an absolute $O((n + |\Phi|)2^{-53})$ to $\mathbb{E}W$ (the remaining inputs are drawn from D exactly, by comparing lazily drawn binary digits of a uniform with those of q_i, so the rejection fills of Section 4.3 are exact and the box estimates rest on exact samples); and the output interval is widened by both and rounded outwards.

4.2 The envelope estimator

For N i.i.d. samples in $[0,1]$ with empirical mean $\hat{\mu}$, the *Hoeffding–KL interval* at level δ is $\{m \in [0,1] : N\,\mathrm{KL}(\hat{\mu}\|m) \le \ln(2/\delta)\}$ with KL the Bernoulli relative entropy; it contains the true mean with probability at least $1-\delta$ [Hoe63, Thm. 1]. (In floating point our implementation excludes a value only when its computed divergence exceeds the bound by more than its rounding error, and allows for the rounding of $\hat{\mu}$, so the computed interval contains the exact one.)

Theorem 4.3 (envelope estimator)**.** *Let $\varphi : \{0,1\}^n \to \mathbb{R}_{\ge 0}$ dominate C, with exact mass Z and an exact sampler for $Q_\varphi = D\varphi/Z$ (for instance an envelope circuit with Lemma 4.2). If $Z = 0$, output $[0,0]$ (valid under domination). Otherwise, for $X \sim Q_\varphi$ let $W = C(X)/\varphi(X)$. Then*

(i) $W \in [0,1]$ and $\mathbb{E}W = p/Z = 1/K$;

(ii) from N independent copies, the Hoeffding–KL interval $[\ell, u]$ for $\mathbb{E}W$ at level δ gives an interval $[Z\ell, \min(Zu, 1)]$ containing p with probability at least $1-\delta$, for every circuit *C as long as φ dominates it;*

(iii) if φ does not dominate C: for integer-valued φ (envelope circuits with literal leaves and integer constants) the same procedure gives a valid interval for $\Pr_D[C(x) = 1 \wedge \varphi(x) > 0] \le p$; for general φ, with W clipped at 1, $Z\ell$ is still a valid lower bound on p;

(iv) for $0 < \varepsilon \le 1$, run sequentially from $N_0 = \lceil c \log(2/\delta_1)/\varepsilon^2 \rceil$ samples, doubling N and using level $\delta_j = 6\delta/(\pi^2 j^2)$ at stage j: all intervals are valid with probability $1-\delta$, and if $p > 0$ the procedure reaches $u \le (1+\varepsilon)\ell$ after $N = O\big(K(\log(1/\delta) + \log\log K)/\varepsilon^2\big)$ samples with probability $1-\delta$ (so both hold with probability $1-2\delta$). With a sample cap it always halts, returning its current, still valid, interval. Each sample costs $O(|C| + |\Phi|)$ arithmetic operations with the sampler of Lemma 4.2.

Proof. (i) If $Q_\varphi(x) > 0$ then $\varphi(x) > 0$; if moreover $C(x) = 1$ then $\varphi(x) \ge 1$, so $W \le 1$. $\mathbb{E}W = \sum_x D(x)\varphi(x)C(x)/(Z\varphi(x)) = p/Z$. (ii) The KL interval is valid for i.i.d. $[0,1]$ variables, and $p \le 1$. (iii) For integer-valued φ, $\varphi(x) > 0$ implies $\varphi(x) \ge 1$, so $W \in [0,1]$ and $\mathbb{E}W = \Pr[C = 1, \varphi > 0]/Z$. For real-valued φ, clipping only lowers the mean, so $\mathbb{E}\min(W,1) \le \Pr[C = 1, \varphi > 0]/Z \le p/Z$. (iv) Let $\mu = 1/K$. By the multiplicative Chernoff bound, once $N \ge c\log(1/\delta_j)/(\mu\varepsilon^2)$ the empirical mean $\hat{\mu}$ is within a factor $1 \pm \varepsilon/8$ of μ with probability $1-\delta_j$. Since $\mathrm{KL}(a\|b) \ge (a-b)^2/(2\max(a,b))$, every m in the KL interval with $m > \hat{\mu}$ satisfies $(m-\hat{\mu})^2 \le am$ with $a = 2\log(2/\delta_j)/N$, hence $m - \hat{\mu} \le a + \sqrt{a\hat{\mu}}$, and similarly below; for a suitable constant c both are at most $(\varepsilon/4)\hat{\mu}$, so $u/\ell \le (1+\varepsilon/4)/(1-\varepsilon/4) \le 1+\varepsilon$. With the stated N_0, the first stage at which this holds has $j = O(\log K)$, whence the $\log\log K$ term; the union bound over stages makes all intervals simultaneously valid. □

So the cost of an envelope is governed by one number. Monte Carlo is the trivial envelope $\varphi \equiv 1$, with $K = 1/p$; exact computation is $\varphi = C$ (when C is itself a d-DNNF), with $K = 1$. In between:

Lemma 4.4 (what K measures)**.** *For $p > 0$, $K = \mathbb{E}_D[\varphi \mid C = 1] + \mathbb{E}_D[\varphi\,\mathbf{1}_{C=0}]/p$. For an envelope with literal leaves and unit constants whose positivity set $\{\varphi > 0\}$ equals $C^{-1}(1)$ (e.g. a DNNF for C), K is the expected number of satisfied derivations of $x \sim D \mid C = 1$, its* expected ambiguity*. For a DNF with m terms, $K \le m$.*

The first term is overcounting (union bounds); the second is mass that the deduction failed to exclude. K also bounds the relative variance $\rho = \frac{Z}{p}\mathbb{E}_D[1/\varphi \mid C = 1] - 1 \le K - 1$ of ZW. The KL procedure of Theorem 4.3(iv) ignores the variance and pays $\Theta(K)$ samples whatever ρ is; a variance-adaptive interval (empirical Bernstein, or DKLR's algorithm $\mathcal{AA}$) pays $O((\rho/\varepsilon^2 + K/\varepsilon)\log(1/\delta))$, so a low dispersion of the derivation count helps only in the ε^{-2} term. The K/ε term cannot be avoided by a sound procedure that uses the weights as a black box, and for tilting envelopes K can exceed ρ exponentially (Proposition 5.22). A d-DNNF has $\varphi = \mathbf{1}_C$, $K = 1$ and $\rho = 0$. For $E = \bigwedge_{i \le m}(x_i \vee y_i)$ under the uniform distribution, used as its own envelope, $K = (4/3)^m$, and the relative variance is at least $(10/9)^m - 1$ under every reweighting of its $\vee$-edges (with equality for unit weights); both are exponential although $D(E) = (3/4)^m$ is known exactly. Theorem 6.4 removes this overcounting altogether: its looseness is $D(E)/p$ rather than $\mathbb{E}_D[\varphi]/p$.

In the spirit of surprise accounting [Hil24]: of the $\log_2(1/p)$ bits of surprise in the event $C(x) = 1$, an envelope with $Z \le 1$ explains $\log_2(1/Z)$ bits by deduction and exact computation (once domination is certified, syntactically or by a proof), and leaves $\log_2 K$ bits to sampling, at a cost of about K samples (up to $\varepsilon^{-2}\log(1/\delta)$ factors). If $Z \ge 1$ the envelope is worse than Monte Carlo and is replaced by $\varphi \equiv 1$. Monte Carlo explains nothing; exact computation explains everything, with $|\Phi|$ possibly exponential in $|C|$. Surprise accounting also charges the explanation's own description length: zero for envelopes computed from C, and $|\pi|$ for prover-supplied ones (Theorem 6.12).

4.3 Dense leaves

Products of independent events that are individually not rare are the most common source of rarity, and they need not have small envelopes: an $\wedge$ of k independent random sub-circuits B_j, each true with probability $1/2$, has $p = 2^{-k}$ but its literal envelope may lose a constant factor per block. Boxes repair this: a box is a sub-function used as its own envelope, its mass estimated by Monte Carlo and its conditional distribution sampled by rejection. Estimation errors in the masses bias the proposal, but the bias on each derivation is a product of per-box ratios that we can bound exactly.

Theorem 4.5 (envelopes with dense leaves)**.** *Let Φ dominate C, with literal leaves and box leaves $B_1, \dots, B_r$ (PROD gates decomposable with respect to supports), where $m_j = D(B_j) \ge 1/R$ and every derivation contains at most L box leaves; let $0 < \varepsilon \le 1$. Estimate each m_j by Monte Carlo to an interval $[\ell_j, u_j]$ at level $\delta/(2r)$ with point value $\hat{m}_j$; compute masses $\hat{Z}$ and sample $\hat{Q}$ as in Lemma 4.2 with $\hat{m}_j$ in place of m_j, filling each box by rejection sampling from D. For $\rho(\mathcal{T}) = \prod_{B_j \in \mathcal{T}} \hat{m}_j/m_j$ let $\rho_{\min}, \rho_{\max}$ be its extreme values over derivations when each m_j ranges over $[\ell_j, u_j]$ (computed by a min-/max-product pass over Φ). Then, with $[\ell, u]$ a valid KL interval for $\mathbb{E}_{\hat{Q}}W$, $W = C/\varphi$, at total level $\delta/2$ (fixed or sequential),*

$$p \in [\hat{Z}\ell/\rho_{\max},\ \hat{Z}u/\rho_{\min}]$$

with probability at least $1 - \delta$. If each box interval satisfies $u_j \le (1 + \varepsilon/(8L))\ell_j$ and importance sampling is run to ratio $1 + \varepsilon/4$, the output is tight with probability $1 - 2\delta$, after $O(K(\log(1/\delta) + \log\log K)/\varepsilon^2)$ samples of $\hat{Q}$, each costing $O(|C| + |\Phi| + \sum_j |B_j| + LR\max_j |B_j|)$ in expectation; estimating the boxes costs $O(rL^2R\log(r/\delta)/\varepsilon^2)$ evaluations.

Proof. The sampler picks derivation $\mathcal{T}$ with probability $\hat{w}(\mathcal{T})/\hat{Z}$ where $\hat{w}$ uses the $\hat{m}_j$, and then draws x exactly from D conditioned on $\mathcal{T}$'s leaves (rejection sampling is exact; decomposability

makes the leaves' conditionings independent), whose density is $D(x)\mathbf{1}[x \models \mathcal{T}]/w(\mathcal{T})$ with the true masses. Hence $\hat{Q}(x) = \frac{D(x)}{\hat{Z}} \sum_{\mathcal{T} \ni x} \rho(\mathcal{T})$, and

$$\hat{Z}\, \mathbb{E}_{\hat{Q}} W = \sum_x D(x)C(x) \cdot \frac{1}{\varphi(x)} \sum_{\mathcal{T} \ni x} \rho(\mathcal{T}) \;\in\; [\rho_{\min}, \rho_{\max}] \cdot p,$$

because $\varphi(x)$ is the number of derivations satisfied by x, so the middle factor is an average of $\rho(\mathcal{T})$. On the event (probability $\geq 1 - \delta/2$) that all $m_j \in [\ell_j, u_j]$, $\rho_{\min}$ and $\rho_{\max}$ bound this average; conditionally on the box estimates the KL interval fails with probability at most $\delta/2$. For tightness: $\rho_{\max}/\rho_{\min} \leq (1 + \varepsilon/(8L))^{2L} \leq e^{\varepsilon/4}$ and $(1 + \varepsilon/4)e^{\varepsilon/4} \leq 1 + \varepsilon$ for $\varepsilon \leq 1$. Since $\hat{Z} = \sum_{\mathcal{T}} \rho(\mathcal{T})w(\mathcal{T}) \leq \rho_{\max} Z$, the mean of W under $\hat{Q}$ is at least $\rho_{\min} p/(\rho_{\max} Z) \geq e^{-\varepsilon/4}/K$, so Theorem 4.3(iv) (which only needs i.i.d. $[0,1]$ samples of that mean) gives the sample count. Computing $\varphi(X)$ evaluates every box, hence the $\sum_j |B_j|$ term. □

With boxes, a product of k independent dense blocks has $K = 1$ and costs $O(k^3 R \log(k/\delta)/\varepsilon^2)$ evaluations, against $2^{\Theta(k)}$ for Monte Carlo; a DNF whose terms are products of dense blocks has an envelope (a SUM of PRODs of boxes) whose positivity set is $C^{-1}(1)$ and with $\varphi \leq m$, so $K \leq m$ by Lemma 4.4.

4.4 Soft envelopes: Chernoff bounds as certificates

Envelopes need not be integer-valued. If $C(x) = [S(x) \geq t]$ with $S = \sum_{j=1}^{k} g_j(x_{B_j})$ a sum of $\{0,1\}$-valued sub-circuits on disjoint blocks (since S is an integer and the comparison is exact, we take t to be an integer, replacing it by $\lceil t \rceil$), then for $\theta \geq 0$

$$\varphi_\theta(x) = e^{-\theta t} \prod_{j=1}^{k} \left([g_j = 0] + e^{\theta}[g_j = 1]\right) = e^{\theta(S(x)-t)} \;\geq\; \mathbf{1}[S(x) \geq t], \quad Z_\theta = e^{-\theta t} \prod_{j=1}^{k} (1 - p_j + p_j e^{\theta}),$$

where $p_j = D(g_j = 1)$. Z_θ is the Chernoff bound, a deductively valid upper bound, and Q_{φ_θ} is the exponentially tilted product distribution. To sample it block by block we assume that the p_j are known exactly and D conditioned on $g_j = b$ can be sampled, e.g. blocks small enough to enumerate (as in our implementation), or boxes with $\min(p_j, 1 - p_j) \geq 1/R$, in which case Theorem 4.5 applies with $L = k$ (with weighted averages of ρ). This is the classical efficient importance sampler for large deviations of sums [Sie76, Buc04]; here is a clean uniform bound on its looseness.

Proposition 4.6 (tilting is polynomially tight)**.** *Let the weights be* 1, $\mathbb{E}_D S \leq t < k_+$, *where* $k_+ = |\{j : p_j > 0\}|$ *and* t *is an integer. There is* $\theta \geq 0$ *with* $\mathbb{E}_{Q_\theta} S = t$, *and for it* $K = Z_\theta/p \leq \frac{4}{3}(2\sqrt{k} + 1)$. *The same bound holds for any* $\theta \geq 0$ *with* $|\mathbb{E}_{Q_\theta} S - t| < 1/(k+2)$, *so* θ *may be found by bisection and* e^θ *taken rational.*

Proof. $\theta \mapsto \mathbb{E}_{Q_\theta} S = \sum_j p_j e^\theta/(1 - p_j + p_j e^\theta)$ is continuous and nondecreasing, at most t at $\theta = 0$ and tends to $k_+ > t$, so θ exists. Under Q_θ the g_j are independent Bernoulli(p_j^θ), so S is Poisson-binomial with mean μ_θ. By Darroch's rule [Dar64] the mode is t when $\mu_\theta = t$, and also when $|\mu_\theta - t| < 1/(k+2)$ (the rule makes k' the unique mode for $k' \leq \mu < k' + 1/(k'+2)$ and for $k' - 1/(n - k' + 2) < \mu \leq k'$). By Chebyshev, $Q_\theta(|S - \mu_\theta| \leq 2\sigma) \geq 3/4$ with $\sigma^2 = \mathrm{Var}_{Q_\theta} S \leq k/4$, and this window contains at most $4\sigma + 1$ integers, so, S being unimodal (log-concave), $Q_\theta(S = t) \geq \frac{3}{4(4\sigma+1)}$. Finally $p = Z_\theta\, \mathbb{E}_{Q_\theta}[e^{-\theta(S-t)} \mathbf{1}_{S \geq t}] \geq Z_\theta\, Q_\theta(S = t)$. □

Our implementation, which also allows nonnegative integer block weights w_j, finds θ by bisection on $\mathbb{E}_{Q_\theta} S$ to within $1/(2(k+2))$ of t, subject to $\theta \max_j w_j \le 699$ (so that every $e^{\theta w_j}$ is a finite double), and compares S with $\lceil t \rceil$ in integer arithmetic, so $\varphi_\theta \ge 1$ wherever $C = 1$.

Unit weights matter. For integer weights under a product distribution the best tilt can be loose by a factor $p^{-\Theta(1)}$: with $S = (k+1)y + \sum_{i \le k} x_i$, $q_y = \eta$, $q_x = 1/2$ and $t = k+1$, so $p = \eta$, the optimal Chernoff envelope has $K = 2^{25.7}$ at $k = 20$, $p = 2^{-40}$. When the p_j are known exactly, the unit-weight case is also computable exactly by dynamic programming over S (pseudo-polynomial in general weights), and arbitrary weights have a deterministic FPTAS under product distributions via read-once branching programs [GKM10] (see also [GKM+11]). The point of Proposition 4.6 is that the Chernoff bound, a one-line deduction from the syntax "C is a threshold of independent blocks", explains all but $O(\log k)$ bits of the surprise, and sampling pays for the rest. When the blocks overlap, the tilted proposal is a Gibbs measure; Section 5.7 certifies its sampler and normaliser, and Remark 5.21 records that the $O(\sqrt{k})$ bound is tight and that $K \to 1$ at the top threshold.

4.5 Certified presumption of independence

The *presumption of independence* [CNX22] estimates $\Pr[\bigwedge_i A_i]$ by $\prod_i \Pr[A_i]$. For monotone events it can be certified: correlation inequalities bound the error in terms of the probability carried by dependent pairs, which can be computed or estimated without ever conditioning on the rare event.

Theorem 4.7 (Harris–Janson sandwich)**.** *Let $C = \bigwedge_{i=1}^{m} \mathrm{cl}_i$ be a CNF in which every variable occurs with only one sign, $B_i = [\mathrm{cl}_i = 0]$ (a subcube), D a product distribution, $b_i = D(B_i)$, $M = \prod_i (1 - b_i)$, $\varepsilon_0 = \max_i b_i < 1$, and for an ordering of the clauses $d_i = \sum_{j<i,\, \mathrm{var}(\mathrm{cl}_j) \cap \mathrm{var}(\mathrm{cl}_i) \neq \emptyset} D(B_i \wedge B_j)$, $\Delta = \sum_i d_i$ (the sum over unordered dependent pairs, as in Theorem 5.10). Then*

$$M \;\le\; p \;\le\; \prod_i (1 - b_i + d_i) \;\le\; M \exp\Big(\sum_i \frac{d_i}{1 - b_i}\Big) \;\le\; M \exp\Big(\frac{\Delta}{1 - \varepsilon_0}\Big),$$

and all quantities are computable with $O(\sum_i |\mathrm{cl}_i|^2 \cdot d)$ arithmetic operations, d the maximum number of clauses sharing a variable. (If some $b_i = 1$ then $p = M = 0$.)

Proof. Flip the variables that occur positively; then every B_i is a principal up-set $\{y : y_{S_i} = \mathbf{1}\}$ under a product measure. The lower bound is the Harris–Kleitman inequality (the events $\neg B_i$ are decreasing, hence positively correlated) [Har60]. The upper bounds are Janson's inequality [Jan90] in the product form of the Boppana–Spencer proof (Theorem 5.10) and in the form of [AS16, Thm. 8.1.1]. $D(B_i)$ and $D(B_i \wedge B_j)$ are products of marginals over S_i and $S_i \cup S_j$. □

So for unate CNFs with $\varepsilon_0 \le 1/2$ whose dependent pairs carry total probability $\Delta \le \varepsilon$, the geometric mean of the two ends has relative error at most $e^{\Delta} - 1$, however small p is. Without a bound on ε_0 the last form can be arbitrarily loose, but the product form, with clauses ordered by decreasing b_i, is at most $M \exp(2\Delta/(1 - \Delta))$ for $\Delta < 1$: if clause i shares a variable with an earlier clause j, then $d_i \ge D(B_i \wedge B_j) \ge b_i + b_j - 1 \ge 2b_i - 1$, so $1 - b_i \ge (1 - \Delta)/2$; otherwise $d_i = 0$. Example: $m = 6 \cdot 10^4$ random clauses of width 12 over $2 \cdot 10^5$ variables give the certified interval $p \in [4.33, 5.07] \cdot 10^{-7}$ (ratio 1.17) in 50 seconds of pure Python (wall-clock, on a shared 4-core machine; Table 6). The version for unate $\wedge$'s of arbitrary monotone sub-circuits, with b_i and Δ estimated by sampling, is Theorem 5.10.

5 Without a prover: sound procedures and their classes

A procedure without a prover must find the structure itself. Ours look for exactly the structure that Section 4 can exploit: independence (variable-disjoint sub-circuits), forcing (literals implied by $C(x) = 1$), small cutsets, dense residuals, and a few algebraic and analytic patterns. All steps are either exact identities under product distributions or Monte Carlo estimates with valid intervals, so the procedures are sound on every circuit; which circuits they are *complete* on is guaranteed by the size of a canonical "explanation tree".

5.1 Decompose–Force–Sample–Branch

Fix $a = \lfloor c \log_2 |C| \rfloor$ for a constant c (our implementation fixes $a = 12$), and call a node an *atom* if its support has at most a variables. Work with the *canonicalised NNF* N: the NNF of C in which every gate of C whose support has at most a variables has been replaced by its reduced OBDD (rule (R8) of Section 5.2), flattened with duplicate children removed, except that an atom is never merged into a larger parent, and with constants propagated ($x \wedge \neg x \mapsto 0$, $x \vee \neg x \mapsto 1$); restricting a node by a partial assignment ρ simplifies it in the same way. Opaque gates (Section 2) are leaves of this NNF: (F) gets no clauses from them, and (B) branches on their smallest variable. A node h is handled by the first rule that applies:

(T) h is a constant or a literal: exact value.

(T′) h is an atom: exact value, by summing D over the $2^{|\mathrm{var}(h)|} \le |C|^c$ assignments of $\mathrm{var}(h)$.

(F) Unit propagation on $\mathrm{Tseitin}(h) \wedge [h = 1]$, repeated on the restricted node until it derives nothing new, derives a nonempty set of input literals F (at the root only, alternating with failed-literal probing and with Gaussian elimination over $\mathbb{F}_2$, see below): $D(h) = D(F) \cdot D(h|_F)$, since $h \Rightarrow F$ and $h|_F$ does not depend on the variables of F. If it derives a conflict, $D(h) = 0$.

(D) The children of h's top gate fall into at least two variable-disjoint groups: $D(h) = \prod_j D(h_j)$ for $\wedge$ and $D(h) = 1 - \prod_j (1 - D(h_j))$ for $\vee$.

(S) Dense test: a pilot of $N_0 = 64K\lceil \ln(2m/\delta) \rceil$ samples finds at least $N_0/(4K)$ hits; then h is a Monte Carlo leaf.

(E) Envelope test: a syntactic envelope Φ_h (Section 5.2) passes the analogous pilot for $\mathbb{E}_{\hat{Q}}[h/\varphi]$ (at least $N_0/(4K)$ total weight in N_0 samples of its proposal $\hat{Q}$); then h is an envelope leaf. If Φ_h has boxes, at most L on a derivation and r_h in all, the pilot is run twice: first with the rough masses found by (R7) (failing it fails the test), then with masses estimated, each at level $\delta/(4mr_h)$ and with at most $N(16K, 1/(2L), \delta/(4mr_h))$ samples (defined below), to intervals with $u_j \le (1 + \frac{1}{2L})\ell_j$ and $\ell_j \ge 1/(16K)$; if these estimates fail, or the (truncated) rejection sampler of a box fails, so does the test. If Φ_h has no boxes and mass 0, then $D(h) = 0$.

(B) Otherwise branch on the variable shared by the most children (ties: smallest index): $D(h) = (1 - q_x)D(h|_{x=0}) + q_x D(h|_{x=1})$.

The root's elimination uses the affine equations implied by the conjuncts of C, the gates reached from the output through $\wedge$-gates that are not atoms: an atom conjunct contributes the equations of the affine hull of its satisfying assignments, and a conjunct built by $\oplus$ and $\neg$ from inputs and

from atoms with affine functions contributes its equation. Like the other rules, it reads atoms only through their functions.

Phase 1 builds the tree with a node budget m; nodes beyond the budget become "loose" leaves. Phase 2 estimates every leaf with fresh samples at confidence δ/ℓ, where ℓ is the number of leaves and d the largest number of leaves on one derivation (all children at (D)-$\wedge$, one child elsewhere), and combines the leaf intervals through the exact node operations, all of which are monotone (our implementation rounds outwards and also carries log-domain bounds, which stay tight for p below the floating-point range). Let $\eta = \varepsilon/(3d)$. Every estimator is the doubling rule of Theorem 4.3(iv) run towards ratio $1+\eta'$ at level δ' and capped at $N(K', \eta', \delta') = c_0 K' \eta'^{-2}(\log(1/\delta') + \log\log K')$ samples, a size by which it has stopped with probability $1-\delta'$ whenever its mean is at least $1/K'$ (c_0 an absolute constant); at the cap it outputs its current interval, which is still valid. (S)-leaves (Monte Carlo) and box-free (E)-leaves use $(16K, \eta, \delta/\ell)$. An (E)-leaf with boxes runs the procedure of Theorem 4.5 (importance sampling towards ratio $1+\eta/4$ at level $\delta/(2\ell)$, each box towards $1+\eta/(8L)$ at level $\delta/(2\ell r_h)$), with the cap $\max\{N(64K, \eta/4, \delta/(2\ell)),\ N(16K, \eta/(8L), \delta/(2\ell r_h))\}$ for every stage; each box fill of its rejection sampler is truncated after $T_{\mathrm{f}} = \lceil 16K \ln(8N_{\mathrm{f}}/\delta)\rceil$ draws, where N_{f}, the sum over these leaves of L times the cap, bounds the number of fills, and a truncated leaf outputs $[0,1]$. A loose leaf is estimated by Monte Carlo with any cap up to $N(16K, \eta, \delta/\ell)$. So DFSB runs in time $\mathrm{poly}(|C|, K, m, B, 1/\varepsilon, \log(1/\delta))$ on every circuit, with B the rule budget of Section 5.2.

Definition 5.1 (ideal explanation tree)**.** $T^*_K(C)$ is the tree built by the same rules with the pilot test (S) replaced by the exact predicate $D(h) \geq 1/K$ and without (E). $m_K(C) = |T^*_K(C)|$.

Theorem 5.2 (DFSB)**.** *For every circuit C and product distribution D, the output interval contains p with probability at least $1-\delta$. If $m_K(C) \leq m$, then with probability at least $1-2\delta$ the output is tight ($U \leq (1+\varepsilon)L$), and the number of evaluations of nodes of the canonicalised NNF N is*

$$O\big(mK\log(m/\delta)\big) + \tilde{O}\Big(\ell \cdot K\, d^2\, \varepsilon^{-2} \log(\ell/\delta)\Big), \qquad \ell \leq m,\ d \leq n,$$

plus $s_7 B$ for the (R7) pilots of each (E)-test, and $\tilde{O}(r_h L^2 K d^2 \varepsilon^{-2} \log(m r_h/\delta))$ box evaluations for each (E)-test or (E)-leaf whose envelope has boxes. An evaluation costs $O(|N|)$, and $O(|N|+|\Phi_h|)$ at an (E)-leaf plus $O(LK \max_j |B_j|)$ in expectation for rejection sampling; here $|N| = O(|N^\circ|^2)$ for the NNF N° before flattening, and $|N^\circ| = O(2^a|C|)$ for $\{\wedge, \vee, \neg, \oplus\}$-circuits (threshold counters add their size). The remaining work is $O(2^a|C|^2)$ for (R8) and the atoms' affine hulls, and $O\big(m(n+2^a+B)|N| + n^2(a|C|+|N|) + n^3\big)$ arithmetic operations.

Proof. Soundness. (T), (T′), (F), (D), (B) are exact identities under a product distribution (independence of variable-disjoint sub-circuits, the law of total probability, and $h \Rightarrow F$), and an envelope of mass 0 forces $D(h) = 0$ because $\varphi \geq \mathbf{1}_h$. Phase 2 runs on a fixed tree with fresh samples; each leaf interval is valid with probability $1-\delta/\ell$ whatever its cap (Theorem 4.3(iv), Theorem 4.5); a truncated rejection sampler agrees with an untruncated one until it stops, and then $[0,1]$ is output. A union bound over the ℓ leaves and interval arithmetic through monotone operations give a valid root interval. Phase 1 only shapes the tree.

Completeness. Let X be a pilot's hit count and $\mu = \mathbb{E}X$. If $D(h) \geq 1/K$ then $\mu \geq N_0/K$ and $\Pr[X < N_0/(4K)] \leq e^{-(3/4)^2\mu/2} \leq (\delta/2m)^{18}$. If $D(h) < 1/(16K)$ then $\mu < N_0/(16K)$ and, by the Chernoff bound $\Pr[X \geq t] \leq e^{-\mu}(e\mu/t)^t$, $\Pr[X \geq N_0/(4K)] \leq (e/4)^{N_0/(4K)} \leq (\delta/2m)^6$. The same two bounds hold for the envelope pilot, whose summands lie in $[0,1]$. So with probability at least $1 - 3m(\delta/2m)^6 \geq 1-\delta/8$ every pilot (at most three per node) accepts every node of density (or $\mathbb{E}_{\hat{Q}}W$) $\geq 1/K$ and none of density $< 1/(16K)$, and with probability at least $1-\delta/4$

all pilot box intervals of (E) are valid. On these events the actual tree is a pruning of $T^*_K(C)$ (where the ideal tree stops, the actual one stops; where it continues, the actual one continues identically or stops early at an (S)- or (E)-leaf; a failed or truncated (E)-test leaves the node to the rules of T^*_K), so it has at most m nodes and no loose leaves. Every (S)-leaf is $16K$-dense; a box-free (E)-leaf has looseness $K_h = Z/p_h$ at most $16K$; and for an (E)-leaf with boxes, the proof of Theorem 4.5 gives $\mathbb{E}_{\hat{Q}} W \le (\rho_{\max}/\rho_{\min})/K_h$ for the pilot proposal, where every derivation has $\rho \in [(1+\frac{1}{2L})^{-L}, (1+\frac{1}{2L})^{L}]$, so $K_h \le 16eK$ and the boxes are $16K$-dense. In Phase 2, with probability $1-\delta/2$ the box intervals are valid, and then the importance-sampling mean of an (E)-leaf with boxes is at least $e^{-\eta/4}/K_h \ge 1/(64K)$ (proof of Theorem 4.5). As the boxes have mass at least $1/(16K)$, each of the at most N_{f} fills exceeds T_{f} draws with probability at most $(1-\frac{1}{16K})^{T_{\mathrm{f}}} \le \delta/(8N_{\mathrm{f}})$, so with probability $1-\delta/8$ no leaf is truncated. Hence, with probability $1-\delta$ more, every leaf estimator stops before its cap, with ratio at most $1+\eta$ (for boxes via Theorem 4.5), which gives the stated counts; and $(1+\varepsilon/(3d))^d \le e^{\varepsilon/3}$, so the root ratio is at most $1+\varepsilon$: products multiply leaf ratios; positive combinations with exact weights do not increase them; and for $f(x) = 1-\prod_j(1-x_j)$, which is monotone and concave along rays with $f(0)=0$, $x' \le bx$ implies $f(x') \le bf(x)$ and $x' \ge ax$ implies $f(x') \ge af(x)$, so (D)-$\vee$ nodes do not increase ratios either. The failure probabilities add up to $\delta/8+\delta/4+\delta/2+\delta/8+\delta = 2\delta$. Costs: each evaluation reads the cone of a node of N, or of a restriction of one; (R8) replaces at most $2|C|$ gates (one per polarity) by OBDDs of $O(2^a)$ nodes, which gives the bound on $|N^\circ|$, and tabulating each costs $O(2^a|C|)$. Flattening copies the children of a shared gate into each parent of the same kind, so a gate of N, or of a restriction, has at most $|N^\circ|$ distinct children, and there are at most $|N^\circ|$ gates; hence $|N| = O(|N^\circ|^2)$, which can be attained. Per tree node, (F) costs $O(n|N|)$ (at most n rounds of unit propagation), (T′) $O(2^a|N|)$, and building Φ_h $O(B|N|)$ besides the pilot evaluations of (R7) (at most B rule applications, s_7 evaluations each). The root's elimination has at most a rows per conjunct, so its probing and elimination cost $O(n^2(a|C|+|N|)+n^3)$. □

With $K = K(|C|)$ and $m = m(|C|)$ fixed polynomials, DFSB therefore beats sampling on $\{C : m_K(C) \le m\}$, which contains circuits with $p = 2^{-\Omega(n)}$. The tree-size condition is sufficient, not necessary: (E)-leaves can stop the tree early, and loose leaves under a rare $\vee$-branch need not spoil tightness. Without (E), on the good event of the proof the actual tree also contains $T^*_{16K}(C)$ as a pruning, so it avoids loose leaves only if $m_{16K}(C) \le m$. Since m_K is nonincreasing in K, one large K suffices, at a cost proportional to K. The class contains:

Corollary 5.3. *Let $s = |C|$.*

(a) Component-dense circuits. *Let the* component tree *of C be the tree built by (T), (T′), (F) and (D) alone. If each of its leaves that is not exact (by (T), (T′) or a refutation) has probability at least $1/K$, then $m_K(C) = O(s)$. This includes read-once $\{\wedge,\vee,\neg\}$-formulas, computed exactly, and read-once compositions $F(\lambda_1,\dots,\lambda_t)$ of non-constant λ_i on disjoint variable sets such that, for each λ_i in the polarity in which it occurs in F (positive if F is monotone) and each set G of literals that λ_i implies, $\lambda_i|_G$ satisfies the hypothesis for the component tree built with (F) restricted to unit propagation (as at every node below the root: probing and elimination act only at the root of C). For instance, λ_i is an atom, or is K-dense and no $\lambda_i|_G$ is (D)-decomposable (forcing does not lower the probability: $D(\lambda_i|_G) = D(\lambda_i)/D(G) \ge D(\lambda_i)$, as $\lambda_i \Rightarrow G$). E.g. an $\wedge$ of t such blocks of probability $1/2$ has $p = 2^{-t}$; it costs no samples if the blocks are atoms, and otherwise $\tilde{O}(Kt^3\varepsilon^{-2}\log(m/\delta))$ evaluations ($d = \ell = t$, plus the pilots).*

(b) Backbones. *If (F) derives a set F with $D(C \mid F) \geq 1/K$ (for instance trigger circuits $R \wedge [x_A = a]$, however the test is written, as long as propagation sees it), then $m_K(C) = 1 + m_K(C|_F)$. If $C|_F$ is an atom or not (D)-decomposable, this is 2; the cost is then 0 samples for an atom and otherwise $O(K\varepsilon^{-2}(\log(1/\delta) + \log\log K) + K\log(m/\delta))$ evaluations (one leaf at $d = 1$, and one pilot), against $2^{|A|}K\varepsilon^{-2}\log(1/\delta)$ for Monte Carlo.*

(c) Cutsets. *Let $\mathrm{Conf}(C)$ be the set of input variables that lie in the supports of two different children of some non-atom gate of the NNF. If the NNF has no opaque gates and $|\mathrm{Conf}(C)| = r$, then $m_K(C) = O(s\,3^r)$ for every K, and with (S) and (E) disabled and a node budget m at least the size of the resulting tree, DFSB is exact cutset conditioning [Pea88]: $O(s\,3^r)$ nodes and no sampling.*

(d) Global variables. *If fixing some set Z of r variables makes C satisfy (a), the variant that first enumerates Z and runs DFSB on each restriction uses $O(2^r s)$ nodes.*

Proof. The formula at a tree node is a restriction of an NNF node, so its support is contained in that node's; and since atoms are never merged into larger parents, two distinct children of a non-atom tree node descend from two different children of a non-atom NNF gate (the gate at which their paths diverge).

(c) Disable (S) and (E). (B) applies only at a non-decomposable non-atom node, on a variable shared by two of its children, which lies in $\mathrm{Conf}(C)$ by the remark above; once $\mathrm{Conf}(C) \cap \mathrm{var}(h)$ is assigned, every non-atom node below h is decomposable, so all leaves are exact ((T), (T′), or 0). For a tree node t let $\sigma_t \in \{0, 1, *\}^{\mathrm{Conf}(C)}$ record the values given to $\mathrm{Conf}(C)$ on the path to t. Call t *terminal* if its formula is not constant but all its children's formulas are. Two distinct nodes with the same σ, each a (B)-node or terminal, are not ancestor and descendant, and their lowest common ancestor is a (D)-node (an (F)-node has one child, and a (B)-node would separate their σ); so their formulas lie in different groups and have disjoint nonempty supports. Hence there are at most s such nodes per σ, $O(s3^r)$ in all. The other nodes are constant leaves (at most two per (B)-node or terminal node), (D)-nodes (fewer than the leaves) and (F)-nodes (each with a non-(F) child, as (F) runs to a fixpoint); so the tree has $O(s3^r)$ nodes, and T_K^*, which stops no later, is a pruning of it.

(a) In T_K^* every leaf of the component tree that is not exact is accepted by (S), so $T_K^*(C)$ is the component tree, which has $O(s)$ nodes by the count in (c) (it has no (B)-nodes, so all σ are empty). In a read-once formula every non-atom gate is decomposable, so all leaves are exact. In a read-once composition the λ_j take both values, independently of each other, so a literal on the variables of λ_i is implied by C only if λ_i, in its polarity, is a top-level conjunct of the NNF of F (the only way a read-once formula fixes one of its variables) and implies that literal. So the root's (F), which derives only literals implied by C, restricts each λ_i by a set G_i of literals that λ_i implies ($G_i = \emptyset$ unless λ_i is such a conjunct). Below the root, (F) is unit propagation, which, like (D), acts on each $\lambda_i|_{G_i}$ as on $\lambda_i|_{G_i}$ alone, as the λ_i share no variables; so the non-exact leaves of the component tree of C are those of the component trees of the $\lambda_i|_{G_i}$ with (F) restricted to unit propagation. (b) and (d) are immediate. □

5.2 Envelope leaves and sound rewriting rules

Nodes that are rare but not decomposable get an envelope from the following rules, each of which produces $\varphi \geq \mathbf{1}_h$ pointwise, so any combination is sound (Theorem 4.3):

(R1) literal $\mapsto$ literal; $\vee(c_1, \ldots, c_k) \mapsto \sum_j \varphi_{c_j}$ (union bound);

(R2) $\wedge$ with forced literals $F \mapsto \prod_{\ell \in F}[\ell] \cdot \varphi_{h|_F}$;

(R3) $\wedge$ with variable-disjoint children $\mapsto \prod_j \varphi_{c_j}$;

(R4) $\wedge$ with a shared variable u (at most a fixed number of times on a path, and only while $|\Phi|$ is below a fixed polynomial budget) $\mapsto \sum_b [u = b] \cdot \varphi_{h|_{u=b}}$ (Shannon expansion);

(R5) $\wedge$ otherwise $\mapsto$ product over a greedy variable-disjoint subset of conjuncts, most informative first (dropping conjuncts);

(R6) anything $\mapsto 1$, used whenever the rules give mass > 1, and for every further node once B rules have been applied (B a fixed polynomial);

(R7) a dense node $\mapsto$ a box (Theorem 4.5), when a pilot of s_7 samples finds hit rate at least τ (s_7 and τ fixed constants; the hit rate is the box's rough mass) and, unless the node is opaque, the other rules give more than twice that mass;

(R8) *small-cone canonicalisation*: every gate whose support has at most a variables is replaced, before anything else, by its reduced OBDD in the order of the variable indices, computed by tabulation; in DFSB these are atoms, exact leaves by (T′). Equal functions have equal reduced OBDDs, which hash-consing identifies, so replacing a sub-circuit computing a gate (whose internal gates feed nothing else) by an equivalent circuit with the same support of at most a variables changes neither the canonicalised NNF nor the gates of C above the atoms nor the atoms' functions, which is all that the root's elimination reads; hence it changes neither the tree of DFSB nor membership in the classes above, as long as a, K and m stay fixed (they depend on $|C|$, which such a rewrite can change); and gadgets such as $(x \oplus r) \oplus r$ become the literal x. Rewritings of cones with more than a variables are not covered: $A \wedge B \mapsto (A \wedge B) \wedge (A \vee B)$ makes the top gate non-decomposable.

Our implementation uses $s_7 = 512$, $\tau = 0.05$, $B = 2 \cdot 10^5$ and at most three Shannon expansions per path.

Proposition 5.4 (DNNF envelopes)**.** *If the NNF of C (after (R8)) is a DNNF, rules (R1)–(R3) produce its own arithmetic circuit φ, whose positivity set is $C^{-1}(1)$; so, unless the budget B runs out, $K \leq \mathbb{E}_D[\varphi \mid C = 1]$, its expected ambiguity, with equality unless (R6) or (R7) replaces a sub-envelope by one of no larger mass. For a DNF with m terms (contradictory terms and repeated literals are removed by the simplification of Section 5.1), $K \leq m$ (Karp–Luby–Madras). Under product distributions with rational marginals, the FPRAS of [MdC26] with the sampler of Theorem 6.4 removes the dependence on the ambiguity (its looseness is $D(E)/p = 1$), in time* $\mathrm{poly}(|C|, n, \max_i \log b_i, 1/\varepsilon, \log(1/\delta))$ *with b_i the denominators of the marginals. So all DNNFs admit polynomial-time relative-error estimation, and so do all circuits that (R1)–(R6) and (R8) turn into a DNNF envelope whose function E has $D(E) \leq Kp$ with $K \leq \mathrm{poly}(|C|)$.*

5.3 Context width: compiling any circuit into a DNNF

DFSB branches on shared variables but treats overlapping $\vee$'s by envelopes. With the #DNNF FPRAS available, non-deterministic $\vee$'s cost nothing, and the only obstacle is $\wedge$-gates whose children share variables. The following parameter measures exactly that, and generalises Darwiche's dtree conditioning [Dar01] from CNFs to arbitrary circuits (a binary $\wedge$-tree over the clauses of a CNF is a dtree). Like the width of a dtree, it is a parameter of how the $\wedge$'s are bracketed, not of the function.

Definition 5.5 (context width)**.** Let N be the NNF of C (Section 2) built gate by gate from the output: an $\wedge$- or $\vee$-gate of C gives one gate for each polarity in which it occurs below the output, with the same children (by De Morgan), $\oplus$- and threshold gates are expanded as described below, nested gates are not merged, distinct gates of C stay distinct even when they have the same children (identifying them can raise cw), and the children of a gate form a set ($h \wedge h = h$). So N consists of the gates reachable from its root; $|N|$ is its number of wires. For an $\wedge$-gate g let $S_g = \bigcup_{a \neq b \text{ children of } g} \mathrm{var}(a) \cap \mathrm{var}(b)$ be its overlap. For a gate h let $\mathrm{Up}(h)$ be the set of $\wedge$-gates from which h is reachable (including h), and define the *context* $\kappa(h) = \mathrm{var}(h) \cap \bigcup_{g \in \mathrm{Up}(h)} S_g$. The context width is $\mathrm{cw}(N) = \max_h |\kappa(h)|$.

For $\{\wedge, \vee, \neg\}$-circuits $|N| \leq 2|C|$ (one copy of each gate per polarity). A fan-in-k $\oplus$-gate becomes a chain with $12(k-1)$ wires for both polarities, so $|N| \leq 12|C|$ when $\oplus$-gates are present, and a threshold gate of fan-in k whose weights w_i, scaled to integers, have total $W = \sum_i |w_i|$ becomes a sequential counter with $O(kW)$ wires (a negative weight acts on the complemented input, $w_i h = |w_i| \neg h - |w_i|$); threshold gates with large W are covered only as frontier gates (see below). Merging an $\vee$-gate into an $\vee$-parent does not increase cw ($\vee$-gates have no overlap, and the sets $\mathrm{Up}(h)$ do not change), but merging $\wedge$-gates can: for $2 \leq w \leq n-2$, a CNF with clauses on the $n - w + 1$ sliding windows of width w of its variable order has $\mathrm{cw} = w$ as a left-deep $\wedge$-chain in window order (for every w, $\mathrm{cw} \leq w$), and $\mathrm{cw} = n - 2$ as a single $\wedge$-gate over its clauses. We write $\mathrm{cw}^*(C)$ for the minimum of cw over the NNFs obtained from N by merging $\wedge$-gates of fan-out 1 into their $\wedge$-parents and replacing $\wedge$-gates by binary trees of $\wedge$-gates over the same children (which at most doubles $|N|$); any bracketing an algorithm chooses, e.g. that of Lemma 5.7, bounds it.

Theorem 5.6 (context-width compilation)**.** *Let N be an NNF of C as above and $\mathrm{cw} = \mathrm{cw}(N)$. Then C is equivalent to a DNNF with at most $(|N|+1)(\mathrm{cw}+3)2^{\mathrm{cw}}$ wires, computable in time $O(|N|(n + (\mathrm{cw}+1)2^{\mathrm{cw}}))$. $\mathrm{cw}(N) = 0$ iff N is a DNNF. If every $\vee$-gate of N is deterministic (those produced by the $\oplus$- and threshold expansions are; those produced from $\wedge$-gates of C under a negation need not be), the result is a d-DNNF, and p is computed exactly in time linear in its size under every product distribution. Consequently (via Theorem 6.4(a) and [MdC26]) p has an FPRAS under every product distribution with rational marginals $q_i = a_i/b_i$, running in time $\mathrm{poly}(|N|2^{\mathrm{cw}}, n, \max_i \log b_i, 1/\varepsilon, \log(1/\delta))$; polynomial when $\mathrm{cw} = O(\log |N|)$.*

Proof. Two facts. (F1) If h is a child of g then $\mathrm{Up}(h) \supseteq \mathrm{Up}(g)$, so $\kappa(g) \cap \mathrm{var}(h) \subseteq \kappa(h)$. (F2) If h_i, h_j are distinct children of an $\wedge$-gate g then $\mathrm{var}(h_i) \cap \mathrm{var}(h_j) \subseteq S_g \cap \mathrm{var}(g) \subseteq \kappa(g)$. Nodes of the DNNF: for each gate h and $\sigma \in \{0,1\}^{\kappa(h)}$ a node $N[h, \sigma]$ for $h|_\sigma$; for each wire $g \to h$ and each assignment π to $\kappa(g) \cap \mathrm{var}(h)$ a node $M[h, \pi] = \bigvee_\tau (\mathrm{lits}(\tau) \wedge N[h, \pi \cup \tau])$, τ ranging over assignments to $\kappa(h) \setminus \mathrm{dom}\,\pi$ (a deterministic Shannon expansion; $\pi \cup \tau$ covers $\kappa(h)$ by F1). Set $N[g, \sigma] = \mathrm{op}_j M[h_j, \sigma|_{\kappa(g) \cap \mathrm{var}(h_j)}]$ for a gate g with children h_j, literal leaves become constants when their variable is in $\kappa(h)$, and the root is $M[\mathrm{root}, \emptyset]$. Correctness is induction on $g|_\sigma = \mathrm{op}_j h_j|_\sigma$. Decomposability: inside M, the literals of τ live on $\kappa(h) \setminus \mathrm{dom}\,\pi$ and $N[h, \cdot]$ on $\mathrm{var}(h) \setminus \kappa(h)$; at an $\wedge$-gate the children's variables lie in $\mathrm{var}(h_j) \setminus \kappa(g)$, pairwise disjoint by F2. Size: a gate h gives $2^{|\kappa(h)|}$ nodes $N[h, \cdot]$ with $\mathrm{fanin}(h)$ wires each, at most $|N|2^{\mathrm{cw}}$ wires together. A wire $g \to h$ gives $2^{|\mathrm{dom}\,\pi|}$ nodes $M[h, \pi]$ with at most $(t+2)2^t$ wires each, $t = |\kappa(h)| - |\mathrm{dom}\,\pi|$, so at most $(\mathrm{cw}+2)2^{\mathrm{cw}}$; the root adds as much, for at most $(|N|+1)(\mathrm{cw}+3)2^{\mathrm{cw}}$ wires in all. Time: supports, overlaps and contexts take $O(n)$ bit operations per wire of N; each child of a node $N[g, \sigma]$ and each of the 2^t terms of a node $M[h, \pi]$ is built in time $O(\mathrm{cw} + 1)$. Since $S_g \subseteq \kappa(g)$, $\mathrm{cw} = 0$ iff every S_g is empty. The d-DNNF claim holds because conditioning preserves disjointness and the M-nodes are deterministic. □

Lemma 5.7 (bracketing along a path decomposition)**.** *Let F be a CNF whose primal graph (variables adjacent when they share a clause) has a path decomposition $B_1, \dots, B_r$ of width k. Let $\ell(c)$ be the index of the first bag containing* $\mathrm{var}(c)$ *(one exists since* $\mathrm{var}(c)$ *is a clique), order the clauses as $c_1, \dots, c_m$ with $\ell(c_1) \le \cdots \le \ell(c_m)$, and bracket F as the left-deep chain $a_1 = c_1$, $a_j = a_{j-1} \wedge c_j$. Then $\kappa(a_j)$ and $\kappa(c_j)$ lie in $B_{\ell(c_j)}$, so* $\mathrm{cw} \le k+1$*. The same bound holds for an $\vee$ of such CNFs whose chains share no $\wedge$-gate. For constant k, a path decomposition of width k, if one exists, is found in linear time [Bod96, BK96], and the order from it in linear time.*

Proof. Write $\ell_j = \ell(c_j)$. The bags containing a variable form an interval, so if $v \in \mathrm{var}(c_i) \cap \mathrm{var}(c_{i'})$ with $i \le j \le i'$ then $v \in B_{\ell_i} \cap B_{\ell_{i'}} \subseteq B_{\ell_j}$. The $\wedge$-gates in $\mathrm{Up}(a_j)$ are among $a_j, \dots, a_m$, and $S_{a_{j'}} \subseteq \mathrm{var}(c_{j'})$. So $v \in \kappa(a_j)$ means $v \in \mathrm{var}(c_i)$ for some $i \le j$ and $v \in \mathrm{var}(c_{j'})$ for some $j' \ge j$, hence $v \in B_{\ell_j}$. Further $\kappa(c_j) \subseteq \mathrm{var}(c_j) \subseteq B_{\ell_j}$, a literal has $|\kappa| \le 1$, and bags have at most $k+1$ elements. In an $\vee$ of such CNFs, $\vee$-gates add no overlap and $\mathrm{Up}(a_j)$ lies in the chain of a_j; a clause or literal shared between chains still has κ inside its own variables. For the running time: clauses have at most $k+1$ variables, so the primal graph has $O(k|F|)$ edges, and $\ell(c)$ is the largest index of a first bag of a variable of c. □

Example 5.8 ($\vee$'s are easy, $\wedge$'s are hard)**.** Let $C = \bigvee_{i \le t} F_i$ where each F_i is a CNF with clauses on sliding windows of width w along its own variable order, bracketed as a left-deep $\wedge$-chain in window order. The windows are a path decomposition of width $w-1$ of each F_i, so $\mathrm{cw} \le w$ by Lemma 5.7, and an FPRAS exists when $w = O(\log n)$, while p can be $2^{-\Theta(n)}$. Exact computation is #P-hard already for $\mathrm{cw} = 0$, for ORs of at most 5 *read-once* CNFs: greedily edge-colour a graph of maximum degree 3 with 5 colours, write its vertex-cover CNF as $\bigwedge_{c \le 5} \varphi_c$ with each φ_c a matching (hence read-once) CNF, and use $\#\bigwedge_c \varphi_c = \sum_{\emptyset \ne S \subseteq [5]} (-1)^{|S|+1} \# \bigvee_{c \in S} \varphi_c$; counting vertex covers (equivalently, independent sets) is #P-complete in graphs of maximum degree 3 [Gre00].

The same construction works with a *frontier* of gates that are compiled otherwise: gates of small support by tabulation (as the implementation does), and gates whose cone is a circuit of treewidth k into d-SDNNFs of size $2^{O(k)}$ times that of the cone [ACMS20]. Context width is then computed with frontier gates (one per polarity in which they occur) as leaves; a frontier gate f compiled to a DNNF D_f gives the nodes $N[f, \sigma] = D_f|_\sigma$, so the size becomes $\sum_f 2^{|\kappa(f)|} |D_f|$ plus the bound of Theorem 5.6 for the remaining gates, and the result is a d-DNNF if, in addition to the hypothesis of Theorem 5.6, every D_f is one.

5.4 Modules and quotients

A gate g in the cone of the output is a *module* if every path from an input in $\mathrm{var}(g)$ to the output passes through g. Then $C(x) = Q_g(g(x_{\mathrm{var}(g)}), x_{\mathrm{rest}})$ for the *quotient* Q_g obtained by replacing g with a fresh input, x_{rest} being the inputs outside $\mathrm{var}(g)$, and under a product distribution $g(x)$ is independent of x_{rest}. The supports of modules form a laminar family: if $v \in \mathrm{var}(g) \cap \mathrm{var}(h)$, a path from x_v through g to the output passes through h, so one of g, h is in the cone of the other and their supports are nested. Modules are found in linear time [DR96]. For a module g, choose one module for each support that is maximal among the module supports properly inside $\mathrm{var}(g)$. By laminarity the chosen $g_1, \dots, g_k$ have disjoint supports; every path from $\mathrm{var}(g_j)$ to g passes through g_j (extended to the output it meets g_j, and g_j is not above g since $\mathrm{var}(g_j) \subsetneq \mathrm{var}(g)$); so $g = R_g(g_1, \dots, g_k, x')$, where x' are the remaining inputs of $\mathrm{var}(g)$ and R_g is obtained from the cone of g by replacing each g_j with a fresh input. This is the module tree.

Lemma 5.9 (sensitivity of quotients)**.** *Let* $F(q) = \Pr_{y\sim\bigotimes_j \mathrm{Bern}(q_j)}[Q(y) = 1]$ *and* $q, \hat{q} \in [0,1]^k$*. If* $\hat{q}_j \in e^{\pm\xi_j} q_j$ *and* $1-\hat{q}_j \in e^{\pm\xi_j}(1-q_j)$ *for all* j*, then* $F(\hat{q}) \in e^{\pm\sum_j \xi_j} F(q)$*. If* Q *is monotone (resp. antitone) in* y_j*, the condition on* q_j *(resp.* $1-q_j$*) alone suffices for that coordinate. If* $q_j \in [L_j, U_j]$ *with* $U_j \le \frac{1}{2}$*, then* $\hat{q}_j = \sqrt{L_j U_j}$ *satisfies both conditions with* $\xi_j = \frac{1}{2}\log(U_j/L_j)$ *(and symmetrically, for* $1-q_j$*, if* $L_j \ge \frac{1}{2}$*).*

Proof. F is multilinear: $F = q_j A + (1-q_j)B$, with $A, B \ge 0$ when the other coordinates lie in $[0,1]$, so scaling either term by a factor in $e^{\pm\xi_j}$ scales F within $e^{\pm\xi_j}$. If Q is monotone in y_j then $A \ge B$, and for $c \ge 1$ with $cq_j \le 1$, $F \le F(cq_j) = F + (c-1)q_j(A-B) \le F + (c-1)q_j A \le cF$; symmetrically for $c \le 1$. Change one coordinate at a time; every intermediate point lies in $[0,1]^k$. For the last claim let $r = \sqrt{U_j/L_j}$, so $\hat{q}_j = rL_j = U_j/r$ and $\hat{q}_j/q_j \in [1/r, r]$. The ratio $(1-\hat{q}_j)/(1-q_j)$ lies between $(1-\hat{q}_j)/(1-L_j)$ and $(1-\hat{q}_j)/(1-U_j)$, both in $[1/r, r]$ because $r(1-\hat{q}_j) - (1-L_j) = (r-1)(1-L_j-\hat{q}_j) \ge 0$ and $r(1-U_j) - (1-\hat{q}_j) = (r-1)(1-U_j-\hat{q}_j) \ge 0$ when $U_j \le \frac{1}{2}$. □

Let g be a module with submodules $g_1, \ldots, g_k$ as above, let $q_j = D(g_j) \in [L_j, U_j] \subseteq [0,1]$, and let Q be R_g or $\neg R_g$, according to whether $D(g)$ or $D(\neg g)$ is wanted. For any $\hat{q}_j \in [0,1]$, the least ξ_j for which the conditions of Lemma 5.9 hold for every $q_j \in [L_j, U_j]$ is the largest of $\log\frac{U_j}{\hat{q}_j}$, $\log\frac{\hat{q}_j}{L_j}$ (omitted if Q is antitone in y_j), $\log\frac{1-L_j}{1-\hat{q}_j}$ and $\log\frac{1-\hat{q}_j}{1-U_j}$ (omitted if Q is monotone in y_j), reading $\log\frac{0}{0}$ as 0; we use it, rounded upward, for the $\hat{q}_j$ actually computed, and if it is infinite the bound below is $[0,1]$. If $L_j = U_j$, take $\hat{q}_j = L_j$, so that $\xi_j = 0$. Otherwise three rules choose $\hat{q}_j$: $\sqrt{L_j U_j}$ if $U_j \le \frac{1}{2}$ or Q is monotone in y_j, for which $\xi_j = \frac{1}{2}\log(U_j/L_j)$ by Lemma 5.9; symmetrically $1 - \sqrt{(1-L_j)(1-U_j)}$ if $L_j \ge \frac{1}{2}$ or Q is antitone in y_j (if both rules apply, the one with the smaller ξ_j); otherwise $\sqrt{L_j U_j}$, for which $\xi_j = \max\{\frac{1}{2}\log\frac{U_j}{L_j}, \log\frac{1-L_j}{1-\hat{q}_j}, \log\frac{1-\hat{q}_j}{1-U_j}\}$ (all up to the rounding of $\hat{q}_j$). By Lemma 5.9, an interval $[L_F, U_F]$ for $\Pr[Q=1]$ under $\bigotimes_j \mathrm{Bern}(\hat{q}_j)$ and D on x', from any sound method (exact for small quotients, the threshold dynamic program, DFSB, Monte Carlo), gives $\Pr_D[Q(g_1, \ldots, g_k, x') = 1] \in [e^{-\sum_j \xi_j} L_F, \min\{1, e^{\sum_j \xi_j} U_F\}]$; the clipping at 1 keeps every interval inside $[0,1]$, which the rules need (else $\hat{q}_j$ could leave $[0,1]$). Walking the module tree bottom-up and making each of the M quotient estimates at confidence δ/M gives an interval for p that is valid with probability $1-\delta$ (union bound; each estimate is valid conditionally on the earlier ones).

Which polarity of a submodule is rare is not known before it is estimated, so every module g below the output is estimated in both polarities, with $Q = R_g$ and with $Q = \neg R_g$ (two of the M estimates, from the same submodule intervals), and for each input the parent uses whichever of the two intervals for q_j, or their intersection, gives the smallest ξ_j. For completeness, call a polarity *used*: the positive one of the output and, for a module used with quotient Q and each j, the positive (negative) polarity of g_j if Q is monotone (antitone) in y_j, and otherwise the rarer one, of probability at most $\frac{1}{2}$. Let ρ_j be the log-ratio of the interval for g_j in its used polarity. If Q is monotone or antitone in y_j, or that interval lies in $[0, \frac{1}{2}]$, the first two rules give $2\xi_j \le \rho_j$; otherwise that interval contains $\frac{1}{2}$ and the rules give $2\xi_j \le (1 + O(\rho_j))\rho_j$. So the log-ratio $\log(U/L)$ of the final interval is at most the sum S of the log-ratios of the quotient estimates in used polarities when the third rule is never needed, and at most $e^{O(dS)}S$ in general, for dS below a constant, d the depth of the module tree. Completeness thus needs only the quotient estimates in used polarities to have small relative error. Thresholds, majorities and other tractable functions of independent sub-circuits are the typical case.

5.5 The presumption of independence, certified

Write $C = \bigwedge_{i=1}^{m} A_i$, where $A_1, \dots, A_m$ are the distinct nodes of C (gates, inputs or constants) that are not $\wedge$-gates and are reached from the output through $\wedge$-gates only (the output itself if it is not an $\wedge$-gate); as $\wedge$ is idempotent, repeats are dropped, so $m \le |C|$. The A_i are arbitrary sub-circuits, with supports S_i; let $B_i = \neg A_i$, $b_i = D(B_i)$, $\mu = \sum_i b_i$, and G the dependency graph ($i \sim j$ iff $i \neq j$ and $S_i \cap S_j \neq \emptyset$). The presumption of independence estimates p by $M = \prod_i (1 - b_i)$.

Call C *unate* if the following top-down propagation of polarities from the output reaches no input variable with both polarities: $\wedge$ and $\vee$ pass on the polarity they receive and $\neg$ flips it; a $\oplus$-gate passes both polarities to each input; a threshold gate passes its polarity to the inputs of nonnegative weight and the flipped one to the inputs of negative weight (a negative weight counts as a negation), whatever the size of the weights; any other gate disqualifies C. This is a syntactic check: one top-down pass computes the set of polarities with which each gate is reached, and together with the supports S_i it takes $O(|C|n)$ time. If C is unate, flip the variables that occur positively ($x_v \mapsto 1 - x_v$, $q_v \mapsto 1 - q_v$). By induction from the inputs, every gate reached only positively (negatively) now computes a decreasing (increasing) function, and a gate reached with both polarities has no variable in its cone; so every A_i is decreasing and every B_i increasing, and D is still a product distribution. (Harris's inequality needs only that all B_i be monotone in the same direction.)

Theorem 5.10 (certified independence, first order)**.** *Let C be unate with $\max_i b_i < 1$ (if some $b_i = 1$ then $p = M = 0$). For any ordering of the events and $d_i = \sum_{j<i,\, j\sim i} D(B_i \wedge B_j)$,*

$$M \le p \le \prod_i (1 - b_i + d_i) \le M \exp\Big(\sum_i \frac{d_i}{1 - b_i}\Big) \le M \exp\Big(\frac{\Delta}{1 - \max_i b_i}\Big),$$

where $\Delta = \sum_i d_i = \sum_{\{i,j\}:\, i\sim j} D(B_i \wedge B_j)$ is a sum over unordered dependent pairs (the convention throughout). Let $0 < \varepsilon \le 1$, $\varepsilon' = \ln(1 + \varepsilon) \ge \varepsilon \ln 2$ and $0 < \delta < 1$. The following procedure is sound on every circuit:

(1) if C is not unate, output $[0, 1]$;

(2) pilot: *draw $N_0 = \lceil 8 \ln(4/\delta) \rceil$ samples of x_{S_i} for each i; let $\hat{\mu}_0$ be the sum over i of the empirical frequencies of B_i;*

(3) pre-test on Δ: *for each edge $\{i, j\}$ of G draw $N_\Delta = \lceil 26 \ln(4/\delta)/\varepsilon' \rceil$ samples of $x_{S_i \cup S_j}$; let $\hat{\Delta}$ be the sum over edges of the empirical frequencies of $B_i \wedge B_j$; if $\hat{\Delta} > \varepsilon'/16$, output $[0, 1]$;*

(4) main stage and pre-test on b: *in each of $R = \lceil 8 \ln(4/\delta) \rceil$ rounds r draw $N = \lceil 560(2\hat{\mu}_0 + 1)/\varepsilon'^2 + 72 \ln(4mR/\delta) \rceil$ fresh samples of x_{S_i} for each i, giving frequencies $\hat{b}_i^{(r)}$; if some $\hat{b}_i^{(r)} > 1/3$, output $[0, 1]$; otherwise let $\hat{M} = \exp\big(\mathrm{median}_r \sum_i \log(1 - \hat{b}_i^{(r)})\big)$ and output $[\hat{M} e^{-\varepsilon'/4}, \min\{1, \hat{M} e^{3\varepsilon'/4}\}]$.*

It is complete, with probability $1 - \delta/2$, on unate circuits with $\max_i b_i \le 1/4$ and $\Delta \le \varepsilon'/64$. Its expected cost is $O\big(\sum_i |A_i| \big[((\mu+1)\varepsilon^{-2} + \log(m/\delta)) \log(1/\delta) + \deg_G(i)\, \varepsilon^{-1} \log(1/\delta)\big]\big)$ gate operations after the syntactic checks. As $\mu \le \log(1/p) + \Delta/(1 - \max_i b_i)$, this is linear in μ, hence in $\log(1/p)$ on the complete class, with no dependence on $1/p$ (which can be $e^{\Theta(\mu)}$).

Proof. Sandwich. Flip as above. Write $p = \prod_i (1 - c_i)$ with $c_i = \Pr[B_i \mid F_{<i}]$, $F_X = \bigwedge_{j \in X} \neg B_j$ (by Harris $\Pr[F_{<i}] \ge \prod_{j<i} (1 - b_j) > 0$). B_i is increasing and $F_{<i}$ decreasing, so $c_i \le b_i$ (Harris),

giving $p \geq M$. Split the earlier events into neighbours N and non-neighbours R of i (Boppana–Spencer [BS89]): B_i is independent of F_R (disjoint supports), so $c_i \geq \Pr[B_i F_N \mid F_R] \geq b_i - \sum_{j\in N} \Pr[B_i B_j \mid F_R] \geq b_i - d_i$, the last step by Harris ($B_i B_j$ increasing, F_R decreasing). Then $1 - b_i + d_i \leq (1-b_i)e^{d_i/(1-b_i)}$. (If $b_i = 0$ then $c_i = d_i = 0$ and the i-th factors are 1.)

Procedure. $\hat{M} \leq 1$, so the output lies in $[0,1]$, and when it is not $[0,1]$ its ratio is at most $e^{\varepsilon'} = 1 + \varepsilon$. Four failure events have probability at most $\delta/4$ each. (a) *Pilot.* $N_0\hat{\mu}_0$ is a sum of independent Bernoulli variables with mean $N_0\mu$. If $N_0\mu \geq 8\ln(4/\delta)$, Chernoff gives $\Pr[\hat{\mu}_0 \leq \mu/2] \leq e^{-N_0\mu/8} \leq \delta/4$; otherwise $\mu < 1$. So $\mu \leq 2\hat{\mu}_0 + 1$, and hence $N \geq 560\mu/\varepsilon'^2$, except with probability $\delta/4$. (b) *Pre-test on* Δ. $N_\Delta\hat{\Delta}$ is a sum of independent Bernoulli variables with mean $\lambda = N_\Delta\Delta$. If $\Delta > \varepsilon'/4$, then $\Pr[\hat{\Delta} \leq \varepsilon'/16] \leq \Pr[N_\Delta\hat{\Delta} \leq \lambda/4] \leq e^{-9\lambda/32} \leq e^{-9N_\Delta\varepsilon'/128} \leq \delta/4$. If $\Delta \leq \varepsilon'/64$, then with $\lambda' = N_\Delta\varepsilon'/64 \geq \lambda$, $\Pr[\hat{\Delta} > \varepsilon'/16] \leq e^{-\lambda'}(e/4)^{4\lambda'} = e^{-(1+4\ln(4/e))\lambda'} \leq \delta/4$, as $26(1 + 4\ln(4/e)) > 64$. (c) *Pre-test on* b. If some $b_i > 1/2$, Hoeffding gives $\Pr[\hat{b}_i^{(1)} \leq 1/3] \leq e^{-N/18} \leq \delta/4$. If $\max_i b_i \leq 1/4$, the probability that some $\hat{b}_i^{(r)} > 1/3$ is at most $mRe^{-N/72} \leq \delta/4$. (d) *Main stage.* Let $\max_i b_i \leq 1/2$ and $N \geq 560\mu/\varepsilon'^2$; fix a round, and let F be the event that all its $\hat{b}_i \leq 1/3$. With $Y_i = \hat{b}_i - b_i$, $\Lambda = -\sum_i Y_i/(1-b_i)$ and $Q = \sum_i Y_i^2$ we have $\operatorname{Var}\Lambda = \sum_i b_i/(N(1-b_i)) \leq 2\mu/N$ and $\mathbb{E}Q \leq \mu/N$. On F, Taylor's theorem with $\hat{b}_i, b_i \leq 1/2$ gives $|\log(1-\hat{b}_i) - \log(1-b_i) + Y_i/(1-b_i)| \leq 2Y_i^2$, so $|\sum_i \log(1-\hat{b}_i) - \log M| \leq |\Lambda| + 2Q$, and by Chebyshev and Markov (with $\varepsilon' \leq \ln 2$)

$$\Pr\Big[F \wedge \Big|\sum_i \log(1-\hat{b}_i) - \log M\Big| > \frac{\varepsilon'}{4}\Big] \leq \frac{2\mu/N}{(\varepsilon'/8)^2} + \frac{2\mu/N}{\varepsilon'/8} \leq \frac{140\,\mu}{N\varepsilon'^2} \leq \frac{1}{4}.$$

If every round passes the pre-test and the median is off by more than $\varepsilon'/4$, at least $R/2$ of the independent rounds have this property, which has probability at most $e^{-R/8} \leq \delta/4$. Outside these events, a non-trivial output has $\max_i b_i \leq 1/2$, $\Delta \leq \varepsilon'/4$ and $|\log\hat{M} - \log M| \leq \varepsilon'/4$, so by the sandwich $\hat{M}e^{-\varepsilon'/4} \leq M \leq p \leq Me^{2\Delta} \leq \hat{M}e^{3\varepsilon'/4}$, and $p \leq 1$. On the complete class only (b) and (c) can produce $[0,1]$. Cost: $\mathbb{E}\hat{\mu}_0 = \mu$, so $\mathbb{E}N \leq 560(2\mu+1)/\varepsilon'^2 + 72\ln(4mR/\delta) + 1$; a sample of x_{S_i} costs $O(|A_i|)$, and the pre-test on Δ uses N_Δ samples per edge. Finally $\mu \leq -\log M \leq \log(1/p) + \Delta/(1 - \max_i b_i)$. The constants are not optimised. □

For a unate CNF all quantities are products of marginals and the sandwich is deterministic (Theorem 4.7). For triangle-freeness of $G(N,q)$ (B_i: the i-th triangle is present) at $N = 10^4$ and $\mu = 10$, $\Delta = 0.023$ and the window is $[M, Me^{0.023}]$. The bound needs only *syntactic* structure (unateness, supports); semantic monotonicity or independence is coNP-hard to check and invisible to sampling.

Second order. Correcting by covariances, $M_2 = \prod_i(1-b_i+C_i)$ with $C_i = \sum_{j<i, j\sim i} \operatorname{Cov}(B_i, B_j)$, the same Harris/Boppana–Spencer argument together with Bonferroni inequalities gives, when $\max_i b_i < 1$ and every $\beta_i = \sum_{j<i, j\sim i} b_j < 1$,

$$|\log p - \log M_2| \leq E_3 = \sum_i \frac{\Theta_i + \Theta_i' + \beta_i C_i}{(1-\beta_i)(1-b_i)},$$

where $\Theta_i' = \sum D(B_iB_jB_k)$ and $\Theta_i = b_i \sum D(B_jB_k)$, both over the triples $\{i,j,k\}$ with $j,k < i$ that are connected in G (Theorem A.1). So $\sum_i \Theta_i'$ is the total probability of connected triples, whatever the order. On triangle-freeness of $G(10^4, q)$ with $\mu = 100$ the first-order window is $e^{1.07}$, while $E_3 \leq 0.0184$ for every order, so $\log p = -98.93 \pm 0.02$. Here the symmetric local lemma holds ($e\,b\,(d+1) \approx 5 \cdot 10^{-5}$, $d = 3(N-3)$ the maximum degree of G), but the counting

conditions of the local-lemma module of Section 5.9, which with b in place of 2^{-k} allow each variable to lie in about $b^{-1/4.82} \approx 80$ events, fail (each edge lies in $N-2$ triangles), and so does the condition of Proposition 5.12 ($r = N-2$). All numbers in this section are computed by `experiments/check_independence_numbers.py`. For principal up-sets Mousset, Noever, Panagiotou and Samotij [MNPS20] give a full hierarchy of such bounds in terms of joint cumulants; the statement for arbitrary monotone sub-circuits and its proof are in Appendix A.

Proposition 5.11 (pairwise data is not enough)**.** *Let all variables be Bernoulli(q), $q \le 1/2$. The "stars" $B_i = y \wedge z_i$ and the "triangle" $B_1 = z_1 z_2, B_2 = z_2 z_3, B_3 = z_3 z_1$ have the same dependency graph, the same $b_i = q^2$ and the same $D(B_i \wedge B_j) = q^3$; taking M' disjoint copies, $\log(p_{\text{stars}}/p_{\text{triangles}}) = M' \log\left(1 + \frac{q^3(1-q)}{1-3q^2+2q^3}\right) \ge 0.29\,\Delta$. So no certificate computed from single and pairwise probabilities can have a window narrower than $e^{0.29\Delta}$: the window $e^{\Delta/(1-\max_i b_i)}$ of Theorem 5.10 is optimal up to the constant in the exponent.*

Proof. $p_{\text{stars}} = (1-q) + q(1-q)^3$ and $p_{\text{triangles}} = (1-q)^3 + 3q(1-q)^2 = 1 - 3q^2 + 2q^3$; their difference is $q^3(1-q)$. Each copy contributes $3q^3$ to Δ, and 0.29 is the minimum over $q \in (0, 1/2]$ of $\log(1 + q^3(1-q)/(1-3q^2+2q^3))/(3q^3)$, which is 0.2939, attained at $q = 0.279$. □

Beyond pairs. Two further certified routes need no monotonicity. (i) *Interface dynamic programming.* If Z contains every variable shared by two of the A_i, then

$$p = \sum_{z \in \{0,1\}^Z} D_Z(z) \prod_i a_i(z_{I_i}), \qquad a_i(y) = \Pr[A_i \mid x_{I_i} = y],\ I_i = S_i \cap Z,$$

computed by variable elimination along a tree decomposition of the interface hypergraph $\{I_i\}$ with Monte Carlo (or recursively estimated) factor tables. The expression is a polynomial with nonnegative coefficients in the $\sum_i 2^{|I_i|}$ table entries, so interval arithmetic on confidence intervals for all entries (with a union bound) is sound for every circuit. If every entry's interval has ratio at most $e^{\varepsilon'/m}$, the output's ratio is at most $e^{\varepsilon'} = 1+\varepsilon$; an entry then costs $O(m^2\varepsilon^{-2}\log(\sum_i 2^{|I_i|}/\delta)/a_i(y))$ samples, so the route is complete and efficient when all entries are at least $1/\mathrm{poly}(|C|)$ and $\max_i |I_i|$ and the width of the decomposition are $O(\log|C|)$. It handles strong, extensive correlations: on a chain of 300 modules with $p \approx 3 \cdot 10^{-115}$, where independence is off by e^{64}, the certified interval for $\log p$ (at $\delta = 0.05$) has width 7.3 with 10^5 samples per table entry and 0.073 with 10^9. This is DFSB's cutset rule organised by a tree decomposition. (ii) *The Möbius expansion.* Let $Z_U = \Pr[\bigwedge_{i \in U} A_i]$ and, when all $Z_U > 0$, $\kappa(T) = \sum_{U \subseteq T} (-1)^{|T \setminus U|} \log Z_U$. Then $\log p = \sum_{T \subseteq [m]} \kappa(T)$, and $\kappa(T) = 0$ unless T is connected in G (the Z_U factorize over the components of T). The terms $\kappa(\{i\}) = \log(1-b_i)$ are the presumption of independence, and the higher ones are log-interaction corrections computed from non-rare probabilities. Let $d \ge 1$ be the maximum degree of G and r the maximum number of supports S_i containing a variable ($r \le d+1$).

Proposition 5.12 (cluster expansion)**.** *Let $B_1, \dots, B_m$ be arbitrary events with supports S_i under a product distribution, with $\max_i b_i \le (2e^3 d)^{-r}$. Then $Z_U \ge e^{-|U|/e}$ for every U, and $\sum_{|T| \ge s} |\kappa(T)| \le m e^{-s}$ for every s. Hence the following is sound on every circuit: certify $\max_i b_i \le (2e^3 d)^{-r}$ by upper confidence bounds (else output $[0,1]$); compute confidence intervals for $\log Z_U$ for all $U \subseteq T$, T connected with $|T| < s = \lceil \ln(4m/\varepsilon') \rceil$; combine them by interval arithmetic and widen by $m e^{-s} \le \varepsilon'/4$. If every $\log Z_U$ is known within $\pm\varepsilon'/(4m(2ed)^s)$, the interval for $\log p$ has width at most ε'. For bounded d this is an FPRAS on the instances with $\max_i b_i \le (2e^3 d)^{-r}/2$, using $(m/\varepsilon)^{O(1+\log d)} \log(1/\delta)$ samples.*

Proof sketch. Expanding $\prod_i(1-\mathbf{1}_{B_i})$, $p=\sum_T(-1)^{|T|}\Pr[\bigwedge_{i\in T}B_i]$, and $\Pr[\bigwedge_T B_i]$ factorizes over the connected components of $G[T]$. So p is the partition function of the polymer model whose polymers are the connected sets γ of G, with weights $w(\gamma)=(-1)^{|\gamma|}\Pr[\bigwedge_{i\in\gamma}B_i]$, two polymers being compatible when they are disjoint and non-adjacent. By Finner's generalized Hölder inequality for product measures [Fin92], as every variable lies in at most r supports, $|w(\gamma)|\le\prod_{i\in\gamma}b_i^{1/r}\le\beta^{|\gamma|}$ with $\beta=(\max_i b_i)^{1/r}\le 1/(2e^3d)$. A polymer incompatible with γ meets one of the at most $(d+1)|\gamma|$ vertices of γ and its neighbourhood, and at most $(ed)^{s-1}$ connected sets of size s contain a given vertex, so

$$\sum_{\gamma'\not\sim\gamma}|w(\gamma')|e^{2|\gamma'|}\le(d+1)|\gamma|\sum_{s\ge1}(ed)^{s-1}(e^2\beta)^s\le(d+1)|\gamma|\,e^2\beta\sum_{s\ge1}2^{1-s}\le\frac{d+1}{ed}|\gamma|\le|\gamma|.$$

This is the Kotecký–Preiss condition [KP86] with $a(\gamma)=|\gamma|$ and decay $|\gamma|$. It gives an absolutely convergent cluster expansion $\log p=\sum_X\Phi(X)$ over clusters X (multisets of polymers with connected incompatibility graph) with $\sum_{X\ni i}|\Phi(X)|e^{\|X\|}\le 1$ for every i, where $X\ni i$ means $i\in\operatorname{supp}X=\bigcup_{\gamma\in X}\gamma$ and $\|X\|=\sum_{\gamma\in X}|\gamma|\ge|\operatorname{supp}X|$. The condition holds for the sub-model on every U, so $Z_U>0$ and $\log Z_U=\sum_{\operatorname{supp}X\subseteq U}\Phi(X)$; by Möbius inversion $\kappa(T)$ is the sum of $\Phi(X)$ over the clusters with support exactly T. Hence $\sum_{|T|\ge s}|\kappa(T)|\le\sum_i\sum_{X\ni i,\|X\|\ge s}|\Phi(X)|\le me^{-s}$ and $|\log Z_U|\le\sum_{i\in U}\sum_{X\ni i}|\Phi(X)|\le|U|/e$. With $s=\lceil\ln(4m/\varepsilon')\rceil$ the truncation error is at most $\varepsilon'/4$; the connected T with $|T|<s$ have at most $m(2ed)^s$ subsets U in all, so the estimation intervals add up to a width of at most $\varepsilon'/2$. Each $Z_U\ge e^{-s/e}$ needs $O(e^{s/e}m^2(2ed)^{2s}\varepsilon^{-2}\log(m(2ed)^s/\delta))$ samples, and $(2ed)^s=(m/\varepsilon)^{O(1+\log d)}$; certifying $\max_i b_i$ costs $O((2e^3d)^r\log(m/\delta))$ samples per event, and $(2e^3d)^r=O(1)$ for bounded d. □

5.6 Certified telescoping with Las Vegas samplers

Annealing estimators write $p=\prod_k r_k$ with $r_k=\Pr_{\pi_{k-1}}[\neg B_k]$ and $\pi_{k-1}=D\mid F_{<k}$, and estimate each ratio by sampling from π_{k-1}. They are sound only if the samples really come from π_{k-1}; Markov chains give no certificate of that. *Exact* Las Vegas samplers do, if their running time is certified empirically:

Proposition 5.13 (certified truncation)**.** *Let $\mathcal{A}$ halt with probability 1 and output exactly π, with running time τ. Truncated at T and conditioned on halting, its output law π_T satisfies $d_{\mathrm{TV}}(\pi_T,\pi)\le\Pr[\tau>T]$. If at most $\eta R/2$ of $R=\lceil 8\ln(1/\delta)/\eta\rceil$ truncated runs fail, then $\Pr[\tau>T]\le\eta$ except with probability δ.*

Proof. $\pi=(1-\theta)\pi_T+\theta\pi_{>T}$ with $\theta=\Pr[\tau>T]$, so $\pi_T-\pi=\theta(\pi_T-\pi_{>T})$. If $\theta>\eta$, then by Chernoff at most $\eta R/2$ failures occur with probability at most $e^{-\eta R/8}\le\delta$. □

Let C be unate (Section 5.5), flipped so that the events B_i are increasing and q_v are the flipped marginals. Then every $F_X=\bigwedge_{i\in X}\neg B_i$ is a down-set, and $D(F_X)\ge\prod_{i\in X}(1-b_i)>0$ by Harris when all $b_i<1$. Fix $F=F_X$ with $D(F)>0$ and let $\phi_v(x)=1$ iff no event B_i, $i\in X$, with $v\in S_i$ occurs at $x^{v\to1}$; ϕ_v is decreasing. The heat-bath Glauber dynamics for $D\mid F$ picks a uniform site v and a uniform $u\in[0,1]$ and sets $x_v\leftarrow[u<q_v]\wedge\phi_v(x)$. It is anti-monotone in the sense of Häggström and Nelander: $\Pr[x_v=1\mid x_{-v}]$ is decreasing in x_{-v}. Their bounding-chain coupling from the past [HN98, PW96] runs, with common randomness, a lower chain L and an upper chain U from $L=0$, $U=\mathbf{1}$ at time $-T$, updating $L_v\leftarrow[u<q_v]\wedge\phi_v(U)$ and $U_v\leftarrow[u<q_v]\wedge\phi_v(L)$. Then $L\le X\le U$ for every chain X started in F, so if $L=U$ at time 0 that state is an exact sample of $D\mid F$; otherwise T is doubled, reusing the randomness of times $-1,\dots,-T$. It halts

almost surely: in any window of n steps, with positive probability the sites with $q_v < 1$ are updated first, each with $u \geq q_v$, and then the sites in $W = \{v : q_v = 1\}$; after the first phase both chains are $\leq \mathbf{1}_W$, and $\mathbf{1}_W \in F$ (as $D(F) > 0$ and F is a down-set), so every update of the second phase sets 1 in both chains, and $L = U = \mathbf{1}_W$ at the end. Disjoint windows are independent.

Theorem 5.14 (certified telescoping)**.** *Let $C = \bigwedge_{i=1}^{m} A_i$ as in Section 5.5, $0 < \varepsilon \leq 1$, $\varepsilon' = \ln(1+\varepsilon)$, $0 < \delta < 1$, $\Lambda_{\max} \geq 1$, and let $T = 2^J T_0$ be a truncation time ($J \geq 0$, $T_0 \geq 1$). For unate C, flipped as above, let $b_i^+ \geq b_i$ be upper bounds: $b_i^+ = b_i$, a product of marginals over the distinct literals of A_i, if C is a CNF; otherwise the upper end of the Hoeffding–KL interval (Section 4) at level $\delta/(3m)$ for b_i from $N_b = \lceil 192\, m \ln(6m/\delta) \rceil$ samples of x_{S_i}.*

(0) If C is not unate, output $[0,1]$; if $m = 0$, output $[1,1]$. Otherwise flip C as above.

(1) If some $b_i^+ = 1$, output $[0,0]$ if C is a CNF and $[0,1]$ otherwise.

(2) Stages: *scan the events in a fixed order, adding each to the current stage unless that brings $\rho_k = \prod_{i \in I_k}(1 - b_i^+)$ below $1/2$, in which case it opens a new stage. Let $I_1, \ldots, I_K$ be the stages, $\rho = \min_k \rho_k$, $\Lambda = \sum_k 1/\rho_k$ and $\eta = \varepsilon' \rho/(16K)$. If $\Lambda > \Lambda_{\max}$, output $[0,1]$. Let $\mathcal{A}_k$ be the sampler above for $\pi_{k-1} = D \mid F_{<k}$, $F_{<k} = F_{I_1 \cup \cdots \cup I_{k-1}}$, with the horizons $T_0, 2T_0, \ldots, 2^J T_0 = T$, and $\theta_k = \Pr[\mathcal{A}_k$ does not halt$]$.*

(3) Certificates: *for each k, run the test of Proposition 5.13 on $\mathcal{A}_k$ at level η, with $\delta/(3K)$ in place of δ; if one fails, output $[0,1]$.*

(4) Estimates: *in each of $R = \lceil 8 \ln(3/\delta) \rceil$ rounds and for each k, make $2N$ runs of $\mathcal{A}_k$, $N = \lceil 256\Lambda/\varepsilon'^2 \rceil$; if fewer than N halt, output $[0,1]$; otherwise let $\hat{r}_k$ be the fraction of the first N halted outputs at which no event of I_k occurs. Output $[\hat{M} e^{-\varepsilon'/2}, \min\{1, \hat{M} e^{\varepsilon'/2}\}]$, with $\hat{M}$ the median over the rounds of $\prod_k \hat{r}_k$.*

This procedure is sound on every circuit, for every T_0, J and $\Lambda_{\max}$; unless it outputs $[0,1]$, its output is exact or has $U \leq (1+\varepsilon)L$. If C is unate, $\Lambda \leq \Lambda_{\max}$ and $\theta_k \leq \eta/16$ for all k, it outputs $[0,1]$ with probability at most $\delta/3 + RKe^{-3N/4}$; the second condition holds when $\gamma = 1 - q_{\max} d_{\max}(k_{\max} - 1) > 0$ and $T \geq (n/\gamma)\ln(16n/\eta)$, where $q_{\max} = \max_v q_v$, every variable lies in at most $d_{\max}$ supports and $k_{\max} = \max_i |S_i|$. It uses mN_b samples of the x_{S_i} if C is not a CNF, $K\lceil 8 \ln(3K/\delta)/\eta \rceil$ runs for the certificates and at most $2RKN$ runs for the estimates, each of fewer than $2T$ updates of two chains. As $K \leq \Lambda \leq \Lambda_{\max}$ and $1/\rho \leq \Lambda$, these are $O(\Lambda_{\max}^3 \log(\Lambda_{\max}/\delta)/\varepsilon + \Lambda_{\max}^2 \log(1/\delta)/\varepsilon^2)$ runs. If $\max_i b_i^+ \leq 1/2$, then $\rho \geq 1/2$, $\Lambda \leq 2K$ and $K \leq 4\mu^+ + 1$ with $\mu^+ = \sum_i b_i^+$, so this is $O(K^2 \log(K/\delta)/\varepsilon + K(\mu^+ + 1)\log(1/\delta)/\varepsilon^2)$ runs, and $T = O((n/\gamma)\log(nK/\varepsilon))$ suffices when $\gamma > 0$. This holds for a CNF with $\max_i b_i \leq 1/2$ and, except with probability $\delta/6$, for any unate C with $\max_i b_i \leq 1/8$; in both cases $\Lambda \leq 24\mu + 3 \leq 24m + 3$, so the cut-off $\Lambda_{\max} = 24m + 3$ costs no completeness there.

Proof. Soundness. In step (0), $p = 1$ if $m = 0$; in step (1) for a CNF, $b_i = 1$ and $p = 0$. Otherwise C is unate and all $b_i^+ < 1$, so $\rho > 0$. Let G be the event that $b_i^+ \geq b_i$ for all i (certain for a CNF, of probability at least $1 - \delta/3$ otherwise) and that every $\mathcal{A}_k$ whose certificate passes has $\theta_k \leq \eta$ (by Proposition 5.13 and a union bound, given the stages, this fails with probability at most $\delta/3$). Assume G and that step (4) reaches its output. By Harris, as the $\neg B_i$ are decreasing, $r_k = D(F_{\leq k})/D(F_{<k}) \geq D(F_{I_k}) \geq \prod_{i \in I_k}(1 - b_i) \geq \rho_k$. The outputs of halted runs of $\mathcal{A}_k$ are i.i.d. from a law π'_{k-1} with $d_{\mathrm{TV}}(\pi'_{k-1}, \pi_{k-1}) \leq \theta_k$ (Proposition 5.13), independently of which runs halt, so the N outputs kept in step (4) are i.i.d. from π'_{k-1}, and $r'_k = \pi'_{k-1}(F_{I_k})$ has $|r'_k - r_k| \leq \eta \leq \rho_k/16$.

Bias: $Z' = \prod_k r'_k$ satisfies $Z'/p \in [(1-\eta/\rho)^K, (1+\eta/\rho)^K] \subseteq [1-\varepsilon'/16, e^{\varepsilon'/16}]$. *Variance:* in a round the $\hat{r}_k$ are independent with $\mathbb{E}\hat{r}_k = r'_k$ and relative variance $(1-r'_k)/(Nr'_k) \le (1-\rho_k+\eta)/((\rho_k-\eta)N) \le 2/(\rho_k N)$, so $\hat{Z} = \prod_k \hat{r}_k$ has $\mathbb{E}\hat{Z} = Z'$ and $\mathbb{E}\hat{Z}^2/Z'^2 \le e^{2\Lambda/N} \le e^{\varepsilon'^2/128} \le 1+\varepsilon'^2/64$; by Chebyshev, $\Pr[|\hat{Z}/Z'-1| > \varepsilon'/4] \le 1/4$. By Hoeffding, at least half of the R independent rounds deviate with probability at most $e^{-R/8} \le \delta/3$; otherwise $\hat{M}/Z' \in [1-\varepsilon'/4, 1+\varepsilon'/4]$. As $\varepsilon' \le \ln 2$, $(1-\varepsilon'/4)(1-\varepsilon'/16) \ge e^{-\varepsilon'/2}$ and $(1+\varepsilon'/4)e^{\varepsilon'/16} \le e^{\varepsilon'/2}$, so $p \in [\hat{M}e^{-\varepsilon'/2}, \hat{M}e^{\varepsilon'/2}]$, and $U/L \le e^{\varepsilon'} = 1+\varepsilon$. The three failure events have total probability at most δ.

Certificates. If $\theta_k \le \eta/16$, the number X of failures among the $R_k = \lceil 8\ln(3K/\delta)/\eta \rceil$ runs has $\mathbb{E}X \le \eta R_k/16$, so by Chernoff $\Pr[X > \eta R_k/2] \le (e/8)^{\eta R_k/2} \le (e/8)^{4\ln(3K/\delta)} \le \delta/(3K)$. In step (4), as $\theta_k \le \eta/16 < 1/16$, more than N of $2N$ runs fail with probability at most $e^{-N(1-2\theta_k)^2} \le e^{-3N/4}$ (Hoeffding), so step (4) outputs $[0,1]$ with probability at most $RKe^{-3N/4}$.

Coalescence. Let D_t be the set of sites where L and U differ. An update of v leaves $L_v \ne U_v$ only if $u < q_v$ and $\phi_v(L) \ne \phi_v(U)$, which needs a site $w \in D_t \setminus \{v\}$ sharing a support with v; each w has at most $d_{\max}(k_{\max}-1)$ such v. So $\mathbb{E}[|D_{t+1}| \mid D_t] \le (1-1/n)|D_t| + (q_{\max}/n)d_{\max}(k_{\max}-1)|D_t| = (1-\gamma/n)|D_t|$, and $\theta_k \le \mathbb{E}|D_0| \le ne^{-\gamma T/n} \le \eta/16$ for the chains started at $-T$. (Coalescence from $-T'$ implies coalescence from $-T \le -T'$: the updates preserve $L' \le L \le U \le U'$ for the pairs (L,U) started at $-T$ and (L',U') started at $-T'$.)

Cost. The horizons $T_0, \dots, 2^J T_0 = T$ cost fewer than $2T$ updates per run. As all $\rho_k \le 1$, $K \le \Lambda$ and $1/\rho \le \Lambda$, so $\eta \ge \varepsilon'/(16\Lambda^2)$. A stage accepts a further event only if ρ_k stays $\ge 1/2$, so an event with $b_i^+ > 1/2$ is a stage of its own and every other stage has $\rho_k \ge 1/2$. If $\max_i b_i^+ \le 1/2$, then $\rho \ge 1/2$ and $\Lambda \le 2K$; with $s_i = -\ln(1-b_i^+) \le 2\ln 2\, b_i^+$ (convexity on $[0,1/2]$), stage $k < K$ was closed because $\sum_{i \in I_k} s_i + s_j > \ln 2$ for the first event j of I_{k+1}, and summing over $k < K$ gives $(K-1)\ln 2 < 2\sum_i s_i \le 4\ln 2\, \mu^+$. In general the same count gives $\Lambda \le 8\mu^+ + 2 + 2\sum_{i:\, b_i^+ > 1/2} (1-b_i^+)^{-1}$.

The bounds b_i^+. If C is not a CNF, let $\hat{b}_i$ be the empirical frequency and $\ell = \ln(6m/\delta)$, so $N_b \mathrm{KL}(\hat{b}_i \| b_i^+) \le \ell$. As $\mathrm{KL}(a\|c) \ge (a-c)^2/(2\max\{a,c\})$, $t = b_i^+ - \hat{b}_i$ satisfies $t^2 \le 2(\hat{b}_i + t)\ell/N_b$; if $t > \hat{b}_i$ this gives $t < 4\ell/N_b \le 1/(48m)$, so $b_i^+ \le 2\hat{b}_i + 1/(16m)$. Chernoff's bound with $a = b_i/2 + 1/(32m)$, for which $b_i + a \le 3a$, gives $\Pr[\hat{b}_i \ge b_i + a] \le e^{-N_b \mathrm{KL}(b_i+a\|b_i)} \le e^{-N_b a/6} \le \delta/(6m)$. So, except with probability $\delta/6$, $b_i^+ \le 3b_i + 1/(8m)$ for all i; then $\max_i b_i \le 1/8$ gives $\max_i b_i^+ \le 1/2$ and $\mu^+ \le 3\mu + 1/8$, and $\Lambda \le 2K \le 8\mu^+ + 2 \le 24\mu + 3$ (for a CNF with $\max_i b_i \le 1/2$, directly $\Lambda \le 8\mu + 2$). □

This covers, for instance, hard-core-type instances where Janson's Δ is extensive and the local-lemma conditions fail; beyond the provable regime the method stays sound and decides empirically whether it can answer. The implementation (`cftp.py`, for unate CNFs) runs the chains of $\mathcal{A}_k$ only on the variables of the events in $F_{<k}$ (the others are independent of $F_{<k}$ and are drawn from their marginals), starts at $T_0 = 4|A|$ for the set A of these variables, and otherwise changes only the bookkeeping: it takes N rather than ε and makes N runs per stage and round, keeping all that halt (so its cost is RKN runs and it needs no $\Lambda_{\max}$); it takes the least odd R for which a majority of R independent rounds, each failing with probability $1/4$, fails with probability at most $\delta/2$; it pools the estimation runs of stage k into a Hoeffding–KL upper bound θ_k^+ on θ_k instead of the test of Proposition 5.13, requires $\theta_k^+ < \rho_k$, and widens the interval by the bias factors $e^{\pm a_k}$, $a_k = \theta_k^+/(\rho_k - \theta_k^+)$ (as $r_k/r'_k \in [1/(1+\theta_k/r_k), 1/(1-\theta_k/r_k)]$), and by a Chebyshev bound computed from the per-stage relative variances $(1-\rho_k+\theta_k^+)/((\rho_k-\theta_k^+)n_k)$, with n_k the least number of halted runs of stage k in a round; and it raises the lower end to the Harris bound $\prod_i (1-b_i)$. The same argument applies, and `telescoping_estimate` states the resulting guarantee, including the effect of floating-point rounding. Karp–Luby–Madras sampling

should extend it to ORs of unate $\wedge$-circuits with different polarities, given estimates of each disjunct's probability from Theorem 5.14 and certified samplers for each disjunct's conditional law (at truncation level $O(\varepsilon/t)$ for t disjuncts, as the KLM summand is at least $1/t$); we have not worked out that error budget.

5.7 Local threshold circuits

The modules above use decomposability, small width, monotone correlations or independent blocks. A natural class has none of these: counts of local detectors whose inputs overlap.

Definition 5.15 (local threshold circuit)**.** A *local threshold circuit* (LTC) with arity $k \geq 2$ and degree d is $C = [S \geq t] \wedge H$ with

$$S(x) = \sum_{j=1}^{m} g_j(x_{A_j}), \qquad |A_j| \leq k, \qquad |\{j : i \in A_j\}| \leq d \text{ for every } i,$$

t an integer, the g_j arbitrary Boolean sub-circuits and H an arbitrary circuit ($H \equiv 1$ for a pure LTC). It is recognised syntactically: the output gate is a threshold gate with equal positive weights, or an $\wedge$ with such a child, whose children have supports of size at most k; the truth tables of the g_j cost $O(2^k|g_j|)$ evaluations, and t is the least number of true children at which the threshold gate outputs 1 (with weight w and threshold t', $t = \lceil t'/w \rceil$, computed exactly as in Section 2), so the recognised function is that of C.

Examples are "at least t of m detectors fire" for convolutional detectors on a grid ($k = d = r^2$ for $r \times r$ patches) and networks of bounded receptive field with a count readout. When the scopes overlap along an expander, for instance $g_e(x_u, x_v, y_e)$ for the edges $e = uv$ of a random cubic graph with a private input y_e per edge, the treewidth is linear in m with high probability (random cubic graphs are expanders [Fri08], and a tree decomposition of width w yields a balanced separator of $w + 1$ vertices), so variable elimination takes exponential time; the circuit is neither unate nor an $\wedge$ of local events, and S is not a sum of independent blocks, so Theorems 5.10 and 5.14 and Proposition 4.6 do not apply.

Tilting. For $\theta \geq 0$ let $D_\theta(x) = D(x)e^{\theta S(x)}/Z(\theta)$ with $Z(\theta) = \mathbb{E}_D e^{\theta S}$, and $s(\theta) = \mathbb{E}_{D_\theta} S$. The function $e^{\theta(S-t)}$ is the soft envelope of Section 4.4, but its proposal D_θ is now a Gibbs measure rather than a product, and its mass $Z(\theta)e^{-\theta t}$ has no closed form. Since D_θ has full support, for every circuit of this form

$$p = Z(\theta)\, e^{-\theta t}\, v(\theta), \qquad v(\theta) = \mathbb{E}_{D_\theta} U, \qquad U = C(X)\, e^{-\theta(S(X)-t)} \in [0, 1], \tag{1}$$

and $U \geq e^{-\theta(s'-t)}\mathbf{1}[S = s', H = 1]$ for every integer $s' \geq t$, so

$$v(\theta) \;\geq\; e^{-\theta(s'-t)} \Pr_{D_\theta}[S = s',\, H = 1]. \tag{2}$$

The looseness of the envelope is $K = 1/v(\theta)$. Three things are needed: samples from D_θ, the normaliser $Z(\theta)$, and a lower bound on $v(\theta)$.

A locally checkable mixing certificate. Write $q_i = D(x_i = 1) \in (0,1)$, $a_i = \ln(q_i/(1-q_i))$, $\sigma(z) = 1/(1+e^{-z})$, and $\Delta_i(x) = S(x^{i\leftarrow 1}) - S(x^{i\leftarrow 0})$, which depends only on the neighbourhood $N(i)$ of i (the variables sharing a scope with it; $|N(i)| \le d(k-1)$). Let $n_S \le km$ be the number of variables that lie in some scope; the others do not affect S, are independent of the rest under every D_θ, and are resampled exactly in every sweep. The conditional law of x_i under D_θ is $\Pr_{D_\theta}[x_i = 1 \mid x_{-i}] = \sigma(a_i + \theta\Delta_i(x))$. Let $c_{il}(\theta)$ be the largest change of this probability when only x_l changes, and

$$\kappa_{il}(\theta) = \max_x \ \theta\,|\Delta_i(x) - \Delta_i(x^{l\,\text{flipped}})| \cdot \max\{\sigma'(z) : z \in \text{hull}(a_i,\, a_i + \theta\Delta_i(x),\, a_i + \theta\Delta_i(x^{l\,\text{flipped}}))\},$$

computed by enumerating the $2^{|N(i)|}$ configurations of $N(i)$. A *Dobrushin certificate* at θ is a vector $w > 0$ and $\alpha < 1$ with $\sum_l \kappa_{il}(\theta) w_l \le \alpha w_i$ for every i; it is checked in time $O(n\,d(k-1)2^{d(k-1)})$. The *systematic scan* updates the colour classes of a greedy colouring of the conflict graph ($i \sim l$ iff they share a scope) in a fixed order, resampling all sites of a class simultaneously from their conditional laws (they are conditionally independent).

Lemma 5.16 (certified sampler)**.** *(a) $c_{il}(\theta') \le \kappa_{il}(\theta)$ for $0 \le \theta' \le \theta$; κ_{il} is nondecreasing in θ; and $\kappa_{il}(\theta) \le \theta n_{il}/2$, where n_{il} is the number of scopes containing both i and l, and $\sum_l n_{il} \le d(k-1)$. Hence $w = \mathbf{1}$, $\alpha = \theta d(k-1)/2$ is a certificate for every LTC whenever this α is below 1; in particular $\alpha \le 1-\gamma$ for $\theta \le \theta_0 := 2(1-\gamma)/(d(k-1))$, whatever the g_j are.*

(b) If (w,α) is a certificate at θ, then for every $\theta' \in [0,\theta]$ and every starting state, after τ sweeps the systematic scan at θ' is within total variation $F\alpha^\tau$ of $D_{\theta'}$, where $F = \sum_i w_i / \min_i w_i$.

(c) If moreover $\sum_i \kappa_{il}(\theta) \le 1-\gamma$ for every l, then $\mathrm{Var}_{D_{\theta'}}(f) \le \frac{1}{4\gamma}\sum_i \mathrm{osc}_i(f)^2$ for all $\theta' \in [0,\theta]$ and all f, where $\mathrm{osc}_i(f)$ is the largest change of f under a change of x_i. In particular $\mathrm{Var}_{D_{\theta'}}(S) \le n_S d^2/(4\gamma)$.

Part (b) is Dobrushin's uniqueness condition [Dob70] in the weighted (matrix-norm) form of [DGJ09]; we prove it by the standard coupling with shared uniforms. Part (c) uses path coupling [BD97] and contraction of the random-scan chain [LP17, Section 13.1]. Proofs are in Appendix B. For monotone g_j the change of each factor's contribution is at most 1 instead of 2, which doubles θ_0; the enumerated bound κ and the weights capture such gains automatically.

The normaliser. Since $\frac{d}{d\theta}\ln Z(\theta) = s(\theta)$ and $s'(\theta) = \mathrm{Var}_{D_\theta}(S) \ge 0$, thermodynamic integration [GM98] with a monotone bracket needs no smoothness: on the grid $\theta_l = l\theta/L$, $h = \theta/L$, $h\sum_{l=0}^{L-1} s(\theta_l) \le \ln Z(\theta) \le h\sum_{l=1}^{L} s(\theta_l)$, and the two sums differ by $h(s(\theta) - s(0)) \le h(S_{\max} - s(0))$, where $S_{\max} = \sum_j \max g_j$, $S_{\min} = \sum_j \min g_j$, and $s(0) = \mathbb{E}_D S$ is computed exactly by enumerating each scope.

Lemma 5.17 (normaliser interval)**.** *Let L be even, and let $Y_1, \dots, Y_L$ be independent random variables with values in $[S_{\min}, S_{\max}]$ and $\mathbb{E}Y_l = s(\theta_l)$ (for example $Y_l = S(X_l)$ with $X_l \sim D_{\theta_l}$). Put $R = S_{\max} - S_{\min}$ and $\hat{T} = h\sum_l Y_l$, let $\Sigma = \frac{1}{2}\sum_{r \le L/2}(Y_{2r-1} - Y_{2r})^2$, and*

$$\bar{V} = \min\Big\{2\Big(\tfrac{1}{2}a + \sqrt{\tfrac{1}{4}a^2 + \Sigma}\Big)^2,\ \tfrac{1}{4}LR^2\Big\}, \qquad a = R\sqrt{\ln(1/\delta_2)},$$
$$u = \tfrac{1}{3}hR\ell + \sqrt{(\tfrac{1}{3}hR\ell)^2 + 2h^2\bar{V}\ell}, \qquad \ell = \ln(2/\delta_1).$$

Then with probability at least $1 - \delta_1 - \delta_2$, $\ln Z(\theta) \in \big[\hat{T} - u - h(S_{\max} - s(0)),\ \hat{T} + u\big]$.

Here $\bar{V}$ is an upper confidence bound on $\sum_l \mathrm{Var}(Y_l)$ from pairs of neighbouring grid points (Maurer's inequality for nonnegative variables [Mau03]), and u is Bernstein's bound for the sum. Any unbiased Y_l with values in $[S_{\min}, S_{\max}]$ will do. The implementation uses the Rao–Blackwellised statistic $\frac{1}{|\mathcal{P}|}\sum_{I\in\mathcal{P}} \mathbb{E}_{D_{\theta_l}}[S \mid x_{-I}]$, averaged over the colour classes $\mathcal{P}$ (computable exactly because the sites of a class are conditionally independent), which reduced the variance by factors 2.7–4.8 on the Glauber instances of Table 1 and 1.5–2.4 on its CFTP chain (measured at four tilts in $(0,\theta]$). It also clips Y_l to windows $[c_l - w, c_l + w]$ fixed by a pilot run and bounds the clipped mass by a KL upper bound on the mean exceedance; this replaces R by $2w$ in u and adds $\pm\theta M\bar{z}$ to the interval, where M bounds the exceedances and $\bar{z}$ is the KL upper bound (Lemma B.2).

The procedure TG.

(T0) Recognise $C = [S \ge t] \wedge H$, tabulate the g_j, compute $k, d, S_{\min}, S_{\max}, s(0)$. If $t > S_{\max}$ output $[0,0]$; if $t \le S_{\min}$ and $H \equiv 1$ output $[1,1]$.

(T1) Choose $\theta \ge 0$ with a Dobrushin certificate (w,α), by any rule (we use a pilot run aiming at $s(\theta) \approx t$; $\theta = 0$ if $s(0) \ge t$). Fix L, the sample cap N_{IS}, and $\eta = \delta/(5N_{\mathrm{tot}})$ with $N_{\mathrm{tot}} = L + N_{\mathrm{IS}}$. (Soundness needs only these samples coupled. The analysed pilot of Appendix B also draws its samples by (T2), and there N_{tot} counts them too.)

(T2) Every sample is the state of a fresh systematic scan from $x = 0$ after τ sweeps, where $F\alpha^\tau \le \eta/2$. (In floating point a site update, coupled with the exact one through its uniform, differs from it with probability at most $\eta_{\mathrm{fl}} = \max\{2^{-48}, 2^{-53}(\frac{5}{2}\max_i|a_i| + \frac{1}{2}\theta\Delta_{\max} + 8)\}$, $\Delta_{\max} = \max_{i,x}|\Delta_i(x)|$, and our implementation gives no verdict unless $F\alpha^\tau + \tau n\eta_{\mathrm{fl}} \le \eta$.)

(T3) Draw one sample at each θ_l and apply Lemma 5.17 at levels $\delta/5, \delta/5$ (and $\delta/5$ for the window of Lemma B.2, if used): $\ln Z(\theta) \in [z_-, z_+]$.

(T4) Run the sequential KL procedure of Theorem 4.3(iv) on U with fresh samples at θ, level $\delta/5$, cap N_{IS}: $v(\theta) \in [v_-, v_+]$.

(T5) Output $[e^{z_- - \theta t}v_-,\ \min(1, e^{z_+ - \theta t}v_+)]$.

With $\theta = 0$ the samples are draws from D, $Z = 1$, and TG is Monte Carlo (our implementation draws each coordinate by comparing a 53-bit uniform with q_i and charges $N_{\mathrm{tot}}n2^{-53}$ to the coupling share $\delta/5$).

For completeness we use three hypotheses, with parameters $\beta, \nu \in (0, \frac{1}{2}]$:

(H1) $q_i \in [\beta, 1-\beta]$ for every variable i that lies in some scope.

(H2) (private pivots) there are $\ge \nu m$ factors j, each with a *private* variable $i_j \in A_j$ (in no other scope) such that $\Pr_D[g_j(x^{i_j \leftarrow 1}) \ne g_j(x^{i_j \leftarrow 0})] \ge \nu$.

(H3) $t \le s(\theta_0)$, with $\theta_0 = 2(1-\gamma)/(d(k-1))$ as in Lemma 5.16.

(H1) and (H2) are checked by enumerating scopes; (H3) says that the threshold is reachable by the crude certified tilt θ_0.

Theorem 5.18 (tilted Glauber estimator)**.** *(i) (Soundness.) For every LTC $C = [S \ge t] \wedge H$, every product distribution with $q_i \in (0,1)$ and every choice in (T1), the output of TG contains p with probability at least $1 - \delta$.*

(ii) (Completeness.) Fix k, d, γ, β, ν and $0 < \varepsilon \le 1$, and let

$$b = 3 + \tfrac{d}{\sqrt{\gamma}}, \quad c_1 = \tfrac{4e^5 b^2 k}{\nu^2 \beta}, \quad m_0 = \max\left\{\tfrac{4e^4 k d^2}{\gamma \nu^4}, \tfrac{64 e^{10} b^2 k}{\nu^4 \beta^2}\right\}, \quad K_0 = 16 e^{c_1}\sqrt{n_S}.$$

Suppose $H \equiv 1$, (H1)–(H3) hold and $m \ge m_0$, and run TG without windows (so that (T3) applies Lemma 5.17 alone), with the pilot of Appendix B,

$$L = \Theta\Big(\Big(\frac{n_S}{\gamma(k-1)^2\varepsilon^2} + \frac{m}{d(k-1)\varepsilon}\Big)\ln\frac{10}{\delta}\Big),$$

and a cap N_{IS} at least the sample count that Theorem 4.3(iv) needs for $K = K_0$, accuracy $\varepsilon/3$ and level $\delta/5$, which is $O(K_0(\ln\frac{1}{\delta} + \ln\ln K_0)/\varepsilon^2)$. Then TG outputs an interval of ratio at most $1+\varepsilon$ with probability at least $1 - 3\delta$, using $O((n_S + e^{c_1}\sqrt{n_S})\varepsilon^{-2}\log(n_S/\delta))$ samples plus $O(d^3\sqrt{n_S}\log(n_S/\delta)/(\gamma^2(k-1)))$ pilot samples, each costing $O(\gamma^{-1}\log(n/(\varepsilon\delta)))$ sweeps of n table look-ups.

(iii) (Rarity.) Assume (H1) and (H2), and let $v_{\min} = e^{-4}\beta\nu^2 m/2$ and $V_{\max} = n_S d^2/(4\gamma)$. Then $s(\theta_0) - s(0) \ge \theta_0 v_{\min}$; for $t = \lfloor s(\theta_0)\rfloor$, $p \le \exp(\theta_0 - \theta_0^2 v_{\min}/2)$; and for every integer $t \in [s(0), s(\theta_0)]$, $p \le \exp\big(-(t - s(0))^2 v_{\min}/(2V_{\max}^2)\big)$.

Proof of (i). Consider the idealised procedure TG* in which every sample is exact. By (1), $U \in [0,1]$ has mean $v(\theta)$ under D_θ for every C of this form, so the interval of (T4) fails with probability at most $\delta/5$ (Theorem 4.3(iv) needs only i.i.d. $[0,1]$ samples); (T3) fails with probability at most $3\delta/5$ by Lemmas 5.17 and B.2. The choices in (T1) are made before the counted samples are drawn (or from independent pilot samples), so these bounds hold conditionally on them. On the complement, $p = Z(\theta)e^{-\theta t}v(\theta)$ lies in the output. TG differs from TG* only through its samplers: by Lemma 5.16(b) (and, in floating point, the condition in (T2)) each sample can be coupled with an exact one, failing with probability at most η, and the total over at most N_{tot} samples is at most $\delta/5$. No hypothesis on C beyond the syntax was used. □

The proofs of (ii) and (iii) are in Appendix B. The key step of (ii) is anticoncentration: $\Pr_{D_\theta}[S = t] \ge e^{-c_1}/(16\sqrt{n_S})$ whenever $|s(\theta) - t| \le 2\sqrt{n_S}$ (Lemma B.4), obtained without a local central limit theorem from the private pivots, which given the rest of x make S a shifted sum of independent Bernoulli variables, and from the log-concavity of such sums. By (iii), $p = e^{-\Omega(m)}$ for every threshold $t \in [s(0) + cm,\ s(\theta_0)]$ with $c > 0$ fixed (a range of length $\Omega(m)$ for small c), so there Monte Carlo needs $e^{\Omega(m)}$ samples while TG needs $\mathrm{poly}(n)/\varepsilon^2$; thresholds $t \le s(0) + 2\sqrt{n_S}$ are not rare when $m \ge m_0$ ($p \ge e^{-c_1}/(16\sqrt{n_S})$ by Lemma B.4 at $\theta = 0$). The constant e^{c_1} is very pessimistic: in our Glauber experiments $v(\theta)\sqrt{m}$ is between 1.7 and 2.3. Tilting with a local CLT has been used to count independent sets and matchings of a given size [DP21, JPSS21]; here the statistic is an arbitrary sum of local sub-circuits, the regime is certified by the procedure itself, and the output is sound on every input.

Corollary 5.19 (side conditions)**.** *For $C = [S \ge t] \wedge H$ with an arbitrary circuit H, TG is sound (Theorem 5.18(i)). Assume (H1)–(H3), $m \ge m_0$, and $\pi_H > 0$ with $\Pr_{D_\vartheta}[H \mid S = t] \ge \pi_H$ for every $\vartheta \in [0, \theta_0]$ if $s(0) < t$, and $\Pr_D[H \mid S = \lceil s(0)\rceil] \ge \pi_H$ if $s(0) \ge t$. Then TG is complete as in Theorem 5.18(ii) with K_0 replaced by K_0/π_H (in the cap N_{IS} and in the importance-sampling cost). For instance $\pi_H = D(H)$ when H reads only variables outside all scopes.*

Proof. The pilot and the normaliser do not involve H. In the proof of Theorem 5.18(ii) the importance-sampling step uses $v(\theta) \ge \Pr_{D_\theta}[S = s']$, where $s' = t$ if $s(0) < t$ and $s' = \lceil s(0)\rceil$ (and $\theta = 0$) if $s(0) \ge t$; with H, (2) gives $v(\theta) \ge \Pr_{D_\theta}[S = s']\Pr_{D_\theta}[H \mid S = s'] \ge \Pr_{D_\theta}[S = s']\,\pi_H$. Variables outside all scopes are independent of S under every D_ϑ, with law D. □

The condition must be imposed at $\lceil s(0) \rceil$ when $s(0) \geq t$: for $H = [S \leq t]$ and suitable $t \leq s(0) - cm$, $\Pr[H \mid S = t] = 1$ while $p = \Pr_D[S = t] = e^{-\Omega(m)}$ (by (iii) applied to the tables $1 - g_j$), and TG at $\theta = 0$ is plain Monte Carlo.

Beyond the Dobrushin regime. For supermodular statistics the certificate can be dropped altogether.

Proposition 5.20 (self-certifying sampler)**.** *Suppose every Δ_i is nondecreasing in every x_l, $l \neq i$ (that is, S is supermodular; checked by enumerating each neighbourhood). Replace the sampler (T2) by monotone coupling from the past [PW96] for the systematic scan driven by shared uniforms, with sweeps as time steps, and make TG output "no verdict" as soon as any run fails to coalesce within $T_{\max}$ sweeps. The result is sound for every $\theta \geq 0$, with no certificate, and complete whenever $\theta \leq \mathrm{poly}(n)$, coalescence takes $\mathrm{poly}(n)$ sweeps with high probability (e.g. under a certificate: $\Pr[$no coalescence after τ sweeps$] \leq F\alpha^{\tau}$) and $v(\theta) \geq 1/\mathrm{poly}(n)$.*

Proof. With a shared uniform u, the update $x_i \leftarrow \mathbf{1}[u < \sigma(a_i + \theta\Delta_i(x))]$ is nondecreasing in x when $\theta \geq 0$ and Δ_i is nondecreasing, so a sweep is a monotone random map that preserves D_θ; the chains from $\mathbf{0}$ and $\mathbf{1}$ sandwich all others, and when they coalesce the common state at time 0 is an exact sample [PW96]. Couple the procedure with the one that never times out (which uses exact samples and is sound by the proof of Theorem 5.18(i) without the η term): the two outputs agree whenever the capped procedure gives a verdict. (Our implementation, in floating point, also gives no verdict once the counted runs exceed $\delta/(5\eta_{\mathrm{fl}})$ site updates, so that all of them are coupled with exact updates except with probability $\delta/5$.) Variance bounds are not needed, since Lemma 5.17 uses only the data. Completeness: without windows, Lemma 5.17 with $\bar{V} \leq LR^2/4$ and $R \leq m$ gives width $O(\theta m\sqrt{\ln(1/\delta)/L})$ for $L \geq \ln(10/\delta)$, and the gap term is at most $\theta m/L$, so $L = O((\theta^2 m^2/\varepsilon^2 + 1)\log(1/\delta))$ suffices, which is $\mathrm{poly}(n)/\varepsilon^2$ for $\theta \leq \mathrm{poly}(n)$; (T4) needs $O((\log(1/\delta) + \log\log(1/v(\theta)))/(\varepsilon^2 v(\theta)))$ samples (Theorem 4.3(iv)). □

This differs from certified telescoping (Theorem 5.14) in two ways. There the targets are conditional laws $D \mid F_{<k}$ under hard constraints, sampled by bounding chains for anti-monotone dynamics, and failed runs are discarded, so the truncation bias has to be certified empirically (Proposition 5.13). Here the targets are soft tilts, the dynamics is monotone, and aborting the whole procedure on any timeout needs no certificate (the same abort rule would also make Theorem 5.14 sound without Proposition 5.13, at the price of losing completeness when timeouts have small but non-negligible probability). On a chain of monomials $z_{j-1}z_j y_j$ the sampler needs about 26 sweeps per exact sample at $\theta = 2.5$ (counting both bounding chains and all restarts), where the row sums of the influence bound κ are 2.5 and the exact Dobrushin row sums $\max_i \sum_l c_{il}$ are 1.70 (Table 1).

Remark 5.21 (independent blocks)**.** For blocks on disjoint variables (Section 4.4) the tilted law is a product and none of the above is needed; three points complement Proposition 4.6. (a) By (2), $K \leq e^{\theta(s'-t)}/\Pr_{D_\theta}[S = s']$ for every $\theta \geq 0$ and every integer $s' \geq t$, for arbitrary integer weights. (b) The bound $K = O(\sqrt{k})$ is tight: for k i.i.d. integer-valued blocks g whose law has span 1, and integers $t = ak$ with a fixed and $\mathbb{E}g < a < \max g$, every $\theta \geq 0$ has $K \geq K(\theta_a) = \Theta(\sqrt{k})$ as $k \to \infty$, where $s(\theta_a) = t$ minimises $Z(\theta)e^{-\theta t}$, by the lattice form of the Bahadur–Rao theorem [BRR60]; so tilting is logarithmically efficient but does not have bounded relative error. (c) At the top threshold $t = k_+$, $K \to 1$ as $\theta \to \infty$: the tilted law tends to D conditioned on every block firing. The proof of Proposition 4.6 extends to blocks with log-concave integer laws on contiguous supports, with σ the standard deviation of S under the tilt that makes t a mode: $K \leq \frac{4}{3}(4\sigma + 1)$.

The certified ratio, not the variance, sets the cost. Theorem 4.3 pays $\Theta(K)$ samples whatever the relative variance ρ of the weights is, and a variance-adaptive interval pays $O((\rho/\varepsilon^2 + K/\varepsilon)\log(1/\delta))$. The K/ε term is unavoidable for any sound procedure that uses the weights as a black box, and for tilting envelopes K can exceed ρ exponentially.

Proposition 5.22 (certification gap)**.** *(a) Let $\mathcal{A}$ draw i.i.d. samples of a $[0,B]$-valued variable, stop adaptively, and output an interval that contains the mean with probability at least $1-\delta$ for every $[0,B]$-valued law. If for some law P_0 with mean $p \le B/3$ the interval has ratio at most $1+\varepsilon$ ($\varepsilon \le 1/2$) with probability at least $1-\delta$, then $\mathbb{E}_{P_0}[N] \ge (1-3\delta)B/(3\varepsilon p)$.*

(b) Let $x_1, \dots, x_{n+M}$ be uniform with $M \ge n^2$ even, $g = x_1 \wedge \cdots \wedge x_n$, $S = 2Mg + \sum_{i>n} x_i$ and $t = 3M/2$. Then $C = [S \ge t] = g$ and $p = 2^{-n}$. For every $\theta \ge 0$ the tilting envelope $e^{\theta(S-t)}$ has $K \ge 2^{n/2}$, while at $\theta = n\ln 2/(2M)$ its weights have relative second moment at most 2.07.

So a sound user of these weights needs $\Omega(2^{n/2}/\varepsilon)$ samples, although $O(1/(\varepsilon^2\delta))$ would suffice by Chebyshev's inequality if the variance were known. The fix is deductive: the one-step deduction $C \Rightarrow g$ yields the envelope $\prod_{i\le n}[x_i]$ with $K = 1$. This is also why the counterexamples of Glasserman and Wang [GW97], in which tilting has large variance, cannot cause wrong answers here, only slow ones. The proof is in Appendix B.

Experiments (Table 1; `experiments/exp_local.py`). We ran TG with $\varepsilon = 0.3$, $\delta = 0.05$ and margin $\gamma = 0.25$ (Python with numpy, two runs at a time on a shared 4-core machine, 18–31 ns per site update, pilot included) on four families. (1) A chain LTC $g_j(z_{j-1}, z_j, y_j)$ with random truth tables, each y_j private and pivotal with probability $\ge \frac{1}{2}$, $m = 1600$, with the threshold set so that $p \approx 10^{-20}$; exact p from a transfer matrix (treewidth 2; used only for checking). (2) A random cubic graph on $N = 1500$ vertices with one random table $g_e(x_u, x_v, y_e)$ per edge ($m = 2250$, $n = 3750$) and t at the mean of the most tilted certified measure; no exact or decomposition method applies. (3) The same family at $N = 100$ and $p \approx 10^{-3}$, against 10^7 Monte Carlo samples. (4) The CFTP variant on a chain of 400 monomials at $\theta = 1.5$ and 2.5, where the row sums of the influence bound κ are 1.5 and 2.5 (exact Dobrushin row sums 1.27 and 1.70). The κ row sums of the Glauber instances (0.98 for the chain, 1.34 and 1.37 for the cubic graphs; exact Dobrushin row sums 0.97, 1.32 and 1.35) exceed $1-\gamma$, so the weighted certificate was used. The implementation replaces the pilot of Appendix B by a heuristic one (10 tilts, one mean of 256 chains each, 128 in the CFTP and coverage runs, Glauber chains run only to total variation 10^{-3}, linear interpolation); the pilot fixes only θ and the window, so it affects the width but not the validity of the output, and Theorem 5.18(ii) is a statement about the analysed pilot. Every interval with a reference value contained the exact p or overlapped the Monte Carlo interval. With about $7 \cdot 10^4$ chain samples the certified ratios were 1.66 at $p = 10^{-20.19}$ (point estimate within 1.2%) and 1.84 at $p \approx 10^{-23.7}$, where Monte Carlo would need about $2 \cdot 10^{22}$ and $3 \cdot 10^{25}$ samples for the same ratio. Most of the width comes from the normaliser ($\ln Z$ to ± 0.15 and ± 0.19, of which the window term is 0.06 and 0.07); the importance-sampling interval contributes a ratio of about 1.25, with $v(\theta)\sqrt{m} \approx 2.1$ and 2.2. In the CFTP runs the ratios were 2.13 and 2.88 at $p = 10^{-71.8}$ and $p = 10^{-169.8}$. In 16 repeated runs at $\delta = 0.1$ on a chain with $m = 270$, $q_i \in [0.3, 0.7]$ and exact $p = 10^{-4.45}$ (threshold at the mean of the most tilted certified measure), no interval missed p; the median ratio was 2.05 and the median error of the point estimate 2.2%.

instance	n	θ	$\log_{10} p$ (exact or MC)	certified $\log_{10} p$	ratio	samples	sweeps	s	MC
chain, $m = 1600$	3201	0.492	-20.19	$[-20.31, -20.08]$	1.66	72,560	80	441	$1.8 \cdot 10^{22}$
cubic, $N = 1500$	3750	0.447	–	$[-23.88, -23.61]$	1.84	72,560	95	618	$3.3 \cdot 10^{25}$
cubic, $N = 100$	250	0.457	$[-3.017, -2.992]$	$[-3.051, -2.958]$	1.23	72,560	90	29	$6.7 \cdot 10^{5}$
CFTP, $m = 400$	801	1.5	-71.80	$[-71.99, -71.65]$	2.13	81,280	22^*	45	–
CFTP, $m = 400$	801	2.5	-169.76	$[-170.00, -169.53]$	2.88	81,280	26^*	50	–

Table 1: TG on local threshold circuits ($\varepsilon = 0.3$, $\delta = 0.05$, $\gamma = 0.25$ in every row; `exp_local.py`). "Samples" counts pilot, normaliser and importance samples; each normaliser and importance sample is a fresh chain run for "sweeps" sweeps, and pilot chains run fewer sweeps (to total variation 10^{-3}) (*: CFTP, sweeps per counted exact sample, counting both bounding chains and restarts). "s": wall-clock seconds on a shared 4-core machine, two runs at a time; intervals rounded outward. "MC": samples a Monte Carlo KL interval needs for the same ratio at the same δ, at the exact p, at the upper end of TG's interval when p is unknown, and at the Monte Carlo estimate for $N = 100$, whose $\log_{10} p$ column is the interval from 10^7 Monte Carlo samples (109 s).

5.8 Affine structure and linear thresholds

Parity checks, linear hashes and CRCs make rare events with $p = 2^{-\mathrm{rank}}$, which Gaussian elimination computes exactly under the uniform distribution. They are also where decomposition fails. Take $m \le n/6$ random parity checks and an event inside their solution set that holds on at least a $1/\mathrm{poly}(n)$ fraction of it. With high probability over the checks, the envelope estimator with decomposable envelopes whose leaves have at most $2n/3$ variables, context-width compilation and the node budget behind DFSB's completeness guarantee then cost Monte Carlo's price up to a polynomial factor, and every use of a DNNF envelope costs at least its square root (Theorem 5.26 and Corollary 5.29). This subsection makes affine structure usable soundly, proves that barrier, and maps where tractability ends: under biased product distributions and in combination with thresholds. We write $U = \{i : q_i = 1/2\}$ for the uniform coordinates of D and $D_i(b) = \Pr_D[x_i = b]$.

Certified affine gates. The algebraic normal form (ANF) of a Boolean function is its unique representation as a multilinear polynomial over $\mathbb{F}_2$. We propagate ANFs through the circuit ($\wedge \mapsto ab$ with $x_i^2 = x_i$, $\vee \mapsto a + b + ab$, $\neg \mapsto 1 + a$, $\oplus \mapsto a + b$), store the linear part as a bit vector, and abort a gate when its nonlinear part exceeds T monomials; threshold gates are opaque. A gate whose ANF is computed is computed exactly, because the rules are identities and the ANF is unique; if its degree is at most one the gate is *certified affine*, $g(x) = a_g \cdot x + c_g$ for all x. A chain of two-input XOR gadgets $(a \vee b) \wedge \neg(a \wedge b)$ on k inputs is certified with $T = k - 1$ and $O(k^2)$ work per gate (our implementation also caps the work of each product); a balanced tree of such gadgets needs $T \ge \lfloor k/2 \rfloor \lceil k/2 \rceil$, because its last $\vee$ multiplies two disjoint linear forms. A gate is in *solved form* on a coordinate i, $g(x) = x_i \oplus G(x_{-i})$, if the ANF computed with x_i as one symbol and every gate whose support avoids i as an opaque symbol contains x_i only in the monomial x_i. Substituting Boolean functions for symbols is a homomorphism of Boolean rings, so this is a certificate as well, with $G = g|_{x_i = 0}$. Certification has to be deductive. A gate computing $\ell \oplus \mathbf{1}_{x^*}$ agrees with the affine form ℓ except at a hidden point, which T evaluations miss with probability $1 - T2^{-n}$; yet in $C = [g = 1] \wedge [Ax = b]$ with $\ell \equiv 0$ on $\{Ax = b\} \ni x^*$ it decides between $p = 2^{-n}$ and $p = 0$ (compare Theorem 3.2).

Exact presolve. The $\wedge$-*spine* of C is the set of pairs (g, b) reached from $(\mathrm{out}, 1)$ through $\wedge$-gates with $b = 1$, $\vee$-gates with $b = 0$ and negations, without entering certified affine or solved-form

gates; then $C(x) = \bigwedge_{(g,b)}[g(x) = b]$ identically.

Proposition 5.23 (exact affine and triangular presolve)**.** *Suppose $C(x) = \bigwedge_{j\le k}[x_{p_j} = \pi_j(x)] \wedge R(x)$ for all x, where the p_j are distinct and π_j does not read $x_{p_j}, \dots, x_{p_k}$. Let $P = \{p_1, \dots, p_k\}$, and for $x \in \{0,1\}^n$ let $\tilde{x}$ agree with x off P and satisfy $\tilde{x}_{p_j} = \pi_j(\tilde{x})$, fixed for $j = 1, \dots, k$ in turn; let $\tilde{R}(x) = R(\tilde{x})$, which does not read P. Then, for every product distribution D,*

$$p = M\,\mathbb{E}_D[W], \qquad W(x) = \tilde{R}(x)\prod_{j\le k}\frac{D_{p_j}(\tilde{x}_{p_j})}{\max(q_{p_j}, 1-q_{p_j})} \in [0,1], \qquad M = \prod_{j\le k}\max(q_{p_j}, 1-q_{p_j}).$$

In particular $p = 2^{-k}D(\tilde{R})$ if $P \subseteq U$; and if $|q_{p_j} - \frac{1}{2}| \le \eta$ for all j, then $\mathbb{E}_D W \ge \rho^{-k}D(\tilde{R})$ with $\rho = \frac{1+2\eta}{1-2\eta}$.

Proof. For fixed x_{-P} the constraints determine $x_{p_1}, \dots, x_{p_k}$ in turn, since π_j reads only coordinates off P and earlier pivots. So exactly one completion, $\tilde{x}$, satisfies them, and $p = \sum_{x_{-P}} D_{-P}(x_{-P})D_P(\tilde{x}_P)R(\tilde{x}) = M\,\mathbb{E}_D W$, as W does not depend on x_P. If $P \subseteq U$, then $D_P \equiv 2^{-k} = M$ and $W = \tilde{R}$; with $|q_{p_j} - \frac{1}{2}| \le \eta$ every ratio in W is at least ρ^{-1}. □

Two kinds of spine conjuncts have this form. Certified affine conjuncts $[a_g \cdot x = b \oplus c_g]$ of rank r, in reduced row echelon form $x_P = b' + A'x_{[n]\setminus P}$, have every π_j reading only non-pivots; we choose pivots in U and, when possible, outside the support of the remaining conjuncts. A solved-form conjunct $[x_i \oplus G(x_{-i}) = b]$ gives $\pi = b \oplus G$; the presolve adds such conjuncts greedily, keeping an order in which no earlier G and no affine row reads a later pivot. It then builds $\tilde{R}$ as a circuit (the pivots are replaced by $\oplus$-gates of free inputs and by copies of the cones of the G's) and returns M times any sound interval for $\mathbb{E}_D W$: by default the KL procedure of Theorem 4.3(iv) on W, and, when $P \subseteq U$, any sound method applied to $\tilde{R}$, such as DFSB or the meta-procedure. The certified bound $p \le M$ always holds. So the presolve is exact and sound on every circuit, explains k bits of surprise at polynomial cost, and leaves $\log_2(1/D(\tilde{R}))$ bits to the methods that follow; under biased pivots each pivot costs a factor up to ρ in samples. Rule (F) of DFSB also applies Gaussian elimination to affine conjuncts at the root, but it keeps only the literals the system forces, that is, rows of the reduced system with a single variable. A rank-r system without such rows explains nothing there, and r bits here. Examples, under the uniform distribution (or whenever the pivots can be chosen uniform): the event $[\mathrm{CRC}(x) = 0] \wedge R(x)$ has $p = 2^{-r}D(\tilde{R})$ with r the rank of the CRC map, and a Feistel round $[L \oplus f(R) = t]$, whose pivots are the coordinates of L, has probability $2^{-|L|}$ for every round function f (E4 in Table 2). More generally, under the uniform distribution a map F composed of coordinate permutations and triangular flips $\tau(x) = x \oplus g(x_{-i})e_i$ with $i \in U$ preserves D (each flip is a measure-preserving involution of every fibre $\{x_{-i} = \text{const}\}$), so $D(R \circ F) = D(R)$ for every R, whatever the round functions; our implementation detects only the one-round case. Gaussian elimination is standard in SAT and model-counting tools, and eliminating variables defined by others is a known preprocessing step for model counting [LLM16]. What the presolve adds is its use as an exact surprise-removal step that composes with every sound estimator, together with the biased-pivot version.

Affine-union envelopes. Decomposition fails on parity checks, but abstract interpretation over unions of affine subspaces succeeds, and the result is an envelope.

Theorem 5.24 (affine envelopes)**.** *For every gate g compute lists $E^+(g), E^-(g)$ of affine subspaces of $\mathbb{F}_2^n$. A certified affine gate gets $E^+ = \{a_g \cdot x = 1 + c_g\}$ and $E^- = \{a_g \cdot x = c_g\}$; $\neg$ swaps the lists;*

for $g = g_1 \wedge g_2$ let $E^+(g)$ be the nonempty intersections $V_1 \cap V_2$, $V_j \in E^+(g_j)$, if there are at most L of them, and otherwise the operand list of smaller mass; let $E^-(g) = E^-(g_1) \cup E^-(g_2)$ if it has at most L entries, and otherwise $\{\mathbb{F}_2^n\}$; $\vee$ is dual, $\oplus$ uses $E^+(a \oplus b) = E^+(a) \sqcap E^-(b) \cup E^-(a) \sqcap E^+(b)$ (and similarly for E^-), wider gates are folded, and other gates get $\{\mathbb{F}_2^n\}$ for both lists. Duplicate and contained subspaces may be removed.

(a) For every circuit and every gate, $g^{-1}(1) \subseteq \bigcup E^+(g)$ and $g^{-1}(0) \subseteq \bigcup E^-(g)$. Hence the function $\varphi = \sum_{V \in E^+(\mathrm{out})} \mathbf{1}_V$ dominates C, and $E^+(\mathrm{out}) = \emptyset$ certifies $p = 0$. The lists cost $\mathrm{poly}(|C|, n, L, T)$ time.

(b) If every constraint of V is on uniform coordinates, $D(V) = 2^{-\mathrm{codim} V}$ and $D \mid V$ is sampled exactly by drawing $x \sim D$ and re-solving the pivots; for other V, Proposition 5.30 below gives $D(V)$ and exact samples along a given linear order of the coordinates, in time exponential in its width (our implementation uses the input order here, and orders supplied by a prover in Section 6.3); if the width exceeds a cap, V is replaced by its implied constraints on U (a superset). With these, the envelope estimator (Theorem 4.3) is sound on every circuit, with $K = \sum_{V \in E^+(\mathrm{out})} D(V)/p$; it is the Karp–Luby–Madras estimator for the union.

(c) If $C = \bigvee_{j \le L}([A_j x = b_j] \wedge R_j)$ with certified affine conjuncts and opaque R_j (for instance threshold gates), then $K \le L / \min_j \Pr_D[R_j \mid A_j x = b_j]$ under the uniform distribution.

Proof. (a) Induction over the gates. A certified gate equals its affine form. $\neg$ swaps the one-set and the zero-set. If $g = g_1 \wedge g_2$ and $g(x) = 1$, then x lies in some $V_1 \cap V_2$, and both coarsenings are supersets; if $g(x) = 0$ then $x \in g_1^{-1}(0) \cup g_2^{-1}(0)$. The other gates are dual or are unions of such cases, and multiplicities only increase φ. (b) With constraints on U only, the free uniform coordinates, the pivots solved from them, and the other coordinates drawn from D give each $x \in V$ probability $D(x) 2^{\mathrm{codim} V}$. The rest is Theorem 4.3 with the integer-valued φ. (c) Opaque R_j have $E^+ = \{\mathbb{F}_2^n\}$, so $E^+(\mathrm{out}) \subseteq \{V_j\}$ with $V_j = \{A_j x = b_j\}$, and $\sum_j D(V_j) \le L \max_j D(V_j)$ while $p \ge D(V_{j^*}) \Pr[R_{j^*} \mid V_{j^*}]$ for the maximiser j^*. □

On the 120-input instance E1 of Table 2 (45 gadget parity checks and a 3-DNF) the envelope has six pieces and $K = 1.39$. Decomposability can be generalised to linear independence, which keeps exact masses and samplers under the uniform distribution.

Proposition 5.25 (XOR-DNNF envelopes)**.** *Let Φ be an arithmetic circuit with nonnegative weights whose leaves are nonnegative constants and functions $f(\ell(x))$ of a single nonconstant affine form, and in which, at every PROD gate, the spaces $\mathcal{L}(c)$ spanned by the linear parts of the leaf forms below the children c form a direct sum. Under the uniform distribution the bottom-up pass (SUM $\mapsto$ weighted sum, PROD $\mapsto$ product, leaf $\mapsto (f(0) + f(1))/2$) computes $\mathbb{E}\varphi$ exactly, and the following samples exactly from $D\varphi/\mathbb{E}\varphi$: choose a derivation top-down as in Lemma 4.2, give each of its leaves the value v with probability proportional to $f(v)$, and draw x uniformly from the affine subspace on which every leaf form takes its value.*

The proof is in Appendix C. DNNF envelopes (forms x_i) and unions of affine subspaces (products of leaves $[\ell = b]$ with independent forms) are special cases; $\mathbf{1}_{\{Hx=s\}}$ is an XOR-DNNF with m leaves and $K = 1$, while DNNFs whose leaves have at most $2n/3$ variables need at least $2^{\lambda_{\mathrm{bal}}}/(n+2) - 2$ wires by the next theorem (part (b) with $C = E = \mathbf{1}_{\{Hx=s\}}$). Affine decision trees [KLMT13] use affine decision nodes for exact model counting; XOR-DNNF envelopes also allow nondeterministic sums and arbitrary leaf functions, at the price of looseness.

The parity barrier for decomposable envelopes. For an affine subspace $V \subseteq \mathbb{F}_2^n$ and a cut (Y, Z) of $[n]$ let $\lambda_V(Y) = \dim \pi_Y(V) + \dim \pi_Z(V) - \dim V$, with π_Y the coordinate projection. For $V = \{Hx = s\} \neq \emptyset$ this is $\operatorname{rank} H_Y + \operatorname{rank} H_Z - \operatorname{rank} H$, the connectivity function of the binary matroid of H. Let $\lambda_{\mathrm{bal}}(V) = \min\{\lambda_V(Y) : n/3 < |Y| \leq 2n/3\}$. Call an envelope circuit or DNNF *smooth* if every SUM (or $\vee$) child has the support of its parent and the root has support $[n]$; leaves may be arbitrary nonnegative (Boolean) functions of their declared supports. For a circuit Φ let $G(\Phi)$ be the set of its nodes v with $n/3 < |\mathrm{var}(v)| \leq 2n/3$.

Theorem 5.26 (parity barrier)**.** *Let D be uniform, V a nonempty affine subspace, and C a circuit with $\emptyset \neq C^{-1}(1) \subseteq V$; let $c = |C^{-1}(1)|/|V| = \Pr[C \mid V]$.*

(a) Let Φ be a smooth envelope circuit with fan-in-two PROD gates whose leaves have supports of size at most $2n/3$. If $\varphi \geq 1$ on $C^{-1}(1)$ and $K = \mathbb{E}\varphi/p$, then $|G(\Phi)| \cdot K \geq c\, 2^{\lambda_{\mathrm{bal}}(V)}$.

(b) Let E be a smooth DNNF with fan-in-two $\wedge$-gates whose leaves have supports of size at most $2n/3$, with $C \Rightarrow E$. Then $|G(E)|^2\, D(E)/p \geq c\, 2^{\lambda_{\mathrm{bal}}(V)}$, and $|G(E)| \geq c\, 2^{\lambda_{\mathrm{bal}}(V)}$ if moreover $E \Rightarrow [x \in V]$ (for instance if $E \equiv C$).

Every envelope circuit or DNNF with S wires computes the same function as a smooth one with fan-in-two products and at most $(n+2)(S+2)$ nodes.

The proof (Appendix C) walks down each derivation, following the larger child at products, to its first node with at most $2n/3$ variables. This writes φ as a sum, over balanced nodes g, of products $f_g(x_Y)h_g(x_Z)$. A Cauchy–Schwarz argument over the $2^{\lambda_V(Y)}$ blocks $A_i \times B_i$ into which V splits across the cut then shows that such a product covers at most a $2^{-\lambda_V(Y)/2}$ fraction of V per unit of mass. Part (b) with $E \equiv C$ is the rectangle-cover bound behind known DNNF lower bounds for linear codes [ĎHJ+04, BCMS16]. Here the bound is for envelopes, trades size against looseness, applies to every subset of V, and is parametrised by the connectivity function.

Corollary 5.27. *If $H \in \mathbb{F}_2^{m \times n}$ has rank m and row space R, then for every nonempty $V = \{Hx = s\}$, $\lambda_V(Y) = m - \dim(R \cap \mathbb{F}_2^Y) - \dim(R \cap \mathbb{F}_2^Z)$, where $\mathbb{F}_2^T$ denotes the vectors supported in T; hence $\lambda_{\mathrm{bal}}(V) \geq m - 2j^*$ with $j^* = \max\{r : d_r(R) \leq 2n/3\}$ and d_r the r-th generalised Hamming weight [Wei91].*

Corollary 5.28 (random parity checks)**.** *Let H be uniform in $\mathbb{F}_2^{m \times n}$ with $m \leq n/6$. With probability at least $1 - 2^{1-n/6}$, H has rank m and $\dim(R \cap \mathbb{F}_2^T) \leq 6$ for every $|T| \leq 2n/3$; then $\lambda_{\mathrm{bal}}(V) \geq m - 12$ for every nonempty $V = \{Hx = s\}$. On this event, for every circuit C with $C^{-1}(1) \subseteq V$ and $c = \Pr[C \mid V]$ (so that $p = c2^{-m}$), every Φ as in (a) has $|G(\Phi)|K \geq c\, 2^{m-12} = c^2/(4096p)$ and every E as in (b) has $|G(E)|^2 D(E)/p \geq c^2/(4096p)$.*

Corollary 5.29 (what the barrier rules out)**.** *On the event of Corollary 5.28, for $C^{-1}(1) \subseteq V = \{Hx = s\}$:*

(i) Envelopes. *The envelope estimator with any envelope circuit Φ whose leaves have supports of at most $2n/3$ variables (for instance literal and constant leaves, the (R8) atoms of Section 5.2 when $a \leq 2n/3$, (R7) boxes on at most $2n/3$ variables, and the leaves of Theorem 5.6) evaluates φ at every sample and needs $\Omega(K)$ samples, so its work is $\Omega(c^2/(np))$: Monte Carlo's number of samples up to a factor $O(n/c^2)$. Any implementation reads Φ to compute the mass and draws $\Omega(K)$ samples, so it does $\Omega(c/\sqrt{np})$ work: at most a quadratic speed-up.*

(ii) The #DNNF route. *Theorem 6.4 and Proposition 5.4 with a DNNF $E \supseteq C$ (small leaves) read E at every sample and need $\Omega(D(E)/p)$ samples, hence do $\Omega(c/(n\sqrt{p}))$ work.*

(iii) Context width. *Let N be an NNF of C as in Theorem 5.6, without frontier gates (threshold gates are expanded by counters, Section 5.3; $|N| \le 12|C|$ for the gate-by-gate NNF of a $\{\wedge, \vee, \neg, \oplus\}$-circuit). The DNNF that Theorem 5.6 builds from N has literal and constant leaves, is equivalent to C, and has at most $(|N|+1)(\mathrm{cw}+3)2^{\mathrm{cw}}$ wires, so $\mathrm{cw} \ge m - 14 - \log_2\left((n+2)(|N|+1)(\mathrm{cw}+3)/c\right)$.*

(iv) DFSB. *Every tree built with the exact rules (T), (T′), (F), (D), (B) of Section 5.1 in which every leaf h with more than $2n/3$ variables, other than a refuted one, has $D(h) \ge 1/K'$, where $K' < 2^{m-6}$, has at least $c\,2^{m-12}/(n+6)^2$ nodes. This applies to the ideal tree $T^*_{K'}(C)$ when the atom size a is at most $2n/3$, since its leaves are constants, atoms and (S)-leaves with $D(h) \ge 1/K'$. So the completeness guarantee of Theorem 5.2, which needs $m_{K'}(C)$ to be at most the node budget, requires a budget of at least $c^2/(4096(n+6)^2 p)$ unless $K' \ge 2^{m-6} = c/(64p)$. (Runs that stop early at (E)-leaves are not covered.)*

So for random parity checks with $m \le n/6$ and events with $c \ge 1/\mathrm{poly}(n)$, the costs in (i)–(iv) are exponential in m: within a polynomial factor of Monte Carlo for (i), (iii) and (iv), and at least its square root for any use of a DNNF envelope. For smaller c the bounds weaken; for $c = 2^{-m/2}$, for instance, $c^2/p = p^{-1/3}$. Meanwhile the presolve computes the factor 2^{-m} exactly, and the affine envelope of $[Hx = s] \wedge R$ with opaque R has $K = 1/c$. Conversely, bounded connectivity along a whole tree suffices.

Proposition 5.30 (bounded width; any product distribution)**.** *Let $V = \{Ax = b\}$ and let $\mathcal{T}$ be a rooted binary tree with leaf set $[n]$ such that $\lambda_V(X_t) \le w$ for the leaf set X_t of every node t. Then $\mathbf{1}_V$ has a smooth deterministic DNNF, structured by $\mathcal{T}$, with at most $2n(4^w+1)$ gates; for every product distribution, $D(V)$ and exact samples from $D \mid V$ are computed with $O(n4^w)$ arithmetic operations after $\mathrm{poly}(n,m)$ linear algebra; and $\lambda_{\mathrm{bal}}(V) \le w$.*

The construction (Appendix C) is a dynamic program over *live partial syndromes*: at node t the states are the $s = A_{X_t} x_{X_t}$ with $s \in b + \mathrm{col}(A_{[n]\setminus X_t})$, a coset of $\mathrm{col}(A_{X_t}) \cap \mathrm{col}(A_{[n]\setminus X_t})$, which has $2^{\lambda_V(X_t)}$ elements. Our implementation uses linear orders (caterpillar trees). A narrow tree can be found by the fixed-parameter algorithms of Hliněný and Oum [HO08] or by heuristics, or it can be supplied by a prover; the dynamic program is exact for any tree, and the width only affects time. Let $\mathrm{bw}(V)$ be the least w for which a tree as in Proposition 5.30 exists; for $n \ge 2$ this is the branch-width of the binary matroid of A minus one, since the usual connectivity function [HO08] is $\lambda_V + 1$. Together with Theorem 5.26(b) for $C = E = \mathbf{1}_V$ ($c = 1$), the smallest smooth DNNF for $\mathbf{1}_V$ with fan-in-two $\wedge$-gates and leaves on at most $2n/3$ variables (for instance literal leaves) has between $2^{\lambda_{\mathrm{bal}}(V)}$ and $2n(4^{\mathrm{bw}(V)}+1)$ gates. The two ends are polynomially related when $\mathrm{bw}(V) = \lambda_{\mathrm{bal}}(V) + O(1)$, but not in general: for $V = V_1 \times V_2$ with V_1, V_2 on the two halves of $[n]$, $\lambda_{\mathrm{bal}}(V) = 0$, while $\mathrm{bw}(V) \ge \mathrm{bw}(V_1)$ can be linear in n.

Biased product distributions. Linear bijections preserve the uniform measure, but a triangular flip on coordinate i (with $g \not\equiv 0$) preserves a product distribution with marginals in $(0,1)$ only if $q_i = \frac{1}{2}$. Under a biased product distribution the same tiny circuits become hard.

Theorem 5.31 (biased parity events)**.** *Fix a rational $q \in (0,1) \setminus \{\frac{1}{2}\}$. For $H \in \mathbb{F}_2^{m\times n}$, $s \in \mathbb{F}_2^m$ and $k \ge 1$ let C be the circuit on $N = nk$ inputs $x_{i,c}$ ($i \le n$, $0 \le c < k$)*

$$C = \bigwedge_{i\le n,\ 1\le c<k} [x_{i,0} \oplus x_{i,c} = 0] \ \wedge\ [Hx_{\cdot,0} = s],$$

an $\wedge$ of parity checks of total size $O(N+mn)$. Under $D=\mathrm{Bern}(q)^N$,

$$p=(1-q)^N\sum_{y\in\mathbb{F}_2^n:\,Hy=s}\lambda^{|y|},\qquad \lambda=\Big(\frac{q}{1-q}\Big)^k.$$

(a) If $q<\frac12$ and $k\ge(n+2g+3)/\log_2\frac{1-q}{q}$, any interval $[L,U]\ni p$ with $U\le 2^gL$ decides whether the coset $\{Hy=s\}$ contains a vector of weight at most t, which is NP*-complete [BMvT78].*

(b) Consequently, for every fixed $\gamma>0$, a sound procedure that is complete on these circuits within a factor $2^{N^{1-\gamma}}$ would give NP $\subseteq$ BPP*, and an argument system (Definition 2.3) complete on them within a factor 2 would give* coNP $\subseteq$ MA*, hence* PH $=\Sigma_2^p$ *[BHZ87].*

For $q>\frac12$ complement all inputs, which replaces s by $s+H\mathbf{1}$.

The proof is in Appendix C. The same circuit has $p=2^{-\mathrm{rank}}$ or 0 under the uniform distribution, and its probability is computed exactly at bounded width under every product distribution (Proposition 5.30). So algebraic tractability is a property of the circuit and the distribution together. For $s=0$ and all $q_i\le\frac12$, p is a weight enumerator of $\ker H$, that is, a ferromagnetic Ising partition function of a binary matroid, which has an FPRAS for graphic [JS93] and regular [GJ13b] matroids and, under a complexity-theoretic assumption, none for binary matroids in general [GJ13a]. In E6 of Table 2, banded LDPC checks under Bern(1/3) ($p\approx 2.6\cdot10^{-31}$) are estimated to 4% in 0.2 seconds by the affine envelope with exact dynamic-programming masses, whereas the biased-pivot presolve only certifies $p\le 1.3\cdot10^{-22}$ within its budget.

Linear thresholds. A single linear threshold function (LTF) is tractable under every product distribution, which makes unions of LTF pieces tractable, but conjunctions are generally not.

Proposition 5.32 (single LTFs [GKM+11])**.** *For $L=[w\cdot x\le\theta]$ with integer weights and a product distribution with marginals of β bits, $D(L)$ has a deterministic FPTAS using $O(n^3\beta/\varepsilon)$ exact operations on rationals of* $\mathrm{poly}(n,\beta,1/\varepsilon)$ *bits, and $D\mid L$ has an exact sampler using $O(n\log(n\beta/\varepsilon))$ expected operations per sample after the same preprocessing.*

Gopalan, Klivans, Meka, Štefankovič, Vempala and Vigoda note that their branching-program method counts knapsack solutions under all product and symmetric distributions [GKM+11, Sec. 3.2]. Appendix C gives a self-contained dynamic program, with values rounded to powers of $1+\varepsilon/(2n)$, and a rejection step that makes the sampler exact. With the rounded masses as certified lower bounds, the Karp–Luby–Madras estimator remains sound: if $C=\bigvee_{j\le T}P_j$ where each P_j is an LTF $\wedge$ a subcube, or a variable-disjoint $\wedge$ of such pieces (and of affine subspaces on uniform coordinates), the estimator has mean in $[e^{-\varepsilon/3}p,p]$ and looseness at most $e^{\varepsilon/3}T$ (Appendix C).

Proposition 5.33 (an affine system and one LTF [GKM+11])**.** *Let $C=[Ax=b]\wedge[w\cdot x\le\theta]$ with integer w, and let $x_1,\dots,x_n$ be ordered so that $\lambda_t=\mathrm{rank}A_{\le t}+\mathrm{rank}A_{>t}-\mathrm{rank}A\le w_0$ for every t. Then $D(C)$ has a deterministic FPTAS using $O(n^3 2^{w_0}\beta/\varepsilon)$ operations under every product distribution with β-bit marginals, and $D\mid C$ has an exact sampler.*

This is an instance of [GKM+11, Thm. 3.6 with Claim 3.4]: for $V=\{Ax=b\}$, $D\mid V$ is a small-space source of width 2^{w_0} in their sense, whose states are the live syndromes of Proposition 5.30 and whose transition probabilities are proportional to D_t times the masses of the suffixes, $[w\cdot x\le\theta]$ is a monotone read-once branching program in every variable order, and $D(C)=D(V)\Pr_{D|V}[w\cdot$

$x \le \theta]$. Their construction takes time quadratic in the source width. The direct dynamic program of Appendix C, which runs the threshold recursion inside each live syndrome, is linear in 2^{w_0} and also gives the exact sampler. Without such structure, conjunctions are hard.

Proposition 5.34 (non-closure)**.** *Under the uniform distribution, deciding $p > 0$ is* NP*-complete for each of the following classes of polynomial-size circuits: (a) $[w \cdot x \ge \theta] \wedge [w \cdot x \le \theta]$ (two LTFs; subset sum); (b) $[Hx = s] \wedge [|x| \le t]$ (an affine system and one LTF; coset weight [BMvT78]); (c) $[\sum_{uv \in E}(x_u \oplus x_v) \ge k]$ (a threshold of parities, i.e. a quadratic threshold; maximum cut). For (d) $[|Hx| \le t]$, approximating p within any factor below 2 is* NP*-hard [Var97]. Hence no sound procedure is complete on any of these classes unless* NP $\subseteq$ BPP*, and no argument system is complete for upper bounds on any of them unless* coNP $\subseteq$ MA*.*

Proof. (a)–(c) are the stated NP-complete problems, and a sound procedure complete on the class decides them as in Proposition 3.1(a); an argument system complete for upper bounds gives MA proofs of the complements as in Theorem 3.4. For (d), $x \mapsto Hx$ maps the uniform distribution onto the uniform distribution on the code $\mathcal{C} = \mathrm{col}(H)$, so $p = |\{y \in \mathcal{C} : |y| \le t\}|/|\mathcal{C}|$, which is $1/|\mathcal{C}|$ if the minimum distance of $\mathcal{C}$ exceeds t and at least $2/|\mathcal{C}|$ otherwise; deciding which is NP-hard for codes given by generator matrices [Var97] (take H to be the transposed generator matrix). □

The tractable compositions are thus: $\vee$ (Karp–Luby–Madras), variable-disjoint $\wedge$, $\wedge$ with an affine system whose pivots can be chosen uniform and outside the other conjunct's support (Proposition 5.23, $p = 2^{-r}D(L)$), and affine $\wedge$ LTF at bounded width (Proposition 5.33).

Fourier and degree structure. Rare events with simple Fourier structure are affine.

Proposition 5.35. *Let D be uniform and $C \not\equiv 0$. (a) If C has k nonzero Fourier coefficients then $p \ge 1/k$. (b) If C has degree d as a multilinear polynomial over $\mathbb{F}_2$ or over $\mathbb{R}$ then $p \ge 2^{-d}$. (c) For every affine V and quadratic polynomial Q over $\mathbb{F}_2$, $\Pr_{x \in V}[Q(x) = 0] \in \{0\} \cup [\frac{1}{4}, 1]$.*

Proof. (a) For x_0 with $C(x_0) = 1$, $1 \le \sum_S |\hat{C}(S)| \le \sqrt{k \sum_S \hat{C}(S)^2} = \sqrt{kp}$ by Parseval. (b) We show $\Pr_x[f(x) \ne 0] \ge 2^{-\deg f}$ for every nonzero multilinear polynomial f over $\mathbb{F}_2$ or $\mathbb{R}$, by induction on the number of variables, and apply it to $f = C$. Write $f = x_n G + H$ with G, H free of x_n. If $G \equiv 0$, then $f = H$ has fewer variables. Otherwise $\deg G \le \deg f - 1$, and wherever $G(x_{<n}) \ne 0$ at least one of $f(x_{<n}, 0) = H$ and $f(x_{<n}, 1) = H + G$ is nonzero, so $\Pr[f \ne 0] \ge \frac{1}{2}\Pr[G \ne 0] \ge 2^{-\deg f}$ (the base case is a nonzero constant). (c) Parametrise V affinely; Q becomes a quadratic q on $\mathbb{F}_2^{\dim V}$ with associated alternating form $B(z, z') = q(z + z') + q(z) + q(z') + q(0)$ of rank $2h$. If $h = 0$, q is affine and $\Pr[q = 0] \in \{0, \frac{1}{2}, 1\}$. Otherwise $(\mathbb{E}(-1)^q)^2 = \mathbb{E}_{z'}(-1)^{q(z')+q(0)}\mathbb{E}_z(-1)^{B(z,z')} \le 2^{-2h}$, since the inner expectation vanishes off the radical of B; so $\Pr[q = 0] = \frac{1}{2}(1 + \mathbb{E}(-1)^q) \in [\frac{1}{4}, \frac{3}{4}]$. □

So an event with $p = 2^{-50}$ needs Fourier sparsity at least 2^{50} and degree at least 50, yet 50 independent parity checks have spectral norm 1, and Boolean functions of bounded spectral norm are $\pm$-combinations of boundedly many coset indicators [GS08]. By (c), one quadratic equation adds at most two bits of surprise to an affine envelope; affine systems with one quadric are counted exactly by Dickson's normal form [EK90], while systems of quadrics contain SAT. Sampling on the Fourier side does not help either: for $C = [Ax = b]$ the unbiased estimator $Y = (-1)^{b \cdot y} \prod_{i:(A^\top y)_i = 1}(1 - 2q_i)$, y uniform in $\mathbb{F}_2^m$, has $\mathbb{E}Y^2 \ge 2^{-m}$ (from $y = 0$), a relative variance of about $1/p$ when $p \approx 2^{-m}$.

instance (Monte Carlo samples)	n	p	method	U/L	samples	sec.
E1 gadget checks ($1.2 \cdot 10^{18}$)	120	$1.54 \cdot 10^{-14}$	presolve	1.045	$3.3 \cdot 10^4$	0.27
			affine envelope, $K = 1.39$	1.042	$1.6 \cdot 10^4$	0.46
E2 random checks ($4.1 \cdot 10^{34}$)	600	$4.30 \cdot 10^{-31}$	presolve	1.045	$3.3 \cdot 10^4$	2.3
			affine envelope, $K = 1.38$	1.041	$1.6 \cdot 10^4$	3.7
E4 Feistel ($4.8 \cdot 10^{16}$)	92	$3.69 \cdot 10^{-13}$	triangular presolve	1.042	$6.6 \cdot 10^4$	0.16
			affine envelope	$U{=}5.0 \cdot 10^{-5}$	10^5	0.11
E5 two of three ($3.6 \cdot 10^{12}$)	90	$4.89 \cdot 10^{-9}$	affine envelope	$U{=}5.0 \cdot 10^{-5}$	10^5	0.19
			XPT, 7-node argument	1.040	$6.6 \cdot 10^4$	0.43
E6a banded ($6.9 \cdot 10^{34}$)	300	$2.59 \cdot 10^{-31}$	affine envelope + DP	1.041	$4.1 \cdot 10^3$	0.24
			biased-pivot presolve	$U{=}1.3 \cdot 10^{-22}$	$2 \cdot 10^5$	2.7
E6b copies	216	$1.47 \cdot 10^{-43}$	affine env., width $\le 2^6$	$U{=}5.1 \cdot 10^{-5}$	$2 \cdot 10^5$	0.67
			affine env., width $\le 2^{16}$	1.006	$1.0 \cdot 10^3$	1.9
E3 dense checks ($3.7 \cdot 10^7$)	36	$8.77 \cdot 10^{-5}$	presolve	1.083	$1.6 \cdot 10^4$	0.02
			DFSB, 400 nodes	28.7	$1.5 \cdot 10^6$	79

Table 2: Affine methods ($\varepsilon = 0.05$, $\delta = 0.01$; E3: $\varepsilon = 0.1$, $\delta = 0.05$). E1: 45 checks built from $\wedge, \vee, \neg$ gadgets $\wedge$ a 3-DNF; E2: 100 dense random checks $\wedge$ a 3-DNF; E4: $[L \oplus f(R) = t]$ with $k = 40$ and a nonlinear f, $\wedge$ a side condition; E5: two of three blocks of 14 checks through a threshold gate, $\wedge$ a 3-DNF; E6: Bern(1/3) inputs, 100 banded checks (a) and the copies gadget of Theorem 5.31 with $n = 36$, $m = 12$, $k = 6$ (b). Every interval contained the exact p; "$U{=}\ldots$" marks a run that stopped at its sample cap with the valid interval $[0, U]$.

Experiments. Table 2 reports our implementation (module `affine.py`, script `exp_affine.py`; $\varepsilon = 0.05$, $\delta = 0.01$; wall-clock seconds on a shared 4-core machine). The exact values are computed without the module: by closed forms for E1–E4 (for E1–E3 the checks restricted to the inputs outside the residual have full rank, so $p = 2^{-m} \Pr[R]$), by Gaussian elimination on the raw check rows and inclusion–exclusion for E5, by an exact rational sweep over the bands for E6a, by enumerating the 2^{24}-element coset for E6b, and by enumeration in rational arithmetic for E7; the module's dynamic program agrees with the E6 values to $2 \cdot 10^{-14}$. Monte Carlo is charged the number of samples its KL interval needs for the same ratio. E3 illustrates DFSB on a dense parity event at a size where it can run; with $m = 12 > n/6$ it lies outside Corollary 5.28, so it is not an instance of the barrier. On 36 inputs with 12 dense checks ($p = 8.8 \cdot 10^{-5}$), DFSB with a 400-node budget spent 79 seconds and $1.5 \cdot 10^6$ samples and returned an interval of ratio 29, while the presolve needed $1.6 \cdot 10^4$ samples and 0.02 seconds; at this size plain Monte Carlo is cheap too ($3.7 \cdot 10^7$ bit-parallel evaluations, about 0.2 seconds). In 600 estimates on random 16-input circuits under uniform and product distributions (E7), no interval missed the exact value. The whole experiment takes about 111 seconds.

The implementation is sound in floating point as well. Threshold gates are evaluated and propagated exactly, in integer arithmetic after scaling the weights (Section 2; the weights of a repeated child are added), so the threshold rule (X1) of Section 6.3 is a valid inference about C. Masses are carried as a mantissa and a binary exponent with proven relative-error bounds (at most γ_{5n+5} for the dynamic program, where $\gamma_k = ku/(1-ku)$ and $u = 2^{-53}$), every interval is rounded outward, and an upper end below the range of doubles becomes the smallest positive double rather than 0. The Hoeffding–KL intervals exclude a value only when its divergence certainly exceeds the bound, allowing for the rounding of the empirical mean.

Open questions. (1) Lower bounds for XOR-DNNF envelopes: an explicit rare event with $p = 2^{-m}$ for which every XOR-DNNF envelope has size times looseness $2^{\Omega(m)}$. A Fourier version

of the coset lemma reduces this to quadratic systems whose nonzero combinations have large cross-rank across every balanced linear cut. (2) Is the quadratic gap in Corollary 5.29(i)–(ii) real? (3) For $s = 0$ and $q_i \le \frac{1}{2}$, coNP-hardness does not apply: can an MA prover help, for instance with a certificate of rapid mixing? (4) Detection: certify affineness of sub-circuits whose intermediate ANFs are dense (parities computed through adders or lookup tables), for instance from $\mathbb{F}_2$-polynomial-calculus proofs supplied by a prover.

5.9 Specialised modules

Some structures are cheaper to detect and use directly. Each module below is sound on every input, recognises its class syntactically, and runs in polynomial time (it declines when its parameter exceeds the stated bound):

- *Bounded treewidth.* Variable elimination on the Tseitin factor graph along an elimination order of width w computes p exactly in time $O(|C|\,2^{w+1})$ [Dar09, SS10]. We use the order of the 5-approximate tree decomposition of [BDD+16] (width at most $5\,\mathrm{tw} + 4$, found in time $2^{O(\mathrm{tw})}|C|$), or a greedy order if that is narrower, and decline if $w > c \log |C|$; so the module is complete when $5\,\mathrm{tw} + 4 \le c \log |C|$. (Our implementation uses only a greedy min-degree order with $w \le 18$, and floating point with a proved relative error bound: each table entry is a mantissa with an integer exponent, so nothing underflows.)

- *Thresholds of independent blocks.* Exponential tilting (Proposition 4.6): $K \le \frac{4}{3}(2\sqrt{k} + 1)$ for k unit-weight blocks; arbitrary weights under product distributions via the branching-program FPTAS of [GKM10] (see also [GKM+11]).

- *Unate circuits with sparse overlaps or fast coalescence.* Theorems 4.7, 5.10 and 5.14, the last with a polynomial truncation time T and the cut-off $\Lambda_{\max} = 24|C| + 3$ (notation of Theorem 5.14); it is then complete when $\Lambda \le \Lambda_{\max}$ and $\gamma = 1 - q_{\max} d_{\max}(k_{\max} - 1) > 0$ with $T \ge (n/\gamma) \ln(256 n \Lambda_{\max}^2/\varepsilon')$, which implies the condition of Theorem 5.14 on T as $\eta \ge \varepsilon'/(16\Lambda^2)$. The condition on Λ holds for CNFs with all $b_i \le 1/2$ and, except with probability $\delta/6$, for other unate circuits with all $b_i \le 1/8$, since then $\Lambda \le 24\mu + 3$ and $\mu \le m \le |C|$. Clauses are canonicalised first: repeated literals merged, tautologies dropped.

- *Small context width.* Theorem 5.6, declining if $\mathrm{cw} > c \log |C|$, with the #DNNF FPRAS or, in practice, the envelope estimator on the compiled DNNF.

- *Local-lemma regime.* Under the uniform distribution, k-CNFs whose variables lie in at most $d \le 2^{c'k}$ clauses, for an absolute constant $c' > 0$, admit FPRASs [Moi19, FGYZ21, JPV21, HWY22]; for k-CNFs, $d \lesssim 2^{k/4.82}$ suffices up to factors polynomial in k [WY24]. More generally, so do local CSPs in which every constraint is violated with probability at most β and shares variables with at most Δ others, if $\beta\Delta^c \lesssim 1$ for an absolute constant c [FHY21, JPV21, HWY22]. The conditions are syntactic (for constraints on $O(\log |C|)$ variables, β is computed exactly).

- *Local thresholds.* Unit-weight thresholds of sub-circuits with overlapping scopes of bounded size and degree, possibly $\wedge$ an arbitrary circuit (declining if $dk > c \log |C|$): tilted Glauber sampling with a locally checked Dobrushin certificate (Theorem 5.18), or monotone coupling from the past for supermodular statistics (Proposition 5.20).

- *Affine and quadratic structure* (Section 5.8). Certified affine and solved-form conjuncts are factored out exactly (Proposition 5.23), unions of affine subspaces serve as envelopes (Theorem 5.24), affine systems of bounded width are counted and sampled exactly under every product distribution (Proposition 5.30), and together with one linear threshold they have an FPTAS and an exact sampler (Proposition 5.33); widths above $c \log |C|$ are declined. Under the uniform distribution an affine system with at most one quadratic equation is counted exactly: Gaussian elimination substitutes the affine system into the quadratic polynomial, and Dickson's normal form of the result gives the exact count and an exact sampler [EK90]; the quadric changes the probability inside the affine space by at most a factor 4 unless it makes it 0 (Proposition 5.35). (Systems of several quadratic equations are NP-hard even to decide.)

Theorem 5.36 (meta-procedure)**.** *Fix the polynomial parameters of the procedures: $K = |C|^{\alpha}$, $m = |C|^{\beta}$ and the rule budget B for DFSB; the cut-offs $c \log |C|$ above; for Theorem 5.14 a truncation time $T = |C|^{c}$ and the cut-off $\Lambda_{\max} = 24|C| + 3$; and the sample caps and list budgets of the other modules. Run DFSB and every applicable module, M procedures in all, each at confidence $\delta/(3M)$ and at an accuracy parameter for which its completeness gives $U \le (1+\varepsilon)L$ (e.g. $\varepsilon' = \ln(1+\varepsilon)$ for Theorem 5.10), with exact modules computing in rational arithmetic or in floating point with certified error bounds, rounded outwards. Output the intersection of their intervals, or $[0,1]$ if it is empty. The result is sound on every circuit, runs in polynomial time, and is complete on the union of the classes above (for these parameters and the given ε).*

Proof. With probability at least $1 - \delta/3$ all M intervals are valid simultaneously, and then so is their intersection, which is nonempty. Each procedure runs in polynomial time (Theorem 5.2, the cut-offs and caps; Theorem 5.14 draws $\mathrm{poly}(|C|, \Lambda_{\max}, 1/\varepsilon, \log(1/\delta))$ samples and runs, each run of at most $2T$ updates). If C lies in the class of one procedure, its interval is tight with probability at least $1 - 2\delta/(3M) \ge 1 - 2\delta/3$, and a nonempty intersection $[L', U'] \subseteq [L, U]$ with $U \le (1+\varepsilon)L$ has $U' \le (1+\varepsilon)L'$; so the output is tight with probability at least $1 - \delta$. □

Skipping the remaining procedures once some interval is tight changes neither property, and neither does returning at once an interval that is valid and tight with certainty. Our implementation does both. In this order it runs variable elimination, the Janson sandwich for unate CNFs, tilting, the presolve (with DFSB on $\tilde{R}$ when all pivots are uniform) and affine envelopes when they explain at least one bit (presolve factor and envelope mass at most $1/2$), TG, context width and DFSB; it returns at once the result of variable elimination, and $p = 0$ or $p = 1$ when the affine structure or the attainable sums of a block threshold decide C; and it omits the other modules.

5.10 What remains out of reach

Proposition 3.1 rules out beating sampling on all circuits. Under the randomized exponential-time hypothesis (no bounded-error randomized algorithm decides 3-SAT in time $2^{o(n)}$; cf. [IP01]), no sound procedure that is complete on all satisfiable 3-CNFs has cost $2^{o(\log(1/p))}\mathrm{poly}(|C|)$ on them: since $p \ge 2^{-n}$ there, running it for $2^{o(n)}\mathrm{poly}(|C|)$ steps and answering "satisfiable" iff $L > 0$, as in the proof of Proposition 3.1(a), would decide 3-SAT in time $2^{o(n)}$. A speed-up exponential in the surprise is still possible: exact model counting by branching on the literals of an unsatisfied clause takes time $O^*(1.84^n)$, against about 2^n samples when $p = 2^{-n}$. Three natural kinds of rarity also defeat the syntactic tools:

- *Distributed surprise.* In a random 3-CNF with $m = \alpha n$ clauses, $\alpha > 1/3$ small enough that it is satisfiable with high probability, every clause is satisfied with probability $7/8$ and the

rarity comes from m overlapping weak constraints. Any drop-conjuncts relaxation keeps at most $n/3$ pairwise-disjoint clauses, so $D(E) \geq (7/8)^{n/3}$, while $\mathbb{E}p = (7/8)^m$, so that by Markov's inequality $p \leq n(7/8)^m$ with probability at least $1-1/n$; then $K \geq (8/7)^{m-n/3}/n$, exponential. Annealing or multilevel-splitting estimators work well in practice here. In fixed-level form their lower confidence bounds are sound with no assumption on the chains (Theorem 6.10), but their upper bounds, and so two-sided intervals, rest on the mixing of Markov chains, for which no polynomial-time certificate is known in this regime; exact Las Vegas samplers (Theorem 5.14 for unate circuits, partial rejection sampling [GJL19] for CNFs) are known to be fast only in restricted regimes, and random 3-CNFs are not unate.

- *Pseudorandom rarity.* Preimages $[G(x) = y]$ of a PRG G: by Proposition 3.1(b), if G has security $2^{(1-o(1))k}$, nothing beats sampling on them by more than a factor $2^{o(k)}$.

- *Algebraic rarity outside the decomposable world.* For events that fill a $1/\mathrm{poly}(n)$ fraction of the solution set of $m \leq n/6$ random parity checks, with high probability the decomposition-based methods of this section (envelopes with small leaves, context width, DFSB's completeness guarantee) cost Monte Carlo's price up to polynomial factors, and any use of a DNNF envelope at least its square root (Corollary 5.29). Gaussian elimination handles such checks under the uniform distribution, but under $\mathrm{Bern}(1/3)$ inputs small $\wedge$'s of parity checks on N inputs admit no sound procedure complete within a factor $2^{N^{1-\gamma}}$ (fixed $\gamma > 0$) unless $\mathsf{NP} \subseteq \mathsf{BPP}$, and no argument system complete within a factor 2 unless $\mathsf{coNP} \subseteq \mathsf{MA}$ (Theorem 5.31).

6 With a prover: proof-carrying importance sampling

By Theorem 3.2 a verifier can certify an upper bound below the sampling scale only by reading the circuit, and by Theorem 3.4 no argument system does so for every circuit unless $\mathsf{coNP} \subseteq \mathsf{MA}$. The system below makes the deductive part as strong as any propositional proof system and leaves everything else to sampling. It is sound on every circuit and every argument, and complete whenever the rarity of $C(x) = 1$ has a short *explanation*: a proof that the event is confined to a tractable region in which it is not rare (Theorem 6.5).

6.1 The argument language

Fix a Cook–Reckhow proof system P for propositional unsatisfiability (resolution with RUP/DRUP certificates; DRAT or LRAT, whose RAT steps p-simulate extended resolution; cutting planes; Frege; extended Frege; ...). For circuits F, G write $F \vdash_P \bot$ for a P-refutation of $\mathrm{Tseitin}(F) \wedge [F = 1]$, and $F \vdash_P G$ for a P-refutation of $\mathrm{Tseitin}(F) \wedge [F = 1] \wedge \mathrm{Tseitin}(G) \wedge [G = 0]$, with disjoint auxiliary variables for F and G, which certifies $F \Rightarrow G$. Tseitin encodes a threshold gate by the clauses of its sequential-counter or adder-tree expansion (weights and threshold scaled to integers, Section 2), or, in systems that state them directly (cutting planes, TC^0-Frege), by its defining inequalities or threshold connective. Leaving the gate unconstrained, as our checker does, is a sound relaxation (it only weakens the formula refuted), but then no refutation can use the gate.

Exact counting needs determinism, which is a semantic property: deciding whether a DNNF is deterministic is coNP-hard (for a CNF F, let R_1 be a d-DNNF for F with every occurrence of a variable replaced by its own copy, and R_2 a d-DNNF stating that all copies of each variable agree; $R_1 \vee R_2$ is deterministic iff F is unsatisfiable), and a counting pass that assumes it can

be wrong ($x_1 \vee x_1$ gets mass $2q_1$). The verifier therefore accepts as *checked-deterministic* only an OBDD; a decision-DNNF (every $\vee$-gate has the form $(x_i \wedge \alpha) \vee (\neg x_i \wedge \beta)$); an SDD whose vtree the verifier checks and whose primes it checks, bottom-up, to be pairwise inconsistent (with the polynomial-time apply of SDDs on a common vtree and a DNNF satisfiability test); or a *certified d-DNNF*: a DNNF together with a refutation $c_a \wedge c_b \vdash_P \bot$ for every pair of children c_a, c_b of every $\vee$-gate. For such R under a product distribution with rational marginals, $D(R)$ is computed exactly by one bottom-up pass and $D \mid R$ is sampled exactly top-down (Lemma 4.2 with $\vee$ read as SUM and $\wedge$ as PROD, which gives $\varphi = \mathbf{1}_R$). A DNNF without such a check is only a DNNF; it enters through the ENVELOPE rule below.

Definition 6.1 (PCIS arguments)**.** An argument is a finite DAG fixed before the verifier flips any coin. Every node v carries a circuit C_v (the root carries C) whose target is $q_v = D(C_v)$, and one rule; $u, u_1, u_2, \ldots$ denote its children, and $q_i = \Pr_D[x_i = 1]$.

Deductive rules (checked by the verifier, no randomness).

REFUTE: $C_v \vdash_P \bot$; then $q_v = 0$.

EXACT: a checked-deterministic R with $C_v \vdash_P R$ and $R \vdash_P C_v$ (or $C_v = R$); then $q_v = D(R)$, computed exactly.

SUM: $C_v \vdash_P \bigvee_i C_{u_i}$, $\bigvee_i C_{u_i} \vdash_P C_v$ and $C_{u_i} \wedge C_{u_j} \vdash_P \bot$ for $i \neq j$; then $q_v = \sum_i q_{u_i}$. (With one child this is an equivalence rule.)

PROD: $C_v = C_{u_1} \wedge C_{u_2}$ syntactically (the output gate of C_v is an $\wedge$ of the outputs of C_{u_1} and C_{u_2}), with $\mathrm{var}(C_{u_1}) \cap \mathrm{var}(C_{u_2}) = \emptyset$; then $q_v = q_{u_1} q_{u_2}$.

SPLIT: on x_i, with children carrying the restrictions $C_v|_{x_i=0}$ and $C_v|_{x_i=1}$, which the verifier forms itself; then $q_v = (1 - q_i)q_{u_0} + q_i q_{u_1}$. With a P-proof $C_v \vdash_P \ell$ for a literal ℓ on x_i, the single child $C_v|_\ell$ suffices: $q_v = D(\ell)q_u$.

MONO: $C_v \vdash_P C_u$; then $q_v \leq q_u$.

Statistical rules (each with a confidence budget δ_v and a sample cap, written in unary).

ENVELOPE: an envelope Φ (Definition 4.1; here its leaves may also be the atoms of Theorem 6.5, which are used like boxes with exact mass and an exact conditional sampler) and a P-proof $C_v \vdash_P G$ for a circuit G that implies $\varphi \geq 1$ by construction: $G = \mathrm{supp}\Phi$ (SUM read as $\vee$, PROD as $\wedge$, atoms as their circuits) when all constants of Φ are nonnegative integers, since then φ is integer-valued and $\mathrm{supp}\Phi = [\varphi \geq 1]$; and $G = [w \cdot x \geq \theta]$ for the tilted envelope $\varphi = \beta^{w \cdot x - \theta}$ with rational $\beta \geq 1$, nonnegative integer weights in unary and an integer $0 \leq \theta \leq \sum_i w_i$ (a larger θ makes G unsatisfiable, a case for REFUTE). It is estimated by Theorem 4.3 (no boxes), by Theorem 4.5 (box leaves), by Theorem 6.4(c) when Φ is a DNNF E, or by the weighted Karp–Luby–Madras estimator of Theorem 6.5 when $\varphi = \sum_j \mathbf{1}_{E_j}$ with some DNNF pieces E_j.

PROPOSAL: a lower bound from a proposal in self-certifying form (Proposition 6.8).

MC: plain Monte Carlo on C_v.

The verifier outputs $[0, 1]$ if $\sum_v \delta_v > \delta$. Otherwise it checks every certificate, evaluates the statistical nodes with fresh coins, combines intervals by interval arithmetic through the deductive rules (all of which are monotone in their arguments), intersects with $[0, 1]$, and outputs the root interval. A deductive node whose certificate is missing or wrong gets $[0, 1]$. Before sampling, the verifier checks the object of every statistical node: decomposability, with supports it recomputes;

the checks of its atoms (those above for checked-deterministic circuits, $q_i = 1/2$ on affine atoms); nonnegative rational constants; and, for the tilted form, a rational $\beta \geq 1$. A node whose object fails gets $[0, 1]$. A statistical node whose domination proof is missing or wrong is downgraded to its lower end (upper end 1). Whenever Φ has a constant that is not an integer (as the tilted envelope does), the verifier clips W at 1 (Theorem 4.3(iii)), which changes nothing under domination. A statistical node that reaches its cap returns its current interval, valid by Theorem 4.3(iv), or $[0, 1]$ if the cap is reached inside a rejection loop. So the verifier runs in time polynomial in $|\pi|$ and $|C|$, plus the counter calls of Theorem 6.4.

Theorem 6.2 (soundness)**.** *For every circuit C, product distribution D with rational marginals and string π,* $\Pr[p \notin V(C, D, \pi)] \leq \delta$.

Proof. Each deductive rule whose certificate checks states a true identity or inequality between the true targets: by soundness of P, by the checked determinism of R for EXACT, and by independence of variable-disjoint sub-circuits under D for PROD and SPLIT; a node with a failed certificate gets $[0, 1] \ni q_v$. Interval arithmetic with nonnegative endpoints is inclusion-monotone. Let B_v be the event that statistical node v returns an interval missing q_v. Conditionally on everything computed before v starts, v's target and sampling law are fixed and it uses fresh coins, so $\Pr[B_v \mid \text{history}] \leq \delta_v$ by the interval guarantee of its estimator: Theorem 4.3(ii), Theorem 4.5, Theorem 6.4(c) or Theorem 6.5 for ENVELOPE, Proposition 6.8 for PROPOSAL, the KL interval for MC. A downgraded node fails only if its lower end exceeds q_v, which has the same bound: by Theorem 4.3(iii), with W clipped at 1 when Φ has non-integer constants; for box leaves, because the proof of Theorem 4.5 gives $\hat{Z}\,\mathbb{E}W \leq \rho_{\max} D(C_v \wedge \varphi > 0) \leq \rho_{\max} q_v$ without domination (with $W \leq 1$, as φ is integer-valued or W is clipped), so $\hat{Z}\ell/\rho_{\max}$ is still a lower bound; and by the lower ends in Theorem 6.4(c) and Theorem 6.5. Hence $\Pr[\bigcup_v B_v] \leq \sum_v \delta_v \leq \delta$, and on the complement every interval, including the root's, contains its target. □

Adaptive choices inside a node (sequential stopping) are allowed; what must be fixed in advance is the set of nodes and their budgets, or a summable schedule $\delta_v = \delta\, 2^{-v}$.

Proposition 6.3 (relative-error calculus)**.** *Let the ratio of an interval $[a, b]$ be b/a, with ratio 1 for $[0, 0]$ and ∞ if $a = 0 < b$. Suppose that the nodes reachable from the root are* SUM, SPLIT *and* PROD *nodes and leaves, all with valid certificates; that every leaf is a* REFUTE *or* EXACT *node or a statistical leaf returning an interval of ratio at most $1 + \eta$; and that every derivation of the root from the leaves (one child at* SUM *and* SPLIT*, all children at* PROD*) uses at most d statistical leaves. Then the root interval has ratio at most $(1 + \eta)^d$. With $\eta = \varepsilon/(2d)$ and $\varepsilon \leq 1$ the ratio is at most $1 + \varepsilon$.*

Proof. Induction from the leaves. If the children's intervals $[a_i, b_i]$ have ratios at most r_i, a sum or a convex combination with exact weights has ratio at most $\max_i r_i$ (children with $[0, 0]$ contribute nothing), and a product has ratio at most $r_1 r_2$ (or is $[0, 0]$); intersecting with $[0, 1]$ does not increase the ratio. So the ratio at a node is at most the largest product of leaf ratios over its derivations, hence at most $(1 + \eta)^d$, and $(1 + \varepsilon/(2d))^d \leq e^{\varepsilon/2} \leq 1 + \varepsilon$ for $\varepsilon \leq 1$. □

6.2 The two-sided workhorse: certified envelopes

When the envelope is an arbitrary DNNF E given by the prover, the derivation-weighted estimator of Theorem 4.3 can be needlessly slow, because it pays for the overcounting of derivations: its looseness is $\mathbb{E}_D[\varphi]/p$ rather than $D(E)/p$. For $E = \bigwedge_{i \leq m}(x_i \vee y_i)$ under the uniform distribution,

$\mathbb{E}_D[\varphi] = 1$ while $D(E) = (3/4)^m$, so the looseness is inflated by $(4/3)^m$ (and the relative variance is at least $(10/9)^m - 1$ under every reweighting of the $\vee$-edges). Combining the Meel–de Colnet FPRAS for DNNF [MdC26] with the self-reducibility sampler of Jerrum, Valiant and Vazirani [JVV86], made exactly unbiased by Knuth-style weights [Knu75], removes the overcounting. A counter call with accuracy α and failure probability γ returns, given everything computed before the call, a value in $[(1-\alpha)T, (1+\alpha)T]$ with probability at least $1-\gamma$, where T is the true value; the verifier runs the counter with fresh coins on every call.

Theorem 6.4 (FPRAS-guided envelope sampling)**.** *Let E be a DNNF on $x_1, \dots, x_n$ and D a product distribution with rational marginals $q_i = a_i/b_i \in (0,1)$ (variables with $q_i \in \{0,1\}$ are first replaced by constants); write $D(E \mid \rho)$ for the D-mass of $E|_\rho$ over the variables that ρ leaves free.*

(a) (Weighted counting.) Smooth E, so that every derivation mentions every variable exactly once; give every variable one block y_i of $\lceil \log_2 b_i \rceil$ fresh bits, read as an integer; and replace every occurrence of x_i by $[y_i < a_i]$ and of $\neg x_i$ by $[a_i \le y_i < b_i]$ (OBDDs of size $O(\log b_i)$). The result E' is a DNNF with $\#E' = (\prod_i b_i) D(E)$, so the FPRAS approximates $D(E \mid \rho)$ for every partial assignment ρ.

(b) (Sampler.) If E is satisfiable (a linear-time test), draw $x_1, \dots, x_n$ in turn, starting from $w = 1$. At step i, with $\rho = (x_1, \dots, x_{i-1})$, obtain counter outputs $A_0, A_1 \ge 0$ for $D(E \mid \rho, x_i = 0)$ and $D(E \mid \rho, x_i = 1)$, where the verifier sets $A_c = 0$ exactly when $E|_{\rho, x_i = c}$ is unsatisfiable and replaces a nonpositive output on a satisfiable branch by 1; set $x_i = c$ with probability $D_i(c)A_c/S$, $S = D_i(0)A_0 + D_i(1)A_1 > 0$, and multiply w by S/A_{x_i}. Then $x \in E$, and for every $f : \{0,1\}^n \to \mathbb{R}_{\ge 0}$ and every law of the counter outputs in which the outputs at step i may depend on the history but not on the coins of steps $\ge i$, $\mathbb{E}[f(x)\, w] = \mathbb{E}_D[f \mathbf{1}_E]$; in particular $\mathbb{E}[C(x)w] = D(C \wedge E)$.

(c) (Interval.) If E is unsatisfiable, output $[0,0]$. Otherwise let $\hat{D}$ be one counter call for $D(E)$ with accuracy $1/4$ and failure probability δ_r, output $[0,1]$ if $\hat{D} \le 0$, and else let $M = 3\hat{D}$. Run walks as in (b), with fresh coins, whose counter calls have accuracy $1/(4n)$ and failure probability $\gamma < 1/(2n)$, let $\eta = 2n\gamma < 1$, and let $[\ell, u]$ be the KL interval at level δ_s (for a fixed number of walks or with the schedule of Theorem 4.3(iv)) computed from the walks' values $W' = C(x)\min(w, M)/M \in [0,1]$. Then for every circuit C and every counter that meets the guarantee above,

$$\Pr\left[D(C \wedge E) \notin [M\ell,\ Mu/(1-\eta)]\right] \le \delta_r + \delta_s,$$

and $\Pr[D(C \wedge E) < M\ell] \le \delta_s$ whatever $\hat{D}$ is.

(d) (Cost.) With $0 < \varepsilon \le 1$, $\gamma = \varepsilon/(16n)$ (so $\eta = \varepsilon/8$) and the schedule run until $u \le (1+\varepsilon/2)\ell$, the interval of (c) has ratio at most $1+\varepsilon$ with probability at least $1 - \delta_r - \delta_s$ after $O(K(\log(1/\delta_s) + \log\log K)/\varepsilon^2)$ walks, $K = D(E)/D(C \wedge E)$, each costing one evaluation of C and $2n$ counter calls.

Proof. (a) Smoothing is polynomial [Dar01], and the substitution keeps $\wedge$-gates decomposable because distinct variables get disjoint blocks. If $y_i \ge b_i$ for some i, both substituted literals of x_i are false, and every derivation contains one of them, so $E'(y) = 0$. If $y_i < b_i$ for all i, then $E'(y) = E(x)$ with $x_i = [y_i < a_i]$, and each x has $\prod_{i:x_i=1} a_i \prod_{i:x_i=0}(b_i - a_i) = (\prod_i b_i) D(x)$ such preimages. Conditioning a DNNF on ρ gives a DNNF, and the same count applies to $E|_\rho$ over the free variables.

(b) By induction $E|_\rho$ is satisfiable at every step, so some $A_c > 0$, and $D_i(c) > 0$ for both c; hence $S > 0$, and only satisfiable branches are taken, so $x \in E$. Let $F(\rho) = \mathbb{E}_D[f\mathbf{1}_E \mid \rho]$ (an average over the free variables) and $M_i = w_i F(\rho_i)$, with w_i and ρ_i the weight and prefix after i steps. Given the history and the outputs at step $i+1$, the fresh coin gives

$$\mathbb{E}[M_{i+1} \mid \cdot] = \sum_{c:A_c>0} \frac{D_{i+1}(c)A_c}{S}\, w_i \frac{S}{A_c} F(\rho_i, c) = w_i \sum_c D_{i+1}(c) F(\rho_i, c) = M_i,$$

since $F(\rho_i, c) = 0$ when $A_c = 0$. So (M_i) is a martingale (Knuth's estimator [Knu75]), and $\mathbb{E}[f(x)w] = \mathbb{E}M_n = M_0 = \mathbb{E}_D[f\mathbf{1}_E]$.

(c) If E is unsatisfiable, $D(C \wedge E) = 0$. Fix $f \geq 0$ and let B_i be the event that a counter call at step i is inaccurate. B_i is determined by the outputs at step i, which given the history make B_i have probability at most 2γ, and given those outputs the remaining steps have conditional mean $\mathbb{E}[f(x)w \mid \cdot] = M_{i-1}$ by (b). Hence $\mathbb{E}[f(x)w\mathbf{1}_{B_i}] \leq 2\gamma\, \mathbb{E}M_{i-1} = 2\gamma\, \mathbb{E}_D[f\mathbf{1}_E]$, and on summing, the event $\mathcal{G}$ that all calls on the path are accurate satisfies

$$\mathbb{E}[f(x)\, w\, \mathbf{1}_{\neg\mathcal{G}}] \leq \eta\, \mathbb{E}_D[f\mathbf{1}_E]. \tag{$*$}$$

On $\mathcal{G}$, with $\alpha = 1/(4n)$, $S \in [1-\alpha, 1+\alpha] \cdot D(E \mid \rho)$ and $A_{x_i} \in [1-\alpha, 1+\alpha] \cdot D(E \mid \rho, x_i)$, and $\prod_i D(E \mid \rho_{i-1})/D(E \mid \rho_i) = D(E)/D(E \mid x) = D(E)$; as $((1+\alpha)/(1-\alpha))^n$ decreases in n from $5/3$ at $n = 1$ (towards $e^{1/2}$), $w \in [\frac{3}{5}, \frac{5}{3}] \cdot D(E)$ on $\mathcal{G}$. Let $\mathcal{A}$ be the event $\hat{D} \in [\frac{3}{4}, \frac{5}{4}] \cdot D(E)$, of probability at least $1 - \delta_r$. Let $\mathcal{F}$ consist of $\hat{D}$ and the earlier walks; the counter guarantee holds given $\mathcal{F}$, so (b) and $(*)$ hold conditionally on $\mathcal{F}$. On $\mathcal{A}$, $M \geq \frac{9}{4}D(E) > w$ on $\mathcal{G}$, so by $(*)$ with $f = C$, $\mathbb{E}[C(x)(w - \min(w, M)) \mid \mathcal{F}] \leq \mathbb{E}[C(x)w\mathbf{1}_{\neg\mathcal{G}} \mid \mathcal{F}] \leq \eta\, D(C \wedge E)$, and with (b) every walk has $\mu_- \leq \mathbb{E}[W' \mid \mathcal{F}] \leq \mu_+$, where $\mu_+ = D(C \wedge E)/M$ and $\mu_- = (1-\eta)\mu_+$; the upper inequality holds whatever $\hat{D}$ is. The walks need not be independent, but the Chernoff bounds behind the KL interval hold for such adapted $[0,1]$-valued sequences, since $\mathbb{E}[e^{\lambda W'} \mid \mathcal{F}] \leq 1 - \mu + \mu e^\lambda$ with $\mu = \mu_+$ for $\lambda \geq 0$ and $\mu = \mu_-$ for $\lambda \leq 0$. So with probability at least $1 - \delta_s$, $\ell \leq \mu_+$, and on $\mathcal{A}$ also $u \geq \mu_-$: then $M\ell \leq D(C \wedge E)$ and, on $\mathcal{A}$, $D(C \wedge E) \leq Mu/(1-\eta)$.

(d) On $\mathcal{A}$, $\mu_- \geq (1-\eta)D(C \wedge E)/(\frac{15}{4}D(E)) > 1/(5K)$. The proof of Theorem 4.3(iv) uses only the lower tail of the empirical mean, so by the same Chernoff bound it applies to the walks with $\mu = 1/(5K)$; with $\varepsilon/2$ in place of ε it gives $u \leq (1 + \varepsilon/2)\ell$ after the stated number of walks with probability at least $1 - \delta_s$, and $(1+\varepsilon/2)/(1-\varepsilon/8) \leq 1 + \varepsilon$ for $\varepsilon \leq 1$. □

Theorem 6.5 (certified envelopes)**.** *Let D be a product distribution with rational marginals. An* atom *is a subcube, a checked-deterministic R (Section 6.1), or an affine subspace $\{x : Ax = b\}$ over $\mathbb{F}_2$ such that $q_i = 1/2$ for every variable that A involves; each has an exact rational mass and an exact sampler for D conditioned on it (an affine atom has mass $2^{-\mathrm{rank}A}$ if consistent, and is sampled by drawing $x \sim D$ and re-solving its pivot coordinates). Let $\mathcal{R}_0$ be the class of decomposable products (conjunctions with pairwise disjoint supports) of atoms, whose masses multiply and whose conditional samplers run independently. Let $\mathcal{K}_P(m, K)$ be the class of circuits C for which there are $E_1, \ldots, E_r$, each in $\mathcal{R}_0$ or a DNNF, and a P-proof $C \vdash_P \bigvee_j E_j$, all of total size at most m, with $\Lambda = \sum_j D(E_j) \leq K\,p$. The PCIS verifier is sound on all circuits, and on $C \in \mathcal{K}_P(m, K)$, with these pieces and proof as an* ENVELOPE *node, its interval has ratio at most $1 + \varepsilon$ ($\varepsilon \leq 1$) with probability at least $1 - \delta$, in time*

$$\mathrm{poly}(m, |C|, \log(1/\delta)) + O\big(K(\log(1/\delta) + \log\log K)/\varepsilon^2\big) \cdot \big(|C| + m + n^2 + n\,T\big),$$

where T is the time of one counter call on a DNNF piece at accuracy $1/(4n)$ and failure probability $\varepsilon/(16n)$ ($T = 0$ if there is no DNNF piece). Without the proof, the lower end of the interval remains valid.

Proof. Soundness is Theorem 6.2, given the interval guarantee of the following estimator. Drop the unsatisfiable DNNF pieces (a linear-time test). For $E_j \in \mathcal{R}_0$ let $\lambda_j = D(E_j)$; for a DNNF piece let λ_j be one counter call for $D(E_j)$ with accuracy $1/4$ and failure probability $\delta/(2r)$, and output $[0,1]$ if some $\lambda_j \le 0$. Let $\hat{\Lambda} = \sum_j \lambda_j$ and $\mathrm{cov}(x) = |\{j : x \in E_j\}|$. Draw j with probability $\lambda_j/\hat{\Lambda}$; draw $x \sim D \mid E_j$ and set $w_j = D(E_j)$ if $E_j \in \mathcal{R}_0$, or draw (x, w_j) by the walk of Theorem 6.4(b) on E_j, with counter calls of accuracy $1/(4n)$ and failure probability $\gamma = \varepsilon/(16n)$, if E_j is a DNNF; and set $Y = C(x)\, w_j \hat{\Lambda}/(\lambda_j \mathrm{cov}(x))$. By Theorem 6.4(b) with $f = C/\mathrm{cov}$ (and directly for the exact pieces),

$$\mathbb{E}Y = \sum_j \mathbb{E}_D[(C/\mathrm{cov})\mathbf{1}_{E_j}] = D\Big(C \wedge \bigcup_j E_j\Big).$$

Let $\mathcal{A}$ be the event that every λ_j of a DNNF piece lies in $[\frac{3}{4}, \frac{5}{4}]\cdot D(E_j)$, of probability at least $1-\delta/2$. On $\mathcal{A}$ and on the good event of the walk, $w_j \le \frac{5}{3} D(E_j) \le \frac{20}{9}\lambda_j$, so $Y \le \frac{20}{9}\hat{\Lambda} < M := 3\hat{\Lambda}$; for exact pieces $Y \le \hat{\Lambda}$. So by $(*)$ with $f = C/\mathrm{cov}$, given the λ_j and the earlier samples, each sample has $\mathbb{E}[Y - \min(Y, M) \mid \cdot] \le \eta\, D(C \wedge \bigcup_j E_j)$ with $\eta = \varepsilon/8$, and the argument of Theorem 6.4(c) (conditional means of $\min(Y,M)/M$ at most $\mu_+ = D(C \wedge \bigcup_j E_j)/M$, and at least $(1-\eta)\mu_+$ on $\mathcal{A}$), with the KL interval at level $\delta/2$, shows that $[M\ell, Mu/(1-\eta)]$ contains $D(C \wedge \bigcup_j E_j)$ with probability at least $1-\delta$, while its lower end fails with probability at most $\delta/2$ whatever the λ_j are. Under the proved domination $D(C \wedge \bigcup_j E_j) = p$; without it, $D(C \wedge \bigcup_j E_j) \le p$, so the lower end is still valid. Without DNNF pieces, $\hat{\Lambda} = \Lambda$ and $Y \le \Lambda$, so no clipping is needed ($M = \Lambda$, $\eta = 0$), and this is Theorem 4.3 for $\varphi = \mathrm{cov} = \sum_j \mathbf{1}_{E_j}$, with $Z = \Lambda$ and $Q_\varphi(x) = D(x)\mathrm{cov}(x)/\Lambda$: the Karp–Luby–Madras estimator [KLM89].

Completeness: on $\mathcal{A}$, $\hat{\Lambda} \le \frac{5}{4}\Lambda$, so the conditional means of $\min(Y,M)/M$ are at least $(1-\eta)p/(\frac{15}{4}\Lambda) > 1/(5K)$, and Theorem 4.3(iv) with $\varepsilon/2$ at level $\delta/2$, as in Theorem 6.4(d), gives ratio at most $(1+\varepsilon/2)/(1-\varepsilon/8) \le 1+\varepsilon$ after $O(K(\log(1/\delta) + \log\log K)/\varepsilon^2)$ samples. Each costs one evaluation of C, the membership tests for $\mathrm{cov}(x)$ ($O(m)$), and one draw from a piece: $O(m+n^2)$ for a product of atoms (back-substitution for affine atoms) or $2n$ counter calls for a DNNF piece. Checking the proof and computing the masses λ_j takes $\mathrm{poly}(m, |C|, \log(1/\delta))$. □

Unions are allowed only at the top. A piece that is neither in $\mathcal{R}_0$ nor a DNNF, such as a DNNF whose leaves are affine atoms instead of literals, falls outside the rules of Theorem 6.5: they give it neither an exact mass nor a counter, and distributing its $\vee$-gates over its $\wedge$-gates turns it into a union of possibly exponentially many $\mathcal{R}_0$ pieces. It can still be used through the derivation-count envelope that reads $\vee$ as SUM and has atoms as leaves (Theorem 4.3, with Lemma 4.2 extended to atoms as in the proof of Theorem 4.5 with exact masses; its mass Z is the Λ of that union), and then the looseness is Z/p with $Z = \mathbb{E}_D\varphi$ at least the mass of the piece, and possibly exponentially larger (for $\bigwedge_{i \le m}(x_i \vee y_i)$, $Z = 1$ while the mass is $(3/4)^m$).

Theorem 6.5 lifts the tractable classes it admits as pieces (subcubes, checked-deterministic circuits, affine atoms, and DNNFs through the FPRAS) from *syntactic* envelopes (Section 5) to *provable* ones. Examples:

- *Hidden triggers.* $C = R \wedge T$ where T is a test that holds only on a subcube $x_A = a$ but is not written as one: a linear hash $[Mx_A = Ma]$ over $\mathbb{F}_2$ with M of full column rank, a test $[\mathrm{SPN}(x_A) = \mathrm{SPN}(a)]$ for a bijective substitution-permutation network, or a multiplier test $[c \cdot x_A = c \cdot a]$ with x_A read as an integer and a constant $c \ge 1$. The argument is the piece $x_A = a$ with a refutation of $C \wedge (x_A \ne a)$; then $K = 1/\Pr_D[R \mid x_A = a]$, and the verifier needs $O(K(\log(1/\delta) + \log\log K)/\varepsilon^2)$ samples where Monte Carlo needs $\Theta(K\log(1/\delta)/(D(x_A = a)\varepsilon^2))$. Whether the refutation is short depends on P. In our runs

(Table 8) the DRUP proofs found by a CDCL solver for dense linear hashes grow from 329 steps at $|A| = 12$ to 32910 at $|A| = 24$; this is evidence, not a lower bound, and we conjecture that resolution needs exponential size. Extended resolution, which can simulate Gaussian elimination, refutes them in polynomial size, and Gaussian elimination in Section 5 handles them exactly without a prover. For a three-round SPN on $|A| = 20$ bits, the verifier checks a proof of 25944 steps and samples in 39.5 s in all (wall-clock, on a shared 4-core machine), with interval $[9.49, 9.54] \cdot 10^{-7} \ni p$, while Section 5 gives no verdict within 900 s; on 16 bits it is the other way round: Section 5 is exact, and the solver finds no proof within 1200 s. Multiplier tests are notoriously hard for CDCL solvers.

- *Planted unions.* $C = \bigvee_j (T_j \wedge H_j)$, T_j subcubes of large codimension, H_j dense. With the pieces T_j, $\sum_j D(T_j) \le \sum_j D(T_j \wedge H_j) / \min_j \Pr_D[H_j \mid T_j]$, so $K \le \omega / \min_j \Pr_D[H_j \mid T_j]$, where $\omega = \sum_j D(T_j \wedge H_j)/p$ is the expected number of j with $x \in T_j \wedge H_j$ for $x \sim D \mid C$, at most $\max_x |\{j : x \in T_j\}|$.

- *Thresholds with side conditions.* A P-proof $C \vdash_P [w \cdot x \ge \theta]$ (in cutting planes, say), with nonnegative integer weights in unary and an integer $0 \le \theta \le \sum_i w_i$, certifies the tilted envelope $\varphi = \beta^{w \cdot x - \theta} = \beta^{-\theta} \prod_i ([x_i = 0] + \beta^{w_i}[x_i = 1])$, $\beta \ge 1$ rational, whose constants have polynomial bit length. If $C = [w \cdot x \ge \theta] \wedge H$, its looseness is $K = K_{\text{tilt}} / \Pr_D[H \mid w \cdot x \ge \theta]$, where $K_{\text{tilt}} = Z_\beta / D(w \cdot x \ge \theta)$ is the looseness on the threshold alone. For unit weights with $\mathbb{E}_D[\sum_i x_i] \le \theta < |\{i : q_i > 0\}|$, Proposition 4.6 gives a rational β (by bisection) with $K_{\text{tilt}} \le \frac{4}{3}(2\sqrt{n} + 1)$. For general integer weights no such bound holds: K_{tilt} can be $p^{-\Theta(1)}$ when one heavy coordinate carries the rarity (Section 4.4), and the method helps only when K_{tilt} happens to be small.

- *Hidden independence.* A proof that $C \Leftrightarrow \bigwedge_j C_j(x_{S_j})$ with disjoint S_j (a one-child SUM node), followed by PROD nodes, turns a product of k moderately rare events into k separate estimations.

Sandwiches. A second certificate, outside Definition 6.1, needs no SAT proof. The verifier binarizes C (fan-in two; with weights and threshold scaled to integers, a threshold gate becomes a counter if its weights are polynomially bounded, Section 2, and an adder tree with a comparator otherwise). The certificate gives, for every gate g, OBDDs $L_g \subseteq g^{-1}(1) \subseteq U_g$ in one fixed variable order, with $L_{x_i} = U_{x_i} = x_i$ at the inputs, $L_g = U_g = g$ at constant gates, and every other gate checked against its children: $L_{a \wedge b} \Rightarrow L_a \wedge L_b$ and $U_a \wedge U_b \Rightarrow U_{a \wedge b}$; $L_{a \vee b} \Rightarrow L_a \vee L_b$ and $U_a \vee U_b \Rightarrow U_{a \vee b}$; $L_{\neg a} \wedge U_a \equiv \bot$ and $\neg L_a \Rightarrow U_{\neg a}$; $L_{a \oplus b} \Rightarrow (L_a \wedge \neg U_b) \vee (\neg U_a \wedge L_b)$ and $(U_a \wedge \neg L_b) \vee (\neg L_a \wedge U_b) \Rightarrow U_{a \oplus b}$. Each check is a constant number of OBDD apply operations, and induction over the gates gives the sandwich at every gate. Then $p \in [D(L_{\text{out}}), D(U_{\text{out}})]$ deterministically, and since $p = D(L_{\text{out}}) + D(U_{\text{out}} \setminus L_{\text{out}}) \Pr_D[C \mid U_{\text{out}} \setminus L_{\text{out}}]$, exact samples of $D \mid (U_{\text{out}} \setminus L_{\text{out}})$ estimate the gap with $O(\frac{D(U_{\text{out}} \setminus L_{\text{out}})}{p} \varepsilon^{-2} \log \frac{1}{\delta})$ samples. The certificate is sound on its own: if a check fails the verifier outputs $[0, 1]$, and otherwise the interval $D(L_{\text{out}}) + D(U_{\text{out}} \setminus L_{\text{out}})[\ell, u]$, with $[\ell, u]$ a KL interval for $\Pr_D[C \mid U_{\text{out}} \setminus L_{\text{out}}]$, fails only when $[\ell, u]$ does. With this interval guarantee it can also serve as a statistical leaf of Definition 6.1, and the proof of Theorem 6.2 covers it unchanged.

6.3 Affine arguments

Theorem 6.5 admits affine subspaces on uniform coordinates as atoms of envelope pieces, with domination proved in any Cook–Reckhow system. For parity reasoning resolution appears to be

the wrong system: the refutations of dense linear hashes that a CDCL solver finds grow rapidly, and we conjecture that resolution needs exponential size (Section 6.1). The following arguments check domination with gate propagation and Gaussian elimination. This amounts to a tree-like case analysis over affine forms and gates, in the spirit of parity decision trees, and it extends affine pieces to product distributions.

Definition 6.6 (XOR-propagation trees)**.** Fix an ANF cap T, a parameter of the verifier given in unary (our implementation uses $T = 64$). An *XPT argument* is a binary tree. Each internal node splits on an affine form $\ell(x) = \sum_{i \in I} x_i + c$ with $I \subseteq [n]$ and $c \in \{0, 1\}$ (the children add $\ell(x) = 0$ and $\ell(x) = 1$) or on a gate g (they add $g = 0$ and $g = 1$). Each leaf is labelled *closed* or *open*, and an open leaf may carry a linear order of the coordinates. The *state* of a node is an affine system S and a partial assignment σ of gate values: at the root $S = \emptyset$ and $\sigma(\text{out}) = 1$, and at every node the branch assumptions are added and the state is closed under

(X1) gate semantics, forwards and backwards (an $\wedge$ with value 1 forces its children to 1, an $\wedge$ with value 0 and all children but one equal to 1 forces that child to 0, and so on; for threshold gates, the range of the weighted sum allowed by the known children, computed exactly as in Section 2);

(X2) if g is certified affine with $\sigma(g) = v$, add $a_g \cdot x = v + c_g$ to S;

(X3) if g is certified affine and $a_g \cdot x$ is constant on the solution set $V(S)$, set $\sigma(g)$ to that constant plus c_g.

A node is *refuted* if S becomes inconsistent or a gate receives both values.

The verifier computes the certified affine gates of Section 5.8 and the state of every node. It drops refuted nodes with their subtrees, whatever their labels. Every remaining leaf, including a closed leaf whose refutation fails and any malformed node (for instance a split whose constant is not a bit or whose index is out of range), contributes the piece $V(S_{\text{leaf}})$. The verifier then runs the estimator of Theorem 5.24(b) on the union of the pieces. Masses are $2^{-\text{codim}}$ for pieces constrained only on uniform coordinates. Otherwise they come from the dynamic program of Proposition 5.30 along the leaf's order (the identity order if the given one is not a permutation of $[n]$), below a width cap fixed in advance, and a piece above the cap is replaced by its constraints on uniform coordinates.

Theorem 6.7 (XPT arguments)**.** *(a) For every circuit C, product distribution D and string π, the verifier's interval contains p with probability at least $1 - \delta$.*

(b) If C has an XPT argument with t leaves whose pieces satisfy $\sum_{\text{leaves}} D(V_{\text{leaf}}) \le Kp$, and whose orders have $\lambda_t \le w$ wherever a piece has constraints off the uniform coordinates (and the width cap is at least w), then with this argument the verifier outputs a $(1 + \varepsilon)$-tight interval with probability $1 - \delta$, in time $\text{poly}(|\pi|, |C|, n, T) + O(tn4^w) + O\big(K\varepsilon^{-2}(\log(1/\delta) + \log\log K)\big) \cdot (|C| + tn^2)$.

Proof. (a) Let x satisfy $C(x) = 1$ and agree with a state: $x \in V(S)$, and the true gate values of x agree with σ where it is defined. Then x agrees with the closed state, since each rule is a valid inference about the true gate values of x ((X2) and (X3) use $g(x) = a_g \cdot x + c_g$, which certification makes an identity). So no such x reaches a refuted node. Following, at each split, the branch that matches $\ell(x)$ or $g(x)$, x agrees with every state on its path, avoids refuted nodes, and ends at a leaf or malformed node whose piece contains x. Hence the pieces dominate C for

every π, and these deterministic checks use no coins. Soundness follows from Theorem 5.24(b) and Theorem 4.3. (b) The honest argument produces the stated pieces, so K bounds the looseness and Theorem 4.3(iv) applies. Each sample costs one evaluation of C, one piece sample ($O(n^2)$, or $O(n)$ with the dynamic program), and the membership tests for φ. □

XPT is an instance of the certified-envelope rule with a particular proof system: a piece constrained only on uniform coordinates is an affine atom of Theorem 6.5, and along a linear order the dynamic program of Proposition 5.30 is an OBDD, which is checked-deterministic. So XPT composes with the other rules of Definition 6.1 and inherits Theorem 6.2. It adds two things. First, parity reasoning in polynomial size. A dense linear hash $[Mx_A = Ma]$ needs no split at all, because (X1), (X2) and Gaussian elimination derive $x_A = a$. Splits on gates expose affine structure behind gates that the abstract interpretation of Theorem 5.24 treats as opaque. For example, if C requires two of three parity-check blocks through a threshold gate, splitting on two of the block gates gives three affine pieces with looseness close to that of the residual condition, while the Part-1 envelope stays at the trivial bound (E5 in Table 2: $p = 4.9 \cdot 10^{-9}$, an argument with 7 nodes, of which one is a closed and three are open leaves, ratio 1.04 from $6.6 \cdot 10^4$ samples; the checker visits 6 of the nodes, as propagation refutes the closed leaf when its split is applied). Second, affine pieces under product distributions: the prover supplies narrow linear orders and the verifier recomputes their widths. An order that is not a permutation is ignored, and one wider than the cap only relaxes its piece, so a bad order costs tightness or time, never soundness. The limits are those of Theorem 5.31 and Proposition 5.34: no argument system, XPT or any other, is complete for parity events under Bern(1/3), or for $[Hx = s] \wedge [|x| \le t]$ under the uniform distribution, unless $\mathsf{coNP} \subseteq \mathsf{MA}$. So the barrier of Theorem 3.4 already applies to explicit, sparse circuits with very simple algebraic structure.

6.4 Lower bounds need no deduction

Proposition 6.8 (trace-reconstructible proposals)**.** *The prover supplies a seed law P_R on $\{0,1\}^r$ in* self-certifying *form: circuits $q_i(r_{<i})$, $i \le r$, that output a numerator and a denominator in binary (an autoregressive law, $\Pr[r_i = 1 \mid r_{<i}] = q_i$; the verifier reads a zero denominator as 0 and clamps every value to $[0,1]$); or a smooth decomposable probabilistic circuit with nonnegative rational weights, which the verifier normalises itself (it computes partition functions bottom-up and samples a sum child c with probability proportional to $w_c Z_c$, as in Lemma 4.2); or the law $D\varphi/Z$ of an envelope (Lemma 4.2); if its total mass is 0, the lower bound is 0. The verifier samples $R \sim P_R$ with its own coins and computes $P_R(R)$ from the same object and the same clamped or normalised values, so sampler and pmf agree by construction and need no check, and all numbers have polynomial bit length. The prover also supplies circuits $\psi : \{0,1\}^r \to \{0,1\}^n$ (generator) and $\phi : \{0,1\}^n \to \{0,1\}^r$ (inverter), and a clip level $\tau > 0$. For $R \sim P_R$ and $X = \psi(R)$ let*

$$W = \mathbf{1}[\phi(X) = R]\, C(X) \min\left(D(X)/P_R(R),\, \tau\right) \in [0, \tau].$$

(a) (Soundness.) For every C and every such P_R, ψ, ϕ, τ,

$$\mathbb{E}W = \sum_{x \in C^{-1}(1)\colon \psi(\phi(x)) = x} \min\left(D(x),\, \tau P_R(\phi(x))\right) \le p,$$

so τ times the lower end of a KL interval for W/τ at level δ is a lower bound on p with probability at least $1 - \delta$.

(b) (Completeness.) If $\tau \le Kp$ and the inputs $x \in C^{-1}(1)$ with $\psi(\phi(x)) = x$ and $D(x) \le \tau P_R(\phi(x))$ have total D-mass at least $(1-\varepsilon/2)p$, then after $N = O(K\varepsilon^{-2}\log(1/\delta))$ samples this lower bound is at least $(1-\varepsilon)p$ with probability at least $1-\delta$.

Proof. (a) ψ is injective on $R^* = \{r : \phi(\psi(r)) = r\}$ (if $\psi(r) = \psi(r') = x$ then $r = \phi(x) = r'$) and maps it onto $\{x : \psi(\phi(x)) = x\}$. Hence $\mathbb{E}W = \sum_{r\in R^*} P_R(r)\, C(\psi(r)) \min(D(\psi(r))/P_R(r), \tau)$, and the substitution $x = \psi(r)$ gives the identity; each term is at most $D(x)$ and each x occurs once. (Without the trace check each x would be counted once per preimage and $\mathbb{E}W$ could exceed p.) (b) The inputs named in the hypothesis contribute $D(x)$ each, so $\mathbb{E}W \ge (1-\varepsilon/2)p$. Their inverter images are distinct, so their mass is at most $\tau \sum_r P_R(r) = \tau$; hence $\tau \ge (1-\varepsilon/2)p$ and $\mu = \mathbb{E}[W/\tau] \ge (1-\varepsilon/2)/K \ge 1/(2K)$. By the multiplicative Chernoff bound for $[0,1]$-valued variables, after $N = cK\varepsilon^{-2}\log(1/\delta)$ samples the KL lower end for W/τ is at least $(1-\varepsilon/2)\mu$ with probability $1-\delta$, and $(1-\varepsilon/2)^2 \ge 1-\varepsilon$. □

Special cases: autoregressive models and probabilistic circuits ($r = n$, $\psi = \phi = \mathrm{id}$), an envelope used without a domination proof (whose importance weights then give the lower end of Theorem 4.3(iii)), witness lists (the verifier checks $C(x^{(i)}) = 1$ for distinct $x^{(i)}$: an exact lower bound $\sum_i D(x^{(i)})$, no sampling), and generators with inverters driven by an autoregressive seed law. What must *not* be accepted is a pmf and a sampler supplied separately: their agreement is a semantic property the verifier cannot check, and a sampler that over-produces a seed whose claimed pmf is tiny inflates $\mathbb{E}W$ arbitrarily.

6.5 Multilevel splitting needs no mixing for lower bounds

Annealing and multilevel-splitting estimators work well in practice on events whose rarity is spread over many overlapping constraints (Section 5.10). Their upper tails depend on how well Markov chains mix, and by Theorem 3.2 no amount of their output certifies an upper bound below the sampling scale. Their *mean*, however, is exact in the fixed-level form below whatever the chains do, so their lower confidence bounds need no mixing certificate at all. The unbiasedness of the particle estimate of a normalising constant is classical [DM04, CDMFG12] (for adaptive levels see [BGG+16]); what we record is that it needs only invariance, which a verifier checks syntactically.

Definition 6.9 (fixed-level splitting)**.** Fix sets $\{0,1\}^n = A_0 \supseteq A_1 \supseteq \cdots \supseteq A_K$ with $A_K \subseteq C^{-1}(1)$ and decidable membership (e.g. $A_k = \{x : s(x) < t_k\}$ for a score s computed exactly and levels $t_1 > \cdots > t_K$), a number $N \ge 1$ of particles, and Markov kernels $M_1, \dots, M_{K-1}$ on $\{0,1\}^n$ such that M_k leaves the restriction of D to A_k invariant:

$$\sum_{x\in A_k} D(x) M_k(x,y) = D(y)\mathbf{1}_{A_k}(y) \qquad \text{for all } y. \tag{3}$$

Draw $\xi_0^1, \dots, \xi_0^N$ i.i.d. from D. For $k = 1, \dots, K$: let $S_k = \{i : \xi_{k-1}^i \in A_k\}$ and $N_k = |S_k|$; if $N_k = 0$, output $Z = 0$ and stop; if $k = K$, output $Z = \prod_{j=1}^K N_j/N$; otherwise draw indices $I_1, \dots, I_N$ i.i.d. uniformly from S_k and, independently, $\xi_k^i \sim M_k(\xi_{k-1}^{I_i}, \cdot)$.

Kernels satisfying (3) include the identity; under the uniform distribution, flipping a set of coordinates drawn from a fixed law (independently of x) and accepting iff the result lies in A_k (Metropolis with a symmetric proposal); under a product distribution, the same with acceptance probability $\min(1, D(y)/D(x))$, drawn with exact coins; the heat bath on a block B chosen independently of x, which redraws x_B from D conditioned on the result lying in A_k; and products and fixed mixtures of these. Each of the first four is reversible with respect to D on A_k and

never leaves A_k from a point of A_k, which gives (3) by summing the detailed-balance identity over $x \in A_k$.

Theorem 6.10 (splitting is unbiased without mixing)**.** *For every circuit C, every distribution D on $\{0,1\}^n$, and all sets, N and kernels as in Definition 6.9, fixed before the run or chosen with randomness independent of it, $\mathbb{E}Z = D(A_K) \le p$. If an implementation reports Z' with $0 \le Z' \le Z$ on every run (for instance $Z' = 0$ whenever it cannot decide a membership with certainty, or Z rounded down), then $\mathbb{E}Z' \le p$.*

Proof. Condition on the randomness that chose the sets and kernels; then they are fixed. Write $\eta_k = \frac{1}{N}\sum_i \delta_{\xi_k^i}$ for the empirical measure of the particles after stage k, $Z_0 = 1$, $Z_k = \prod_{j \le k} N_j/N$, and set $Z_k\eta_k = 0$ when the run stops before stage k. For a function ϕ on $\{0,1\}^n$ let $\gamma_k(\phi) = \mathbb{E}[Z_k\eta_k(\phi)]$, and let $\mathcal{F}_{k-1}$ be the σ-field generated by $\xi_0, \dots, \xi_{k-1}$. Since $N_k/N = \eta_{k-1}(\mathbf{1}_{A_k})$, the product $Z_k = Z_{k-1}\eta_{k-1}(\mathbf{1}_{A_k})$ is $\mathcal{F}_{k-1}$-measurable. If $N_k > 0$, then given $\mathcal{F}_{k-1}$ the new particles are i.i.d. with law $N_k^{-1}\sum_{j \in S_k} M_k(\xi_{k-1}^j, \cdot)$, so

$$\mathbb{E}[\eta_k(\phi) \mid \mathcal{F}_{k-1}] = \frac{\eta_{k-1}(\mathbf{1}_{A_k} M_k\phi)}{\eta_{k-1}(\mathbf{1}_{A_k})} \qquad \text{and} \qquad \mathbb{E}[Z_k\eta_k(\phi) \mid \mathcal{F}_{k-1}] = Z_{k-1}\,\eta_{k-1}(\mathbf{1}_{A_k} M_k\phi),$$

which also holds when $N_k = 0$ or the run stopped earlier (both sides are 0). Taking expectations, $\gamma_k(\phi) = \gamma_{k-1}(\mathbf{1}_{A_k} M_k\phi)$ for $1 \le k \le K-1$, and $\mathbb{E}Z = \mathbb{E}[Z_{K-1}\eta_{K-1}(\mathbf{1}_{A_K})] = \gamma_{K-1}(\mathbf{1}_{A_K})$. Now $\gamma_0(\phi) = \mathbb{E}_D\phi$, and if $\gamma_{k-1}(\psi) = \mathbb{E}_D[\mathbf{1}_{A_{k-1}}\psi]$ for all ψ, then by $A_k \subseteq A_{k-1}$ and (3) (multiplied by $\phi(y)$ and summed over y),

$$\gamma_k(\phi) = \mathbb{E}_D[\mathbf{1}_{A_{k-1}}\mathbf{1}_{A_k} M_k\phi] = \mathbb{E}_D[\mathbf{1}_{A_k} M_k\phi] = \mathbb{E}_D[\mathbf{1}_{A_k}\phi].$$

So $\mathbb{E}Z = \gamma_{K-1}(\mathbf{1}_{A_K}) = \mathbb{E}_D[\mathbf{1}_{A_{K-1}}\mathbf{1}_{A_K}] = D(A_K) \le D(C^{-1}(1)) = p$. All quantities are bounded ($0 \le Z_k \le 1$), so the conditional expectations are well defined; with one particle this is the identity behind annealed importance sampling [Nea01]. The second claim follows from $\mathbb{E}Z' \le \mathbb{E}Z$. □

Corollary 6.11 (sound lower bounds from splitting)**.** *Let $Z_1, \dots, Z_R$ be the outputs of independent runs with one fixed plan, and let L be any lower confidence bound at level δ for the mean of i.i.d. nonnegative variables, valid whenever that mean is at most the parameter it bounds: for instance τ times the lower end of a KL interval for $\min(Z_i, \tau)/\tau$ with τ fixed in advance, as in Proposition 6.8, or the betting bound of Proposition 8.3, which needs no clipping and is valid at all sample sizes simultaneously. Then $\Pr[L > p] \le \delta$ on every circuit, with no assumption on how the kernels mix.*

Proof. By Theorem 6.10 the Z_i are i.i.d. given the plan, nonnegative, and have mean at most p; apply the guarantee of the bound conditionally on the plan. □

Mixing governs only the variance of Z, hence completeness. If $M_k(x, \cdot)$ is exactly D conditioned on A_k for every $x \in A_k$, the particles after stage k are i.i.d. from that law whatever came before, so the N_k are independent binomials with parameters N and $\rho_k = D(A_k)/D(A_{k-1})$, and

$$\frac{\operatorname{Var} Z}{(\mathbb{E}Z)^2} = \prod_{k=1}^{K}\Big(1 + \frac{1-\rho_k}{N\rho_k}\Big) - 1 \le \exp\Big(\frac{K(1-\rho)}{N\rho}\Big) - 1$$

for $\rho = \min_k \rho_k$ (as $\mathbb{E}(N_k/N)^2 = \rho_k^2 + \rho_k(1-\rho_k)/N$). With $\rho_k \ge \rho$ and $N = \lceil K/\rho \rceil$ the relative variance is at most $e - 1$, at a cost of $NK \approx K^2/\rho$ draws per replicate. This is $O(\log^2(1/p))$ for fixed ρ when there are $K = O(\log(1/p))$ levels, for instance when $\rho \le \rho_k \le \frac{1}{2}$ for every k

and $A_K = C^{-1}(1)$, since then $p = \prod_k \rho_k \le 2^{-K}$; the lower bound $\rho_k \ge \rho$ alone does not bound K, as levels with ρ_k close to 1 can be added at will. A certificate that the kernels sample their conditional laws accurately would therefore bring the upper half within reach as well; that is open problem 2 of Section 9. Section 8 runs this procedure on neural networks.

6.6 Surprise accounting

Write $\sigma = \log_2(1/p)$ for the surprise of the event. An envelope argument explains $\sigma_e = \log_2(1/Z)$ bits deductively and leaves $\sigma_u = \log_2(Z/p)$ bits to sampling, where Z is the mass of the envelope actually sampled: $Z = D(E)$ for a deterministic envelope or one sampled by Theorem 6.4, $Z = \mathbb{E}_D\varphi$ for Theorem 4.3, and $Z = \Lambda = \sum_j D(E_j)$ for a union (Theorem 6.5). Under completeness,

$$\text{verifier time} \;\le\; \text{poly}(|\pi|, |C|) \;+\; O\big(2^{\sigma_u}\varepsilon^{-2}(\log(1/\delta) + \log(1+\sigma_u))\big) \cdot c_s,$$

where $|\pi|$ is the size of the deductive part of the argument (envelopes or pieces, proofs, circuits), not counting the unary sample caps, which an honest prover sets to the samples needed so that reading them costs no more than the second term; and c_s is the cost of one sample: $O(|C| + |\Phi|)$ for Theorem 4.3 (with $\Phi = E$ for a deterministic envelope), $|C|$ plus $2n$ counter calls for Theorem 6.4, and $O(|C|+m+n^2+nT)$ for a union (Theorem 6.5). For a skeleton of t independent components (PROD at the top) estimated separately, each leaf needs ratio $1+\varepsilon/(2t)$ and confidence δ/t (Proposition 6.3), so the second term becomes $O(t^2\varepsilon^{-2}(\log(t/\delta) + \log(1+\sigma_u)))\sum_\lambda 2^{\sigma_u(\lambda)}\, c_s$: unexplained surprise spread evenly over t independent components costs about $t^3 2^{\sigma_u/t}$ rather than 2^{σ_u}. Both kinds of bits have a sharp meaning in black-box models. Say that a verifier *accesses* a function f *only through* T *evaluations* if it computes f at T points, chosen adaptively by any randomized rule, and uses f in no other way.

Theorem 6.12 (proof bits and surprise bits)**.** *Let D be uniform.*

(a) (Lower bounds.) Let a verifier read an argument of exactly ℓ bits and access C only through T evaluations. Fix a set I of k coordinates and let $S_b = [x_I = b]$ for $b \in \{0,1\}^k$. If the verifier is δ-sound for lower bounds on all circuits and, for each b, some argument makes it output $L > 0$ on $\mathbf{1}_{S_b}$ with probability at least $1-\delta$, then $T \cdot 2^\ell \ge (1-2\delta)2^k$. This is tight up to the factor $\varepsilon^{-2}\log(1/\delta)$: an argument consisting of j bits of b lets the verifier sample x uniformly with those bits fixed, which certifies $L \ge (1-\varepsilon)p$ (Proposition 6.8) after $T = O(2^{k-j}\varepsilon^{-2}\log(1/\delta))$ evaluations.

(b) (Two-sided intervals inside a certified envelope.) Let the inputs be pairs (y, x) with at least $2k$ x-coordinates, and $E = E(y)$ any function of y with $D(E) > 0$. Let a verifier be given E explicitly (as after a checked domination proof), read an argument of any length, and estimate $p(E \wedge G)$ for a function G that it accesses only through T evaluations. Suppose it is δ-sound for upper bounds on all circuits $E \wedge G$, i.e. on all C with $C \Rightarrow E$, and that for $G_0 = S = [x_I = b]$ with $|I| = k$, so that $p_0 = D(E \wedge S) = D(E)2^{-k}$, some argument π_0 makes it output with probability at least $1-\delta$ an interval $[L, U] \ni p_0$ with $U \le (1+\varepsilon)L$, $\varepsilon < 1/2$. Then $T \ge (1-2\delta)D(E)/p_0 = (1-2\delta)2^{\sigma_u}$, where $\sigma_u = \log_2(D(E)/p_0)$ is the surprise left by E. Conversely, if $D(E)$ is computable and $D \mid E$ can be sampled exactly (as for the atoms of Theorem 6.5), the estimator of Theorem 4.3 with $\varphi = \mathbf{1}_E$, which reads no argument and evaluates G only at its samples, is sound on all these circuits and meets the hypothesis with $T = O(2^{\sigma_u}\varepsilon^{-2}(\log(1/\delta) + \log(1+\sigma_u)))$.

Proof. (a) Couple the run on $\mathbf{1}_{S_b}$ with argument π_b and the run on the zero function with the same argument and the same coins; they differ only if a query lands in S_b. Soundness on the zero function bounds $\Pr[L > 0]$ there by δ, so under π_b the run on the zero function hits S_b with probability at least $1 - 2\delta$. For a fixed argument π that run does not depend on b, and since the S_b partition the cube each query lies in exactly one of them: $\sum_b \Pr_\pi[\text{hit } S_b] \le T$. Summing over b and over the 2^ℓ arguments gives $2^k(1 - 2\delta) \le 2^\ell T$. (b) Fix $i_0 \in I$ and a set J of k x-coordinates outside I, and for uniform $b'' \in \{0,1\}^k$ plant $S' = [x_{i_0} = 1 - b_{i_0}] \wedge [x_J = b'']$, which is disjoint from S, has $D(S') = 2^{-k-1}$, and contains each fixed point with probability at most 2^{-k}. Then $G_1 = S \vee S'$ gives $p_1 = D(E \wedge G_1) = 1.5\, p_0$ for every b'' (as E depends only on y), and $E \wedge G_1 \Rightarrow E$, so soundness applies to it. Run the verifier with π_0 and the same coins on G_0 and on G_1; the runs differ only if a query lands in S'. On G_0, with probability at least $1 - \delta$, $U \le (1 + \varepsilon)L \le (1 + \varepsilon)p_0 < p_1$; on G_1, soundness gives $U \ge p_1$ with probability at least $1 - \delta$. So the runs differ with probability at least $1 - 2\delta$. The run on G_0 does not depend on b''; averaging over b'', the expected number of its queries in S' is at most $T2^{-k}$. Hence $T2^{-k} \ge 1 - 2\delta$. Conversely, $\mathbf{1}_E$ dominates every $C \Rightarrow E$, so Theorem 4.3(ii) gives soundness; on $E \wedge G_0$ the looseness is $D(E)/p_0 = 2^{\sigma_u}$, so Theorem 4.3(iv) at level $\delta/2$ gives the count; and each sample $X \in E$ costs one evaluation, $C(X) = G(X)$. □

So, up to polynomial factors, certifying a rare event costs $|\pi| + 2^{\sigma_u(\pi)}$ for the argument π used: a two-part, MDL-like cost in the spirit of surprise accounting [Hil24] (which adds bits, $|\pi| + \sigma_u$, where here the unexplained bits are paid exponentially). Theorem 6.12 gives the two halves a precise sense: each bit of argument reduces the number of evaluations needed for a lower bound by at most a factor 2; and once an envelope E is certified, a verifier that accesses the rest of C only through evaluations needs about 2^{σ_u} of them for a two-sided interval, whatever further argument it reads, and the envelope estimator achieves this. (Theorem 3.2 does not give (b): its planted circuits do not imply E.) Neither part bounds white-box verifiers, which may need no argument and no sampling at all (an explicit subcube is counted exactly).

6.7 Gibbs envelopes

The soft envelopes of Section 6.1 (thresholds with side conditions) need a product proposal. When the statistic has overlapping scopes the proposal is a Gibbs measure, and its sampler and normaliser can only be certified statistically, as in Section 5.7. This gives one more statistical rule for PCIS arguments.

Definition 6.13 (Gibbs-envelope node)**.** A node with circuit C_v may carry: a statistic $S = \sum_j g_j(x_{A_j})$ given by scopes and truth tables; an integer t' and a tilt $\theta \ge 0$; either a weight vector $w > 0$ and $\alpha < 1$, or the claim that S is supermodular; a P-proof of $C_v \vdash_P [S \ge t']$, with $[S \ge t']$ encoded by a fixed counting circuit (e.g. a totalizer); budgets L, N_{IS} and a sweep cap $T_{\max}$, written in unary; and δ_v. The verifier returns $[0,1]$ for the node unless $d(k-1) \le c \log(|\pi| + |C_v|)$, for a fixed constant c, so that every neighbourhood has at most $(|\pi| + |C_v|)^c$ configurations. It checks the proof, recomputes $\kappa(\theta)$ and checks $\sum_l \kappa_{il}(\theta) w_l \le \alpha w_i$ for all i (or checks supermodularity and samples by monotone CFTP, aborting with $[0,1]$ when a run exceeds $T_{\max}$ sweeps, Proposition 5.20), returns $[0,1]$ if the τ of (T2) exceeds $T_{\max}$, and otherwise runs steps (T2)–(T5) of TG at θ, without windows, with $U = C_v(X) e^{-\theta(S(X) - t')}$.

Proposition 6.14 (Gibbs envelopes)**.** *For every circuit and every argument, a Gibbs-envelope node returns an interval containing $q_v = D(C_v)$ with probability at least $1 - \delta_v$; if its proof is missing or wrong, the same computation with $U' = \min(1, U)$ still returns a valid lower bound.*

The verifier runs in time polynomial in $|\pi|$ and $|C_v|$. If C_v has such a node with a proof of size $\mathrm{poly}(|C_v|)$, $d(k-1) \le c \log |C_v|$, *a certificate with* $\alpha \le 1-\gamma$ *and* $v_C(\theta) = \mathbb{E}_{D_\theta}[C_v e^{-\theta(S-t')}] \ge 1/\mathrm{poly}(n)$, *then for* $0 < \varepsilon \le 1$ *the interval has ratio at most* $1+\varepsilon$ *with probability* $1-3\delta_v$ *after* $L + O((\log(1/\delta_v) + \log\log(1/v_C(\theta)))/(\varepsilon^2 v_C(\theta)))$ *samples (with* N_{IS} *at least the second term and* $T_{\max}$ *at least the* τ *of (T2), which is* $O(\gamma^{-1}\log(FN_{\mathrm{tot}}/\delta_v))$*), where* $L = O((\theta^2 m^2/\varepsilon^2 + 1)\log(1/\delta_v))$ *always suffices, and* $L = O((\theta^2 n_S d^2/(\gamma\varepsilon^2) + \theta m/\varepsilon + 1)\log(1/\delta_v))$ *when the column sums of* $\kappa(\theta)$ *are also at most* $1-\gamma$.

Proof. For every C_v, $\mathbb{E}_{D_\theta}[C_v e^{-\theta(S-t')}] = e^{\theta t'} q_v/Z(\theta)$, because D_θ has full support; the proved implication gives $U \le 1$. The proof of Theorem 5.18(i) used nothing else, and the certificate is recomputed by the verifier, so no argument can make it false. Without the implication, $U' \le U$ has mean at most $e^{\theta t'} q_v/Z(\theta)$, so the lower ends of the intervals for $\mathbb{E}U'$ and $\ln Z$ still give a lower bound. Declining with $[0,1]$ is always valid. The enumeration visits at most $n(|\pi| + |C_v|)^c$ configurations, and the budgets are in unary, so the verifier runs in polynomial time. Completeness is the argument of Theorem 5.18(ii) with the anticoncentration step replaced by the assumption on $v_C(\theta)$ and the tilt supplied, so no pilot is run. Without a variance bound, $\bar{V} \le LR^2/4$ in Lemma 5.17, so with $\ell = \ln(10/\delta_v)$, $u \le \frac{2}{3}\theta R\ell/L + \theta R\sqrt{\ell/(2L)}$, and with $R \le m$ and $\theta m/\varepsilon \le (\theta m/\varepsilon)^2 + 1$ this gives the first value of L (which also makes $h(S_{\max} - s(0)) \le \theta m/L \le \varepsilon/10$). With column sums at most $1-\gamma$, Lemma 5.16(c) gives $\mathrm{Var}_{D_{\theta'}} S \le V_{\max} = n_S d^2/(4\gamma)$ for all $\theta' \le \theta$, and the bound $u \le 2\theta\sqrt{8V_{\max}\ell/L} + 6\theta R\ell/L$ (for $L \ge \sqrt{\theta R}$) from the proof of Theorem 5.18(ii), which does not use $\theta \le \theta_0$, gives the second; for $\theta \le \theta_0$ it is at most the L of Theorem 5.18(ii). □

Compared with Theorem 5.18, the prover contributes three things. (1) The domination is proved, so C_v need not be syntactically an LTC: an obfuscated or rewired implementation of a count, or any circuit that implies one, is handled through the proof system. (2) The tilt and the weights are supplied and only checked, which removes the pilot run and can certify regimes where the unweighted bound fails. (3) The threshold t' and the statistic itself are part of the argument, and by Proposition 5.22 they, not the variance of the weights, set the cost: in its instance the one-line proof $C \vdash g$ replaces $K \ge 2^{n/2}$ by $K = 1$.

Remark 6.15 (block certificates)**.** Sites may be replaced by blocks of b variables, each resampled exactly from its conditional law by enumerating 2^b assignments. The influence of block B' on block B is the largest total variation distance between the conditional laws of x_B when x_{-B} changes inside B', computable by enumerating $2^{O(dkb)}$ configurations, and the proof of Lemma 5.16(b) goes through with blocks in place of sites. This lets an argument certify strongly coupled clusters of size $O(\log n)$. (Not implemented.)

Remark 6.16 (lower bounds need no mixing certificate)**.** For any schedule $0 = \theta_0 < \cdots < \theta_L$ and any transitions T_l that preserve D_{θ_l} (heat-bath sweeps, for instance), annealed importance sampling [Nea01] is unbiased for p whether or not the chains mix: draw $x_0 \sim D$, $x_l \sim T_l(x_{l-1}, \cdot)$, and weight $\prod_{l=1}^{L} e^{(\theta_l - \theta_{l-1})S(x_{l-1})} \cdot C(x_L) e^{-\theta_L S(x_L)}$. By Markov's inequality the median of r independent estimates exceeds cp with probability at most $(4/c)^{r/2}$ for $c > 4$, so the median divided by c is a lower bound valid on every circuit, without a certificate. As in Section 6.4, lower bounds need only a proposal; upper bounds need the certificate, or exact samples.

6.8 How much stronger is the prover?

A universal verifier. Fix a consistent theory T with a polynomial-time decidable set of axioms that proves every true $\Delta_0(\exp)$ sentence (an arithmetic sentence with exponentiation all of whose

quantifiers are bounded): for instance any such extension of $I\Delta_0 + \exp$, or ZFC through its standard interpretation of arithmetic. A V_T-argument is a randomized program M, a time bound t written in unary, a budget $\delta_M \le \delta$, and a T-proof of the $\Delta_0(\exp)$ sentence "at most $\delta_M 2^t$ strings $r \in \{0,1\}^t$ make M, run for t steps on (C, D) with coins r, fail to output an interval containing $\sum_x D(x)C(x)$". V_T checks the proof and runs M for t steps (outputting $[0,1]$ if M has not halted), in time polynomial in the length of the argument. It is sound: if the sentence were false, its negation would be a true $\Delta_0(\exp)$ sentence, which T also proves, and T would be inconsistent. It p-simulates every argument system V' with a polynomial time bound t' for which T proves, as one universally quantified sentence, that V' clocked at t' is sound: in polynomial time an argument π' for V' becomes the V_T-argument made of $V'(\cdot,\cdot,\pi',\varepsilon,\delta)$, $t = t'(|C|, |\pi'|, 1/\varepsilon, \log(1/\delta))$, δ and the instance of that T-proof at $(C, D, \pi', \varepsilon, \delta)$ (of polynomial size, with numerals in binary), on which V_T has the output law of V' clocked at t'. So V_T can use any FPRAS, compiler, mixing-time bound or combinatorial theorem as a certified rule. This is the analogue, for estimation, of proof systems that are optimal relative to a theory [KP89].

Lower bounds: the prover helps unless $\mathsf{NP} \subseteq \mathsf{BPP}$. Under the uniform distribution a uniquely satisfiable circuit has $p = 2^{-n}$, and one witness certifies the tight lower bound $L = p$ (a witness list, Section 6.4). If a sound efficient procedure without a prover outputs some $L > 0$ with probability at least $1 - \delta$ on every uniquely satisfiable circuit, then $\mathsf{NP} \subseteq \mathsf{BPP}$: run it with δ of order $1/n$ on $O(n)$ independent Valiant–Vazirani restrictions of a formula [VV86], which include a uniquely satisfiable one with probability at least $3/4$ if the formula is satisfiable and are all unsatisfiable otherwise, and answer "satisfiable" iff some run gives $L > 0$. The matching upper bound needs a proof of uniqueness: a system certifying p within a factor below 2 on every uniquely satisfiable circuit would put coNP in MA, since pad_0 (below) maps unsatisfiable circuits to uniquely satisfiable ones with $p = 2^{-n-1}$ and satisfiable ones to circuits with $p \ge 2^{-n}$ (argue as for Theorem 3.4).

One-way permutations give an average-case version. Let f be a one-way permutation of $\{0,1\}^k$ and $C_y(x, r) = [(f(x), r, \langle x, r\rangle) = y]$ on $2k$ inputs. For $y = (f(x_0), r_0, b)$ with x_0, r_0, b uniform, $p(C_y) = 2^{-2k}[b = \langle x_0, r_0\rangle]$, and on positive instances the preimage (x_0, r_0) certifies $L = p$. A sound polynomial-time procedure without a prover outputs $L > 0$ on a random positive instance $(f(x_0), r_0, \langle x_0, r_0\rangle)$ with probability at most $\delta + \mathrm{negl}(k)$: otherwise, run on $C_{(f(x_0), r_0, b)}$ for a random bit b, answering b if $L > 0$ and $1 - b$ otherwise, it would predict the Goldreich–Levin bit $\langle x_0, r_0\rangle$ from $(f(x_0), r_0)$ [GL89]. Two-sided certificates on these instances also need a proof that $f^{-1}(f(x_0)) = \{x_0\}$ (for V_T, e.g. the instance of a T-proof that f is injective).

Upper bounds on refutable circuits. For a proof system P and a polynomial m let Z_P^m be the set of circuits C with a refutation $C \vdash_P \bot$ (Section 6.1) of size at most $m(|C|)$, and SAT the set of satisfiable circuits; (Z_P^m, SAT) is a slice of the canonical disjoint NP pair of P [Raz94]. For a set $\mathcal{Z}$ of unsatisfiable circuits, $(\mathcal{Z}, \mathrm{SAT})$ is *separable in randomized polynomial time* if a randomized polynomial-time algorithm accepts every $C \in \mathcal{Z}$ and rejects every $C \in \mathrm{SAT}$, each with probability at least $2/3$. For $a \ge 0$ let

$$\mathrm{pad}_a(C)(\beta, x, w) = (\beta \wedge x = 0^n \wedge w = 0^a) \vee (\neg\beta \wedge C(x))$$

on $1 + n + a$ inputs and $\mathrm{pad}_a(\mathcal{Z}) = \{\mathrm{pad}_a(C) : C \in \mathcal{Z}\}$; under the uniform distribution $p(\mathrm{pad}_a(C)) = 2^{-n-a-1} + s\,2^{-n-1}$ if C has s satisfying inputs.

Proposition 6.17. *Let $\mathcal{Z}$ be a set of unsatisfiable circuits and $a \ge 0$.*

(a) *A sound efficient procedure without a prover that is complete on $\mathcal{Z}$ exists iff $(\mathcal{Z}, \mathrm{SAT})$ is separable in randomized polynomial time.*

(b) *If $(\mathcal{Z}, \mathrm{SAT})$ is separable in randomized polynomial time, some sound efficient procedure without a prover is complete on $\mathrm{pad}_a(\mathcal{Z})$. Conversely, if some sound efficient procedure without a prover is complete on $\mathrm{pad}_a(\mathcal{Z})$ under the uniform distribution for some $\varepsilon < 2^a$, then $(\mathcal{Z}, \mathrm{SAT})$ is separable in randomized polynomial time.*

(c) *If $\mathcal{Z} \subseteq Z_P^m$ and P is resolution, cutting planes or a Frege-type system (TC^0-Frege, Frege, extended Frege), the PCIS verifier with P outputs $[p, p]$ on every circuit of $\mathcal{Z}$ and of $\mathrm{pad}_a(\mathcal{Z})$, with arguments of size $m(|C|) + \mathrm{poly}(|C| + a)$ ($p = 0$ on $\mathcal{Z}$, and $p = 2^{-n-a-1}$ on $\mathrm{pad}_a(\mathcal{Z})$ under the uniform distribution).*

Proof. (a) Given a separator, amplify it to error δ by a majority vote, and output $[0, 0]$ if it accepts C and $[0, 1]$ otherwise. This is sound for every C and D: if $p_D(C) > 0$ then C is satisfiable and $[0, 0]$ is output with probability at most δ, and if $p_D(C) = 0$ both outputs contain p. On $\mathcal{Z}$ it outputs $[0, 0]$ with probability at least $1 - \delta$. Conversely, run the procedure under the uniform distribution with $\varepsilon = 1$ and $\delta = 1/8$, and accept iff $U = 0$. On $C \in \mathcal{Z}$, $p = 0$, and a tight interval containing p has $U \le 2L \le 2p = 0$, so this happens with probability at least $1 - 2\delta$. On $C \in \mathrm{SAT}$, $p \ge 2^{-n}$, so $U = 0$ means $p \notin [L, U]$, which has probability at most δ.

(b) Given a separator, output $[0, 1]$ unless the circuit has the syntactic form $\mathrm{pad}_a(C)$; then output $[q, q]$ with $q = D(\beta = 1, x = 0^n, w = 0^a)$ if the amplified separator accepts C, and $[0, 1]$ otherwise. Since $p = q + D(\beta = 0)p_D(C)$, $p \ne q$ only if C is satisfiable, so this is sound, and it is complete on $\mathrm{pad}_a(\mathcal{Z})$. Conversely, run the procedure on $\mathrm{pad}_a(C)$ under the uniform distribution with $\delta = 1/8$, and accept iff $U < \tau = 2^{-n-1}(1 + 2^{-a})$. On $C \in \mathcal{Z}$, $p = 2^{-n-a-1}$, and with probability at least $1 - 2\delta$, $U \le (1 + \varepsilon)L \le (1 + \varepsilon)p < (1 + 2^a)2^{-n-a-1} = \tau$. On $C \in \mathrm{SAT}$, $p \ge \tau$, so $U < \tau$ has probability at most δ.

(c) A REFUTE node with the given refutation outputs $[0, 0]$ on C. For $\mathrm{pad}_a(C)$, SPLIT on β: the restriction to $\beta = 1$ is equivalent to the subcube $x = 0^n, w = 0^a$, an EXACT node whose two implications have proofs with $O(n + a)$ steps; the restriction to $\beta = 0$ gets a REFUTE node whose refutation derives $[C = 1]$ from the gadget in $O(1)$ steps and then uses C's refutation. The root interval is $[q, q]$ with q as in (b). □

A proof system P is *weakly automatable* if for some proof system Q an algorithm finds Q-refutations of every unsatisfiable F in time polynomial in $|F|$ and the size of the shortest P-refutation of F. Pudlák [Pud03] shows that P is weakly automatable iff its canonical pair, formulas with a P-refutation of size at most k versus satisfiable formulas (each paired with 1^k), is separable in *deterministic* polynomial time. Every slice (Z_P^m, SAT) reduces to it, by $C \mapsto (\mathrm{Tseitin}(C) \wedge [C = 1], 1^{m(|C|)})$. Conversely, for the systems of Proposition 6.17(c) it reduces to a single slice: a P-refutation of F gives one of $\mathrm{Tseitin}(C_F) \wedge [C_F = 1]$, for a circuit C_F computing F gate by gate, with polynomial overhead (derive the clauses of F, or for Frege-type systems the equivalence of each gate variable with its subformula), so mapping $(F, 1^k)$ to C_F padded to size at least k reduces the pair to $(Z_P^{m_0}, \mathrm{SAT})$ for a fixed polynomial m_0. Hence, if P is weakly automatable, the prover gives no advantage on the tasks of Proposition 6.17 with $\mathcal{Z} = Z_P^m$; and, for the systems of Proposition 6.17(c), it strictly helps on them whenever some slice is not separable in randomized polynomial time. For these systems the two conditions differ only by derandomisation: P is weakly automatable iff $(Z_P^{m_0}, \mathrm{SAT})$ is separable in deterministic polynomial time, so if $\mathrm{prBPP} = \mathrm{prP}$ the prover strictly helps on these tasks exactly when P is not weakly automatable; without that, a randomized separator does not give Pudlák's automating system,

whose proofs must be checked deterministically. Whether weak automatability of P removes the prover's advantage on larger classes, such as the classes $\mathcal{K}_P(m, K)$ of Theorem 6.5, is open: there the argument also supplies the envelope pieces and the shape of the DAG, which a procedure that can only test given formulas for short refutations does not obviously find.

Hard slices. Let (A, B) be a disjoint NP pair, $A = \{u : \exists y\, \varphi_{|u|}(u, y)\}$ and $B = \{u : \exists z\, \psi_{|u|}(u, z)\}$ with polynomial-size formulas φ_ℓ, ψ_ℓ, such that $\varphi_\ell(u, y) \wedge \psi_\ell(u, z)$, with u, y, z all variables, has P-refutations of polynomial size, the formulas being in the language of P. Suppose P is closed under restrictions and passes from a refutation of a formula to one of its Tseitin encoding with polynomial overhead (as TC^0-Frege, Frege and extended Frege do, by deriving the equivalence of each gate variable with its subformula). Here the Tseitin encoding defines every gate: for TC^0-Frege, a threshold gate by its threshold connective (Section 6.1); for Frege and extended Frege, the threshold connectives of a TC^0-Frege formula are first replaced by polynomial-size NC^1 formulas. The relaxation that leaves threshold gates unconstrained is not meant here. For $u \in B$ with witness z_0, restricting such a refutation by (u, z_0) (the true closed formula $\psi_\ell(u, z_0)$ has a short proof) shows that the formula $F_u = \varphi_{|u|}(u, \cdot)$, as a circuit, lies in Z_P^m for a polynomial m; for $u \in A$ it is satisfiable. So $u \mapsto F_u$ reduces (A, B) to $(\mathcal{Z}, \mathrm{SAT})$ with $\mathcal{Z} = \{F_u : u \in B\} \subseteq Z_P^m$ (cf. [Raz94, Pud03]), and a randomized polynomial-time separator of $(\mathcal{Z}, \mathrm{SAT})$ separates (A, B) in randomized polynomial time, hence, by Adleman's argument [Adl78] (amplify the success probability, then fix coins that are good for all u of each length), by polynomial-size circuits. Bonet, Pitassi and Raz [BPR00] construct such a pair, based on the Diffie–Hellman function, with polynomial-size TC^0-Frege refutations, which no polynomial-size circuits separate unless Blum integers can be factored by polynomial-size circuits (their failure of feasible interpolation); Krajíček and Pudlák [KP98] had constructed one for extended Frege from RSA. Frege and extended Frege p-simulate TC^0-Frege, so Proposition 6.17 gives:

Corollary 6.18. *Suppose Blum integers cannot be factored by polynomial-size circuits, and let P be TC^0-Frege, Frege or extended Frege. Then there are a polynomial m and a family $\mathcal{Z} \subseteq Z_P^m$ such that for every $a \geq 0$ the PCIS verifier with P certifies p exactly, with polynomial-size arguments, on $\mathcal{Z}$ and on $\mathrm{pad}_a(\mathcal{Z})$, while no sound efficient procedure without a prover is complete on $\mathcal{Z}$, nor on $\mathrm{pad}_a(\mathcal{Z})$ under the uniform distribution for any $\varepsilon < 2^a$.*

Proof. Take $\mathcal{Z} = \{F_u : u \in B\}$ for the pair of [BPR00]. Certification is Proposition 6.17(c). An excluded procedure would, by Proposition 6.17(a,b), separate $(\mathcal{Z}, \mathrm{SAT})$ in randomized polynomial time, and hence (A, B) by polynomial-size circuits. □

If LWE is hard for quantum polynomial-time algorithms, Arteche, Carenini and Gray [ACG24] show that no polynomial-time quantum algorithm weakly automates TC^0-Frege. We do not claim the quantum analogue of Corollary 6.18: it would need that no bounded-error quantum algorithm separates a slice (Z_P^m, SAT), which does not obviously follow, since a separator need not output proofs that can be checked deterministically.

Finding the short proofs is a different matter. No algorithm finds resolution refutations of unsatisfiable CNFs F in time polynomial in $|F|$ and the size of the shortest refutation of F unless $\mathsf{P} = \mathsf{NP}$ [AM20]. This concerns automatability; Proposition 6.17 needs only separability of the slices, which would follow from weak automatability, and whether resolution is weakly automatable is, as far as we know, open, so no separation for resolution follows. For preimage circuits $C_y = [g(x) = y]$ of suitable generators g, it is conjectured that strong proof systems have no short refutations of C_y for $y \notin \mathrm{range}(g)$ (proof complexity generators [ABSRW04, Kra04]); if

so, the PCIS verifier with such P cannot certify $p = 0$ on them by REFUTE. Excluding certified small positive upper bounds there would need the stronger hypothesis that there are no short proofs $C_y \vdash_P G$ for the circuits G of ENVELOPE nodes (Section 6.1) whose envelopes have mass $n^{-\omega(1)}$.

Two-sided, exact. Let $\Lambda \in \mathsf{UP} \cap \mathsf{coNP}$, with a polynomial-time verifier $R(u, v)$, $v \in \{0,1\}^k$ (k polynomial in $|u|$), that has exactly one witness if $u \in \Lambda$ and none otherwise, and a verifier R' for the complement. Let $C_u = \mathrm{pad}_0(R(u, \cdot))$, with $R(u, \cdot)$ written as a circuit on k inputs, so that $p(C_u) = 2^{-k-1}(1 + [u \in \Lambda])$ under the uniform distribution. If T proves that R has at most one witness and that no u has witnesses for both R and R', then V_T certifies both values exactly, from the witness of R or of R' and the instances of these two theorems. A sound efficient procedure without a prover that is complete on $\{C_u\}$ for some $\varepsilon < 1$ decides Λ with error at most 2δ, by answering $u \in \Lambda$ iff $L > 2^{-k-1}$ (the two values differ by a factor 2). For the bits of the prime factorisation, encoded canonically (prime factors in increasing order with multiplicity, padded to a fixed length, primality checked by AKS [AKS04] rather than by certificates, so that the witness is unique), ZFC proves both theorems, and this gives a two-sided separation under the worst-case assumption "factoring $\notin \mathsf{BPP}$", with V_T complete on both sides.

Why not NP-hardness? Suppose SAT reduces, by a randomized many-one reduction, to the gap problem "$p \geq t_1$ versus $p \leq t_0$" on circuits C, with $t_1 \geq \gamma t_0$ and $\gamma \geq 1 + 1/\mathrm{poly}(|C|)$, where $t_0 = 0$ is allowed (smaller gaps do not matter: a tight interval decides the gap problem when $1 + \varepsilon < \gamma$, and an efficient procedure reaches such ε in polynomial time only if $1/(\gamma - 1)$ is polynomial). If unsatisfiable formulas go to the small side and some MA system certifies $U < t_1$ there (for instance, is complete there with $1 + \varepsilon < \gamma$), then $\mathsf{coNP} \subseteq \mathsf{AM}$: Arthur samples the reduction (repeated in parallel), Merlin sends arguments, and Arthur accepts if most of them certify $U < t_1$; this is an AMA protocol, and $\mathsf{AMA} = \mathsf{AM}$ [Bab85]. If unsatisfiable formulas go to the large side, the set lower bound protocol [GS86], applied to the conjunction of $O(1/(\gamma - 1))$ independent copies of the circuit (which raises the gap to at least 2), already puts the complement of SAT in AM. A reduction that maps one side into a band of p and the other outside it is handled by combining the two protocols, with Merlin naming the side of the band when the unsatisfiable formulas lie outside it. In each case PH collapses [BHZ87]. So upper-bound and two-sided separations cannot be derived from NP-hardness by randomized many-one reductions (adaptive reductions to the gap problem are not covered); here they come from structured assumptions, such as the hardness of factoring, while lower-bound separations may come from $\mathsf{NP} \not\subseteq \mathsf{BPP}$.

Universality. Upper bounds: no argument system is complete for all circuits unless $\mathsf{coNP} \subseteq \mathsf{MA}$ (Theorem 3.4). Lower bounds, under the uniform distribution: an argument system complete for lower bounds within a constant factor on all circuits exists iff $\mathrm{prSBP} \subseteq \mathrm{prMA}$, for the class SBP of [BGM06]. If V is such a system, with factor c, and a promise problem has a randomized polynomial-time machine that accepts with probability above 2^{-j} on yes-instances and below 2^{-j-1} on no-instances, the conjunction C of $k \geq \max(1, \lceil \log_2 c \rceil)$ runs with independent coins has $p(C) > 2^{-kj}$, respectively $p(C) < 2^{-kj}/c$, so accepting iff V outputs $L \geq 2^{-kj}/c$ is an MA protocol. Conversely, "$p(C) > 2^{-j}$ versus $p(C) < 2^{-j-1}$" is in prSBP; given an MA protocol for it, the argument consisting of $j = \lfloor \log_2(1/p) \rfloor + 1$ and Merlin's proof, with output $L = 2^{-j-1}$ if Arthur accepts and $L = 0$ otherwise, is sound and complete within the factor 4. A constant factor improves to $1 + \varepsilon$ by applying the system to the conjunction of $O(1/\varepsilon)$ independent copies of C and taking the corresponding root of L, rounded down to a rational of polynomial bit length (still

at most p, as $L \le p^k$ for k copies). The inclusion fails relative to an oracle (one with $\mathsf{SBP} \not\subseteq \Sigma_2^p$ [BGM06], while $\mathsf{MA} \subseteq \Sigma_2^p$ relativizes). It holds if $\mathrm{prAM} = \mathrm{prNP}$, since $\mathrm{prSBP} \subseteq \mathrm{prAM}$ by the set lower bound protocol [GS86]; the hitting-set derandomisation of Miltersen and Vinodchandran [MV05], which applies to promise problems, gives this if some language in $\mathsf{NE} \cap \mathsf{coNE}$ needs nondeterministic circuits of size $2^{\Omega(n)}$, and the verifier's soundness then rests on that hypothesis.

7 Experiments

The package `rareprob` (Python, numpy) implements the procedures above: bit-parallel circuit evaluation, NNF conversion with small-cone canonicalisation, the envelope compiler and estimator with boxes, DFSB and the meta-procedure, context-width compilation, variable elimination, exponential tilting, the Harris–Janson sandwich, certified telescoping, the tilted Glauber estimator, the affine presolve and affine envelopes, Gaussian-elimination forcing, a two-watched-literal DRUP checker, the PCIS verifier for envelope arguments with an honest prover that calls a CDCL solver with proof logging (Glucose 4), and the XPT verifier. Threshold gates are evaluated exactly, as defined in Section 2, and constants are normalised to 0 and 1; all threshold gates in the experiments have integer weights and thresholds. The implementation simplifies some procedures: it fixes the atom size at $a = 12$ and the (R7) constants in DFSB, uses greedy elimination orders, replaces the pilot of the tilted Glauber estimator by a heuristic one, runs a subset of the modules in the meta-procedure (Section 5.9), and expands a threshold gate only when the absolute weights of its integer form sum to at most 4096 (others are opaque); none of this affects soundness. Intervals are computed as in the proofs (Hoeffding–KL intervals combined by union bounds, and the Bernstein and Maurer bounds of Section 5.7). Monte Carlo is not run for tiny p: we report the number of samples its KL interval needs, at the same (ε, δ), when the empirical mean equals p (computed for $p \ge 10^{-6}$ and extrapolated as c/p below; this favours Monte Carlo). Times are wall-clock seconds on a shared 4-core machine on which several experiments ran at once, so they indicate orders of magnitude only. Scripts are in `experiments/`. Tables 3 to 8 and the figures are generated by `make_tables.py` and `make_figures.py` from the raw output of the script named in each caption; Tables 1 and 2 are transcribed from `results/local.json` and `results/affine.json`.

Soundness (Table 3). We ran each procedure 100 to 300 times at $\delta = 0.1$ on random circuits and on families with known p, and counted the intervals that miss p. The reference values are exact rationals: computed by enumeration (random circuits with random marginals, DNFs, unate CNFs, and the circuits of the arguments), or exact products of dyadic block probabilities, or, for thresholds of blocks, exact Poisson-binomial tails over the dyadic block probabilities. Intervals are compared with them without slack, so floating-point errors of the procedures would count as misses. The arguments, on obfuscated DNFs, were of five kinds: the honest envelope with a DRUP proof from the solver; a wrong envelope (one term dropped) with the honest proof of the right one; the wrong envelope with ten random unit clauses and the empty clause as "proof"; an envelope disjoint from most of C's mass, with no proof; and a malformed envelope, the honest one plus a non-decomposable product $x_i \wedge \neg x_i$ whose second literal declares an empty support, with the honest proof and with none. No interval missed p in the 2,250 trials, 750 of them with dishonest arguments, nor in the 600 affine and 16 local-threshold coverage runs of Sections 5.7 and 5.8; the intervals are conservative (union bounds, KL intervals, outward rounding), so miss rates far below δ are expected.

procedure	trials	misses	miss rate
envelope (auto) on random circuits	300	0	0.000
DFSB on random circuits	150	0	0.000
KLM envelope on DNFs	300	0	0.000
envelope with boxes on block products	100	0	0.000
exponential tilting on thresholds	200	0	0.000
Harris-Janson sandwich (deterministic)	300	0	0.000
argument: honest envelope + DRUP proof	150	0	0.000
argument: wrong envelope + stale proof	150	0	0.000
argument: wrong envelope + bogus proof	150	0	0.000
argument: disjoint envelope, no proof	150	0	0.000
argument: malformed envelope (lied supports)	300	0	0.000

Table 3: Empirical miss rates at $\delta = 0.1$, with intervals compared to the reference values without slack (`exp_coverage.py`).

Cost versus rarity (Figure 1, Table 4). Six families with exactly known p under the uniform distribution, at $\varepsilon = 0.1$, $\delta = 0.05$, all run through DFSB except the last:

- *obfuscated DNF* (`dnf-obfuscated`): 16 random terms of width $w = 8, \ldots, 56$ over 64 variables, each literal routed through up to two $(x \oplus r) \oplus r$ gadgets, r a random other input variable, terms built as balanced trees, the $\vee$ written as $\neg \wedge \neg$. Small-cone canonicalisation recovers the literals, and for $w \geq 16$ the root is an envelope leaf whose envelope is the DNF itself ($K \approx 1$): 1.24 to $1.65 \cdot 10^5$ evaluations and ratio 1.004 at every w, against $1.33 \cdot 10^7$ to $1.46 \cdot 10^{19}$ for Monte Carlo. At $w = 8$ ($p = 0.06$) the root is a dense leaf, and the accuracy split (the leaf is run to ratio $1 + \varepsilon/3$) and the pilots make DFSB 22 times costlier than Monte Carlo;

- *triggers* (`trigger`): $R(x_{\text{body}}) \wedge [x_S = a]$ with $k = 10, \ldots, 80$ trigger bits, the test written as $\neg(x_i \oplus a_i)$ gates and R a random 40-gate circuit drawn on 16 body variables (of probability 0.5 to 0.78), whose output depends on only 4 to 9 of them. Forcing finds $x_S = a$, the restricted body is then an atom (at most 12 variables), and DFSB computes p exactly, with no samples, down to $p = 4 \cdot 10^{-25}$ (Corollary 5.3(b));

- *triggers with a dense body* (`trigger-cycle20`): an $\wedge$ of $k = 20, 40, 80$ literals and the body $[\sum_i (x_{v_i} \oplus x_{v_{i+1}}) \geq 12]$ around a cycle of 20 variables (probability 0.32; not decomposable, and above the atom size). After forcing, the body is one dense leaf: $1.8 \cdot 10^5$ evaluations and ratio at most 1.033 for every k, down to $p = 2.7 \cdot 10^{-25}$;

- *products of atoms* (`blocks`): $\bigwedge_{j \leq t} B_j$ with B_j random 12-gate circuits on disjoint 6-variable blocks, $t = 4, \ldots, 32$; every block is an atom, so DFSB computes p exactly by (F), (D) and (T′), with no samples (Corollary 5.3(a));

- *products of dense blocks* (`blocks-cycle16`): $\bigwedge_{j \leq d} B_j$, $d = 2, \ldots, 6$, each B_j the cycle threshold above on its own 16 variables (threshold 12, probability 0.059). Each block is a dense leaf estimated to accuracy $\varepsilon/(3d)$ (Theorem 5.2), so the cost grows like d^3 (Corollary 5.3(a)): from $1.7 \cdot 10^7$ evaluations at $d = 2$ to $4.0 \cdot 10^8$ at $d = 6$. Monte Carlo is cheaper for $d \leq 3$; the curves cross between $d = 3$ and $d = 4$ (speed-up 1.96 at $d = 4$), and at $d = 6$ DFSB is 187 times cheaper;

- *thresholds of blocks* (`threshold`): $[\sum_j B_j \geq t]$ over $k = 50, \ldots, 800$ random 8-gate blocks on 5 variables each, with t three eighths of the way from the mean to k, by exponential tilting (Proposition 4.6): $3.3 \cdot 10^4$ to $1.3 \cdot 10^5$ evaluations, with looseness K from 4.7 to 16.6, growing like $\sqrt{k}$.

The cost of Monte Carlo grows like 2^σ in the surprise $\sigma = \log_2(1/p)$. Ours is zero when every leaf is exact, flat in σ for DNFs, for triggers with a dense body and for thresholds of blocks, and polynomial in the number of sampled leaves, with a factor d^3 that lets Monte Carlo win on small products of dense blocks. Each run took at most 1.2 seconds, except the products of dense blocks (8.9 to 528 seconds).

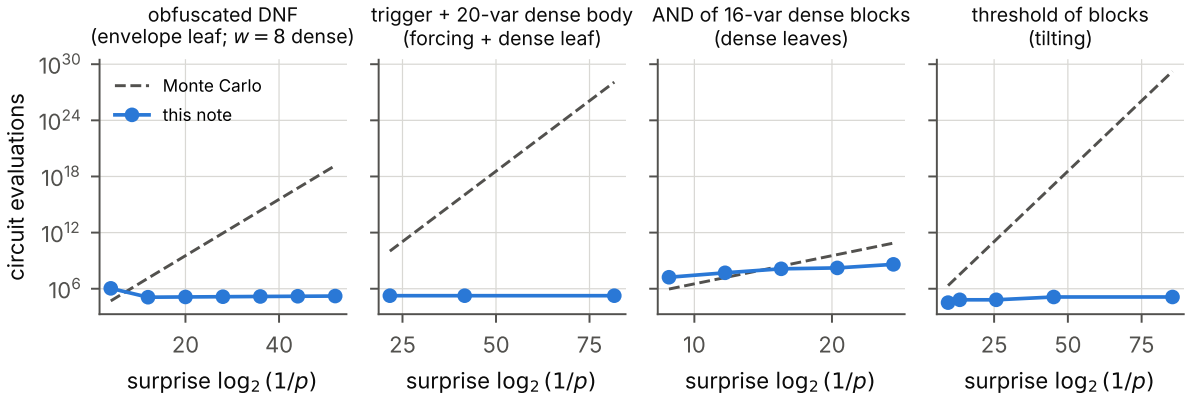


Figure 1: Circuit evaluations to reach an interval of ratio ≤ 1.1 ($\varepsilon = 0.1$, $\delta = 0.05$) against the surprise $\log_2(1/p)$, compared with Monte Carlo's requirement, for the four families of Table 4 that sample; triggers with an atom body and products of atoms are computed exactly with no samples (`exp_speedup.py`).

Context width (Table 5). ORs of t CNFs with clauses on sliding windows of width w, each CNF along its own random variable order, built as left-deep $\wedge$-chains in window order (the bracketing of Lemma 5.7, kept by the unflattened NNF). The compiler of Theorem 5.6 finds context width w, the bound of Example 5.8, and produces an equivalent DNNF 1.5 to 2.6 times the size of the circuit; the envelope estimator on it stops at the minimum batch of 1,024 samples, and for the rare instances the top $\vee$'s children are almost disjoint ($K \approx 1$, ratio 1.004). At $n = 800$ and $t = 8$ this gives $p \in [3.93, 3.95] \cdot 10^{-46}$ in 11 seconds, where Monte Carlo would need about $8 \cdot 10^{48}$ samples. Instances with $n \leq 24$ agree with enumeration.

Certified independence (Table 6). Unate random k-CNFs with up to $2 \cdot 10^5$ clauses on up to $2 \cdot 10^6$ variables: the Harris–Janson interval is deterministic and took from under a second to about 3 minutes in Python, and on 200 small instances it contained the exact p (computed in rational arithmetic) every time. At $n = 2 \cdot 10^6$ it gives $p \in [6.18, 7.35] \cdot 10^{-22}$ (ratio 1.19) in 71 seconds; Monte Carlo would need about $1.6 \cdot 10^{24}$ evaluations for the same ratio. The second-order sandwich of Appendix A was checked against enumeration on 382 random instances with no violations.

family	param	p	interval ratio	cost	Monte Carlo	speed-up
dnf-obfuscated	8	$6.08 \cdot 10^{-2}$	1.033	$1.10 \cdot 10^{6}$	$5.02 \cdot 10^{4}$	$4.58 \cdot 10^{-2}$
dnf-obfuscated	16	$2.44 \cdot 10^{-4}$	1.004	$1.24 \cdot 10^{5}$	$1.33 \cdot 10^{7}$	$1.07 \cdot 10^{2}$
dnf-obfuscated	24	$9.54 \cdot 10^{-7}$	1.004	$1.32 \cdot 10^{5}$	$3.41 \cdot 10^{9}$	$2.58 \cdot 10^{4}$
dnf-obfuscated	32	$3.73 \cdot 10^{-9}$	1.004	$1.40 \cdot 10^{5}$	$8.73 \cdot 10^{11}$	$6.22 \cdot 10^{6}$
dnf-obfuscated	40	$1.46 \cdot 10^{-11}$	1.004	$1.48 \cdot 10^{5}$	$2.23 \cdot 10^{14}$	$1.50 \cdot 10^{9}$
dnf-obfuscated	48	$5.68 \cdot 10^{-14}$	1.004	$1.57 \cdot 10^{5}$	$5.72 \cdot 10^{16}$	$3.65 \cdot 10^{11}$
dnf-obfuscated	56	$2.22 \cdot 10^{-16}$	1.004	$1.65 \cdot 10^{5}$	$1.46 \cdot 10^{19}$	$8.88 \cdot 10^{13}$
trigger	10	$7.63 \cdot 10^{-4}$	1.000	0	$4.26 \cdot 10^{6}$	$4.26 \cdot 10^{6}$
trigger	20	$5.36 \cdot 10^{-7}$	1.000	0	$6.06 \cdot 10^{9}$	$6.06 \cdot 10^{9}$
trigger	30	$4.66 \cdot 10^{-10}$	1.000	0	$6.98 \cdot 10^{12}$	$6.98 \cdot 10^{12}$
trigger	40	$5.68 \cdot 10^{-13}$	1.000	0	$5.72 \cdot 10^{15}$	$5.72 \cdot 10^{15}$
trigger	60	$4.34 \cdot 10^{-19}$	1.000	0	$7.50 \cdot 10^{21}$	$7.50 \cdot 10^{21}$
trigger	80	$4.14 \cdot 10^{-25}$	1.000	0	$7.86 \cdot 10^{27}$	$7.86 \cdot 10^{27}$
trigger-cycle20	20	$3.09 \cdot 10^{-7}$	1.029	$1.80 \cdot 10^{5}$	$1.05 \cdot 10^{10}$	$5.84 \cdot 10^{4}$
trigger-cycle20	40	$2.94 \cdot 10^{-13}$	1.033	$1.80 \cdot 10^{5}$	$1.10 \cdot 10^{16}$	$6.12 \cdot 10^{10}$
trigger-cycle20	80	$2.68 \cdot 10^{-25}$	1.033	$1.80 \cdot 10^{5}$	$1.21 \cdot 10^{28}$	$6.73 \cdot 10^{22}$
blocks	4	$6.59 \cdot 10^{-2}$	1.000	0	$4.60 \cdot 10^{4}$	$4.60 \cdot 10^{4}$
blocks	8	$4.13 \cdot 10^{-3}$	1.000	0	$7.83 \cdot 10^{5}$	$7.83 \cdot 10^{5}$
blocks	16	$6.27 \cdot 10^{-5}$	1.000	0	$5.19 \cdot 10^{7}$	$5.19 \cdot 10^{7}$
blocks	24	$1.36 \cdot 10^{-7}$	1.000	0	$2.39 \cdot 10^{10}$	$2.39 \cdot 10^{10}$
blocks	32	$1.52 \cdot 10^{-9}$	1.000	0	$2.15 \cdot 10^{12}$	$2.15 \cdot 10^{12}$
blocks-cycle16	2	$3.51 \cdot 10^{-3}$	1.025	$1.69 \cdot 10^{7}$	$9.23 \cdot 10^{5}$	$5.47 \cdot 10^{-2}$
blocks-cycle16	3	$2.08 \cdot 10^{-4}$	1.027	$5.05 \cdot 10^{7}$	$1.56 \cdot 10^{7}$	0.31
blocks-cycle16	4	$1.23 \cdot 10^{-5}$	1.026	$1.34 \cdot 10^{8}$	$2.64 \cdot 10^{8}$	1.96
blocks-cycle16	5	$7.29 \cdot 10^{-7}$	1.033	$1.68 \cdot 10^{8}$	$4.46 \cdot 10^{9}$	26.5
blocks-cycle16	6	$4.32 \cdot 10^{-8}$	1.029	$4.03 \cdot 10^{8}$	$7.53 \cdot 10^{10}$	$1.87 \cdot 10^{2}$
threshold	50	$1.49 \cdot 10^{-3}$	1.088	$3.28 \cdot 10^{4}$	$2.18 \cdot 10^{6}$	66.6
threshold	100	$9.53 \cdot 10^{-5}$	1.075	$6.55 \cdot 10^{4}$	$3.41 \cdot 10^{7}$	$5.21 \cdot 10^{2}$
threshold	200	$1.82 \cdot 10^{-8}$	1.091	$6.55 \cdot 10^{4}$	$1.79 \cdot 10^{11}$	$2.73 \cdot 10^{6}$
threshold	400	$2.47 \cdot 10^{-14}$	1.078	$1.31 \cdot 10^{5}$	$1.32 \cdot 10^{17}$	$1.01 \cdot 10^{12}$
threshold	800	$1.87 \cdot 10^{-26}$	1.093	$1.31 \cdot 10^{5}$	$1.74 \cdot 10^{29}$	$1.33 \cdot 10^{24}$

Table 4: The runs of Figure 1 and the two exact families. "Param" is w, k, k, t, d and k for the six families in the order of the text. "Cost" counts sampled circuit evaluations (pilots, leaf and box sampling); exact tabulation of atoms and small cones is not counted. "Speed-up" is Monte Carlo's requirement divided by the cost; for the exact rows (cost 0) it shows Monte Carlo's requirement.

Certified telescoping (Table 7). Hard-core ladders: $x \sim \mathrm{Bern}(0.3)^n$ and $C =$ "x is an independent set of the $2 \times L$ ladder", $L = 20, 40, 60$. Janson's Δ is extensive, so the sandwich is useless (its window spans a factor 25 to $2.6 \cdot 10^4$); bounding-chain CFTP gives exact samples, the truncation certificates contribute a bias factor below $e^{0.02}$, and the certified intervals (ratios 1.29 to 1.61) contain the exact p (a rational, from a transfer matrix), as do the Harris–Janson windows. The number of CFTP runs grows linearly in L, with the number of stages, from $2.3 \cdot 10^5$ to $6.8 \cdot 10^5$, while $1/p$ grows exponentially. In pure Python the constants are poor (a run for a later stage makes at least $4|A|$ single-site updates of both bounding chains, A the variables of the earlier stages, and the three instances took 22 to 294 seconds), so for these moderate p Monte Carlo is still faster; the point is that the cost grows polynomially in L and that the method is sound without any mixing-time proof.

Local thresholds (Table 1). Section 5.7 reports the tilted Glauber estimator on chain and random-cubic-graph local threshold circuits with random local tables: $p = 10^{-20.19}$ and $p \approx 10^{-23.7}$ certified to ratios 1.66 and 1.84 with about $7 \cdot 10^4$ chain samples, where Monte Carlo would need about $2 \cdot 10^{22}$ and $3 \cdot 10^{25}$, and $p = 10^{-169.8}$ to ratio 2.88 with coupling from the past beyond the Dobrushin regime. The values are checked against exact transfer-matrix values for chains and 10^7 Monte Carlo samples for a small cubic instance (`exp_local.py`).

n	t	w	cw	$\lvert C\rvert$	DNNF size	interval for p	samples	time (s)
20	4	3	3	249	640	[0.389, 0.417] ($p = 0.393$)	1024	0.1
24	4	3	3	304	627	[0.209, 0.221] ($p = 0.215$)	1024	0.1
100	4	3	3	1375	2350	$[9.50 \cdot 10^{-6}, 9.55 \cdot 10^{-6}]$	1024	0.3
200	4	3	3	2757	4575	$[1.57 \cdot 10^{-11}, 1.58 \cdot 10^{-11}]$	1024	0.8
400	4	3	3	5633	8927	$[2.91 \cdot 10^{-23}, 2.93 \cdot 10^{-23}]$	1024	1.8
800	8	3	3	22300	33927	$[3.93 \cdot 10^{-46}, 3.95 \cdot 10^{-46}]$	1024	11.1
400	4	5	5	7115	15137	$[1.45 \cdot 10^{-5}, 1.47 \cdot 10^{-5}]$	1024	5.4

Table 5: Context-width compilation of ORs of window-CNFs ($\varepsilon = 0.1$, $\delta = 0.05$; `exp_ctx.py`). Time is compilation plus estimation.

n	clauses	width	certified interval	ratio	time (s)
10000	2000	10	[0.141, 0.148]	1.039	0.8
100000	20000	12	$[7.57 \cdot 10^{-3}, 7.84 \cdot 10^{-3}]$	1.035	11.0
200000	60000	12	$[4.33 \cdot 10^{-7}, 5.07 \cdot 10^{-7}]$	1.167	50.3
1000000	200000	14	$[4.99 \cdot 10^{-6}, 5.15 \cdot 10^{-6}]$	1.030	194.2
2000000	200000	12	$[6.18 \cdot 10^{-22}, 7.35 \cdot 10^{-22}]$	1.187	71.0

Table 6: Harris–Janson sandwich for random unate CNFs (`exp_janson.py`).

Affine structure (Table 2). Section 5.8 reports the presolve, affine envelopes, XPT arguments and DFSB on parity-check families under uniform and biased product distributions, with exact values computed without the module under test (`exp_affine.py`). For instance, 100 dense random parity checks on 600 inputs and a 3-DNF give $p = 4.30 \cdot 10^{-31}$, which the presolve certifies to ratio 1.045 with $3.3 \cdot 10^4$ samples, where Monte Carlo would need $4.1 \cdot 10^{34}$.

What the prover adds (Table 8, Figure 2). Hidden triggers $R \wedge T(x_S)$, with R a random 40-gate circuit on 12 body variables and a test that accepts only $x_S = a$: *linear-hash* tests $[Mx_S = Ma]$ with a dense $(k+4)\times k$ matrix of rank k over $\mathbb{F}_2$, and *SPN* tests $[\mathrm{SPN}(x_S) = \mathrm{SPN}(a)]$ with random bijective 4-bit S-boxes, dense invertible linear layers and random round constants. Without a prover, Gaussian-elimination forcing recognises linear hashes: Part 1 fixes all k trigger bits and computes p exactly. SPN tests are reached only through small-cone canonicalisation: a test on at most $a = 12$ variables is an atom whose OBDD exposes $x_S = a$ to forcing ($k = 8, 12$: exact, with no samples), and DFSB reduces larger tests to atoms by branching, with about 2^{k-a} leaves. At $k = 16$ it computed p exactly after $1.1 \cdot 10^8$ evaluations in its pilots (659 seconds); at $k = 20$ it gave no answer within 900 seconds. With a prover, the argument is the subcube $x_S = a$ plus a DRUP proof from the solver that $C \Rightarrow (x_S = a)$; the verifier checks it with its own RUP checker and samples only R, reaching ratio at most 1.093. The proofs found by the solver grow exponentially with k even for linear hashes (329 to 32,910 steps for $k = 12$ to 24; CDCL solvers do not perform Gaussian elimination); this is evidence, not a resolution lower bound, that the class of Theorem 6.5 depends on the proof system. The solver found no proof for the 16-bit SPN instance within 1200 seconds, but one of 25,944 steps for the 20-bit instance, on which the verifier (proof check and sampling) took 39.5 seconds.

Surprise accounting (Figure 3). For a 40-bit trigger on a body R with $\Pr[R] = 1/8$, an argument that reveals j of the bits (a subcube envelope on those bits, with a DRUP proof of domination) leaves $\sigma_u = (40 - j) + 3$ bits to sampling. At $\varepsilon = 0.2$, $\delta = 0.05$ the verifier's sample count grows from $1.6 \cdot 10^4$ with every bit revealed to $1.7 \cdot 10^7$ with ten unrevealed, a factor 2^{10} over ten bits; with twelve unrevealed it stopped at its cap of $2 \cdot 10^7$ samples with a valid interval of

L	n	p (exact)	Harris–Janson window	certified telescoping	CFTP runs
20	40	$2.31 \cdot 10^{-2}$	$[4.21 \cdot 10^{-3}, 0.107]$	$[2.07 \cdot 10^{-2}, 2.69 \cdot 10^{-2}]$	$2.34 \cdot 10^{5}$
40	80	$4.90 \cdot 10^{-4}$	$[1.46 \cdot 10^{-5}, 1.18 \cdot 10^{-2}]$	$[4.15 \cdot 10^{-4}, 6.06 \cdot 10^{-4}]$	$4.42 \cdot 10^{5}$
60	120	$1.04 \cdot 10^{-5}$	$[5.12 \cdot 10^{-8}, 1.31 \cdot 10^{-3}]$	$[8.36 \cdot 10^{-6}, 1.35 \cdot 10^{-5}]$	$6.76 \cdot 10^{5}$

Table 7: Certified telescoping on hard-core ladders (`exp_cftp.py`).

family	k	$\lvert C\rvert$	p	Part 1 ratio	proof steps	check (s)	Part 2 ratio
linear hash	12	78	$1.53 \cdot 10^{-4}$	1	329	0.0	1.088
linear hash	16	86	$3.81 \cdot 10^{-6}$	1	1269	0.2	1.079
linear hash	20	82	$5.96 \cdot 10^{-7}$	1	5717	2.1	1.093
linear hash	24	96	$2.98 \cdot 10^{-8}$	1	32910	81.4	1.085
SPN, 2 rounds	8	246	$3.85 \cdot 10^{-3}$	1	278	0.0	1.027
SPN, 2 rounds	12	334	$1.37 \cdot 10^{-4}$	1	697	0.2	1.074
SPN, 3 rounds	16	609	$6.68 \cdot 10^{-6}$	1	no proof within 1200 s		
SPN, 3 rounds	20	756	$9.54 \cdot 10^{-7}$	none in 900 s	25944	39.5	1.004

Table 8: Hidden triggers with and without a prover (`exp_arguments.py`). Part 1 ratio is that of the DFSB interval; Part 2 ratio is that of the verifier's interval after checking the proof; "check" is the verifier's time (proof check and sampling). For the 16-bit SPN instance the solver found no proof within 1200 seconds, and Part 1 was exact there.

ratio 1.42. This is the $O(2^{\sigma_u})$ count of the envelope estimator, and it is consistent with the lower bound of Theorem 6.12(b) for verifiers that use the rest of the circuit only through evaluations; that bound is a worst case over planted circuits and says nothing specific about this instance.

8 Case study: networks trained to output a constant

This section tests the framework on the case that motivates it: a ReLU network trained to always output 1 on uniformly random inputs $x \in \{-1, +1\}^d$, and the rare inputs on which it fails. The answer is sharply divided.

- *Sound lower bounds work very well wherever p could be checked.* Fixed-level splitting (Theorem 6.10) combined with a betting confidence bound (Proposition 8.3) certifies L/p of 0.920–0.942 on the exact suite (for $p \geq 10^{-5}$), and 0.916–0.973 of a reference value p_{ref} on the 8 nets at $d = 32$–128 whose reference was checked independently (exactly at $d = 32$, by Monte Carlo down to $p \approx 6.3 \cdot 10^{-9}$), with no assumption on mixing (Sections 8.4 and 8.5). On 7 more nets, with p_{ref} down to $5.2 \cdot 10^{-21}$, it reaches 0.865–0.945 of p_{ref}; there p_{ref} is the mean of more replicates of the same estimator, which cannot reveal a failure mode that the chains never reach (Section 8.2). These runs cost $4.1 \cdot 10^{6}$–$5.7 \cdot 10^{7}$ network evaluations (19–550 s).

- *Exact values and certificates of $p = 0$ work at $d \lesssim 30$, or when the margin is large.* Certified partitions (or plain enumeration, where cheaper) count the failures of every suite net with $d \leq 28$ exactly, and certificates of $p = 0$ are cheap for long-trained nets (Section 8.6).

- *For squared-loss nets with $p > 0$ at $d \geq 40$, no sound two-sided method we found beats Monte Carlo.* Section 8.7 explains why: trained failures are small-ball events centred at the middle of the box, and three barriers (box, cube, decoupling) block every bound we could make cheap.

What remains open is listed at the end (Section 8.8).

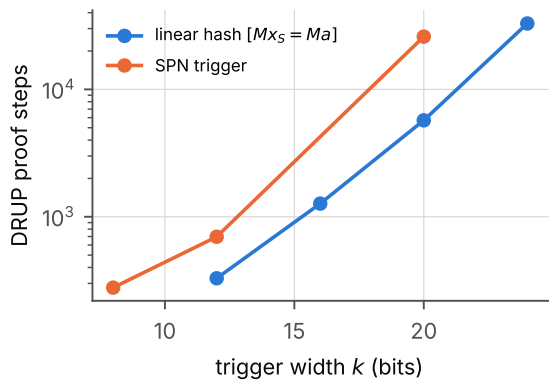


Figure 2: DRUP proof size for hidden triggers (SPN: 2 rounds for $k \le 12$, 3 for $k = 20$; the 16-bit instance, with no proof within 1200 seconds, is omitted).

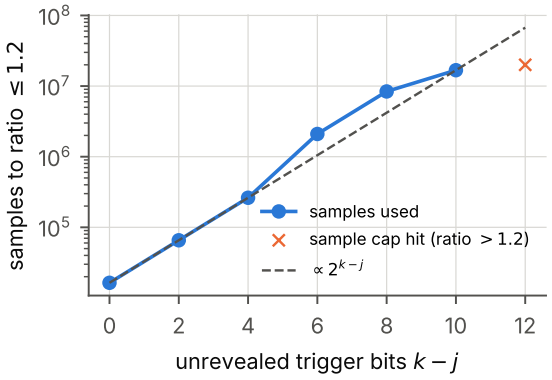


Figure 3: Surprise accounting: samples to reach ratio ≤ 1.2 when an argument reveals all but $k - j$ bits of a 40-bit trigger, against $k - j$ ($\sigma_u = k - j + 3$). The cross marks a run stopped at its cap of $2 \cdot 10^7$ samples.

Test bed and costs. A net has float64 weights W_ℓ, b_ℓ and computes

$$f(x) = w_L \cdot \mathrm{relu}(W_{L-1} \cdots \mathrm{relu}(W_1 x + b_1) \cdots + b_{L-1}) + b_L;$$

we write F for the same expression in exact real arithmetic on the exact binary values of the weights (as for threshold gates in Section 2). It was trained with Adam on fresh uniform inputs towards the constant target 1, either with the squared loss, where the rare event is $C(x) = [F(x) < t]$ with $t = \frac{1}{2}$, or with the logistic loss on a logit, where $t = 0$. D is uniform on $\{-1, +1\}^d$ and $p = \Pr_D[F(x) < t]$. Since the inputs are ± 1 and the weights dyadic, C is a polynomial-size circuit in the sense of Section 2 (exact integer arithmetic on the scaled weights), so all of Sections 3 to 6 applies; the procedures below use only the weights and evaluations of F. The suite (in `experiments/nets`) has 30 nets, $d = 24, 28, 64, 128$, with one or two hidden layers of width 32 to 128 and both losses; a name such as `mse_d24_w32_s1500` gives the loss, d, the widths and the number of training steps. p is exact (by enumeration) for the 21 nets with $d \le 28$; $z = (\mathbb{E}f - t)/\mathrm{sd}(f)$ measures how far into the tail training has pushed the event. Costs are reported in forward-pass equivalents (fpe: one evaluation of the net on one input; a verifier's arithmetic is divided by that of one evaluation) and in wall-clock seconds on a *shared 4-core machine* on which other jobs ran at the same time, so seconds indicate orders of magnitude only. Monte Carlo with $3\ln(2/\delta)/(\varepsilon^2 p)$ samples achieves relative error ε with probability $1 - \delta$ (multiplicative Chernoff bound; an interval of ratio about $1 + 2\varepsilon$); at $\varepsilon = 0.1$, $\delta = 0.05$ this is about $1107/p$ samples. The count is sufficient, not necessary, and the direction favours Monte Carlo. Monte Carlo seconds use one time per sample for each net, the median of the measurements made in the experiments, which differ by up to 2.7 times on the shared machine.

8.1 Certified evaluation

Every method below decides $[F(x) < t]$ for many x, but evaluates f in floating point. An *admissible evaluation* uses correctly rounded binary64 additions, subtractions, multiplications or fused multiply–adds (rounding to nearest, gradual underflow), computes relu exactly as $\max(z, 0)$,

and may sum each dot product in any order. Let $u = 2^{-53}$, $\eta = 2^{-1075}$, $\gamma_n = nu/(1-nu)$, and Ω the largest finite double. Define, entrywise, $H_0 = \mathbf{1}$, $e_0 = 0$ and for each layer ℓ of fan-in n_ℓ

$$A_\ell = |W_\ell| H_{\ell-1} + |b_\ell|, \qquad e_\ell = \gamma_{n_\ell+1} A_\ell + (2n_\ell + 2)\eta\,\mathbf{1} + |W_\ell| e_{\ell-1}, \qquad H_\ell = A_\ell + e_\ell,$$

computed in exact rational arithmetic, and let $e = e_L$ (rounded up to a double).

Lemma 8.1 (certified evaluation)**.** *If every fan-in is at most 2^{52} and every entry of every H_ℓ is at most Ω, then no admissible evaluation overflows, and $|f_{\mathrm{fl}}(x) - F(x)| \le e$ for every $x \in \{-1,+1\}^d$ and every admissible evaluation f_{fl}. Consequently, for the computed value $v = f_{\mathrm{fl}}(x)$: if $v < t - e$ then $F(x) < t$; if $v \ge t + e$ then $F(x) \ge t$; and if $v < t - 2e$ then every admissible evaluation of f at x is below t. Points with $t - e \le v < t + e$, or with v not finite, are undecided: the verifiers re-evaluate them in exact rational arithmetic, and the splitting procedure, which flags every value within $2e$ of a level, reports $Z = 0$ for the replicate that met them.*

Proof. As long as nothing overflows, a multiplication or fused multiply–add with exact result y returns $y(1+\epsilon) + \eta'$ with $|\epsilon| \le u$ and $|\eta'| \le \eta$, and an addition or subtraction returns $y(1+\epsilon)$, since a sum whose exact value is subnormal is computed exactly [Hig02, Chap. 2]. Consider $s = b + \sum_{i \le n} w_i h_i$ computed in any order, with floating-point operands. Each of the $n+1$ terms passes through at most $n+1$ roundings (its product, or the fused multiply–add that forms it, and at most n additions), so it is multiplied by some $1+\theta$ with $|\theta| \le \gamma_{n+1}$ [Hig02, Lemma 3.1]; each of the at most n terms η' is multiplied by at most n factors $1+\epsilon$, hence by at most $(1+u)^n \le 2$ (as $nu \le \frac{1}{2}$). So $|\hat{s} - s| \le \gamma_{n+1}(|b| + \sum_i |w_i||h_i|) + 2n\eta$, and every partial sum, computed or exact, is at most $(1+\gamma_{n+1})(|b| + \sum_i |w_i||h_i|) + 2n\eta$ in absolute value.

Let h_ℓ and $\hat{h}_\ell$ be the exact and computed outputs of layer ℓ (after relu for hidden layers), and suppose $|h_{\ell-1}|, |\hat{h}_{\ell-1}| \le H_{\ell-1}$ and $|\hat{h}_{\ell-1} - h_{\ell-1}| \le e_{\ell-1}$ entrywise (true for $\ell = 1$, where both are x). The computed pre-activation $\hat{z}$ of a unit satisfies $|\hat{z} - (b + w \cdot \hat{h}_{\ell-1})| \le \gamma_{n+1} A_\ell + 2n\eta$ by the dot-product bound (as $|b| + |w| \cdot |\hat{h}_{\ell-1}| \le A_\ell$), and $|w \cdot (\hat{h}_{\ell-1} - h_{\ell-1})| \le |w| \cdot e_{\ell-1}$; so $|\hat{z} - z| \le e_\ell$. As relu is exact and 1-Lipschitz, $|\hat{h}_\ell - h_\ell| \le e_\ell$, and $|h_\ell| \le A_\ell \le H_\ell$, $|\hat{h}_\ell| \le A_\ell + e_\ell = H_\ell$. Overflow: take the first operation of an admissible evaluation that would overflow. All earlier ones obey the model, so its exact result is a product or a partial sum of layer ℓ, bounded as above by $(1+\gamma_{n_\ell+1})A_\ell + 2n_\ell\eta \le H_\ell \le \Omega$, and rounding to nearest maps a real of absolute value at most Ω to a finite double: a contradiction. The consequences follow from $|v - F(x)| \le e$ and, for the last one, $|v - v'| \le 2e$ for two admissible evaluations v, v'. □

The bound is checked against exact rational evaluation in three evaluation orders (numpy, left to right, fused multiply–add) on 11 nets, among them the adversarial ones below; it exceeds the largest observed error by a factor 4.1 to 1800. Every engine of this section had an edge-case hole of this kind before it was rewritten, found by an adversarial review and now a regression test: products that underflow to subnormal numbers (or to zero) and are then multiplied by huge downstream weights; a fused multiply–add that rounds a subnormal tie differently from a separate product and sum; partial sums that overflow although every weight is finite; a NaN produced by an overflow and then read as a "dead" unit; an output that overflows to $+\infty$ and was then read as surely above t; an exact count $|W|/2^d$ rounded to nearest, hence up, in the subnormal range ($d \ge 1075$); and a unit dropped from a linear bound without charging the rounding of its own coefficient. Each made some version of our code return a wrong certified answer, typically an "exact" $p = 0$ for a net with $p = \frac{1}{2}$ or $p = 1$. None affects the trained nets, whose weights are far from these ranges, but soundness on every net requires Lemma 8.1 or an equivalent outward-rounded interval arithmetic (which the certified-partition verifier of Section 8.6 uses), and treating

every non-finite value as undecided. On the final code, against exact rational counts on small adversarial nets (extreme and subnormal scales, exact ties, crafted cancellation, $p = 0$ and $p = 1$): the certified-partition verifier passed 37,553 bound checks, 260 honest exact certificates and 1,144 dishonest or malformed arguments with no error, and its sampled intervals missed in 13 of 1,300 runs at $\delta = 0.1$; branch and bound passed 12,000 bound checks, 232 complete runs and 796 runs in the overflow window with no violation; the analytic bounds were checked on 168 nets, with 0 misses in 540 interval runs.

8.2 The architecture is as hard as circuits

Proposition 8.2 (hardness inside the architecture)**.** *Identify $x_i = -1$ with "variable i is true".*

(a) For a 3-CNF $\varphi = \bigwedge_{j\le m}(\lambda_{j1} \vee \lambda_{j2} \vee \lambda_{j3})$ let $\ell_{jk}(x) = x_i$ if $\lambda_{jk} = v_i$ and $\ell_{jk}(x) = -x_i$ if $\lambda_{jk} = \neg v_i$, and

$$f_\varphi(x) = \sum_{j=1}^{m} \operatorname{relu}(\ell_{j1}(x) + \ell_{j2}(x) + \ell_{j3}(x) - 2), \qquad t = \tfrac{1}{2}.$$

This is a one-hidden-layer net of width m with integer weights, $f_\varphi(x)$ is the number of clauses that x violates, and $[f_\varphi < t]$ is the set of satisfying assignments. Hence a sound efficient procedure that is complete on the nets f_φ gives $\mathsf{NP} \subseteq \mathsf{BPP}$, and an argument system complete for upper bounds on those with $p = 0$ gives $\mathsf{coNP} \subseteq \mathsf{MA}$.

(b) For a net f, a set I of $k \ge 1$ inputs, $b \in \{-1, +1\}^I$ and a power of two $B > \max_x F(x) - t$ (computable, e.g. from $|w_L|H_{L-1} + |b_L|$), let

$$f_{I,b}(x) = f(x) - B\, g_{I,b}(x), \qquad g_{I,b}(x) = \operatorname{relu}\Big(\sum_{i\in I} b_i x_i - (k-1)\Big),$$

with the extra unit carried to the output through each later hidden layer by a unit with weight 1 from it and 0 to the others ($\operatorname{relu}(g) = g$ as $g \ge 0$). Then $f_{I,b} = f$ outside $S_b = [x_I = b]$ and $F_{I,b} < t$ on S_b, so $[F_{I,b} < t] = C \vee \mathbf{1}_{S_b}$ and $p \le p(f_{I,b}) \le p + 2^{-k}$.

(c) Consequently Theorem 3.2 and Theorem 6.12(a) hold around every net: if a verifier reads an argument and accesses the net only through T evaluations of its real output, and is sound for upper bounds on all nets of the architecture (with one extra unit per hidden layer), then $\Pr[V^f$ outputs $U < 2^{-k}] \le \delta + T2^{-k}$ for every net f; and if it reads ℓ argument bits, is δ-sound for lower bounds on all nets, and for every b some argument makes it output $L > 0$ on $f_{I,b}$ with probability $1 - \delta$, where $p(f) = 0$, then $T2^\ell \ge (1 - 2\delta)2^k$.

Proof. (a) Each $\ell_{jk}(x) \in \{\pm 1\}$, and $\ell_{jk}(x) = -1$ iff the literal is true. If a clause is violated its three literals are $+1$ and the unit outputs $\operatorname{relu}(1) = 1$; otherwise the sum is at most 1 and the unit outputs 0 (a repeated variable merges into one weight of ± 2 or 0, and the argument is the same). So f_φ counts violated clauses and $f_\varphi < \frac{1}{2}$ iff it is 0. All values are small integers, so every evaluation is exact. The map $\varphi \mapsto f_\varphi$ is computable in polynomial time and preserves p under the uniform distributions (with $x_i = (-1)^{y_i}$), so a procedure or argument system for the nets f_φ gives one for 3-CNFs, and Proposition 3.1(a) and Theorem 3.4 apply. (Katz et al. [KBD+17] prove NP-hardness of related verification questions.)

(b) On S_b the sum $\sum_{i\in I} b_i x_i$ equals k and $g_{I,b} = 1$; off S_b some term is -1, the sum is at most $k - 2$, and $g_{I,b} = 0$. So $F_{I,b} = F$ off S_b and $F_{I,b} = F - B < t$ on S_b. The carried units compute $g_{I,b}$ exactly. Then $p(f_{I,b}) = p + D(S_b \setminus C^{-1}(1)) \in [p, p + 2^{-k}]$.

(c) The proofs of Theorem 3.2 and Theorem 6.12(a) use only that the planted functions agree with the original one outside the planted sets, that these sets partition the cube as b varies, and that each planted function has $p \geq 2^{-k}$ (for (a): the original function has $p = 0$). With $f_{I,b}$ in place of $f \vee \mathbf{1}_{H_t}$, resp. $\mathbf{1}_{S_b}$, and f in place of the zero function, all three hold, and the oracle answers (real outputs) of f and $f_{I,b}$ coincide off S_b. So the coupling arguments go through verbatim: for upper bounds, the runs on f and on $f_{I,b}$ with the same coins differ only if a query lands in S_b, which for uniform b has probability at most $T2^{-k}$; for lower bounds, $\sum_b \Pr_\pi[\text{hit } S_b] \leq T$ for each fixed argument π, and soundness on f forces $\Pr[\text{hit } S_b] \geq 1 - 2\delta$ under the argument for b. □

The encoding was checked against direct counting on 40 random CNFs. On a planted net with $d = 40$, $k = 26$ and $p = 1.6 \cdot 10^{-8}$ (a visible failure set of 821 inputs, $p = 7.5 \cdot 10^{-10}$ before planting, plus a hidden subcube), the splitting lower bound of Section 8.4 returned $L = 0$ and the point estimate 0 in all 5 runs (each at most $6.1 \cdot 10^6$ fpe and 6.1 s): sound. The hidden subcube was missed, as expected: averaged over b, T evaluations hit it with probability at most $T2^{-26}$, here at most 0.09 per run (the coupling in the proof of (c)). At this budget the visible part was not reached either, which (c) does not force. So efficiency on trained nets must come from their structure, and soundness may not assume it.

8.3 What trained failure sets look like

Table 9 summarises the failure sets of the suite nets small enough to analyse completely (scripts in `experiments/nets_structure`).

- *Squared loss makes the output nearly even.* $\mathrm{corr}(f(x), f(-x))$ is 0.82–0.99 on the one-hidden-layer squared-loss nets, against −0.41–0.25 for the logistic ones, and the level-1 Fourier weight is 0.3%–0.8% of $\mathrm{Var} f$ against 38%–61%. Training cancels the linear part, so $f \approx c + \frac{1}{2}\sum_j a_j |z_j|$ with $z_j = w_j \cdot x + b_j$, and a failure is a *joint small-ball event* of the units with $a_j > 0$: on `mse_d24_w32_s1500` their even part carries a fraction 0.95 of the drop of f at the failures (the linear part −0.02), and a mean of 8.4 positive units sit within one coordinate flip of zero at a failure, against 3.8 at a typical input. The even parts of the positive- and negative-weight units are correlated −0.84–−0.75.

- *The small ball sits at the centre of the box.* $F(0) - t$ ranges over −0.40–−0.02: $F(0) < t$ on all 13 squared-loss nets with $d \leq 28$, and a point of $[-1, 1]^d$ with $F < t$ exists on 29 of the 30 suite nets (on the sphere $|x|^2 = d$: 30).

- *No direction, many modes, local clumps.* The mean failing input has $|\bar{x}_i|$ of 0.03–0.22 on squared-loss nets; 0 lies in the convex hull of the failure set for all 20 one-hidden-layer squared-loss nets of width at least 32 examined (23 of the 25 of any width), so for them every halfspace containing all failures has mass at least $\frac{1}{2}$, i.e. looseness $K \geq 1/(2p)$ ($5.1 \cdot 10^4$ on `mse_d24_w32_s1500`, $1.5 \cdot 10^6$ on `mse_d28_w64_s1000`). Failing inputs are about $d/2$ apart on average (11.9 at $d = 24$) but clump under single bit flips: the 163 failures of `mse_d24_w32_s1500` form 74 clusters (the largest of 25) in 90 local minima of f.

- $p = 0$ *is the usual end state.* Of 107 one-hidden-layer nets trained for at least 1000 (squared loss) or 300 (logistic loss) steps, 98 are certified to have $p = 0$, 5 have $p > 0$ (counted exactly) and 4 remain open (Section 8.6). Squared-loss nets reach $p = 0$ in 97% of cases when the Gaussian expected number of failures $2^d \Phi(-z)$ is below 10^{-2} and in 0% when it is above 100.

net	p	F	$\mathrm{corr}(f(x), f(-x))$	W_1	$F(0) - t$	$\overline{\lvert\bar{x}_i\rvert}$	halfspace K	minima	clusters
`mse_d24_w32_s1000`	$3.6 \cdot 10^{-4}$	6012	0.93	0.3%	−0.108	0.07	1400	1600	768
`mse_d24_w32_s1500`	$9.7 \cdot 10^{-6}$	163	0.88	0.4%	−0.064	0.11	$5.1 \cdot 10^4$	90	74
`mse_d24_w32_s2000`	$6 \cdot 10^{-8}$	1	0.82	0.3%	−0.022	1.00	1.0	1	1
`mse_d28_w64_s600`	$6 \cdot 10^{-5}$	16155	0.98	0.6%	−0.309	0.06	8300	5684	3244
`mse_d28_w64_s1000`	$3.3 \cdot 10^{-7}$	89	0.96	0.6%	−0.277	0.22	$1.5 \cdot 10^6$	68	66
`mse_d28_w32x32_s500`	$1.3 \cdot 10^{-4}$	36023	0.24	0.6%	−0.240	0.16	3700	10828	5741
`bce_d28_w32_s60`	$1.3 \cdot 10^{-5}$	3543	−0.10	55.4%	0.260	0.46	10^4	916	380
`bce_d28_w64x64_s60`	$1.9 \cdot 10^{-5}$	5071	−0.38	61.8%	0.118	0.52	6400	612	159

Table 9: Failure sets of suite nets (F failing inputs). W_1: level-1 Fourier weight as a fraction of $\mathrm{Var} f$; $\overline{\lvert\bar{x}_i\rvert}$: mean over coordinates of the absolute mean failing input; halfspace K: mass of the best halfspace containing all failures, over p; minima: single-flip local minima of f among failures; clusters: connected components under single flips (`nets_structure/structure.py`, `nets_analytic/barrier_table.py`).

8.4 Sound lower bounds: splitting and betting

The splitting estimator of Theorem 6.10 has nonnegative replicates with mean at most p but an uncontrolled upper tail; clipping them, as Proposition 6.8 does, wastes their concentration. A betting e-value [WSR24] needs only nonnegativity.

Proposition 8.3 (betting lower bound)**.** *Let $Z_1, Z_2, \cdots \geq 0$ satisfy $\mathbb{E}[Z_i \mid Z_1, \ldots, Z_{i-1}] \leq p$ (e.g. i.i.d. with $\mathbb{E} Z_i \leq p$), let $\Lambda \subset (0, 1]$ be finite and $\delta \in (0, 1)$. For $p_0 > 0$ let*

$$\mathcal{E}_R(p_0) = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} \prod_{i=1}^{R} \Big(1 - \lambda + \lambda \frac{Z_i}{p_0}\Big), \qquad L_R = \sup\{p_0 \in (0, 1] : \mathcal{E}_R(p_0) \geq 1/\delta\}$$

($\sup \emptyset = 0$). Then $\mathcal{E}_R$ is nonincreasing in p_0, and

(a) (anytime validity) $\Pr[\exists R : \ L_R > p] \leq \delta$, so the running maximum of L_R and any data-dependent stopping time give valid lower bounds;

(b) (completeness) let $0 < \varepsilon \leq \frac{1}{2}$, $\mathbb{E} Z_i = \mu \geq (1 - \varepsilon/2)p$ i.i.d., and $\mathbb{E} Z_i^2 \leq (1 + v)\mu^2$. If $\Lambda \ni \lambda'$ with $\lambda^*/2 \leq \lambda' \leq \lambda^*$, where $\lambda^* = \min\{\frac{1}{2}, a/(4\sigma^2)\}$, $a = \mu/((1 - \varepsilon)p) - 1$ and $\sigma^2 = \mathbb{E}(Z/((1 - \varepsilon)p) - 1)^2$ (e.g. $\Lambda = \{2^{-1}, \ldots, 2^{-J}\}$ with $2^{-J} \leq \varepsilon/(32(1 + v))$), then $R \geq C(1 + v)\varepsilon^{-2} \log(|\Lambda|/\delta)$ replicates give $L_R \geq (1 - \varepsilon)p$ with probability at least $1 - \delta$, for an absolute constant C.

Proof. Each factor is nonnegative ($\lambda \leq 1$, $Z_i \geq 0$) and nonincreasing in p_0, hence so are the products and $\mathcal{E}_R$. (a) If $p = 0$ then all $Z_i = 0$ almost surely, $\mathcal{E}_R(p_0) = |\Lambda|^{-1} \sum_\lambda (1 - \lambda)^R \leq 1 < 1/\delta$ and $L_R = 0$. If $p > 0$, $M_R = \mathcal{E}_R(p)$ is a nonnegative supermartingale with $M_0 = 1$, since $\mathbb{E}[1 - \lambda + \lambda Z_{R+1}/p \mid Z_{\leq R}] \leq 1$ and averages of supermartingales are supermartingales. By Ville's inequality [Vil39], $\Pr[\sup_R M_R \geq 1/\delta] \leq \delta$. If $L_R > p$ for some R, there is $p_0 > p$ with $\mathcal{E}_R(p_0) \geq 1/\delta$, and by monotonicity $M_R = \mathcal{E}_R(p) \geq 1/\delta$.

(b) Let $p_0 = (1 - \varepsilon)p$ and $Y_i = Z_i/p_0 - 1 \geq -1$, with $\mathbb{E} Y = a \geq (1 - \varepsilon/2)/(1 - \varepsilon) - 1 \geq \varepsilon/2$ and $\mathbb{E} Y^2 = \sigma^2$. As $\mu \leq p$ and $\varepsilon \leq \frac{1}{2}$, $\mu/p_0 \leq 2$ and $a \leq 1$, so $\sigma^2 = \mathrm{Var}(Z)/p_0^2 + a^2 \leq 4v + 1 \leq 4(1 + v)$. It suffices that the term for $\lambda = \lambda'$ alone reaches $|\Lambda|/\delta$, i.e. $\sum_i \log(1 + \lambda' Y_i) \geq \log(|\Lambda|/\delta)$. Truncate: $\tilde{Y}_i = \min(Y_i, m)$ with $m = \sigma^2/(2a)$. Since $\log(1 + \lambda' y)$ increases in y, $\log(1 + \lambda' Y_i) \geq V_i := \log(1 + \lambda' \tilde{Y}_i)$. From $(y - m)_+ \leq y^2/(4m)$ we get $\mathbb{E}\tilde{Y} \geq a - \sigma^2/(4m) = a/2$, and $\tilde{Y}^2 \leq Y^2$. Now $\lambda' \tilde{Y}_i \in [-\frac{1}{2}, \frac{1}{8}]$ (as $\lambda' m \leq \lambda^* \sigma^2/(2a) \leq \frac{1}{8}$), and $\log(1 + w) \geq w - w^2$ on

$[-\frac{1}{2}, 1]$, so $\mathbb{E}V \geq \lambda' a/2 - \lambda'^2\sigma^2 \geq \lambda' a/4$, using $\lambda'\sigma^2 \leq \lambda^*\sigma^2 \leq a/4$. Moreover $|V_i| \leq 2|\lambda'\tilde{Y}_i| \leq 1$, so $\mathrm{Var}V \leq 4\lambda'^2\sigma^2 \leq \lambda' a$. Bernstein's inequality for the i.i.d. V_i (with $|V_i - \mathbb{E}V| \leq 2$), applied to a deviation of $R\mathbb{E}V - R\lambda' a/8 \geq R\lambda' a/8$, gives

$$\Pr\Big[\sum_i V_i \leq \frac{R\lambda' a}{8}\Big] \leq \exp\Big(-\frac{(R\lambda' a/8)^2}{2R\lambda' a + \frac{4}{3}\cdot\frac{R\lambda' a}{8}}\Big) \leq e^{-R\lambda' a/140}.$$

Once $R \geq 140\log(|\Lambda|/\delta)/(\lambda' a)$ this probability is at most δ, and otherwise

$$\sum_i \log(1+\lambda' Y_i) \;\geq\; \sum_i V_i \;>\; \frac{R\lambda' a}{8} \;\geq\; \log(|\Lambda|/\delta).$$

Finally $\lambda' a \geq \lambda^* a/2 = \min\{a/4, a^2/(8\sigma^2)\} \geq \varepsilon^2/(128(1+v))$, using $a \geq \varepsilon/2$, $\sigma^2 \leq 4(1+v)$ and $\varepsilon \leq 1$. For the dyadic grid, $\lambda^* \geq \min\{\frac{1}{2}, \varepsilon/(32(1+v))\} \geq 2^{-J}$ and $\lambda^* \leq \frac{1}{2}$, so some grid point lies in $[\lambda^*/2, \lambda^*]$. □

Unlike the KL route of Proposition 6.8, nothing is clipped and heavy upper tails of Z cost nothing in validity; completeness depends on the relative variance v of Z (Proposition 8.3(b)), that is, on how much of the mean is carried by rare large replicates. The implementation uses the fixed grid $\Lambda = \{0.1, 0.2, \ldots, 0.9, 0.95, 1\}$, which is sound by (a) for any grid; (b) is stated for a dyadic grid. On the suite replicates, betting gives a mean L/p of 0.84 and 0.92 at the two budgets below, against 0.56 and 0.70 for the clipped KL bound and 0.57 and 0.68 for the bound $\delta^{1/R}\min_i Z_i$ (both also sound).

The procedure. A pilot run of adaptive splitting, with randomness independent of everything after it, fixes the plan: levels $t_1 > \cdots > t_K$ near the 20% quantile of the particles' outputs, each placed in a gap between observed output values, ending at $t_K = t_{\mathrm{cert}}$, the largest double at most $t - 2e$; and a flip size per level. Then R replicates of Definition 6.9 run with N particles, $A_k = \{F < t_k\}$ decided by the computed value, and 10 Metropolis steps per level, each flipping r_k random coordinates and accepted iff the point stays in A_k. A replicate in which some computed value is not finite or lies within $2e$ of the level being tested reports $Z = 0$; otherwise it reports the exact rational $\prod_k N_k/N$ rounded down. Every evaluated x with finite $v < t_{\mathrm{cert}}$ joins a witness set W (witnesses supplied from outside are re-evaluated first). The output is $L = \max(L_R, |W|/2^d)$, with $|W|/2^d$ rounded down and L_R from Proposition 8.3 (its grid Λ is checked to lie in $[0,1]$); in sequential mode replicates are added until L_R reaches a target fraction of their running mean.

Corollary 8.4 (soundness on every net)**.** *For every net with finite weights and every δ, $\Pr[L > p] \leq \delta$, also in sequential mode. If some H_ℓ exceeds Ω (so e is undefined), the procedure outputs 0. The same holds with p replaced by $\Pr[f_{\mathrm{fl}}(x) < t]$ for any admissible evaluation f_{fl}.*

Proof. Couple a replicate with the run of Definition 6.9 on the same random numbers in which membership in $A_k = \{F < t_k\}$ is decided exactly. By Lemma 8.1 the two make the same decisions until the first value within e (inclusive) of a level, or not finite; every such value is within $2e$ of the level and raises the flag, so the reported Z is at most the ideal one on every run. The levels decrease strictly, so the A_k are nested; the flip proposal is symmetric and independent of x, so the kernel satisfies (3) under the uniform distribution. Given the pilot, Theorem 6.10 gives i.i.d. replicates with $\mathbb{E}Z_i \leq D(F < t_{\mathrm{cert}}) \leq p$, and Proposition 8.3(a) gives $\Pr[\exists R : L_R > p] \leq \delta$. Every witness has $v < t_{\mathrm{cert}} \leq t - 2e$, so $F(x) < t$ and $|W|/2^d \leq p$ surely; the maximum of the two bounds therefore exceeds p with probability at most δ. For an admissible f_{fl}, $F < t_{\mathrm{cert}}$ implies $f_{\mathrm{fl}} < t_{\mathrm{cert}} + e \leq t - e$, so $D(F < t_{\mathrm{cert}})$ and $|W|/2^d$ also lower-bound $\Pr[f_{\mathrm{fl}} < t]$. □

No step needs the chains to mix. Empirically, on a $d = 12$ net with exactly known $D(A_K)$ the mean of 20,000 replicates with *zero* Metropolis steps is 0.968 times $D(A_K)$ ($z = -0.78$), and all 9 kernel variants tested give $|z| \leq 1.19$. The adversarial campaign ran 13,500 runs on 9 adversarial nets (decoys, needles, extreme scales, exact ties, $p = 0$ and $p = 1$) at tiny budgets: the worst miss rates were 0.127 at $\delta = 0.2$ and 0.287 at $\delta = 0.5$, and no witness bound ever exceeded p; on random untrained nets, 0 of 200 runs missed at $\delta = 0.05$ and 4 of 200 at $\delta = 0.3$; the 9 floating-point attacks of Section 8.1 give $L = 0$ in all 45 runs.

8.5 Results for splitting

Budgets: M is $N = 1{,}000$, $R = 32$; L is $N = 3{,}000$, $R = 64$; $\delta = 0.05$; costs include the pilot. Ratios L/p are means over 5 runs (exact suite) or 3 runs ($d \geq 32$).

Exact suite (Table 10): 315 runs, none with $L > p$, 0 flagged replicates. For $p \geq 10^{-5}$ the mean L/p is 0.860–0.900 at budget M ($3.7 \cdot 10^5$–$2.1 \cdot 10^6$ fpe, 0.42–5.1 s) and 0.920–0.942 at budget L ($2.2 \cdot 10^6$–$1.4 \cdot 10^7$ fpe, 5.4–35 s). On the four nets with $p < 10^{-5}$ the betting bound at budget L reaches 0.784 ($p = 6 \cdot 10^{-8}$, 1 failing input), 0.807 ($p = 7.5 \cdot 10^{-9}$, 2 failing inputs), 0.919 ($p = 3.3 \cdot 10^{-7}$, 89) and 0.924 ($p = 9.7 \cdot 10^{-6}$, 163), at $1.4 \cdot 10^7$–$2.3 \cdot 10^7$ fpe and 18–31 s, which is 95, 840, 140 and 7.2 times fewer evaluations than a one-sided Monte Carlo bound reaching the same ratio (54, 670, 83 and 4.5 times less wall-clock); there the witness sets are exact (every failing input found), so the reported bound equals p. Monte Carlo is cheaper for $p \gtrsim 4 \cdot 10^{-4}$, and splitting is cheaper for $p \lesssim 10^{-4}$. The mean of Z/p over the 160–320 replicates per net and budget is 0.913–1.029, a check of unbiasedness on trained nets. The efficiency $\kappa = \mathrm{relvar}(Z) \times (\text{cost per replicate})$ grows like $(1/p)^{0.44}$ over the 26 (net, budget) pairs with $p \leq 10^{-2}$, against Monte Carlo's $1/p$, so the gain grows like $(1/p)^{0.56}$.

Larger d (Table 11): here p is unknown, and ratios are against a reference p_{ref}, the mean of 100 replicates with $N = 3{,}000$ (unbiased for $D(A_K)$, not a bound). The reference was checked by exact enumeration of all 2^{32} inputs at $d = 32$ (97 failing inputs, $p = 2.26 \cdot 10^{-8}$ against $p_{\mathrm{ref}} = 2.21 \cdot 10^{-8} \pm 7 \cdot 10^{-10}$, 4700 s of enumeration) and by independent Monte Carlo down to $p \approx 6.3 \cdot 10^{-9}$ (e.g. $5.2 \cdot 10^{-6} \pm 2.5 \cdot 10^{-7}$ against $p_{\mathrm{ref}} = 4.88 \cdot 10^{-6}$ on the suite's `mse_d64_w64_s1000`); below that it is unverified. On the 15 nets with a well-estimated reference (p_{ref} ranging over 0.039–$5.2 \cdot 10^{-21}$), the betting bound at budget L reaches 0.865–0.973 of p_{ref} at $4.1 \cdot 10^6$–$5.7 \cdot 10^7$ fpe (19–550 s): 0.916–0.973 on the 8 nets whose reference was checked, and 0.865–0.945 on the 7 whose reference was not (p_{ref} from $2.3 \cdot 10^{-8}$–$5.2 \cdot 10^{-21}$). On the latter the ratio is to the mean of the same estimator, whose mean is $D(A_K) \leq p$: a failure mode that the chains never reach is missing from both L and p_{ref}, as the planted net of Section 8.2 shows, so these ratios cannot reveal one. This includes the suite net `bce_d64_w64_s100`, where a MILP solver had found one failing input ($f = -1.09$, re-evaluated) while Monte Carlo with 10^7 samples found 0: $p_{\mathrm{ref}} = 2.29 \cdot 10^{-8}$, ratio 0.919 at $2 \cdot 10^7$ fpe (80 s), and up to $2.5 \cdot 10^5$ distinct certified failing inputs per run; and `bce_d128_w128x128_s300` with $p_{\mathrm{ref}} = 5.2 \cdot 10^{-21}$: ratio 0.865 at $5.7 \cdot 10^7$ fpe (550 s), about 10^{15} times fewer evaluations than Monte Carlo. In sequential mode with target 0.9, `smc_mse_d64_w64_s1500` ($p_{\mathrm{ref}} = 5.8 \cdot 10^{-9}$) reaches 0.930 at $6.6 \cdot 10^6$ fpe (13 s), `smc_mse_d128_w64_s1000` ($p_{\mathrm{ref}} = 5.6 \cdot 10^{-11}$) reaches 0.924 at $7.7 \cdot 10^6$ fpe (35 s), and `bce_d128_w128x128_s300` reaches 0.887 at $6.7 \cdot 10^7$ fpe (520 s). At a fixed budget the cost is linear in the number of levels, i.e. in $\log(1/p)$ (Figure 4).

Where it stops. On 2 nets trained into the far tail (z from 7.8–8.4) the betting bound reaches only 0.17–0.18 of p_{ref}, raised to 0.42–0.47 by the witness sets; on 5 more (among them the suite's `mse_d64_w64_s3000`, `bce_d64_w64_s300` and `bce_d64_w64_s1000`) no reference replicate reached

net	p	F	Monte Carlo fpe (s)	budget M L/p	budget M fpe (s)	budget L L/p	budget L fpe (s)	wit.	speed-up at L KL fpe (s); task fpe
mse_d24_w32_s300	0.071	1193958	$1.6 \cdot 10^4$ (0.012)	0.877	$3.7 \cdot 10^5$ (0.42)	0.936	$2.2 \cdot 10^6$ (5.5)	0.11	$8.5 \cdot 10^{-3}$ ($2.5 \cdot 10^{-3}$); 0.018
mse_d24_w32_s600	$8.3 \cdot 10^{-3}$	138936	$1.3 \cdot 10^5$ (0.11)	0.874	$7.7 \cdot 10^5$ (0.77)	0.939	$4.5 \cdot 10^6$ (10)	0.62	0.041 (0.014); 0.080
mse_d24_w32_s1000	$3.6 \cdot 10^{-4}$	6012	$3.1 \cdot 10^6$ (2.3)	0.882	$1.5 \cdot 10^6$ (1.1)	0.937	$8.4 \cdot 10^6$ (14)	1.00	0.48 (0.22); 0.92
mse_d24_w32_s1500	$9.7 \cdot 10^{-6}$	163	$1.1 \cdot 10^8$ (94)	0.852	$2.4 \cdot 10^6$ (1.4)	0.924	$1.4 \cdot 10^7$ (18)	1.00	7.2 (4.5); 14
mse_d24_w32_s2000	$6 \cdot 10^{-8}$	1	$1.9 \cdot 10^{10}$ ($1.2 \cdot 10^4$)	0.538	$3.4 \cdot 10^6$ (1.7)	0.784	$1.9 \cdot 10^7$ (23)	1.00	95 (54); 210
bce_d28_w32_s30	$1.9 \cdot 10^{-3}$	507779	$5.9 \cdot 10^5$ (0.36)	0.888	10^6 (1.7)	0.935	$6.1 \cdot 10^6$ (10)	0.27	0.12 (0.043); 0.23
bce_d28_w32_s60	$1.3 \cdot 10^{-5}$	3543	$8.4 \cdot 10^7$ (75)	0.872	$2.1 \cdot 10^6$ (2.6)	0.932	$1.4 \cdot 10^7$ (18)	1.00	7.0 (4.7); 14
bce_d28_w32_s100	$7.5 \cdot 10^{-9}$	2	$1.5 \cdot 10^{11}$ ($1.2 \cdot 10^5$)	0.616	$3.7 \cdot 10^6$ (3.2)	0.807	$2.3 \cdot 10^7$ (23)	1.00	840 (670); 1800
bce_d28_w32_s150	0	0	–	$L = 0$ in all 15 runs (correct); budget L: $2.4 \cdot 10^7$ (24)					
mse_d28_w64_s300	0.013	3468961	$8.6 \cdot 10^4$ (0.098)	0.887	$7.1 \cdot 10^5$ (0.93)	0.942	$4.1 \cdot 10^6$ (12)	0.05	0.032 (0.013); 0.062
mse_d28_w64_s600	$6 \cdot 10^{-5}$	16155	$1.8 \cdot 10^7$ (18)	0.860	$1.8 \cdot 10^6$ (1.6)	0.920	$1.2 \cdot 10^7$ (24)	0.97	1.2 (0.62); 2.4
mse_d28_w64_s1000	$3.3 \cdot 10^{-7}$	89	$3.3 \cdot 10^9$ (3300)	0.803	$3 \cdot 10^6$ (2.3)	0.919	$1.8 \cdot 10^7$ (31)	1.00	140 (83); 280
mse_d28_w64_s1500	0	0	–	$L = 0$ in all 15 runs (correct); budget L: $2.4 \cdot 10^7$ (40)					
bce_d28_w64x64_s20	0.093	25051039	$1.2 \cdot 10^4$ (0.032)	0.885	$3.7 \cdot 10^5$ (1.3)	0.942	$2.2 \cdot 10^6$ (8.1)	0.01	$7.7 \cdot 10^{-3}$ ($5.4 \cdot 10^{-3}$); 0.016
bce_d28_w64x64_s40	$2.3 \cdot 10^{-3}$	613171	$4.8 \cdot 10^5$ (1.2)	0.880	10^6 (3.4)	0.939	$6.1 \cdot 10^6$ (22)	0.26	0.11 (0.075); 0.21
bce_d28_w64x64_s60	$1.9 \cdot 10^{-5}$	5071	$5.9 \cdot 10^7$ (160)	0.870	$2.1 \cdot 10^6$ (5.1)	0.931	$1.2 \cdot 10^7$ (35)	1.00	5.3 (4.8); 10
bce_d28_w64x64_s100	0	0	–	$L = 0$ in all 15 runs (correct); budget L: $2.4 \cdot 10^7$ (47)					
mse_d28_w32x32_s100	0.060	16224649	$1.8 \cdot 10^4$ (0.022)	0.873	$3.7 \cdot 10^5$ (0.59)	0.940	$2.2 \cdot 10^6$ (5.4)	0.01	0.012 ($5.6 \cdot 10^{-3}$); 0.024
mse_d28_w32x32_s200	0.015	3948451	$7.5 \cdot 10^4$ (0.095)	0.863	$7.1 \cdot 10^5$ (0.90)	0.937	$4.1 \cdot 10^6$ (9.6)	0.05	0.023 (0.013); 0.046
mse_d28_w32x32_s300	$3.6 \cdot 10^{-3}$	960130	$3.1 \cdot 10^5$ (0.37)	0.900	10^6 (1.2)	0.934	$6.1 \cdot 10^6$ (11)	0.27	0.061 (0.039); 0.12
mse_d28_w32x32_s500	$1.3 \cdot 10^{-4}$	36023	$8.2 \cdot 10^6$ (11)	0.874	$1.7 \cdot 10^6$ (1.9)	0.921	10^7 (16)	0.92	0.68 (0.55); 1.3

Table 10: Splitting lower bounds on the exact suite ($d \leq 28$, $\delta = 0.05$). Costs are fpe with wall-clock seconds in parentheses; Monte Carlo is the cost of relative error $\varepsilon = 0.1$, $3\ln(2/\delta)/(\varepsilon^2 p)$ samples (an interval of ratio about 1.22), its seconds at the median measured time per sample of the net, as in Table 12. L/p is the mean betting ratio over 5 runs; "wit." the mean witness bound over p (the reported bound is the larger of the two). Speed-ups at budget L: "KL" against the Monte Carlo samples a one-sided KL lower bound needs to reach the same ratio when the empirical mean equals p (which favours Monte Carlo), in fpe and in seconds; "task" against $3\ln(2/\delta)/((1 - L/p)^2 p)$ (`nets_lower/run_suite.py`, `results/tables.json`).

the last level, so every bound is 0: sound, and uninformative about p, which may well be 0. This is the lower-bound face of Proposition 8.2(c).

Wall clock and mixing. One splitting evaluation costs a median of 0.96 times one batched Monte Carlo sample in the same process (10–90% range 0.67–1.58 over 114 cells), so measured in the same process, speed-ups in seconds are close to those in fpe; the tables use the median Monte Carlo time per net described above, so their speed-ups in seconds differ from those in fpe by up to the spread of those measurements. The late-level acceptance rate is 0.00–0.03 at $d \leq 32$ with tiny p, where the final particles collapse onto 1–27 distinct points (the witness sets then cover the gap). On the 4 nets examined at $d = 64$ and 128 the chains appear to mix: the acceptance rate stays at least 0.10 at every level and the runs end with 201–338 distinct final particles. This is necessary for mixing, not sufficient: a mode that no chain reaches leaves no trace in either statistic, nor in p_{ref}. Nothing in the soundness argument depends on mixing.

8.6 Exact counts and certificates of zero: certified partitions

A *certified partition* is a decision tree over input coordinates (so its leaves are subcubes $Q = \{x : x_F = \sigma\}$ with free coordinates R, $|R| = r$) whose leaves are labelled SAFE (with a certificate that $F \geq t$ on Q), FAIL ($F < t$ on Q), ENUM (enumerate Q) or OPEN; the verifier outputs $[(\mathcal{F} + |U|\ell)/2^d, (\mathcal{F} + |U|u)/2^d]$, with $\mathcal{F}$ the number of failing inputs in FAIL and ENUM leaves, U the union of OPEN leaves, and $[\ell, u]$ the KL interval for the failing fraction of uniform samples from U. Certificates come in two kinds, both checked in outward-rounded interval arithmetic. A *linear* certificate fixes, for each unit relaxed from below, a slope $\alpha \in [0, 1]$ (any value is valid;

net	d	z	p_{ref} (check)	L/p_{ref} at L	fpe (s) at L	seq. 0.9: ratio, fpe (s)	KL speed-up fpe (s)
`smc_mse_d32_w64_s600`	32	5.5	$2.21 \cdot 10^{-8}$ (exact $2.26 \cdot 10^{-8}$)	0.916 [0.89]; wit. 1.02	$2 \cdot 10^{7}$ (38)	0.83, $5.9 \cdot 10^{6}$ (10)	1800 (2300)
`mse_d64_w64_s300`	64	1.7	0.0392 (MC $0.035 \pm 2 \cdot 10^{-3}$)	0.953 [0.95]	$4.1 \cdot 10^{6}$ (19)	0.93, $8.7 \cdot 10^{5}$ (2.9)	0.016 (0.014)
`smc_mse_d64_w32x32_s300`	64	3.7	$2.93 \cdot 10^{-5}$ (MC $2.9 \cdot 10^{-5} \pm 5 \cdot 10^{-7}$)	0.935 [0.91]	$1.2 \cdot 10^{7}$ (49)	0.92, $2.8 \cdot 10^{6}$ (8.1)	3.8 (4.2)
`smc_mse_d64_w64_s1000`	64	4.2	$7.36 \cdot 10^{-6}$ (MC $7.2 \cdot 10^{-6} \pm 4 \cdot 10^{-7}$)	0.944 [0.93]	$1.4 \cdot 10^{7}$ (56)	0.94, $3.3 \cdot 10^{6}$ (7.9)	18 (18)
`mse_d64_w64_s1000`	64	4.3	$4.88 \cdot 10^{-6}$ (MC $5.2 \cdot 10^{-6} \pm 3 \cdot 10^{-7}$)	0.940 [0.93]	$1.4 \cdot 10^{7}$ (54)	0.89, $3.1 \cdot 10^{6}$ (11)	23 (23)
`smc_mse_d64_w32x32_s600`	64	5.5	$3.22 \cdot 10^{-8}$ (MC $3.2 \cdot 10^{-8} \pm 3 \cdot 10^{-9}$)	0.973 [0.95]	$2 \cdot 10^{7}$ (66)	0.93, $8.5 \cdot 10^{6}$ (19)	$1.2 \cdot 10^{4}$ ($1.5 \cdot 10^{4}$)
`bce_d64_w64_s100`	64	4.5	$2.29 \cdot 10^{-8}$	0.919 [0.91]	$2 \cdot 10^{7}$ (80)	0.89, $5.3 \cdot 10^{6}$ (18)	1900 (1800)
`smc_mse_d64_w64_s1500`	64	5.5	$5.77 \cdot 10^{-9}$ (MC $6.3 \cdot 10^{-9} \pm 10^{-9}$)	0.964 [0.95]	$2.2 \cdot 10^{7}$ (69)	0.93, $6.6 \cdot 10^{6}$ (13)	$3.6 \cdot 10^{4}$ ($4.1 \cdot 10^{4}$)
`smc2_mse_d64_w64_s1500`	64	6.9	$7.62 \cdot 10^{-13}$	0.945 [0.91]	$3.3 \cdot 10^{7}$ (96)	0.93, $3.3 \cdot 10^{7}$ (58)	$7.6 \cdot 10^{7}$ ($9.8 \cdot 10^{7}$)
`smc_bce_d64_w64x64_s300`	64	5.2	$7.34 \cdot 10^{-16}$	0.891 [0.86]	$4.2 \cdot 10^{7}$ (270)	0.98, $5.2 \cdot 10^{7}$ (140)	$1.5 \cdot 10^{10}$ ($1.9 \cdot 10^{10}$)
`smc2_mse_d64_w64_s2000`	64	8.4	$2.4 \cdot 10^{-18}$	0.168 [0.12]; wit. 0.42	$4.8 \cdot 10^{7}$ (130)	–	$2.7 \cdot 10^{10}$ ($3.9 \cdot 10^{10}$)
`smc_bce_d64_w64x64_s600`	64	6.1	no reference replicate reached the last level ($8.3 \cdot 10^{7}$ fpe, 240 s): every bound is 0				
`bce_d64_w64_s300`	64	7.1	no reference replicate reached the last level ($7.9 \cdot 10^{7}$ fpe, 180 s): every bound is 0				
`bce_d64_w64_s1000`	64	8.8	no reference replicate reached the last level ($7.1 \cdot 10^{7}$ fpe, 160 s): every bound is 0				
`mse_d64_w64_s3000`	64	10.0	no reference replicate reached the last level ($8 \cdot 10^{7}$ fpe, 200 s): every bound is 0				
`smc_mse_d128_w64_s600`	128	3.9	$1.85 \cdot 10^{-5}$ (MC $1.9 \cdot 10^{-5} \pm 9 \cdot 10^{-7}$)	0.939 [0.93]	$1.2 \cdot 10^{7}$ (140)	0.90, $2.9 \cdot 10^{6}$ (16)	7.0 (6.7)
`smc_mse_d128_w64_s1000`	128	5.8	$5.59 \cdot 10^{-11}$	0.934 [0.93]	$2.8 \cdot 10^{7}$ (260)	0.92, $7.7 \cdot 10^{6}$ (35)	$8.6 \cdot 10^{5}$ (10^{6})
`bce_d128_w128x128_s100`	128	5.3	$9.08 \cdot 10^{-12}$	0.928 [0.92]	$3.1 \cdot 10^{7}$ (370)	0.90, $8.9 \cdot 10^{6}$ (80)	$4 \cdot 10^{6}$ ($4.1 \cdot 10^{6}$)
`smc2_mse_d128_w64_s1000`	128	6.8	$2.82 \cdot 10^{-15}$	0.940 [0.93]	$3.9 \cdot 10^{7}$ (300)	0.88, $1.5 \cdot 10^{7}$ (55)	$1.4 \cdot 10^{10}$ ($1.7 \cdot 10^{10}$)
`bce_d128_w128x128_s300`	128	6.5	$5.18 \cdot 10^{-21}$	0.865 [0.78]	$5.7 \cdot 10^{7}$ (550)	0.89, $6.7 \cdot 10^{7}$ (520)	10^{15} ($1.3 \cdot 10^{15}$)
`bce_d128_w128x128_s1000`	128	7.8	$2.64 \cdot 10^{-37}$	0.183 [0.09]; wit. 0.47	$9.9 \cdot 10^{7}$ (1100)	–	$1.3 \cdot 10^{29}$ ($2.9 \cdot 10^{29}$)
`smc2_mse_d128_w64_s1500`	128	8.9	no reference replicate reached the last level ($1.2 \cdot 10^{8}$ fpe, 490 s): every bound is 0				

Table 11: Splitting lower bounds at $d = 32$–128: the suite's nets and nets trained for this experiment (`smc_`, `smc2_`). p_{ref} is the mean of 100 replicates (exact count or independent Monte Carlo in parentheses where available); L/p_{ref} is the mean [minimum] betting ratio at budget L, "wit." the reported ratio when the witness bound is larger; "seq. 0.9" the sequential mode stopped at target 0.9; speed-up against the one-sided KL Monte Carlo yardstick at p_{ref} (`nets_lower/run_scaling.py`, `run_sequential.py`).

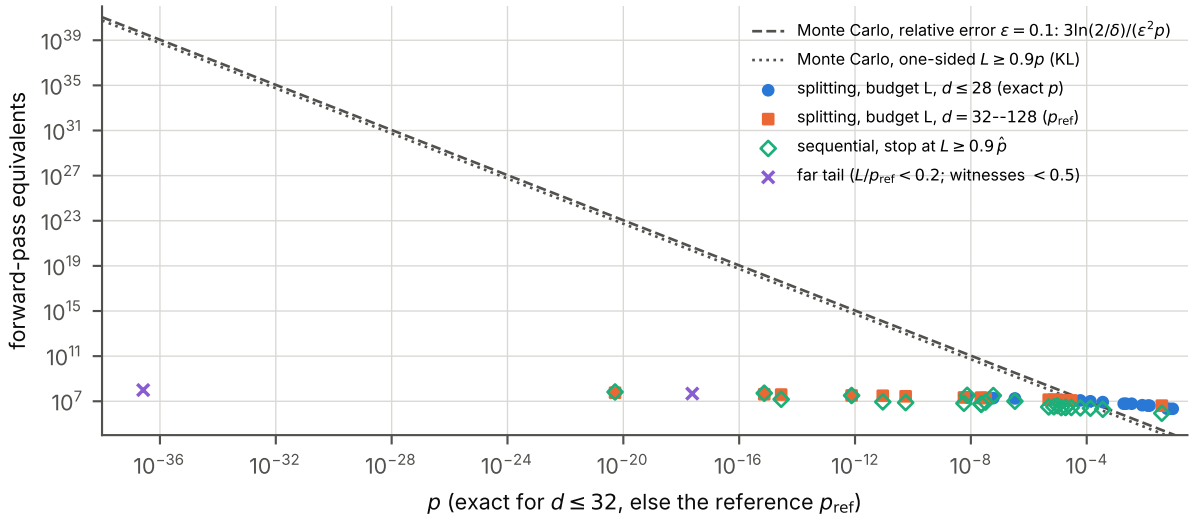


Figure 4: Cost of the splitting lower bound (fpe, budget L or sequential with target 0.9) against p, with the Monte Carlo yardsticks. Ratios reached are in Tables 10 and 11; the crosses are the far-tail nets, whose references are themselves uncertain.

the verifier clips), and relaxes units from above by chords, as in CROWN [ZWC+18]; the verifier recomputes everything else. A *degree-2* certificate fixes, for each first-layer unit, the parameters of a convex quadratic q_j below or above relu on its interval (the verifier computes q_j from them and proves its validity), and a vector μ and shift s for the S-lemma step below.

Proposition 8.5 (certified partitions)**.** *(a) (Soundness.) For every net and every argument, the output contains p with probability at least $1-\delta$; without OPEN leaves it is p exactly. The argument has the structure of a* PCIS *argument (Definition 6.1):* SPLIT *nodes, deterministic leaves whose refutations are linear or degree-2 certificates or enumeration in place of P-refutations, and one envelope node $\varphi = \mathbf{1}_U$ over the union of the OPEN leaves, which lie in different branches of the tree (no rule of Definition 6.1 forms such a node, so Theorem 6.2 does not apply literally). Soundness is proved directly below, and the sampled part costs what Theorem 4.3(iv) charges for the looseness $|U|/(2^d D(C \wedge U))$.*

(b) (Ideal size.) If C has $\mathcal{F} \geq 1$ failing inputs, a tree whose leaves are all pure (failure-free subcubes or single failing points) with at most $2\mathcal{F}(d - \lfloor \log_2 \mathcal{F} \rfloor + 1)$ nodes exists; for $p = 0$ one node suffices. Deciding whether a leaf is pure is coNP*-complete already for the nets of Proposition 8.2(a), so Theorem 3.4 bites through the leaf checks, not through the length of the tree.*

(c) (Relaxation gap, one hidden layer.) Let $f = c + \sum_j a_j \mathrm{relu}(w_j \cdot x + b_j)$ and let lb *be the linear bound on Q with slopes $\alpha_j = \mathbf{1}[U_j \geq -L_j]$ for the unstable units with $a_j > 0$ (CROWN's choice) and chords for those with $a_j < 0$, where $[L_j, U_j]$ is the range of z_j on Q. Then*

$$\min_Q F - \mathrm{lb} \leq \sum_{j \ \text{unstable on } Q} c_j |a_j| \, \|w_{j,R}\|_1, \qquad c_j = 1 \ (a_j > 0), \quad c_j = \tfrac{1}{2} \ (a_j < 0),$$

so a SAFE certificate exists as soon as the margin $\min_Q F - t$ exceeds the right-hand side (plus the verifier's outward-rounding slack, and the prover's acceptance margin of 2^{-30} times a bound on $|f|$).

Proof. (a) The verifier parses the tree itself: every split must be on a free coordinate of its node and have two children, so the leaves partition the cube; their sizes are summed as integers, and a malformed tree yields $[0, 1]$. *Linear certificates.* Work backwards from an affine lower bound $f \geq G \cdot h_\ell + c$ in the post-activations of layer ℓ (initially $G = w_L$, $c = b_L$), valid on Q, given intervals $L \leq z \leq U$ for the pre-activations $z = W_\ell h_{\ell-1} + b_\ell$ valid on Q. For a stable unit, $h_j = z_j$ ($L_j \geq 0$) or $h_j = 0$ ($U_j \leq 0$). For an unstable unit with $G_j \geq 0$ use $h_j \geq \alpha_j z_j$, valid for every real z_j and $\alpha_j \in [0, 1]$; with $G_j < 0$ use $h_j \leq \lambda_j (z_j - L_j)$ with $\lambda_j \geq U_j/(U_j - L_j)$, valid on $[L_j, U_j]$ because relu is convex and below its chord there; if the sign of G_j is not certain, use $0 \leq h_j \leq U_j$, i.e. $G_j h_j \geq \min(G_j, 0) U_j$. Each step keeps an affine lower bound valid on Q, now in $h_{\ell-1}$; at the input layer it is affine in x, and its minimum over the box of Q, attained at a vertex, is its minimum over Q. The intervals of hidden layers are computed by the same recursion, so the induction is over layers. A FAIL certificate is the mirror image for $-f$. Evaluating all this in outward-rounded interval arithmetic (each operation enclosed by the neighbouring doubles of its result, each matrix product by Higham's bound with the underflow term carried through later products) makes the computed bound a bound on the exact one. *Degree-2 certificates.* The layers after the first are relaxed linearly first, exactly as in a linear certificate; G_j is the resulting (enclosed) coefficient of $h_j = \mathrm{relu}(z_j)$ for a first-layer unit j, and a unit whose G_j has uncertain sign gets $q_j = 0$ and the charge $G_{\mathrm{lo},j} U_j$, as above. For $G_j \geq 0$ the verifier needs $q_j \leq$ relu on

$[L_j, U_j]$ and for $G_j \le 0$ $q_j \ge$ relu there. It accepts only convex q_j (curvature $\kappa_j \ge 0$), and checks, in interval arithmetic: for a minorant, $q_j(L_j) \le 0$, $q_j(0) \le 0$ and $q_j(U_j) \le U_j$, which suffice because a convex function on an interval is at most the larger of its two endpoint values (apply this to q_j on $[L_j, 0]$ and to $q_j - z$ on $[0, U_j]$); for a majorant, $q_j(L_j) \ge 0$ and $q_j'(L_j) \ge 0$ (so $q_j \ge 0$ on $[L_j, U_j]$ by convexity), and either $(q_j - z)(U_j) \ge 0$ and $(q_j - z)'(U_j) \le 0$, or $(q_j - z)(0) \ge 0$ and $(q_j - z)'(0) \ge 0$ (so $q_j \ge z$ on $[0, U_j]$, again by convexity). A unit whose check fails gets the always valid $q_j = 0$ from below or the chord from above. Then $f \ge Y^\top M Y$ on Q for $Y = (1, x_R)$ and a symmetric M the verifier computes. If $M - \operatorname{diag}(\mu) - sI \succeq 0$ then, as $Y_i^2 = 1$ for every coordinate of Y, $Y^\top M Y \ge \sum_i \mu_i Y_i^2 + s|Y|^2 = \sum_i \mu_i + (r+1)s$. Positive semidefiniteness of every matrix in the entrywise enclosure is checked by one Cholesky factorisation of $A - cI$, A the midpoint: if it succeeds, $A - cI + \Delta A = \hat{R}^\top \hat{R} \succeq 0$ with $\|\Delta A\|_2 \le \gamma_{r+2} \operatorname{tr}(A)/(1 - \gamma_{r+2})$ for $c \ge 0$ [Hig02, Thm. 10.3] (plus underflow terms, and the rounding of forming $A - cI$, at most $2u$ times its largest diagonal entry in absolute value), and every matrix within Frobenius distance ρ of A has smallest eigenvalue at least $c - \|\Delta A\|_2 - \rho$ minus these terms by Weyl's inequality, which the verifier checks to be positive. *Other leaves.* ENUM leaves are counted exactly by Lemma 8.1 with rational re-evaluation of undecided points; a leaf whose certificate fails or is non-finite is demoted to OPEN. So $p = \mathcal{F}/2^d + D(C \wedge U)$, and the envelope $\mathbf{1}_U$ has exact mass $|U|/2^d$ and an exact uniform sampler (a uniform integer below $|U|$ selects a leaf and an offset), so Theorem 4.3(ii) gives the interval with probability $1 - \delta$. Enumeration makes the proof system complete; checking is polynomial in the size of the certificates and of the enumerated subcubes.

(b) Split the coordinates in a fixed order and stop at a node when its subcube contains no failing input or is a single point. Internal nodes are prefixes of length $k < d$ of failing inputs, at most $\min(2^k, \mathcal{F})$ of each length, so there are at most $\sum_{k=0}^{d-1} \min(2^k, \mathcal{F}) \le (2^{\lambda+1} - 1) + (d - 1 - \lambda)\mathcal{F} \le \mathcal{F}(d - \lambda + 1) - 1$ of them, with $\lambda = \lfloor \log_2 \mathcal{F} \rfloor$; a binary tree has one more leaf than internal nodes. Deciding that a subcube contains no failing input is in coNP (a failing input, re-evaluated exactly, refutes it) and coNP-hard already for the whole cube of the nets f_φ, by Proposition 8.2(a).

(c) For one hidden layer the bound is $\mathrm{lb} = \min_Q g$ with $g = c + \sum_j a_j \ell_j(z_j)$, ℓ_j the relaxation of unit j, and the minimum is attained at a point $x^* \in Q$. Then $\min_Q F - \mathrm{lb} \le F(x^*) - g(x^*) = \sum_j a_j(\mathrm{relu}(z_j(x^*)) - \ell_j(z_j(x^*)))$, where stable units contribute 0. On $[L, U]$ with $L < 0 < U$: for $a_j > 0$, $\mathrm{relu}(z) - \alpha z \le \max((1 - \alpha)U, -\alpha L)$, which is $\min(U, -L) \le (U - L)/2$ for CROWN's α; for $a_j < 0$, the chord minus relu is at most its value at 0, $-LU/(U - L) \le (U - L)/4$. Since z_j ranges over the centre $\pm\|w_{j,R}\|_1$, $U_j - L_j = 2\|w_{j,R}\|_1$. □

Heuristically, a unit is unstable on a random r-dimensional subcube with probability proportional to $\|w_{j,R}\|_1 \propto r$, and at the minimiser of the relaxation its pre-activation deviates from 0 by about $\|w_{j,R}\|_2 \propto \sqrt{r}$, so the gap grows like $r^{3/2}$ (bound (c) is the worst case, r^2). We measure exponents of 1.7–1.9 on the one-hidden-layer squared-loss nets, between the heuristic 3/2 and the worst case 2 and closer to the latter, so the heuristic is at best a rough guide (the gap scan was computed with the previous verifier, Figure 5). The margin $\min_Q F - t$ shrinks as r grows, so certified leaves have a bounded number r^* of free inputs and the tree has about 2^{d-r^*} nodes *whatever p is*.

Results (Tables 12 and 13, `experiments/nets_prover`, `nets_verify`). All 21 suite nets with $d \le 28$ are counted exactly (every count equals the enumerated one). The wins are the logistic nets trained past the point where the margin dominates: `bce_d28_w32_s100` ($p = 7.5 \cdot 10^{-9}$, two failing inputs) is certified with 20,789 nodes and $4.1 \cdot 10^4$ verifier fpe, 4.2 s end to end (prover and verifier), against 69 s for enumeration and about 33 hours ($1.5 \cdot 10^{11}$ samples) for Monte Carlo;

`bce_d28_w32_s150` is certified to have $p = 0$ with 2,061 nodes, 4100 fpe, in 1.4 s, which Monte Carlo can never do. For squared loss at $d = 24$ the certificate is only 1.0–4.2 times cheaper than enumerating the cube in fpe, and in seconds enumeration (5.1–5.8 s) beats prover plus verifier (18–110 s) on every $d = 24$ net: the numpy verifier costs 5.4–12 μs per fpe, against 0.62–2.7 μs per Monte Carlo sample. Branch and bound without a prover (CROWN bounds, `nets_verify`) counts exactly on 11 of the 13 one-hidden-layer nets (the other two end with 9 and 1 undecided points, and their intervals contain p), at 0.71–3.6 times the cost of enumeration for squared loss and $8.3 \cdot 10^{-3}$–0.12 for logistic loss. These ratios are in branch and bound's own cost model, in which enumerating the cube costs $2 \cdot 10^6$ fpe at $d = 24$ and $2.8 \cdot 10^7$ at $d = 28$; the enumeration column of Table 12 is in the certified-partition verifier's cost model, which counts fewer flops per point. Branch and bound spends 0.9–1.9 times the wall-clock of a Monte Carlo sample per fpe, in the same process. Degree-2 leaves make trees 2.0–2.1 times smaller but cost 1.5–3.4 times more fpe and 9.5–17 times more seconds to verify. Two hidden layers defeat both: no certificate is planned cheaper than enumeration (190–500 s), and branch and bound leaves upper ends of 0.78–0.91 after its time limit. A perfect engine, deciding every pure subcube at once, would need 10–4800 times fewer nodes than CROWN's tree (for the 163 failures of `mse_d24_w32_s1500`: 3,841 nodes, within the bound 5,868 of (b), against $5.2 \cdot 10^5$), and greedily grown failing subcubes have dimension at most 4: the obstacle is deciding SAFE, not the geometry of the failures.

Certificates of $p = 0$. Of 107 one-hidden-layer nets trained for at least 1000 (squared loss) or 300 (logistic loss) steps, 98 are certified to have $p = 0$ (Table 13), 5 have $p > 0$ and are counted exactly, and 4 at $d = 40$–64 remain open. Longer training makes certification nearly free (about 5 fpe and 10^{-3} s, at the root, for some nets up to $d = 32$ with squared and $d = 40$ with logistic loss); at $d = 48$ and 64 it costs $3.2 \cdot 10^7$ and $3.7 \cdot 10^7$ fpe (40 and 49 s). Monte Carlo with 10^7 samples only shows $p \leq 3 \cdot 10^{-7}$. For the suite net `bce_d64_w64_s1000` we attempted the whole certificate within 110 minutes on 4 worker processes: 691 of its 1,024 top-level pieces, chosen in random order, were certified exactly, all with no failing input, so 67.5% of the cube is certified failure-free ($3 \cdot 10^7$ nodes, $6 \cdot 10^7$ verifier fpe; $2.4 \cdot 10^4$ s of proving and 2000 s of verifying summed over the workers, 6700 s of wall-clock). Extrapolating from these uniformly sampled pieces, the whole certificate has $4.4 \cdot 10^7 \pm 10^6$ nodes and costs $8.8 \cdot 10^7 \pm 2 \cdot 10^6$ verifier fpe; as many Monte Carlo samples would only show $p \leq 3.4 \cdot 10^{-8}$. For the other $d = 64$ suite nets the estimated certificates cost $4.3 \cdot 10^{10}$ (`bce_d64_w64_s300`) to $9.5 \cdot 10^{17}$ fpe (`mse_d64_w64_s300`, more than the $5.7 \cdot 10^{17}$ of enumeration).

8.7 Why upper bounds fail on trained squared-loss nets

By Theorem 3.2 an upper bound below the sampling scale needs deduction about the weights. For squared-loss nets three barriers explain why every deduction we could make cheap fails, and Table 14 measures them on all 30 suite nets. Extend F to the box $[-1, 1]^d$ by the same formula.

Proposition 8.6 (box barrier)**.** *Suppose $F(0) < t$.*

(a) Every function $B \leq F$ on $[-1, 1]^d$ has $B(0) < t$, so no bound valid on the box (interval arithmetic, CROWN, any LP or convex relaxation, MILP over the box, Lipschitz bounds) certifies $F \geq t$ on the cube; the same holds for a subcube whose centre has $F < t$. More generally, the same holds, without assuming $F(0) < t$, as soon as $F(y) < t$ at some point y of the box (of the subcube's box).

(b) Let $\Phi : [-1, 1]^d \to \mathbb{R}_{\geq 0}$ be convex in each coordinate separately (e.g. a positive combination of exponentials of affine functions, or a product of such with factors $1 + e^{\alpha_j z_j}$) with $\Phi \geq 1$

net	p	F	argument	verifier fpe (s)	prover s	B&B fpe (s)	enumeration fpe (s)	Monte Carlo fpe (s)
mse_d24_w32_s300	0.071	1193958	42227 nodes	$1.3 \cdot 10^6$ (7.2)	110	$7.3 \cdot 10^6$ (10)	$1.3 \cdot 10^6$ (5.4)	$1.6 \cdot 10^4$ (0.012)
mse_d24_w32_s600	$8.3 \cdot 10^{-3}$	138936	177329 nodes	$1.2 \cdot 10^6$ (7.5)	76	$5.7 \cdot 10^6$ (7.2)	$1.3 \cdot 10^6$ (5.1)	$1.3 \cdot 10^5$ (0.11)
mse_d24_w32_s1000	$3.6 \cdot 10^{-4}$	6012	305973 nodes	$8.7 \cdot 10^5$ (8.0)	49	$3.7 \cdot 10^6$ (4.3)	$1.3 \cdot 10^6$ (5.2)	$3.1 \cdot 10^6$ (2.3)
mse_d24_w32_s1500	$9.7 \cdot 10^{-6}$	163	253127 nodes	$5.4 \cdot 10^5$ (5.0)	29	$2.4 \cdot 10^6$ (2.6)	$1.3 \cdot 10^6$ (5.8)	$1.1 \cdot 10^8$ (94)
mse_d24_w32_s2000	$6 \cdot 10^{-8}$	1	160647 nodes	$3.2 \cdot 10^5$ (2.4)	16	$1.4 \cdot 10^6$ (1.5)	$1.3 \cdot 10^6$ (5.2)	$1.9 \cdot 10^{10}$ ($1.2 \cdot 10^4$)
mse_d28_w64_s300	0.013	3468961	[ENUM]	$1.9 \cdot 10^7$ (150)	1.3	$1.1 \cdot 10^8$ (120)*	$1.9 \cdot 10^7$ (150)	$8.6 \cdot 10^4$ (0.098)
mse_d28_w64_s600	$6 \cdot 10^{-5}$	16155	[ENUM]	$1.9 \cdot 10^7$ (150)	1.1	$9 \cdot 10^7$ (98)	$1.9 \cdot 10^7$ (150)	$1.8 \cdot 10^7$ (18)
mse_d28_w64_s1000	$3.3 \cdot 10^{-7}$	89	[ENUM]	$1.9 \cdot 10^7$ (150)	1.5	$6.8 \cdot 10^7$ (73)	$1.9 \cdot 10^7$ (150)	$3.3 \cdot 10^9$ (3300)
mse_d28_w64_s1500	0	0	4983651 nodes, 32 pieces	10^7 (130)	800	$4.4 \cdot 10^7$ (47)	$1.9 \cdot 10^7$ (140)	cannot certify 0
bce_d28_w32_s30	$1.9 \cdot 10^{-3}$	507779	1201597 nodes	$3.4 \cdot 10^6$ (39)	210	$1.4 \cdot 10^7$ (16)*	$1.9 \cdot 10^7$ (71)	$5.9 \cdot 10^5$ (0.36)
bce_d28_w32_s60	$1.3 \cdot 10^{-5}$	3543	237621 nodes	$5 \cdot 10^5$ (5.9)	31	$3.5 \cdot 10^6$ (3.4)	$1.9 \cdot 10^7$ (65)	$8.4 \cdot 10^7$ (75)
bce_d28_w32_s100	$7.5 \cdot 10^{-9}$	2	20789 nodes	$4.1 \cdot 10^4$ (0.31)	3.9	$8.2 \cdot 10^5$ (0.69)	$1.9 \cdot 10^7$ (69)	$1.5 \cdot 10^{11}$ ($1.2 \cdot 10^5$)
bce_d28_w32_s150	0	0	2061 nodes	4100 (0.032)	1.4	$2.3 \cdot 10^5$ (0.17)	$1.9 \cdot 10^7$ (74)	cannot certify 0
two hidden layers (8 nets)	0–0.093		[ENUM]	$1.5 \cdot 10^8$–$1.9 \cdot 10^8$ (190–500)		B&B: upper end 0.78–0.91 after 940–2600 s (4 nets)		

Table 12: Exact counts at $d \leq 28$ ($\delta = 0.05$, $\varepsilon = 0.1$ for Monte Carlo, whose seconds are at the median measured time per sample of the net, as in Table 10). Argument: the certified partition built by the prover, or [ENUM] when its plan put the certificate above a fixed fraction of the cost of enumeration. B&B: branch and bound with CROWN bounds and no prover; * marks the two runs that ended with 9 and 1 undecided points, whose intervals contain p (`nets_prover/run_suite.py`, `nets_verify/exp_suite.py`, `wallclock.py`).

loss	d	nets	$p = 0$ certified	$p > 0$ exact	open	at the root	median fpe (s)
squared	12	15	15	0	0	0	2800 ($4 \cdot 10^{-3}$)
squared	16	15	14	1	0	2	7900 ($9 \cdot 10^{-3}$)
squared	20	15	15	0	0	4	7800 (0.010)
squared	24	15	13	2	0	2	$2.6 \cdot 10^4$ (0.031)
squared	28	10	8	2	0	3	$1.1 \cdot 10^5$ (0.11)
squared	32	10	10	0	0	3	$2.9 \cdot 10^5$ (0.38)
squared	40	3	2	0	1	0	3300 ($8 \cdot 10^{-3}$)
squared	48	3	2	0	1	0	$3.2 \cdot 10^7$ (40)
squared	64	3	1	0	2	0	$3.7 \cdot 10^7$ (49)
logistic	16	6	6	0	0	3	140 ($2 \cdot 10^{-3}$)
logistic	24	6	6	0	0	2	3600 ($8.5 \cdot 10^{-3}$)
logistic	32	3	3	0	0	2	4.9 (10^{-3})
logistic	40	3	3	0	0	1	$5.2 \cdot 10^6$ (7.0)

Table 13: Frequency and cost of certificates of $p = 0$ (one hidden layer of width 32, trained for at least 1000 squared-loss or 300 logistic-loss steps; branch and bound with CROWN bounds, at most 600 s per net; `nets_verify/exp_zero.py`). "At the root": certified with a single bound.

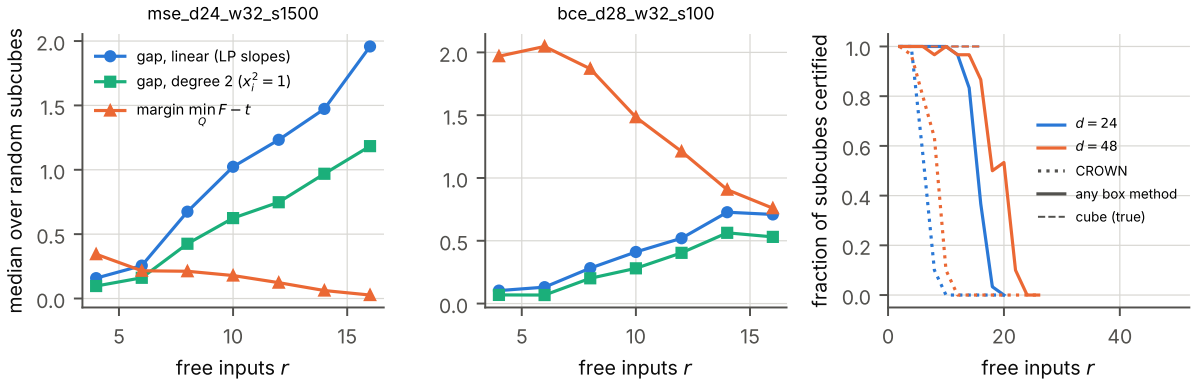


Figure 5: Left and middle: median gap between the exact minimum of F on random r-dimensional subcubes and the certified bound, linear and degree-2, against the margin $\min_Q F - t$ (`nets_prover/gap_scan.py`, computed with the previous verifier, whose bounds agree with the current ones far below the scale shown). Right: fraction of subcubes certified by CROWN, by any method valid on the subcube's box (exact box minimum by MILP, witnesses re-evaluated), and true on the cube, for squared-loss nets of width 32 trained for 1500 steps (`nets_verify/exp_barrier.py`).

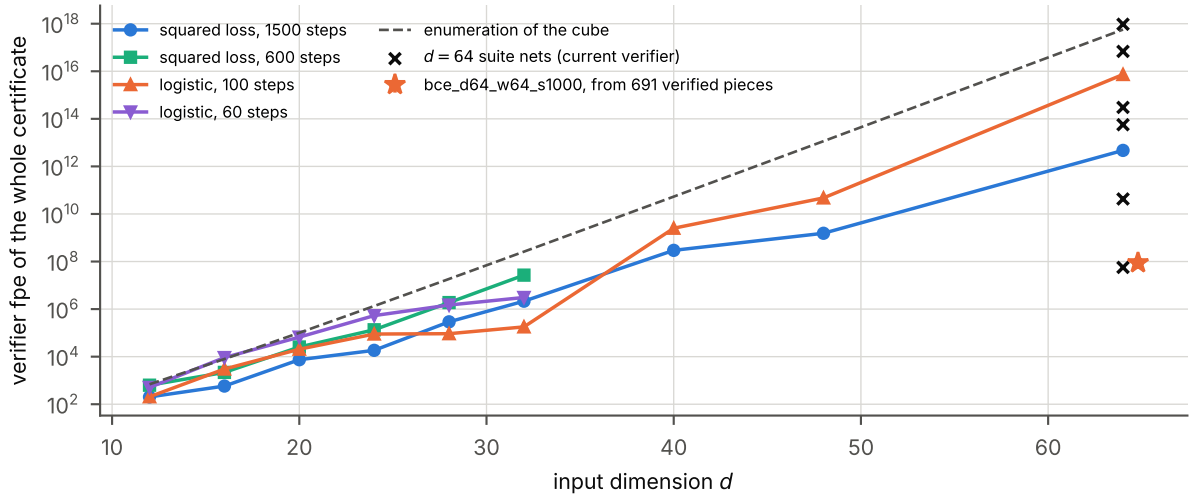


Figure 6: Verifier cost of the whole certified partition against d (Knuth estimates [Knu75] from random root-to-leaf probes of the prover's tree, medians over 3 seeds, in the previous verifier's cost model, which counts fewer flops per leaf than the current one; `nets_prover/scaling.py`), the $d = 64$ suite nets with the current verifier (`knuth_d64.py`), and the extrapolation from the certified pieces of `bce_d64_w64_s1000` (`full_d64.py`). The fitted growth is $2^{(0.68–0.78)d}$ for the two squared-loss families and $2^{(0.63–0.85)d}$ for the two logistic ones.

wherever $F < t$ on the box. Then $Z = \mathbb{E}_D\Phi \geq \Phi(0) \geq 1$, so the envelope Φ has looseness $K \geq 1/p$.

Proof. (a) $B(0) \leq F(0) < t$; for a subcube, its box contains its centre; and in general $B(y) \leq F(y) < t$. (b) For x uniform on $\{\pm 1\}^d$, convexity in x_1 gives $\mathbb{E}_{x_1}\Phi(x_1, y) \geq \Phi(0, y)$ for every y, and inducting over the coordinates, $\mathbb{E}_D\Phi \geq \Phi(0)$; since $F(0) < t$, $\Phi(0) \geq 1$. A product of positive log-convex functions is log-convex, hence convex, which covers the examples. □

The extreme small-ball point is the centre: $F(0) = c + \sum_j a_j \mathrm{relu}(b_j)$ has every $|z_j|$ as small as the biases allow. The barrier covers the tilts and the sign-pattern envelopes we built; it does not cover cell-based envelopes from a dynamic program over a few hidden forms (which are not functions on the box; a constructed small-ball net has one with $Z < 1$), enumeration, exact optimisation over the cube, or arguments using $x_i^2 = 1$, such as the degree-2 certificates above.

Proposition 8.7 (cube barrier)**.** *If $g_0 + \gamma \cdot x \leq F(x)$ for all $x \in \{\pm 1\}^d$, then $g_0 \leq s^* = \min_x \frac{1}{2}(F(x) + F(-x))$. Hence the tilted envelope $\varphi(x) = e^{\theta(t - g_0 - \gamma \cdot x)}$ ($\theta \geq 0$), which dominates $[F < t]$ on the cube, has mass $Z = e^{\theta(t-g_0)} \prod_i \cosh(\theta\gamma_i) \geq e^{\theta(t-s^*)}$, which is at least 1 when $s^* \leq t$, however the minorant was certified.*

Proof. Add the inequalities at x and $-x$: $2g_0 \leq F(x) + F(-x)$. The mass is a product over independent coordinates, and $\cosh \geq 1$. □

Proposition 8.8 (decoupling floor)**.** *Let $F = \sum_{k=1}^m F_k$, $\theta \geq 0$ and $q_1, \ldots, q_m \geq 1$. Every bound of the form $p \leq e^{\theta t} \prod_k \|e^{-\theta F_k}\|_{q_k}$ (the generalised Hölder inequality, with $\sum_k 1/q_k = 1$; Finner's inequality [Fin92], for parts depending on overlapping sets of independent coordinates) is at least $\inf_{\theta \geq 0} e^{\theta t} \prod_k \mathbb{E}e^{-\theta F_k}$, the Chernoff bound the parts would give if they were independent.*

Proof. For $Y \geq 0$ and $q \geq 1$, $\|Y\|_q \geq \|Y\|_1 = \mathbb{E}Y$ (Jensen), so each bound is at least $e^{\theta t} \prod_k \mathbb{E}e^{-\theta F_k}$ at its θ. □

With the hidden units as parts, the independent-parts floor is 0.06–0.47 on the one-hidden-layer squared-loss nets with $d \leq 28$, whose p ranges over 0.071–$6 \cdot 10^{-8}$: the even parts of positive- and negative-weight units cancel (correlation -0.84–-0.75), so the true lower tail is far thinner than independence predicts. This is the opposite of the unate case of Theorem 4.7, where positive correlation makes the independence estimate a certified lower bound. Measured on the suite (Table 14): $F(0) < t$ on all 13 squared-loss nets with $d \leq 28$; $s^* < t$ on 11 of the 12 with $p > 0$ (the exception has $s^* - t = 0.011$); the best certified Hölder bound is 0.94–0.98 and McDiarmid 0.992–0.997 there, and McDiarmid is at least 0.90 on every net; the best box tilt has $\log Z \geq 0$ on all 30 nets (its optimiser ends at $\theta \to 0$, i.e. at Monte Carlo); bounding the positive units alone, even with their exact law, cannot go below 0.983–0.995; and CROWN's median gap on 8-dimensional subcubes exceeds the median margin on 13 of the 13 squared-loss nets with $d \leq 28$, so its r_{50} (the largest subcube dimension at which half of the random subcubes are certified) is small. The ideal Chernoff envelope $e^{\theta(t-F)}$ would have $K = 2.1$–11: the obstacle is computing its mass and sampling it, not information.

Methods with an oracle that optimises over the cube escape the box barrier, at verification cost. Perturb-and-MAP [HJ12] certifies $U = 1.6 \cdot 10^{-3}$ at $p = 9.7 \cdot 10^{-6}$ and $4.8 \cdot 10^{-6}$ at $p = 6 \cdot 10^{-8}$ (1,720 MAP calls each, 520 and 480 s) and $1.8 \cdot 10^{-4}$ at $p = 7.5 \cdot 10^{-9}$ (380 calls, 1100 s); on the suite nets with $p \leq 10^{-4}$ it certifies $U = p^c$ with $c = 0.38$–0.74; the cube-certified tilt exists only when $s^* > t$, and on `mse_d24_w32_s2000` has $Z = 1.3 \cdot 10^{-5}$, $K = 221$ and gives $[5.7 \cdot 10^{-8}, 6.22 \cdot 10^{-8}]$ from $2.1 \cdot 10^6$ samples, against $1.9 \cdot 10^{10}$ for Monte Carlo, after 405 MAP calls (180 s in all). Each

net	p	$F(0)-t$	$\min_{\text{box}} F-t$	s^*-t	floor	$U_{\text{Hölder}}$	$\log Z_{\text{tilt}}$	Z_{sign}	gap_8	margin_8	r_{50}
mse_d24_w32_s300	0.071	−0.196	−1.317	−1.040	0.47	0.974	$4\cdot10^{-5}$	≥ 1	1.01	−0.19	2
mse_d24_w32_s600	$8.3\cdot10^{-3}$	−0.152	−0.827	−0.515	0.38	0.966	$3\cdot10^{-5}$	≥ 1	0.96	0.03	2
mse_d24_w32_s1000	$3.6\cdot10^{-4}$	−0.108	−0.581	−0.240	0.31	0.959	$3\cdot10^{-5}$	≥ 1	0.87	0.17	4
mse_d24_w32_s1500	$9.7\cdot10^{-6}$	−0.064	−0.412	−0.090	0.24	0.951	$3\cdot10^{-5}$	≥ 1	0.80	0.24	4
mse_d24_w32_s2000	$6\cdot10^{-8}$	−0.022	−0.307	0.011	0.18	0.939	$2\cdot10^{-5}$	≥ 1	0.72	0.29	4
mse_d28_w64_s300	0.013	−0.332	−0.859	−0.623	0.28	0.980	$5\cdot10^{-5}$	≥ 1	0.98	0.09	2
mse_d28_w64_s600	$6\cdot10^{-5}$	−0.309	−0.565	−0.230	0.19	0.974	$4\cdot10^{-5}$	≥ 1	0.88	0.21	4
mse_d28_w64_s1000	$3.3\cdot10^{-7}$	−0.277	−0.431	−0.059	0.13	0.968	$4\cdot10^{-5}$	≥ 1	0.82	0.26	4
mse_d28_w64_s1500	0	−0.238	−0.349	0.040	0.063	0.956	$3\cdot10^{-5}$	≥ 1	0.69	0.32	4
mse_d28_w32x32_s100	0.060	−0.397	−0.970	−0.634	-	-	$4\cdot10^{-3}$	-	2.15	−0.17	2
mse_d28_w32x32_s200	0.015	−0.359	−0.909	−0.420	-	-	$4\cdot10^{-3}$	-	1.77	0.03	2
mse_d28_w32x32_s300	$3.6\cdot10^{-3}$	−0.319	−0.765	−0.274	-	-	$3\cdot10^{-3}$	-	1.61	0.12	2
mse_d28_w32x32_s500	$1.3\cdot10^{-4}$	−0.240	−0.576	−0.112	-	-	$3\cdot10^{-3}$	-	1.37	0.19	4
bce_d28_w32_s30	$1.9\cdot10^{-3}$	0.129	−1.286	−0.407	0.020	0.759	$2\cdot10^{-5}$	≥ 1	0.71	0.62	6
bce_d28_w32_s60	$1.3\cdot10^{-5}$	0.260	−0.753	0.100	$1.1\cdot10^{-3}$	0.614	10^{-5}	≥ 1	0.65	1.24	10
bce_d28_w32_s100	$7.5\cdot10^{-9}$	0.418	−0.390	0.603	$2\cdot10^{-5}$	0.453	$7\cdot10^{-6}$	≥ 1	0.68	1.71	12
bce_d28_w32_s150	0	0.591	−0.155	1.011	$6.5\cdot10^{-8}$	0.317	$7\cdot10^{-6}$	≥ 1	0.75	2.25	12
bce_d28_w64x64_s20	0.093	0.033	−1.790	−0.795	-	-	$9\cdot10^{-3}$	-	4.15	−0.14	2
bce_d28_w64x64_s40	$2.3\cdot10^{-3}$	0.074	−1.284	−0.155	-	-	$9\cdot10^{-3}$	-	4.00	0.54	2
bce_d28_w64x64_s60	$1.9\cdot10^{-5}$	0.118	−0.776	0.335	-	-	$9\cdot10^{-3}$	-	4.15	0.91	4
bce_d28_w64x64_s100	0	0.203	−0.424	1.040	-	-	$8\cdot10^{-3}$	-	4.09	1.86	4
mse_d64_w64_s300	?	−0.318	−1.836	$-1.332^{\ddagger}$	0.18	0.991	$8\cdot10^{-5}$	≥ 1	0.44	0.15	4
mse_d64_w64_s1000	?	−0.207	−0.738	$-0.342^{\ddagger}$	0.033	0.973	$5\cdot10^{-5}$	≥ 1	0.29	0.33	8
mse_d64_w64_s3000	?	0.114	−0.133	$0.136^{\ddagger}$	$2.6\cdot10^{-5}$	0.871	$2\cdot10^{-5}$	≥ 1	0.14	0.42	14
bce_d64_w64_s100	?	0.463	−2.017	$0.149^{\ddagger}$	$9.9\cdot10^{-7}$	0.679	$5\cdot10^{-5}$	≥ 1	0.50	2.72	16
bce_d64_w64_s300	?	0.890	−0.786	$1.373^{\ddagger}$	$1.7\cdot10^{-19}$	0.427	$4\cdot10^{-5}$	≥ 1	0.53	4.56	24
bce_d64_w64_s1000	?	1.520	$0.114^{\dagger}$	$2.554^{\ddagger}$	$4\cdot10^{-41}$	0.220	$2\cdot10^{-5}$	$4.9\cdot10^{-3}$	0.65	6.47	24
bce_d128_w128x128_s100	?	0.194	−1.164	$1.144^{\ddagger}$	-	-	0.04	-	1.30	4.28	12
bce_d128_w128x128_s300	?	0.324	−0.711	$2.380^{\ddagger}$	-	-	0.04	-	1.31	6.29	14
bce_d128_w128x128_s1000	?	0.499	−0.203	$4.210^{\ddagger}$	-	-	0.03	-	1.35	9.69	16

Table 14: Barrier diagnostics (`nets_analytic/barrier_table.py`). $\min_{\text{box}} F-t$: value at a re-evaluated box point found by MILP (†: the solver's claimed optimum, not certified); s^*-t: exact for $d\leq 28$, else the best value found (‡, an upper bound on s^*-t); floor: independent-units Chernoff value (Proposition 8.8); $U_{\text{Hölder}}$: best certified Hölder bound; $\log Z_{\text{tilt}}$: best box-relaxed tilt; Z_{sign}: sign-pattern envelope (one hidden layer); gap_8 and margin_8: median CROWN gap and margin on 8-dimensional subcubes; r_{50}: CROWN's certified subcube dimension. "?": p unknown; "-": not computed for two hidden layers.

call is NP-hard; we answered them by enumeration, so these certificates cost more than computing p, and a MILP solver leaves gaps of 22–33 nats on the $d=64$, width-64 suite nets after 300 s. The one large-margin success needs no oracle: on `bce_d64_w64_s1000` ($F(0)-t=1.52$, and the solver's box minimum is $t+0.114$) the sign-pattern envelope, with 12 negative units split, has $Z=4.9\cdot10^{-3}$, and $2\cdot10^7$ samples certify $p\leq 2.4\cdot10^{-9}$, against $1.8\cdot10^{-7}$ for Monte Carlo with the same samples (1000 s in all). No net with $d\leq 30$ is in this regime, so it is not checked against an exact p.

8.8 Regimes, heuristics, and what remains open

Table 15 collects the regime map. Heuristic estimators are useful comparators but unsound (Table 16, `experiments/nets_structure`, 3,075 (net, level) points on 547 nets with exact P): the best, "select" (a quadratic fit to the lower tail of 10^5 samples for one-hidden-layer squared loss, a sphere-input surrogate otherwise), is within a factor 10 on 95% of points, worst off by $10^{3.3}$, and predicts $p=0$ correctly on 98% of nets. The presumption of independence over hidden units overestimates by a median $10^{2.65}$, for the cancellation reason of Proposition 8.8. Nets of the same architecture break all of them: the output Gaussian is off by $10^{15.7}$ on a lattice small ball

at $d = 400$, and the tail fit misses a planted 2^{-16} subcube by $10^{1.6}$.

Tried without gain: soft (tempered) levels, block-Gibbs moves and selection pooled over trajectories; explicit self-certifying proposals (autoregressive and kernel-mixture models fitted to splitting samples); McDiarmid, Esseen/Halász-type and exact-moment bounds; envelopes over a few hidden forms; and quadratic and LP-dual branch-and-bound prototypes.

regime	best sound two-sided	best sound lower bound	exact or $p = 0$ certificate	cost against Monte Carlo
$p \gtrsim 10^{-3}$, any d	Monte Carlo	Monte Carlo	enumeration at $d \lesssim 30$, else none	Monte Carlo 8–130 times cheaper than splitting (fpe)
$d \lesssim 30$, $p > 0$	exact count (enumeration, B&B, certified partition)	witness list (exact when F is small), splitting	certified partition, B&B	verifier 210, $5.8 \cdot 10^4$, $3.6 \cdot 10^6$ times cheaper in fpe at $p = 9.7 \cdot 10^{-6}$, $6 \cdot 10^{-8}$, $7.5 \cdot 10^{-9}$; end to end in seconds 2.8, 660, $2.9 \cdot 10^4$
$d \geq 40$, $p > 0$, large failure set	Monte Carlo only (squared loss: Section 8.7)	splitting, $L/p_{\text{ref}} = 0.87$–0.97	none within reach (whole certificate $5.7 \cdot 10^{13}$–$9.5 \cdot 10^{17}$ fpe at $d = 64$)	lower bound 3.8–10^{15} times cheaper for $p_{\text{ref}} \leq 10^{-4}$ (fpe, KL yardstick)
$p = 0$, or a large margin	certificate of $p = 0$; sign-pattern envelope (one $d = 64$ net)	– ($L = 0$ is right)	98 of 107 nets certified; at $d = 64$, 67.5% of the cube in 110 min	Monte Carlo never certifies $p = 0$
far tail, $z \gtrsim 7.8$	none	witness list (0.42–0.47 of p_{ref}); splitting alone 0.17–0.18	only with a large margin	–

Table 15: Regime map for the suite and the nets trained here. Costs as in Tables 10 to 12.

estimator	median	median $\lvert\cdot\rvert$	$\times 10$	$\times 10$, $P \leq 10^{-6}$	worst	$p = 0$ right
select	0.04	0.18	95%	87%	3.3	98%
tail fit	0.07	0.18	93%	72%	7.5	96%
sphere surrogate	0.24	0.36	77%	71%	5.7	94%
output Gaussian	0.28	0.68	63%	35%	43.1	90%
independence (units)	2.65	2.65	13%	2%	7.6	59%
independence (even form)	1.57	1.57	35%	13%	6.7	70%

Table 16: Unsound heuristic estimators against exact P: median, median absolute and worst $\log_{10}(\text{estimate}/P)$; "$\times 10$": fraction of points within a factor 10 (all points, and those with $P \leq 10^{-6}$); and the fraction of nets on which "$2^d\times$ estimate < 1" predicts $p = 0$ correctly. Tail fit: a quadratic fit to the log lower tail of 10^5 samples, extrapolated; select: the tail fit for one hidden layer and squared loss, the sphere surrogate otherwise; independence: hidden units treated as independent (`nets_structure/heuristics.py`).

Bottom line. Sound lower bounds work very well on trained nets wherever p could be checked: exactly at $d \leq 32$, and by independent Monte Carlo down to $p \approx 6.3 \cdot 10^{-9}$ at $d = 64$ and 128. Theorem 6.10 makes splitting sound without mixing, and at $d = 64$ and 128 the chains appear to mix (at small d with tiny p, where they do not, the witness sets are exact). Below the checked

range the ratio is to the mean of the same estimator, which cannot reveal a missed mode (the planted net of Section 8.2). Exact values and certificates of $p = 0$ work at d up to about 30, or with large margins (long logistic training), where certified partitions are short. For squared-loss nets with $p > 0$ at $d \geq 40$, no sound two-sided method we found beats Monte Carlo: this is unavoidable for verifiers that only evaluate the net (Proposition 8.2(c)), unavoidable in the worst case for all verifiers unless $\mathsf{coNP} \subseteq \mathsf{MA}$ (Proposition 8.2(a)), and, on the trained nets, the box, cube and decoupling barriers rule out every relaxation-based deduction we could make cheap: the certified partitions we could build grow like $2^{(0.68\text{–}0.78)d}$ (Knuth estimates for the squared-loss families of Figure 6, in the previous verifier's cost model). Open:

1. *Cube-aware deduction.* Can certificates that use $x_i^2 = 1$ (degree-4 sums of squares, Lasserre levels) or exact MAP-based arguments certify subcubes whose dimension grows with d on squared-loss nets? Degree 2 removes only 15%–39% of the linear gap (10th–90th percentile over the 27 pairs of a one-hidden-layer squared-loss net and a subcube dimension in the gap scan of Figure 5). A proof-size lower bound for some proof system on trained nets would settle the question the other way.

2. *Why the chains appear to mix.* At $d = 64$ and 128 the late-level acceptance rates and the numbers of distinct final particles look like those of a mixing chain; whether the level sets of trained nets really are well connected under bit flips, and why, is open. A certified mixing bound would give two-sided intervals (Section 6.5).

3. *Two hidden layers and the far tail.* Intermediate-bound gaps defeat every certificate we tried with two hidden layers, and nets trained into the far tail defeat the lower bound; whether targeted proposals with exact corrections help is untested.

4. *Training for verifiability.* Margin or certified training lowers the per-input influence that sets r^*; how far it moves the regimes of Table 15 is open.

9 Discussion and open problems

What the answers are. Without a prover, our procedures beat sampling where the rarity of $C(x) = 1$ is *visible*: built from independent parts, forced literals, small cutsets, unions, $\wedge$'s of small context width, thresholds of independent blocks or of local detectors in a certified regime, affine constraints, or sparse monotone overlaps, possibly disguised by rewriting cones of logarithmic support. DFSB plus the specialised modules is sound on every circuit and complete on the union of their classes (Theorems 5.2 and 5.36). Visibility depends on the method and the distribution: random parity checks are transparent to Gaussian elimination under the uniform distribution, but for events that fill a $1/\mathrm{poly}(n)$ fraction of the solution set of $m \leq n/6$ random checks, with high probability, envelopes with small leaves, context-width compilation and DFSB's completeness guarantee cost Monte Carlo's price up to polynomial factors, and any use of a DNNF envelope at least its square root (Corollary 5.29); and under biased inputs small $\wedge$'s of parity checks admit no sound procedure that is complete on them unless $\mathsf{NP} \subseteq \mathsf{BPP}$ (Theorem 5.31). With a prover, the visible structure may be replaced by *provable* structure: any confinement of the rare inputs to a union of tractable pieces (subcubes, checked-deterministic circuits, affine atoms, DNNFs) of small total mass that a propositional proof system can certify (Theorem 6.5). For upper bounds this helps strictly whenever slices of the proof system's canonical pair are not separable in randomized polynomial time (Proposition 6.17 and Corollary 6.18). In both settings the cost has the same shape, polynomial in the size of the explanation and exponential in the unexplained surprise, and under the uniform distribution, once an envelope is certified, the exponential part is unavoidable on some circuits for verifiers that access the rest of the circuit only through evaluations (Theorem 6.12(b)).

Relation to heuristic arguments. ARC's programme [CNX22, Hil24] asks for estimators that use a model's internals to beat sampling, and measures explanations by surprise accounting. The present results isolate the part of that programme that can be made *sound against adversarial circuits*:

- the presumption of independence is certified where independence is syntactic (variable-disjoint parts, PROD) or where correlation inequalities control the error (Theorems 4.7 and 5.10);
- surprise accounting becomes a cost formula for a sound verifier, $\mathrm{poly}(|\pi|) + 2^{\sigma_u(\pi)}$ up to polynomial factors, with $|\pi|$ the size of the argument's deductive part, and under the uniform distribution both halves have a black-box meaning: in the worst case (planted subcubes), a bit of argument saves at most a factor 2 in evaluations for lower bounds, and once an envelope is certified, a verifier that accesses the rest of C only through evaluations needs about 2^{σ_u} of them for two-sided intervals on some circuits, which the envelope estimator achieves (Theorem 6.12);
- what soundness forbids is equally sharp: upper bounds must come from deduction (Theorem 3.2), and unless $\mathsf{coNP} \subseteq \mathsf{MA}$ no system certifies them with polynomial-size arguments for all circuits (Theorem 3.4). A heuristic estimator that is accurate on random circuits but not sound on adversarial ones is therefore not a counterexample to anything here; it lives in the gap between "no certificate exists" and "the estimate is wrong".

The rarity that defeats certification is *distributed surprise*: many weak, overlapping constraints (random CNFs below the threshold), or pseudorandomness. Strongly overlapping constraints

(large Δ, outside the local-lemma and cluster-expansion regimes) are certified here only where an exact sampler or a locally checkable mixing certificate exists (Theorems 5.14 and 5.18); weakly overlapping ones by the Harris–Janson sandwich, the cluster expansion and the local-lemma FPRASs (Theorems 4.7 and 5.10, Proposition 5.12, and Section 5.9). Annealing and splitting estimators handle many of the strongly overlapping ones in practice, and ARC-style heuristic estimators may handle both kinds on typical instances; neither comes with a certificate that a polynomial-time verifier can check.

Open problems.

1. *Envelope lower bounds for distributed surprise.* Does every polynomial-size DNNF envelope of a random 3-CNF with $1/3 < \alpha < \alpha_{\text{sat}}$ have $D(E)/p = 2^{\Omega(n)}$? Proved here only for drop-conjunct relaxations; for random parity checks the analogous bound is Theorem 5.26.

2. *Certified annealing.* Is there a polynomial-size, MA-checkable certificate that a sampler for $D \mid A_1 \wedge \cdots \wedge A_j$ is accurate (a checkable mixing certificate) beyond exact samplers (Theorem 5.14) and the Dobrushin regime of soft tilts (Lemma 5.16)? By Theorem 3.2 it cannot be purely statistical. Fixed-level multilevel splitting already gives sound lower bounds without one (Theorem 6.10); such a certificate is what would make its upper bounds, and so its two-sided intervals, sound.

3. *Nested unions over approximate leaves.* Can the DNNF FPRAS be run over DNNF skeletons whose leaves are sets given only by approximate counters, approximate samplers and membership tests? This would give unions of unions of products of dense blocks at any depth.

4. *A natural two-sided separation.* Find a $\mathsf{UP} \cap \mathsf{coNP}$ problem not in BPP (under a standard assumption) whose unique-witness property and the disjointness of its two witness relations have polynomial-size extended Frege proofs (the disjointness follows from uniqueness when both relations come from one unique-witness relation, as for factorisation); then the two-sided separation of Section 6.8 holds with a standard proof checker instead of a theory like ZFC.

5. *Weak automatability and envelopes.* If P is weakly automatable, is there a procedure without a prover that is complete on the classes $\mathcal{K}_P(m, K)$ of Theorem 6.5, i.e. that finds certified envelopes and not only tests them?

6. *Optimal arguments.* How well can $\min_\pi \left(|\pi| + 2^{\sigma_u(\pi)}\right)$ be approximated for natural certificate languages? Choosing which conjuncts to keep is weighted set packing on supports; choosing branch variables is related to pseudo-tree depth.

7. *Networks.* Binarised or ReLU networks with rare activation patterns are the case of most interest for alignment. Local threshold circuits (Section 5.7) cover bounded receptive fields with a count readout. For general networks, interval-bound propagation gives gate-wise outer approximations; can it be turned into sandwich certificates (Section 6) with small looseness for trained networks? For ReLU networks trained to output a constant, Section 8 gives a partial answer. For squared-loss training the answer is negative for bounds valid on the box: on every squared-loss suite net a point of the box with $F < t$ exists, so such bounds fail at the root (Proposition 8.6(a)), and the certified partitions we could build grow empirically like $2^{(0.68\text{–}0.78)d}$ (Knuth estimates in the previous verifier's cost model, Figure 6).

Whether case splits combined with certificates that use $x_i^2 = 1$, or exact optimisation over the cube, escape this is open. For logistic-loss training to large margins the answer is positive on the nets we tried, with certificates of $p = 0$ far smaller than enumeration; and lower bounds need no such certificate (Corollary 8.4).

8. *Heuristic to certified.* When a presumption-of-independence estimate is accurate, when can it be compiled into a certificate with polynomial overhead, for instance by proving the independence it presumes, or by bounding the correlations it ignores?

A The second-order independence sandwich

Setting of Theorem 5.10: increasing events $B_1, \dots, B_m$ (after flipping the variables that occur positively, Section 5.5) with syntactic supports S_i, a product distribution, and a fixed order. For each i let $N_i = \{j < i : j \sim i\}$, $R_i = \{j < i : j \not\sim i\}$, $\Gamma(j)$ the neighbourhood of j, $b_i = D(B_i)$ with $\max_i b_i < 1$, $\beta_i = \sum_{j \in N_i} b_j$ (assumed < 1),

$$C_i = \sum_{j \in N_i} \operatorname{Cov}(B_i, B_j) \quad (\geq 0 \text{ by Harris}),$$

$$\Theta_i = b_i \Big[\sum_{j \in N_i} \sum_{k \in R_i \cap \Gamma(j)} D(B_j B_k) + \sum_{j < j' \in N_i} D(B_j B_{j'}) \Big],$$

$$\Theta_i' = \sum_{j \in N_i} \sum_{k \in R_i \cap \Gamma(j)} D(B_i B_j B_k) + \sum_{j < j' \in N_i} D(B_i B_j B_{j'}).$$

Each triple $\{i, j, k\}$ with $j, k < i$ that is connected in G occurs exactly once in these sums (in the first if i is adjacent to only one of j, k, in the second if it is adjacent to both), so $\Theta_i' = \sum D(B_i B_j B_k)$ and $\Theta_i = b_i \sum D(B_j B_k)$ over these triples, and $\sum_i \Theta_i'$ does not depend on the order. If $b_i = 0$ then c_i, C_i, Θ_i and Θ_i' vanish and the i-th factors below equal 1; so we assume $b_i > 0$ whenever we condition on B_i.

Theorem A.1.

$$\prod_i \Big(1 - \min\Big\{b_i,\ b_i - C_i + \frac{\Theta_i'}{1 - \beta_i}\Big\}\Big) \ \leq\ p \ \leq\ \prod_i \Big(1 - \max\Big\{0,\ b_i - \frac{C_i + \Theta_i}{1 - \beta_i}\Big\}\Big).$$

Consequently, with $M_2 = \prod_i (1 - b_i + C_i)$, $|\log p - \log M_2| \leq E_3 := \sum_i \frac{\Theta_i + \Theta_i' + \beta_i C_i}{(1-\beta_i)(1-b_i)}$.

Proof. Fix i, write $N = N_i$, $R = R_i$, $F_X = \bigwedge_{j \in X} \neg B_j$ and $\nu = D(\cdot \mid F_R)$ (by Harris $\Pr[F_R] \geq \prod_{j \in R}(1 - b_j) > 0$). Then $p = \prod_i (1 - c_i)$ with $c_i = \Pr[B_i \mid F_N \wedge F_R]$. Since S_i is disjoint from the supports of the events in R, $\nu(B_i) = b_i$ and

$$c_i = b_i - \frac{b_i\,[\nu(F_N) - \nu(F_N \mid B_i)]}{\nu(F_N)}, \qquad \nu(F_N) - \nu(F_N \mid B_i) = \nu\big(\textstyle\bigcup_N B_j \mid B_i\big) - \nu\big(\textstyle\bigcup_N B_j\big).$$

We use six facts. Each follows from the Harris inequality (an increasing and a decreasing event are negatively correlated under a product measure) and from independence of events with disjoint supports. For $j \in N$ let $R' = R \cap \Gamma(j)$ and $R'' = R \setminus R'$; the events of R'' have supports disjoint from $S_i \cup S_j$, and $\Pr[F_R] \leq \Pr[F_{R''}]$.

(H1) $\nu(B_j \mid B_i) \leq D(B_i B_j)/b_i$ (Harris for $B_i B_j$ versus F_R).

(H2) $\nu(B_j) \le b_j$ (Harris).

(H3) $\nu(B_j) \ge b_j - \sum_{k\in R'} D(B_jB_k)$: $\Pr[B_jF_R] \ge \Pr[B_jF_{R''}] - \sum_{k\in R'} \Pr[B_jB_kF_{R''}] \ge (b_j - \sum_k D(B_jB_k))\Pr[F_{R''}]$ by independence and Harris; divide by $\Pr[F_R] \le \Pr[F_{R''}]$ (trivial if the bracket is negative).

(H4) $\nu(B_jB_{j'}) \le D(B_jB_{j'})$ (Harris).

(H5) $\nu(B_j \mid B_i) \ge [D(B_iB_j) - \sum_{k\in R'} D(B_iB_jB_k)]/b_i$: as (H3), with B_iB_j in place of B_j.

(H6) $\nu(B_jB_{j'} \mid B_i) \le D(B_iB_jB_{j'})/b_i$ (Harris).

Upper bound on p. By Bonferroni, $\nu(\bigcup B_j \mid B_i) \le \sum_j \nu(B_j \mid B_i)$ and $\nu(\bigcup B_j) \ge \sum_j \nu(B_j) - \sum_{j<j'} \nu(B_jB_{j'})$. With (H1), (H3), (H4), $b_i[\nu(F_N) - \nu(F_N \mid B_i)] \le C_i + \Theta_i$; with (H2), $\nu(F_N) \ge 1 - \beta_i$. So $c_i \ge b_i - (C_i + \Theta_i)/(1 - \beta_i)$, and $c_i \ge 0$.

Lower bound on p. By Bonferroni the other way with (H5), (H6), (H2), $b_i[\nu(F_N) - \nu(F_N \mid B_i)] \ge C_i - \Theta'_i$. If $C_i \ge \Theta'_i$ use $\nu(F_N) \le 1$; otherwise use $\nu(F_N) \ge 1 - \beta_i$. In both cases (using $C_i \ge 0$) $c_i \le b_i - C_i + \Theta'_i/(1 - \beta_i)$; and $c_i \le b_i$ by Harris.

The E_3 bound. Put $x_i = 1 - b_i + C_i \ge 1 - b_i$. On the upper side, the factor is at most $x_i + (\Theta_i + \beta_iC_i)/(1-\beta_i)$ (or 1, in which case $1 - x_i < (\Theta_i + \beta_iC_i)/(1-\beta_i)$), and $\log(x+y) - \log x \le y/x$. On the lower side, the factor is at least $\max\{1 - b_i,\ x_i - \Theta'_i/(1 - \beta_i)\}$; if the second term is the larger one, $\log(x - y) - \log x \ge -y/(x - y) \ge -y/(1 - b_i)$, and otherwise $C_i < \Theta'_i/(1 - \beta_i)$ and $\log(1 - b_i) - \log x_i \ge -C_i/(1 - b_i)$, which is no smaller. Summing over i gives the claim. □

We checked the theorem against exact enumeration on 382 random instances (3–8 random monotone events, each a monotone DNF on 2–4 of 8–14 variables, random product distributions and orders; `experiments/check_second_order.py`): no violations, median $E_3 = 0.064$. The min in the lower bound matters: without it a factor can become negative, and a product of two negative factors can exceed p.

B Local threshold circuits: proofs

Notation is that of Section 5.7: $D_\theta \propto De^{\theta S}$, $s(\theta) = \mathbb{E}_{D_\theta} S$, $r_i(x) = \sigma(a_i + \theta\Delta_i(x)) = \Pr_{D_\theta}[x_i = 1 \mid x_{-i}]$, $R = S_{\max} - S_{\min}$.

B.1 The sampler

Proof of Lemma 5.16. (a) Let x, x' differ only at l, $z = a_i + \theta'\Delta_i(x)$, $z' = a_i + \theta'\Delta_i(x')$. By the mean value theorem $|\sigma(z) - \sigma(z')| \le |z - z'| \max_{[z,z']} \sigma'$. For $\theta' \le \theta$ the points $a_i + \theta'\Delta$ lie on the segments from a_i to $a_i + \theta\Delta$, so the hull for θ' is contained in the hull for θ, and $|z - z'| = \theta'|\Delta_i(x) - \Delta_i(x')|$; this gives $c_{il}(\theta') \le \kappa_{il}(\theta)$, and the same argument shows that κ_{il} is nondecreasing. A factor that does not contain l contributes the same amount to $\Delta_i(x)$ and $\Delta_i(x')$; one that contains i and l contributes a difference of two values in $\{-1, 0, 1\}$, so $|\Delta_i(x) - \Delta_i(x')| \le 2n_{il}$, and $\sigma' \le \frac{1}{4}$. Finally $\sum_{l\ne i} n_{il} = \sum_{j\ni i}(|A_j| - 1) \le d(k - 1)$.

(b) A class update resamples sites that share no scope, so given the other sites they are independent with their single-site conditional laws, and the update preserves $D_{\theta'}$. Run two copies X, Y with the same uniforms ($x_i \leftarrow \mathbf{1}[u_i < r_i(x)]$), X from the given state and Y from $D_{\theta'}$, so that Y stays distributed as $D_{\theta'}$. Let $e_i = \Pr[X_i \ne Y_i]$. When site i is updated, given the current states the new values differ with probability $|r_i(X) - r_i(Y)| \le \sum_{l:X_l\ne Y_l} c_{il}(\theta')$ (change the

differing neighbours one at a time); taking expectations, the new e_i is at most $\sum_l \kappa_{il}(\theta) e_l$, where the e_l are current values (no neighbour of i is updated together with i). Let $M = \max_i e_i/w_i$ at the start of a sweep. During the sweep all ratios stay at most M, and each updated site gets $e_i \le \sum_l \kappa_{il} w_l M \le \alpha M w_i$; after the sweep every site has been updated, so the maximum ratio is at most αM. Initially $M \le 1/\min_i w_i$, so after τ sweeps $d_{\mathrm{TV}}(X_\tau, D_{\theta'}) \le \Pr[X_\tau \ne Y_\tau] \le \sum_i e_i \le \alpha^\tau \sum_i w_i / \min_i w_i$.

(c) Consider the random-scan heat-bath chain $P = \frac{1}{n}\sum_i P_i$, $P_i f = \mathbb{E}_{D_{\theta'}}[f \mid x_{-i}]$; each P_i is an orthogonal projection in $L^2(D_{\theta'})$, so P is reversible. For x, y differing only at l, update the same site in both: if it is l, the two conditional laws coincide; otherwise couple maximally, creating a disagreement with probability $c_{il}(\theta') \le \kappa_{il}(\theta)$. The expected Hamming distance after one step is at most $1 - \frac{1}{n} + \frac{1}{n}\sum_{i \ne l} \kappa_{il}(\theta) \le 1 - \gamma/n$. By path coupling [BD97] this contraction extends to all pairs, and a reversible chain whose transportation distance contracts by $1-\gamma/n$ has spectral gap at least γ/n [LP17, Section 13.1]. The Poincaré inequality then gives $\mathrm{Var}(f) \le \frac{n}{\gamma}\langle f, (I-P)f\rangle = \frac{1}{\gamma}\sum_i \mathbb{E}[\mathrm{Var}(f \mid x_{-i})] \le \frac{1}{4\gamma}\sum_i \mathrm{osc}_i(f)^2$. For $f = S$, $\mathrm{osc}_i(S) \le d$ for the n_S variables in some scope and $\mathrm{osc}_i(S) = 0$ for the others. □

B.2 The normaliser

Lemma B.1 (Maurer [Mau03])**.** *For independent nonnegative $X_1, \dots, X_N$ with finite second moments and $\tau > 0$, $\Pr\left[\sum_r (\mathbb{E}X_r - X_r) \ge \tau\right] \le \exp\left(-\tau^2/(2\sum_r \mathbb{E}X_r^2)\right)$.*

Proof. For $y \ge 0$, $e^{-y} \le 1 - y + y^2/2$, so for $\lambda > 0$, $\mathbb{E}e^{-\lambda X_r} \le 1 - \lambda\mathbb{E}X_r + \lambda^2 \mathbb{E}X_r^2/2 \le \exp(-\lambda \mathbb{E}X_r + \lambda^2 \mathbb{E}X_r^2/2)$. By Markov's inequality the probability is at most $\exp(-\lambda\tau + \lambda^2 \sum_r \mathbb{E}X_r^2/2)$; take $\lambda = \tau/\sum_r \mathbb{E}X_r^2$. □

Proof of Lemma 5.17. The function $\ln Z$ is smooth, with $(\ln Z)' = s$ and $s' = \mathrm{Var}_{D_\theta}(S) \ge 0$, so $\ln Z(\theta) = \int_0^\theta s$ lies between the lower and the upper Riemann sum; the upper one, $T = h\sum_{l=1}^{L} s(\theta_l)$, exceeds the lower one by $h(s(\theta) - s(0)) \le h(S_{\max} - s(0))$. It remains to show $|\hat{T} - T| \le u$ with probability $1 - \delta_1 - \delta_2$.

Variance. For $D_r = \frac{1}{2}(Y_{2r-1} - Y_{2r})^2$, independence gives

$$\mathbb{E}D_r = \tfrac{1}{2}(\mathrm{Var}\, Y_{2r-1} + \mathrm{Var}\, Y_{2r}) + \tfrac{1}{2}(\mathbb{E}Y_{2r-1} - \mathbb{E}Y_{2r})^2,$$

so $\sum_l \mathrm{Var}\, Y_l \le 2\mathbb{E}\Sigma$. Since $D_r \in [0, R^2/2]$, $\mathbb{E}D_r^2 \le \frac{R^2}{2}\mathbb{E}D_r$, and Lemma B.1 with $\tau = a\sqrt{\mathbb{E}\Sigma}$ shows that, with probability at least $1 - \delta_2$, $\mathbb{E}\Sigma \le \Sigma + a\sqrt{\mathbb{E}\Sigma}$, that is $\sqrt{\mathbb{E}\Sigma} \le \frac{a}{2} + \sqrt{\frac{a^2}{4} + \Sigma}$; then $\sum_l \mathrm{Var}\, Y_l \le \bar{V}$. (Also $\mathrm{Var}\, Y_l \le R^2/4$ always.)

Bernstein. The variables $h(Y_l - \mathbb{E}Y_l)$ are independent, centred, bounded by hR in absolute value, with total variance $v = h^2 \sum_l \mathrm{Var}\, Y_l$, so by Bernstein's inequality (e.g. [BLM13, Chapter 2]) $\Pr[|\hat{T} - T| \ge u(v)] \le \delta_1$, where $u(v)$ solves $u^2 = 2\ell(v + hRu/3)$, i.e. $u(v) = \frac{1}{3}hR\ell + \sqrt{(\frac{1}{3}hR\ell)^2 + 2v\ell}$. As $u(\cdot)$ is increasing, $\{|\hat{T} - T| > u(h^2\bar{V})\} \subseteq \{\bar{V} < \sum_l \mathrm{Var}\, Y_l\} \cup \{|\hat{T} - T| > u(v)\}$, which has probability at most $\delta_2 + \delta_1$. □

Lemma B.2 (windows)**.** *In Lemma 5.17, let $c_l \in [S_{\min}, S_{\max}]$ and $w > 0$ be fixed before the Y_l are drawn, $\hat{Y}_l = \min(\max(Y_l, c_l - w), c_l + w)$, $E_l = |Y_l - \hat{Y}_l|$, $M = \max_l \max(S_{\max} - c_l - w,\ c_l - w - S_{\min},\ 0)$, and $\bar{z}$ the upper end of the KL interval (level δ_3) for the mean of the $E_l/M \in [0,1]$. Define $\hat{T}, \Sigma, \bar{V}, u$ as in Lemma 5.17 with Y_l replaced by $\hat{Y}_l - c_l$ in Σ, $\hat{T} = h\sum_l \hat{Y}_l$, $R^2/2$ replaced by $2w^2$ in $a = \sqrt{2(R^2/2)\ln(1/\delta_2)}$, and hR by $2hw$ in u. Then with probability at least $1 - \delta_1 - \delta_2 - \delta_3$, $\ln Z(\theta) \in [\hat{T} - u - \theta M\bar{z} - h(S_{\max} - s(0)),\ \hat{T} + u + \theta M\bar{z}]$.*

Proof. $|s(\theta_l) - \mathbb{E}\hat{Y}_l| \le \mathbb{E}E_l$, and $E_l \le M$ because Y_l and c_l lie in $[S_{\min}, S_{\max}]$. Hoeffding's bound holds for independent, not necessarily identically distributed, $[0,1]$ variables [Hoe63], so with probability $1-\delta_3$, $h\sum_l \mathbb{E}E_l = \theta M \cdot \frac{1}{L}\sum_l \mathbb{E}[E_l/M] \le \theta M\bar{z}$. The rest is the proof of Lemma 5.17 applied to $\hat{Y}_l - c_l \in [-w, w]$ (pairs: $\mathbb{E}D_r$ still dominates the average of the two variances; $D_r \in [0, 2w^2]$). □

B.3 Anticoncentration

Lemma B.3 (local bound for Poisson-binomials)**.** *Let $X = \sum_{l\le N}\xi_l$ with independent $\xi_l \sim$ Bernoulli(π_l), $\pi_l \in (0,1)$, $\mu = \mathbb{E}X$, $v = \operatorname{Var}X$. For every integer u with $|u-\mu| \le v/e$,*

$$\Pr[X = u] \;\ge\; \frac{3}{8\sqrt{N}+4}\,\exp\Big(-\frac{e(u-\mu)^2}{2v}\Big).$$

Proof. The law of X is log-concave with support $\{0,\dots,N\}$, and $v \le \min(\mu, N-\mu)$, so $1 \le u \le N-1$. By log-concavity there is a finite λ such that u is a mode of the tilted law $\Pr_\lambda[X = j] \propto \Pr[X=j]e^{\lambda j}$, which is again Poisson-binomial, with variance $\varsigma^2 \le N/4$. By Chebyshev's inequality an open window of length 4ς around its mean carries probability at least $\frac{3}{4}$ and contains at most $4\varsigma+1$ integers, so $\Pr_\lambda[X=u] \ge \frac{3}{4(4\varsigma+1)} \ge \frac{3}{8\sqrt{N}+4}$. Moreover $\Pr[X=u] = \Pr_\lambda[X=u]e^{-(\lambda u - \Lambda(\lambda))} \ge \Pr_\lambda[X=u]e^{-\Lambda^*(u)}$, where $\Lambda(\lambda) = \ln \mathbb{E}e^{\lambda X}$ and Λ^* is its Legendre transform. For $|\lambda| \le 1$, $\pi(\lambda)(1-\pi(\lambda)) = \pi(1-\pi)e^\lambda/(1-\pi+\pi e^\lambda)^2 \ge e^{-|\lambda|}\pi(1-\pi)$, so $\Lambda'' \ge v/e$ on $[-1,1]$. Then $\Lambda'(1) \ge \mu + v/e \ge u$ and $\Lambda'(-1) \le u$, the supremum defining $\Lambda^*(u)$ is attained in $[-1,1]$, and there $\Lambda(\lambda) \ge \lambda\mu + \frac{v}{2e}\lambda^2$; hence $\Lambda^*(u) \le e(u-\mu)^2/(2v)$. □

Lemma B.4 (anticoncentration)**.** *Assume (H1), (H2), $m \ge m_0$, $0 \le \theta \le \theta_0$, and let s' be an integer with $|s' - s(\theta)| \le 2\sqrt{n_S}$. Then $\Pr_{D_\theta}[S = s'] \ge e^{-c_1}/(16\sqrt{n_S})$.*

Proof. Let P be the set of factors in (H2), $I = \{i_j : j \in P\}$ and $y = x_{[n]\setminus I}$. Each i_j lies only in A_j, so $S = R_0(y) + \sum_{j\in P}\Delta_j(y)x_{i_j}$ with $\Delta_j(y) = g_j(\cdot, x_{i_j}{=}1) - g_j(\cdot, x_{i_j}{=}0) \in \{-1,0,1\}$, and given y the x_{i_j} are independent with $\Pr[x_{i_j} = 1 \mid y] = \sigma(a_{i_j} + \theta\Delta_j(y)) \in [\eta_0, 1-\eta_0]$, $\eta_0 = \beta e^{-2}$ (as $\theta \le \theta_0 \le 2$). Let $N(y) = |\{j \in P : \Delta_j(y) \ne 0\}|$, let X_{big} be the sum of the x_{i_j} over the larger of the classes $\{\Delta_j = 1\}$, $\{\Delta_j = -1\}$ (at least $N(y)/2$ terms) and X_{small} the other; given (y, X_{small}), $S = c \pm X_{\text{big}}$ for a constant c. Let $M' = \mathbb{E}[S \mid y, X_{\text{small}}]$.

(E1) By Lemma 5.16(c) (the crude certificate has column sums $\le 1-\gamma$ for $\theta \le \theta_0$), $\operatorname{Var}_{D_\theta}S \le n_S d^2/(4\gamma)$, so $|S - s(\theta)| \le d\sqrt{n_S/\gamma}$ with probability at least $\frac{3}{4}$. (E2) $\Delta_j(y) \ne 0$ is an event on x_B, $B = A_j \setminus \{i_j\}$, of D-probability at least ν. Given all other variables, the density of x_B under D_θ relative to D lies in $[e^{-\theta d(k-1)}, e^{\theta d(k-1)}] \subseteq [e^{-2}, e^2]$, since S varies by at most $d|B|$ with x_B. So $\mathbb{E}N \ge e^{-2}\nu^2 m$. As $\operatorname{osc}_i(N) \le d$ for the $n_S \le km$ variables in scopes and 0 for the others, $\operatorname{Var}N \le kmd^2/(4\gamma)$, and Chebyshev gives $\Pr[N < N_0] \le \Pr[N < \mathbb{E}N/2] \le e^4 kd^2/(\gamma\nu^4 m) \le \frac{1}{4}$, with $N_0 = e^{-2}\nu^2 m/2$. (E3) Given (y, X_{small}), X_{big} has at most $|P| \le n_S$ terms (the private variables i_j are distinct), so $\operatorname{Var}X_{\text{big}} \le n_S/4$ and $|S - M'| \le \sqrt{n_S}$ with conditional probability at least $\frac{3}{4}$.

The event $G = \{N(y) \ge N_0,\ |s' - M'| \le b\sqrt{n_S}\}$ is a function of (y, X_{small}) and contains (E1)$\cap$(E2)$\cap$(E3), since $|s'-M'| \le 2\sqrt{n_S} + d\sqrt{n_S/\gamma} + \sqrt{n_S} = b\sqrt{n_S}$; so $\Pr[G] \ge \frac{1}{4}$. On G, X_{big} has at least $N_0/2$ terms with variances at least $\eta_0/2$, so its conditional variance is $v \ge e^{-4}\nu^2\beta m/8$ and $v/e \ge e^{-5}\nu^2\beta m/8 \ge b\sqrt{km} \ge b\sqrt{n_S}$ by the choice of m_0. Lemma B.3 with $N \le n_S$ terms and $|u - \mathbb{E}X_{\text{big}}| = |s'-M'| \le b\sqrt{n_S}$ gives $\Pr[S = s' \mid y, X_{\text{small}}] \ge \frac{1}{4\sqrt{n_S}}\exp(-eb^2 n_S/(2v)) \ge \frac{e^{-c_1}}{4\sqrt{n_S}}$, using $n_S \le km$. Average over G. □

B.4 Completeness and rarity

The pilot. Let $V_{\max} = n_S d^2/(4\gamma)$, $L_p = \lceil 2\theta_0 V_{\max}/\sqrt{n_S} \rceil$ and $\vartheta_l = l\theta_0/L_p$, $l = 0, \dots, L_p$. If $s(0) \geq t$, set $\theta = 0$. Otherwise, at each ϑ_l take the median $\hat{s}_l$ of $\lceil 8\ln((L_p+1)/\delta) \rceil$ means of $\lceil 4d^2/\gamma \rceil$ samples each, and let $\theta = \vartheta_{l^*}$ with $l^* = \arg\min_l |\hat{s}_l - t|$. The pilot samples are drawn by the sampler of (T2) and, for this analysis, counted in N_{tot} (so the level $\eta = \delta/(5N_{\mathrm{tot}})$ covers them); soundness does not need this. (The implementation uses a cheaper heuristic pilot, which affects only the width of the output.)

Proof of Theorem 5.18(ii). Work on the event that every sample of the run equals an exact sample (probability $\geq 1 - \delta/5$). The crude certificate ($w = \mathbf{1}$, $\alpha = \theta d(k-1)/2 \leq 1 - \gamma$, $F = n$) gives $\tau = O(\gamma^{-1} \log(nN_{\mathrm{tot}}/\delta))$, and $N_{\mathrm{tot}} = \mathrm{poly}(n, 1/\varepsilon, \log(1/\delta))$.

Tilt. By Lemma 5.16(c), $s' = \mathrm{Var} \leq V_{\max}$ on $[0, \theta_0]$. Each group mean has variance at most $n_S/16$, so it is within $\sqrt{n_S}/2$ of $s(\vartheta_l)$ with probability $\geq \frac{3}{4}$, and all medians are, with probability $\geq 1 - \delta$. If $s(0) < t \leq s(\theta_0)$, some ϑ' has $s(\vartheta') = t$, and the nearest grid point has $|s(\vartheta_{l_0}) - t| \leq V_{\max}\theta_0/(2L_p) \leq \sqrt{n_S}/4$. Then $|s(\theta) - t| \leq |\hat{s}_{l^*} - t| + \frac{\sqrt{n_S}}{2} \leq |\hat{s}_{l_0} - t| + \frac{\sqrt{n_S}}{2} \leq \frac{5}{4}\sqrt{n_S}$. By (2) with $s' = t$ and Lemma B.4, $v(\theta) \geq e^{-c_1}/(16\sqrt{n_S}) = 1/K_0$. If $s(0) \geq t$, then $\theta = 0$ and $v(0) = p \geq \Pr_D[S = \lceil s(0) \rceil] \geq 1/K_0$ by Lemma B.4.

Normaliser. Let $\ell = \ln(10/\delta)$; the Bernstein step of Lemma 5.17 runs at level $\delta_1 = \delta/5$, so its $\ln(2/\delta_1)$ equals ℓ, and the other logarithms below, $\ln(5/\delta)$, are at most ℓ. With $D_r \in [0, R^2/2]$ and $\mathrm{Var}\, D_r \leq \frac{R^2}{2}\mathbb{E}D_r$, Bernstein's inequality gives $\Sigma \leq 2\mathbb{E}\Sigma + \frac{4}{3}R^2 \ln(5/\delta)$ with probability $\geq 1 - \delta/5$. Also $\mathbb{E}\Sigma \leq \frac{1}{2}LV_{\max} + \frac{1}{2}hV_{\max}R$, since neighbouring means differ by at most $hV_{\max}$ and these differences sum to at most R. As $\bar{V} \leq 2a^2 + 4\Sigma$ with $a^2 = R^2 \ln(5/\delta)$, we get $\bar{V} \leq 4LV_{\max} + 4hV_{\max}R + 10R^2\ell$, and, since $u \leq \frac{2}{3}hR\ell + h\sqrt{2\bar{V}\ell}$, for $L \geq \sqrt{\theta R}$,

$$u \;\leq\; 2\theta\sqrt{8V_{\max}\ell/L} \;+\; 6\theta R\ell/L.$$

Since $\theta^2 V_{\max} \leq \theta_0^2 n_S d^2/(4\gamma) \leq n_S/(\gamma(k-1)^2)$ and $\theta R \leq \theta_0 m = 2(1-\gamma)m/(d(k-1))$, the stated L makes $u \leq \varepsilon/10$ and $h(S_{\max} - s(0)) \leq \theta_0 m/L \leq \varepsilon/10$.

Importance sampling. With $K = 1/v(\theta) \leq K_0$, Theorem 4.3(iv) reaches ratio $1 + \varepsilon/3$ after $O(K(\log(1/\delta) + \log\log K)/\varepsilon^2)$ samples with probability $\geq 1 - \delta/5$, within the cap N_{IS} by hypothesis.

The output ratio is then at most $e^{2u + h(S_{\max} - s(0))}(1 + \varepsilon/3) \leq e^{3\varepsilon/10}(1 + \varepsilon/3) \leq 1 + \varepsilon$ for $\varepsilon \leq 1$ (the middle expression is convex in ε, equal to 1 at 0 and below 2 at 1). Adding the failure probabilities of the pilot, the coupling, the three tightness events and the soundness events gives at most 3δ. The pilot uses $(L_p + 1)\lceil 4d^2/\gamma \rceil \lceil 8\ln((L_p+1)/\delta) \rceil = O(d^3\sqrt{n_S}\log(n_S/\delta)/(\gamma^2(k-1)))$ samples. □

Proof of Theorem 5.18(iii). In the notation of the proof of Lemma B.4, for $u \in [0, \theta_0]$, $\mathrm{Var}_{D_u} S \geq \mathbb{E}\,\mathrm{Var}(S \mid y) \geq \mathbb{E}[N(y)]\,\eta_0(1 - \eta_0) \geq e^{-4}\nu^2\beta m/2 = v_{\min}$ (this part used only $\theta_0 d(k-1) \leq 2$). Hence $s(\theta_0) - s(0) = \int_0^{\theta_0} \mathrm{Var}_{D_u} S\, du \geq \theta_0 v_{\min}$, and integrating by parts, for every $\vartheta \in [0, \theta_0]$, $\ln Z(\vartheta) - \vartheta s(\vartheta) = -\int_0^{\vartheta} u\, \mathrm{Var}_{D_u} S\, du \leq -\vartheta^2 v_{\min}/2$. With $\vartheta = \theta_0$ and $t \geq s(\theta_0) - 1$, the Chernoff bound $p \leq Z(\theta_0)e^{-\theta_0 t}$ gives the second claim. For the third, let $t \in (s(0), s(\theta_0)]$ (for $t = s(0)$ there is nothing to prove) and, by continuity, $\vartheta \in (0, \theta_0]$ with $s(\vartheta) = t$. By Lemma 5.16(c) with the crude certificate, $t - s(0) = \int_0^{\vartheta} \mathrm{Var}_{D_u} S\, du \leq \vartheta V_{\max}$, so $\ln p \leq \ln Z(\vartheta) - \vartheta t \leq -\vartheta^2 v_{\min}/2 \leq -(t - s(0))^2 v_{\min}/(2V_{\max}^2)$. □

B.5 The certification gap

Proof of Proposition 5.22. (a) Let $\lambda = 2\varepsilon p/(B-p) \le 3\varepsilon p/B \le 1$ and $P_1 = (1-\lambda)P_0 + \lambda\delta_B$, whose mean is $p + \lambda(B-p) = (1+2\varepsilon)p$. An interval of ratio at most $1+\varepsilon$ that contains p lies below $(1+\varepsilon)p$, so it cannot contain the mean of P_1. Couple a draw from P_1 as a draw from P_0 replaced by B with probability λ, independently. The runs on P_0 and P_1 coincide unless some draw that is actually taken is replaced; whether draw i is taken depends only on earlier draws, so this has probability at most $\sum_i \lambda \Pr[N \ge i] = \lambda \mathbb{E}_{P_0} N$. The run on P_0 outputs a tight interval containing p with probability $\ge 1-2\delta$, the run on P_1 an interval containing its mean with probability $\ge 1-\delta$, and these events are disjoint when the outputs coincide. Hence $\lambda \mathbb{E}_{P_0} N \ge 1 - 3\delta$.

(b) $\sum_{i>n} x_i \le M < t \le 2M$, so $C = g$. With $q = 2^{-n}$, $Z(\theta) = (1 - q + qe^{2M\theta})\big(\frac{1+e^{\theta}}{2}\big)^M$, and $\frac{1+e^{\theta}}{2} \ge e^{\theta/2}$ gives $Z(\theta)e^{-\theta t} \ge (1-q)e^{-M\theta} + qe^{M\theta} \ge 2\sqrt{q(1-q)}$, so $K \ge 2\sqrt{(1-q)/q} \ge 2^{n/2}$. The weight $W = C\,Z(\theta)e^{-\theta S}$ under D_θ has $\mathbb{E}W^2 = Z(\theta)\mathbb{E}_D[Ce^{-\theta S}] = Z(\theta)\,q\,e^{-2M\theta}\big(\frac{1+e^{-\theta}}{2}\big)^M$. At $e^{2M\theta} = 1/q$ this is $q^2(2-q)\cosh(\theta/2)^{2M} \le 2q^2\exp(M\theta^2/4) = 2q^2\exp(n^2(\ln 2)^2/(16M)) \le 2.07\,q^2$. □

C Proofs for affine structure

Notation: for $T \subseteq [n]$, $\mathbb{F}_2^T$ is the set of vectors supported in T; A_T is the submatrix of the columns in T; for an affine $V = v_0 + V_0$ and a cut (Y, Z), π_Y is the coordinate projection. For $V = \{Ax = b\} \ne \emptyset$, $\dim \pi_Y(V) = \dim V - \dim(V_0 \cap \{x_Y = 0\}) = (n - \mathrm{rank}A) - (|Z| - \mathrm{rank}A_Z)$ and symmetrically, so $\lambda_V(Y) = \mathrm{rank}A_Y + \mathrm{rank}A_Z - \mathrm{rank}A$.

C.1 XOR-DNNF envelopes (Proposition 5.25)

For a leaf form ℓ write $\bar{\ell}$ for its linear part, and for a derivation $\mathcal{T}$ (one child at each SUM, all children at each PROD) let $\Lambda(\mathcal{T})$ be its occurrences of form leaves. The vectors $\bar{\ell}$, $\ell \in \Lambda(\mathcal{T})$, are linearly independent: by induction, a leaf contributes one nonzero vector, a SUM inherits from its chosen child, and at a PROD the vectors below child c lie in $\mathcal{L}(c)$, where they are independent, while the sum of the spaces $\mathcal{L}(c)$ is direct. For uniform x and independent $\bar{\ell}_1, \dots, \bar{\ell}_r$, the vector $(\ell_1(x), \dots, \ell_r(x))$ is uniform on $\mathbb{F}_2^r$, because $x \mapsto (\bar{\ell}_i \cdot x)_i$ is onto. Hence $\mathrm{val}_{\mathcal{T}}(x) = w(\mathcal{T}) \prod_{\ell \in \Lambda(\mathcal{T})} f_\ell(\ell(x))$, with $w(\mathcal{T})$ the product of the weights and constants of $\mathcal{T}$, has $\mathbb{E}\,\mathrm{val}_{\mathcal{T}} = w(\mathcal{T}) \prod_\ell \frac{f_\ell(0)+f_\ell(1)}{2}$. Since $\varphi = \sum_{\mathcal{T}} \mathrm{val}_{\mathcal{T}}$ and the derivations below a PROD are tuples of derivations below its children, the bottom-up pass computes $\sum_{\mathcal{T}} \mathbb{E}\,\mathrm{val}_{\mathcal{T}} = \mathbb{E}\varphi$. The top-down choice selects $\mathcal{T}$ with probability $\mathbb{E}\,\mathrm{val}_{\mathcal{T}}/\mathbb{E}\varphi$ (the choice probabilities telescope, as in Lemma 4.2). Given $\mathcal{T}$, the leaf values v_ℓ are independent with probabilities $f_\ell(v_\ell)/(f_\ell(0) + f_\ell(1))$, and x is uniform on $\{x : \ell(x) = v_\ell\ \forall \ell\}$, which has 2^{n-r} points by independence. So $\Pr[x \mid \mathcal{T}] = 2^{r-n} \prod_\ell \frac{f_\ell(\ell(x))}{f_\ell(0)+f_\ell(1)} = \frac{\mathrm{val}_{\mathcal{T}}(x)}{2^n \mathbb{E}\,\mathrm{val}_{\mathcal{T}}}$ and $\Pr[x] = \sum_{\mathcal{T}} \frac{\mathrm{val}_{\mathcal{T}}(x)}{2^n \mathbb{E}\varphi} = \frac{2^{-n}\varphi(x)}{\mathbb{E}\varphi}$. □

C.2 The parity barrier (Theorem 5.26)

Smoothing. Let Φ have S wires; then it has at most $S+1$ nodes (all reachable from the root). Add, for each $i \in [n]$, a leaf $\mathbf{1}_i$ computing the constant 1 with declared support $\{i\}$ (in a DNNF, the constant-true function of x_i). Replace each wire from a SUM (or $\vee$) node u to a child c with $\mathrm{var}(c) \subsetneq \mathrm{var}(u)$ by a chain of binary PROD nodes multiplying c by the leaves $\mathbf{1}_i$, $i \in \mathrm{var}(u) \setminus \mathrm{var}(c)$, and do the same above the root for $i \notin \mathrm{var}(\mathrm{root})$. Replace every PROD with $k \ge 3$ children by a

chain of $k-1$ binary ones. The function is unchanged, products remain decomposable ($i \notin \mathrm{var}(c)$), and the result is smooth with at most $(S+1)+n+nS+n+S \le (n+2)(S+2)$ nodes.

Decomposition. Let Φ be smooth with binary products and leaves of support at most $2n/3$, and $n \ge 2$. For a derivation $\mathcal{T}$ walk from the root: at a SUM go to the child chosen by $\mathcal{T}$, at a PROD go to the child with the larger support (ties broken by a fixed rule), and stop at the first node g with $|\mathrm{var}(g)| \le 2n/3$. The root has support $[n]$ and leaves have at most $2n/3$ variables, so g exists. Its parent u has $|\mathrm{var}(u)| > 2n/3$ and is not a SUM, by smoothness. So u is a PROD whose support is the disjoint union of its children's supports, and $|\mathrm{var}(g)| \ge |\mathrm{var}(u)|/2 > n/3$, that is, $g \in G(\Phi)$. The walk is determined by its path P (the nodes visited and the choices at SUM nodes). The derivations with walk P are exactly P together with an arbitrary derivation below g and an arbitrary derivation below each off-path child o of a PROD on P. Hence

$$\varphi(x) = \sum_{g \in G(\Phi)} \varphi_g(x)\,\psi_g(x), \qquad \psi_g = \sum_{P \text{ ending at } g} w(P) \prod_{o \in \mathrm{off}(P)} \varphi_o,$$

where φ_v is the function computed at v and $w(P) \ge 0$ collects the weights on P. An off-path child o is a sibling, under a decomposable PROD, of a node whose support contains $\mathrm{var}(g)$, so $\mathrm{var}(o) \cap \mathrm{var}(g) = \emptyset$. Thus every term is $t_g(x) = f_g(x_{Y_g}) h_g(x_{Z_g})$ with $Y_g = \mathrm{var}(g)$, $Z_g = [n] \setminus Y_g$ and $f_g, h_g \ge 0$. For a DNNF, the same walk gives $E(x) = \bigvee_g (\varphi_g(x) \wedge \psi_g(x))$ with ψ_g an $\vee$ of $\wedge$'s, so every term is the indicator of a rectangle $R_g = A_g \times B_g$ with $R_g \subseteq E^{-1}(1)$.

Lemma C.1 (coset lemma)**.** *Let V be a nonempty affine subspace, (Y, Z) a cut, $\lambda = \lambda_V(Y)$, and $f : \mathbb{F}_2^Y \to \mathbb{R}_{\ge 0}$, $h : \mathbb{F}_2^Z \to \mathbb{R}_{\ge 0}$. Then*

$$\sum_{x \in V} \sqrt{f(x_Y) h(x_Z)} \;\le\; \sqrt{|V|\, 2^{-\lambda} \textstyle\sum_y f(y) \sum_z h(z)}.$$

In particular, a rectangle $R = A \times B$ satisfies $|R \cap V| \le \sqrt{|V| 2^{-\lambda} |R|}$.

Proof. Let $W_Y = \{y : (y,0) \in V_0\}$ and $W_Z = \{z : (0,z) \in V_0\}$. For $y \in \pi_Y(V)$ the fibre $\{z : (y,z) \in V\}$ is a coset of W_Z, and y, y' have the same fibre if and only if $y - y' \in W_Y$. So V is the disjoint union of N blocks $A_i \times B_i$, where the A_i are the W_Y-cosets partitioning $\pi_Y(V)$ and the B_i are distinct W_Z-cosets. Since $W_Y \times \{0\} = V_0 \cap \ker \pi_Z$, we have $\dim W_Y = \dim V - \dim \pi_Z(V)$, so $N = |\pi_Y(V)|/|W_Y| = 2^\lambda$ and $|V| = N |W_Y| |W_Z|$. By Cauchy–Schwarz in each block and then across blocks,

$$\sum_{x \in V} \sqrt{fh} = \sum_i \Big(\sum_{A_i} \sqrt{f} \Big) \Big(\sum_{B_i} \sqrt{h} \Big) \le \sqrt{|W_Y||W_Z|} \sum_i \sqrt{F_i H_i} \le \sqrt{|W_Y||W_Z| \textstyle\sum_i F_i \sum_i H_i},$$

with $F_i = \sum_{A_i} f$ and $H_i = \sum_{B_i} h$. The A_i are disjoint, and so are the B_i, and $|W_Y||W_Z| = |V| 2^{-\lambda}$. For the rectangle take $f = \mathbf{1}_A$, $h = \mathbf{1}_B$. □

Proof of (a). Smooth Φ first (this does not change φ or K) and decompose. For $g \in G = G(\Phi)$ let $a_g = |V|^{-1} \sum_{x \in V} \min(1, t_g(x))$ and $b_g = |V|^{-1} \sum_{x \in \mathbb{F}_2^n} t_g(x)$. Since $\min(1,t) \le \sqrt{t}$, Lemma C.1 gives $a_g \le \sqrt{b_g 2^{-\lambda_V(Y_g)}} \le \sqrt{b_g 2^{-\lambda_{\mathrm{bal}}}}$, as $n/3 < |Y_g| \le 2n/3$. For $x \in C^{-1}(1)$, $1 \le \min(1, \varphi(x)) \le \sum_g \min(1, t_g(x))$, so $\sum_g a_g \ge c$. Also $\sum_g b_g = |V|^{-1} \sum_x \varphi(x) = 2^n \mathbb{E}\varphi / |V| = Kc$, because $p = c|V|2^{-n}$. By Cauchy–Schwarz, $c \le 2^{-\lambda_{\mathrm{bal}}/2} \sum_g \sqrt{b_g} \le 2^{-\lambda_{\mathrm{bal}}/2} \sqrt{|G| Kc}$, that is, $|G| K \ge c 2^{\lambda_{\mathrm{bal}}}$.

Proof of (b). Now $C^{-1}(1) \subseteq \bigcup_g (R_g \cap V)$ with $R_g \subseteq E^{-1}(1)$, so by Lemma C.1 $c|V| \le \sum_g \sqrt{|V| 2^{-\lambda_{\mathrm{bal}}} |R_g|} \le |G| \sqrt{|V| 2^{-\lambda_{\mathrm{bal}}} |E^{-1}(1)|}$. Substituting $|E^{-1}(1)| = (D(E)/p)\, c|V|$ and squaring gives $c \le |G|^2 2^{-\lambda_{\mathrm{bal}}} D(E)/p$. If $E \Rightarrow [x \in V]$ then $R_g \subseteq V$, so $|R_g| = |R_g \cap V| \le \sqrt{|V| 2^{-\lambda_{\mathrm{bal}}} |R_g|}$, i.e. $|R_g| \le |V| 2^{-\lambda_{\mathrm{bal}}}$, and $c|V| \le |G| |V| 2^{-\lambda_{\mathrm{bal}}}$. □

C.3 The corollaries

Corollary 5.27. As H has full row rank, $y \mapsto y^\top H$ is injective, and $y^\top H_Y = 0$ if and only if $y^\top H \in \mathbb{F}_2^Z$. So $\operatorname{rank} H_Y = m - \dim(R \cap \mathbb{F}_2^Z)$ and likewise for Z, and $\lambda_V(Y) = m - \dim(R \cap \mathbb{F}_2^Y) - \dim(R \cap \mathbb{F}_2^Z)$. An r-dimensional subspace of $R \cap \mathbb{F}_2^T$ has support at most $|T|$, so $\dim(R \cap \mathbb{F}_2^T) \le j^*$ whenever $|T| \le 2n/3$; for a balanced Y both $|Y|$ and $|Z|$ are at most $2n/3$.

Corollary 5.28. Fix T with $|T| \le 2n/3$ and a 7-dimensional subspace $\Lambda \subseteq \mathbb{F}_2^m$ with basis $y_1, \dots, y_7$. The columns of H_{T^c} are independent and uniform, and for each column h the vector $(y_i \cdot h)_{i \le 7}$ is uniform on $\mathbb{F}_2^7$. So the probability that $y^\top H_{T^c} = 0$ for all $y \in \Lambda$ is $2^{-7|T^c|} \le 2^{-7n/3}$. There are at most 2^{7m} such subspaces and 2^n sets T, so the probability that some T has $\dim\{y : y^\top H_{T^c} = 0\} \ge 7$ is at most $2^{n+7m-7n/3} \le 2^{-n/6}$ for $m \le n/6$. Also $\Pr[\operatorname{rank} H < m] \le \sum_{y \ne 0} \Pr[y^\top H = 0] < 2^{m-n} \le 2^{-n/6}$. On the complement, H has full rank and $\dim(R \cap \mathbb{F}_2^T) = \dim\{y : y^\top H_{T^c} = 0\} \le 6$ for all $|T| \le 2n/3$, so $\lambda_{\mathrm{bal}} \ge m - 12$ by Corollary 5.27. Finally $|V| = 2^{n-m}$ gives $p = c2^{-m}$ and $c2^{m-12} = c^2/(4096p)$.

Corollary 5.29. Let $\widetilde{\Phi}$ be the smoothed circuit, so that $|G(\widetilde{\Phi})| \le (n+2)(S+2) \le 3(n+2)S$ for an envelope with $S \ge 1$ wires.

First, sample counts. If a KL interval $[\ell, u]$ for the mean μ of a $[0,1]$-valued variable, computed from N samples with empirical mean $\hat{\mu}$ at level δ, is valid and tight ($\ell \le \mu \le u \le (1+\varepsilon)\ell$, $u \le 1/2$), then $\hat{\mu} \in [\ell, u]$, $u - \hat{\mu} \le \varepsilon\hat{\mu}$, and $N\mathrm{KL}(\hat{\mu} \| u) = \ln(2/\delta)$. Since $\mathrm{KL}(a\|b) \le (a-b)^2/(b(1-b)) \le 2\varepsilon^2 a$ for $a \le b \le \min((1+\varepsilon)a, 1/2)$, we get $N \ge \ln(2/\delta)/(2\varepsilon^2(1+\varepsilon)\mu)$. With $\mu = 1/K$ this is $\Omega(K \log(1/\delta)/\varepsilon^2)$ samples.

(i) The estimator evaluates φ on every sample, at cost at least S, so its work is at least $SK \ge |G(\widetilde{\Phi})|K/(3(n+2)) \ge c^2/(12288(n+2)p)$. Any implementation spends at least S on the mass and draws $\Omega(K)$ samples, hence $\Omega(\sqrt{SK}) = \Omega(c/\sqrt{np})$ work. (ii) The sampler of Theorem 6.4 makes $2n$ counter calls per sample, each reading E, and needs $\Omega(D(E)/p)$ samples, because the values $W' \in [0,1]$ of Theorem 6.4(c) have mean $\Theta(1/K_E)$. So its work is $\Omega(|E|K_E)$. As $K_E \ge 1$, $|E|K_E \ge \sqrt{|E|^2 K_E} \ge \sqrt{|G(\widetilde{E})|^2 K_E/(3(n+2))} = \Omega(c/(n\sqrt{p}))$. (iii) For an NNF N without frontier gates, the DNNF E of Theorem 5.6 has literal and constant leaves, satisfies $E \equiv C \Rightarrow [x \in V]$, and has $S \le (|N|+1)(\mathrm{cw}+3)2^{\mathrm{cw}}$ wires. So (b) gives $3(n+2)S \ge |G(\widetilde{E})| \ge c2^{m-12}$; take logarithms and use $\log_2 3 < 2$. (iv) A tree built with (T), (T′), (F), (D), (B) is turned into a DNNF $E \equiv C$: a (T) node is a literal or constant leaf, and a refuted node is the constant-0 leaf with empty support; a (T′) atom and every other leaf h (an (S)-leaf or a node left unexpanded) is a leaf with the function of the restricted node h on its support; (F) is an $\wedge$ of the forced literals and the child, (D) an $\wedge$ or $\vee$ of the children, and (B) on x is $(\neg x \wedge h|_{x=0}) \vee (x \wedge h|_{x=1})$. The rules are identities and the $\wedge$'s are decomposable, because restriction removes assigned variables. A tree with m_T nodes gives at most $(n+7)m_T$ wires, so after smoothing at most $(n+2)((n+7)m_T + 2) \le (n+6)^2 m_T$ nodes. Leaves may now have more than $2n/3$ variables, and a walk may stop at such a leaf B with $Y = \mathrm{var}(B)$. Its term $B(x_Y) \wedge \psi_B(x_Z)$ is a rectangle inside $E^{-1}(1) = C^{-1}(1) \subseteq V$. If it is nonempty, $B^{-1}(1)$ lies in a single coset of $W_Y = \ker H_Y$, so $D(B) \le |W_Y| 2^{-|Y|} = 2^{-\operatorname{rank} H_Y}$. On the event of Corollary 5.28, $|Z| < n/3$ gives $\operatorname{rank} H_Y = m - \dim(R \cap \mathbb{F}_2^Z) \ge m - 6$, so $D(B) \le 2^{6-m} < 1/K'$, contradicting the hypothesis on leaves with more than $2n/3$ variables. So every nonempty term ends in $G(\widetilde{E})$, the proof of (b) applies unchanged, and $(n+6)^2 m_T \ge |G(\widetilde{E})| \ge c2^{m-12}$. □

C.4 Bounded width (Proposition 5.30)

Let a_i be the columns of A and $\sigma_t(x) = \sum_{i \in X_t} a_i x_i$. The live set $S_t = \mathrm{col}(A_{X_t}) \cap (b + \mathrm{col}(A_{\bar{X}_t}))$ is the set of values of σ_t on V: if $x \in V$ then $b - \sigma_t(x) = A_{\bar{X}_t} x_{\bar{X}_t}$, and every $s \in S_t$ is attained by

combining x_{X_t} with $\sigma_t = s$ and $x_{\bar{X}_t}$ with $A_{\bar{X}_t} x_{\bar{X}_t} = b - s$. S_t is empty or a coset of $\mathrm{col}(A_{X_t}) \cap \mathrm{col}(A_{\bar{X}_t})$, whose dimension is $\mathrm{rank}A_{X_t} + \mathrm{rank}A_{\bar{X}_t} - \mathrm{rank}A = \lambda_V(X_t)$. Define gates $N_t[s]$, $s \in S_t$: for a leaf $t = \{i\}$, $N_t[0] = \neg x_i$ and $N_t[a_i] = x_i$ (the constant-true function of x_i if $a_i = 0$); for children t_1, t_2, $N_t[s] = \bigvee_{s_1 \in S_{t_1},\, s+s_1 \in S_{t_2}} (N_{t_1}[s_1] \wedge N_{t_2}[s + s_1])$. By induction $N_t[s](x) = 1$ implies $\sigma_t(x) = s$, and $N_t[\sigma_t(x)](x) = 1$ for $x \in V$; so $N_{\mathrm{root}}[b] = \mathbf{1}_V$. The $\wedge$'s are decomposable, the disjuncts of $N_t[s]$ have distinct s_1 and are therefore disjoint, and every $N_t[s]$ has support X_t. Each internal node has at most $|S_{t_1}||S_{t_2}| \le 4^w$ $\wedge$-gates and 2^w $\vee$-gates, so there are at most $2n(4^w + 1)$ gates. Masses and exact samples under any product distribution then follow from Lemma 4.2 (deterministic, so $K = 1$). Finally, walking from the root to the larger child while $|X_t| > 2n/3$ reaches a node with $n/3 < |X_t| \le 2n/3$, so $\lambda_{\mathrm{bal}}(V) \le w$. □

C.5 Biased parity events (Theorem 5.31)

The copy constraints force $x_{i,c} = x_{i,0} =: y_i$ for all c, an event of probability q^k if $y_i = 1$ and $(1-q)^k$ if $y_i = 0$. Summing over y with $Hy = s$ gives the formula. Let w^* be the minimum weight in the coset and $\theta = (1-q)^N \lambda^t$. The coset has at most 2^n elements and $\lambda < 1$, so $\lambda^{w^*} \le p/(1-q)^N \le 2^n \lambda^{w^*}$. If $w^* \le t$ then $p \ge \theta$; if $w^* \ge t+1$ then $p \le \theta \cdot 2^n \lambda \le \theta 2^{-2g-3}$ by the choice of k. An interval $[L, U] \ni p$ with $U \le 2^g L$ therefore has $U \ge \theta$ in the first case and $U \le 2^g p < \theta$ in the second. θ is a rational of polynomial bit length, so the comparison is efficient. This proves (a).

(b) Given $\gamma > 0$, take $g = n^a$ with $a = \lceil 2/\gamma \rceil$ and the smallest allowed k. Then $N = nk = \Theta(n^{a+1})$ and $N^{1-\gamma} \le g$ for large n, because $(a+1)(1-\gamma) < a$. Running the procedure with $\delta = 1/8$ and answering "weight $\le t$" if and only if $U \ge \theta$ is correct with probability at least $3/4$, which puts an NP-complete problem in BPP. For arguments take $g = 1$. Merlin sends π, and Arthur runs the verifier and accepts if and only if $U < \theta/2$. If $w^* \ge t+1$, the honest π gives, with probability at least $1 - 2\delta$, a valid interval with $U \le 2L \le 2p \le \theta/16$, so Arthur accepts. If $w^* \le t$, accepting means $U < \theta/2 < p$, which soundness bounds by δ. With $\delta \le 1/6$ this is an MA protocol for the complement of the coset-weight problem, which is coNP-complete. Then $\mathsf{coNP} \subseteq \mathsf{MA} \subseteq \mathsf{AM}$ and $\mathsf{PH} = \Sigma_2^p$ [BHZ87]. For $q > \frac{1}{2}$ substitute $x \mapsto 1 - x$; this maps $\mathrm{Bern}(q)$ to $\mathrm{Bern}(1-q)$, keeps the copy constraints, and turns $Hx_{\cdot,0} = s$ into $Hx'_{\cdot,0} = s + H\mathbf{1}$. □

C.6 Linear thresholds

Proposition 5.32. Complementing the coordinates with negative weight puts L in packing form $S = \{y : \sum_i v_i y_i \le \kappa\}$ with $v_i \ge 0$, under the product distribution ν with marginals ν_i. Coordinates with $\nu_i \in \{0, 1\}$ are fixed first. Let $F_i(c) = \nu(\sum_{j \le i} v_j y_j \le c)$, so $F_0 = \mathbf{1}[c \ge 0]$ and $F_i(c) = (1-\nu_i) F_{i-1}(c) + \nu_i F_{i-1}(c - v_i)$. With $Q = 1 + \varepsilon/(2n)$ set $G_0 = F_0$, $H_i(c) = (1-\nu_i) G_{i-1}(c) + \nu_i G_{i-1}(c - v_i)$ and $G_i(c) = Q^{\lfloor \log_Q H_i(c) \rfloor}$ (0 if $H_i(c) = 0$). Since the recurrence is a positive combination, induction gives $H_i \le F_i \le Q^{i-1} H_i$ and $G_i \le H_i < Q G_i$, hence $G_i \le F_i \le Q^i G_i$ and $G_n(\kappa) \le D(L) \le e^{\varepsilon/2} G_n(\kappa)$. Each G_i is a nondecreasing step function. Its positive values are at least π_0/e with $\pi_0 = \prod_j \min(\nu_j, 1-\nu_j) \ge 2^{-n\beta}$, the probability of one assignment, so it takes $O(n(n\beta+1)/\varepsilon)$ distinct values and has as many breakpoints. H_i is a merge of two shifted copies of G_{i-1}, which gives $O(n^3\beta/\varepsilon)$ operations in all. The values Q^j with $j = O(n(n\beta+1)/\varepsilon)$, the $H_i(c)$ and the exponents $\lfloor \log_Q H_i(c) \rfloor$ (found by comparing $H_i(c)$ with powers of Q) are rationals and integers of $\mathrm{poly}(n, \beta, 1/\varepsilon)$ bits, so all of this is exact arithmetic of that size; so is the acceptance probability below. To sample: start from $c_n = \kappa$ (assuming $G_n(\kappa) > 0$; otherwise $D(L) = 0$), and for $i = n, \ldots, 1$ set $y_i = 1$ with probability $\nu_i G_{i-1}(c_i - v_i)/H_i(c_i)$ and $c_{i-1} = c_i - v_i y_i$. Every visited G and H is positive. The output $y \in S$

has probability $\nu(y)\prod_{i<n} r_i/H_n(\kappa)$ with $r_i = G_i(c_i)/H_i(c_i) \in (Q^{-1}, 1]$. Accepting with probability $Q^{-(n-1)}/\prod_{i<n} r_i \le 1$ leaves probability $\nu(y)Q^{-(n-1)}/H_n(\kappa)$ for every $y \in S$, i.e. exactly $\nu \mid S$, with acceptance probability $\nu(S)Q^{-(n-1)}/H_n(\kappa) \ge e^{-\varepsilon/2}$. Each step is a binary search. □

Unions with approximate masses. Let the sets $P_1, \dots, P_T$ have exact samplers and lower approximations G_j of their masses, $e^{-\varepsilon'} D(P_j) \le G_j \le D(P_j)$, and let $C \Rightarrow \bigvee_j P_j$. Pick j with probability $G_j/\hat{G}$, $\hat{G} = \sum_j G_j$, draw $x \sim D \mid P_j$, and let $W = C(x)/\mathrm{cov}(x) \in [0,1]$ with $\mathrm{cov}(x) = |\{j : x \in P_j\}|$. Then

$$\hat{G}\,\mathbb{E}W = \sum_x D(x)C(x) \cdot \frac{1}{\mathrm{cov}(x)} \sum_{j \ni x} \frac{G_j}{D(P_j)} \in [e^{-\varepsilon'} p,\, p],$$

so $p \in [\hat{G}\ell, e^{\varepsilon'}\hat{G}u]$ for a valid interval $[\ell, u]$ for $\mathbb{E}W$. If $C \equiv \bigvee_j P_j$, then $\sum_j D(P_j) \le Tp$ and the looseness is at most $e^{\varepsilon'}T$. For a variable-disjoint $\wedge$ of r pieces, multiply masses approximated within $e^{-\varepsilon'/r}$ and sample the factors independently. Affine subspaces on uniform coordinates have exact masses and samplers. □

Proposition 5.33. Complement as before (this changes b by the columns of the complemented coordinates). For $t \le n$ and a live syndrome $s \in S_t$ (the set of Proposition 5.30 for the prefix $\{1, \dots, t\}$, of size at most 2^{w_0}) let $F_t(s, c) = D(A_{\le t} x_{\le t} = s,\ \sum_{j \le t} v_j y_j \le c)$. Then $F_t(s,c) = (1-\nu_t)F_{t-1}(s,c) + \nu_t F_{t-1}(s + a_t, c - v_t)$, and both states on the right are live at $t-1$ whenever their value is nonzero, since $s + \{0, a_t\} \subseteq b + \mathrm{col}(A_{\ge t})$. So the recurrence closes on live states, and $D(C) = F_n(b, \kappa)$. The rounding, the bound on breakpoints and the rejection-corrected sampler are as for a single threshold, run on 2^{w_0} step functions per position. □

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